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The equation $6x^4 + x + y^3 + y + 1 = 0$ has no integer solutions

Size $h = 109$, length $l = 11.58$

Theorem 1. *The equation*

$$6x^4 + x + y^3 + y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses 119 auxiliary polynomials $H_0, \dots, H_{118}$, 10 of which are constants, and a graph on these vertices with 365 edges; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = 6x^4 + x + y^3 + y + 1$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y, \\ H_1 &= -y - 1, \\ H_2 &= 2x - y, \\ H_3 &= -x - y - 1, \\ H_4 &= 2x - y - 1, \\ H_5 &= x - y - 2, \\ H_6 &= -x - y + 2, \\ H_7 &= -x - y - 2, \\ H_8 &= -2x - y + 1, \\ H_9 &= -2x - y - 1, \\ H_{10} &= 2x - y - 2, \\ H_{11} &= -2x - y - 2, \\ H_{12} &= -2y - 1, \\ H_{13} &= x - 2y + 1, \\ H_{14} &= x - 2y - 1, \\ H_{15} &= -x - 2y - 1, \\ H_{16} &= 2x - 2y - 1, \\ H_{17} &= x - 2y - 2, \\ H_{18} &= -2x - 2y - 1, \\ H_{19} &= 2x^2 - y, \\ H_{20} &= x^2 - y + 1, \\ H_{21} &= x^2 - y - 1, \\ H_{22} &= x^2 - x - y, \\ H_{23} &= x^2 - x + y, \\ H_{24} &= x^2 + y - 1, \\ H_{25} &= x^2 + x + y, \\ H_{26} &= 2x^2 + y, \\ H_{27} &= 2x^2 - y + 1, \\ H_{28} &= 2x^2 - y - 1, \\ H_{29} &= 2x^2 - x - y, \\ H_{30} &= x^2 - x - y + 1, \\ H_{31} &= x^2 - x - y - 1, \\ H_{32} &= x^2 - 2x - y, \\ H_{33} &= x^2 - 2x + y, \\ H_{34} &= x^2 - x + y - 1, \\ H_{35} &= x^2 - x + y + 1, \\ H_{36} &= 2x^2 - x + y, \end{aligned}$$

$$\begin{aligned}
H_{37} &= 2x^2 + y - 1, \\
H_{38} &= 2x^2 + x + y, \\
H_{39} &= 2x^2 + x - y - 1, \\
H_{40} &= 2x^2 - y + 2, \\
H_{41} &= 2x^2 - y - 2, \\
H_{42} &= 2x^2 - x - y - 1, \\
H_{43} &= 2x^2 - 2x - y, \\
H_{44} &= x^2 + 2x - y + 1, \\
H_{45} &= x^2 + x - y - 2, \\
H_{46} &= x^2 - x - y - 2, \\
H_{47} &= x^2 - 2x - y + 1, \\
H_{48} &= x^2 - 2x - y - 1, \\
H_{49} &= x^2 - 2x + y - 1, \\
H_{50} &= x^2 + x + y + 2, \\
H_{51} &= x^2 + 2x + y - 1, \\
H_{52} &= x^2 + 2x + y + 1, \\
H_{53} &= 2x^2 - 2x + y, \\
H_{54} &= 2x^2 - x + y - 1, \\
H_{55} &= 2x^2 - x + y + 1, \\
H_{56} &= 2x^2 + x + y - 1, \\
H_{57} &= 2x^2 + x + y + 1, \\
H_{58} &= 2x^2 + 2x + y, \\
H_{59} &= 2x^2 + 2x - y + 1, \\
H_{60} &= 2x^2 + x - y - 2, \\
H_{61} &= 2x^2 - x - y - 2, \\
H_{62} &= 2x^2 - 2x - y + 1, \\
H_{63} &= 2x^2 - 2x - y - 1, \\
H_{64} &= x^2 + 2x - y - 2, \\
H_{65} &= x^2 - 2x - y + 2, \\
H_{66} &= x^2 - 2x - y - 2, \\
H_{67} &= x^2 - 2x + y - 2, \\
H_{68} &= x^2 - 2x + y + 2, \\
H_{69} &= x^2 + 2x + y - 2, \\
H_{70} &= x^2 + 2x + y + 2, \\
H_{71} &= 2x^2 - 2x + y + 1, \\
H_{72} &= 2x^2 - x + y - 2, \\
H_{73} &= 2x^2 + x + y - 2, \\
H_{74} &= 2x^2 + x + y + 2, \\
H_{75} &= 2x^2 + 2x + y + 1,
\end{aligned}$$

$$\begin{aligned}
H_{76} &= 2x^2 + 2x - y - 2, \\
H_{77} &= 2x^2 - 2x - y + 2, \\
H_{78} &= 2x^2 - 2x - y - 2, \\
H_{79} &= 2x^2 - 2x + y - 2, \\
H_{80} &= 2x^2 - 2x + y + 2, \\
H_{81} &= 2x^2 + 2x + y - 2, \\
H_{82} &= 2x^2 + 2x + y + 2, \\
H_{83} &= x^2 + x - 2y, \\
H_{84} &= x^2 + 2y - 1, \\
H_{85} &= x^2 + 2y + 1, \\
H_{86} &= x^2 + x + 2y, \\
H_{87} &= x^2 - x + 2y - 1, \\
H_{88} &= x^2 - x + 2y + 1, \\
H_{89} &= x^2 + 2y - 2, \\
H_{90} &= x^2 + 2y + 2, \\
H_{91} &= x^2 + x + 2y + 1, \\
H_{92} &= x^2 + 2x - 2y + 1, \\
H_{93} &= x^2 + 2x - 2y - 1, \\
H_{94} &= x^2 + x - 2y + 2, \\
H_{95} &= x^2 + x - 2y - 2, \\
H_{96} &= x^2 - x - 2y + 2, \\
H_{97} &= x^2 - x - 2y - 2, \\
H_{98} &= x^2 - 2x - 2y + 1, \\
H_{99} &= x^2 - 2x + 2y - 1, \\
H_{100} &= x^2 - 2x + 2y + 1, \\
H_{101} &= x^2 - x + 2y - 2, \\
H_{102} &= x^2 - x + 2y + 2, \\
H_{103} &= x^2 + x + 2y - 2, \\
H_{104} &= x^2 + x + 2y + 2, \\
H_{105} &= x^2 + 2x + 2y - 1, \\
H_{106} &= x^2 + 2x + 2y + 1, \\
H_{107} &= x^2 - 2x - 2y - 2, \\
H_{108} &= x^2 + 2x + 2y - 2, \\
H_{109} &= 1, \\
H_{110} &= -1, \\
H_{111} &= 6, \\
H_{112} &= -2, \\
H_{113} &= -3, \\
H_{114} &= 3, \\
H_{115} &= 14,
\end{aligned}$$

$$\begin{aligned}
H_{116} &= 2, \\
H_{117} &= 5, \\
H_{118} &= 10.
\end{aligned}$$

The edges are $\{0, 83\}, \{0, 86\}, \{1, 12\}, \{1, 93\}, \{2, 13\}, \{2, 63\}, \{2, 79\}, \{2, 102\}, \{3, 107\}, \{4, 11\}, \{4, 19\}, \{4, 27\}, \{4, 35\}, \{4, 58\}, \{4, 64\}, \{4, 100\}, \{5, 29\}, \{5, 30\}, \{5, 44\}, \{5, 52\}, \{5, 60\}, \{5, 62\}, \{5, 68\}, \{5, 82\}, \{5, 90\}, \{6, 17\}, \{6, 33\}, \{6, 52\}, \{6, 53\}, \{6, 58\}, \{6, 69\}, \{6, 77\}, \{7, 15\}, \{7, 20\}, \{7, 36\}, \{7, 44\}, \{7, 54\}, \{7, 59\}, \{7, 71\}, \{7, 75\}, \{7, 78\}, \{7, 93\}, \{7, 106\}, \{8, 16\}, \{8, 26\}, \{8, 38\}, \{8, 50\}, \{8, 57\}, \{8, 103\}, \{9, 19\}, \{9, 104\}, \{9, 105\}, \{9, 106\}, \{9, 107\}, \{10, 34\}, \{10, 60\}, \{10, 80\}, \{10, 100\}, \{11, 16\}, \{11, 18\}, \{11, 26\}, \{11, 27\}, \{11, 31\}, \{11, 35\}, \{11, 40\}, \{11, 60\}, \{11, 61\}, \{11, 74\}, \{11, 76\}, \{11, 85\}, \{11, 97\}, \{11, 104\}, \{11, 106\}, \{12, 18\}, \{12, 28\}, \{12, 74\}, \{13, 49\}, \{14, 18\}, \{14, 39\}, \{14, 42\}, \{15, 16\}, \{15, 17\}, \{15, 19\}, \{15, 42\}, \{15, 64\}, \{15, 74\}, \{15, 75\}, \{16, 63\}, \{16, 68\}, \{17, 64\}, \{17, 101\}, \{18, 35\}, \{18, 42\}, \{18, 57\}, \{18, 58\}, \{18, 60\}, \{18, 91\}, \{19, 40\}, \{19, 79\}, \{19, 81\}, \{20, 68\}, \{20, 87\}, \{20, 96\}, \{20, 98\}, \{21, 27\}, \{21, 48\}, \{21, 64\}, \{21, 95\}, \{21, 97\}, \{21, 107\}, \{22, 40\}, \{22, 43\}, \{22, 48\}, \{22, 49\}, \{22, 85\}, \{23, 31\}, \{23, 55\}, \{23, 100\}, \{23, 104\}, \{24, 36\}, \{24, 43\}, \{24, 74\}, \{24, 80\}, \{24, 100\}, \{24, 103\}, \{24, 106\}, \{25, 30\}, \{25, 37\}, \{25, 64\}, \{25, 85\}, \{25, 90\}, \{25, 104\}, \{25, 108\}, \{26, 39\}, \{26, 57\}, \{26, 75\}, \{26, 76\}, \{26, 81\}, \{27, 47\}, \{27, 63\}, \{27, 71\}, \{27, 77\}, \{27, 92\}, \{28, 76\}, \{28, 78\}, \{29, 31\}, \{29, 39\}, \{29, 50\}, \{29, 55\}, \{29, 65\}, \{29, 78\}, \{29, 83\}, \{30, 33\}, \{30, 69\}, \{30, 96\}, \{30, 106\}, \{31, 59\}, \{31, 85\}, \{31, 107\}, \{32, 46\}, \{32, 57\}, \{32, 64\}, \{32, 91\}, \{33, 44\}, \{33, 57\}, \{33, 69\}, \{33, 72\}, \{33, 92\}, \{34, 37\}, \{34, 62\}, \{34, 64\}, \{34, 70\}, \{34, 91\}, \{34, 96\}, \{34, 99\}, \{35, 57\}, \{35, 64\}, \{35, 69\}, \{35, 88\}, \{35, 100\}, \{35, 104\}, \{35, 105\}, \{35, 106\}, \{36, 56\}, \{36, 61\}, \{36, 85\}, \{36, 87\}, \{37, 52\}, \{37, 58\}, \{37, 73\}, \{37, 79\}, \{37, 81\}, \{37, 82\}, \{38, 60\}, \{38, 75\}, \{38, 87\}, \{39, 56\}, \{39, 77\}, \{39, 92\}, \{40, 65\}, \{40, 77\}, \{40, 78\}, \{41, 63\}, \{41, 78\}, \{41, 80\}, \{42, 43\}, \{42, 47\}, \{42, 54\}, \{42, 55\}, \{42, 60\}, \{42, 75\}, \{42, 95\}, \{42, 98\}, \{43, 55\}, \{43, 77\}, \{43, 80\}, \{44, 47\}, \{44, 87\}, \{45, 47\}, \{45, 66\}, \{45, 82\}, \{45, 100\}, \{46, 56\}, \{46, 66\}, \{46, 74\}, \{46, 80\}, \{46, 101\}, \{47, 100\}, \{48, 97\}, \{48, 98\}, \{50, 58\}, \{50, 64\}, \{50, 70\}, \{50, 76\}, \{50, 106\}, \{51, 78\}, \{51, 89\}, \{51, 96\}, \{51, 104\}, \{52, 53\}, \{52, 58\}, \{52, 85\}, \{52, 91\}, \{53, 58\}, \{53, 75\}, \{54, 80\}, \{55, 81\}, \{55, 104\}, \{56, 81\}, \{56, 108\}, \{58, 62\}, \{58, 63\}, \{58, 73\}, \{58, 80\}, \{58, 86\}, \{58, 100\}, \{59, 63\}, \{59, 71\}, \{59, 75\}, \{59, 103\}, \{60, 70\}, \{60, 78\}, \{61, 63\}, \{61, 78\}, \{62, 71\}, \{63, 71\}, \{64, 70\}, \{64, 93\}, \{64, 101\}, \{65, 72\}, \{65, 81\}, \{66, 82\}, \{67, 69\}, \{67, 70\}, \{67, 72\}, \{68, 69\}, \{68, 70\}, \{69, 77\}, \{70, 104\}, \{71, 75\}, \{72, 74\}, \{72, 75\}, \{73, 102\}, \{74, 108\}, \{75, 78\}, \{75, 85\}, \{75, 104\}, \{76, 77\}, \{76, 94\}, \{78, 81\}, \{79, 82\}, \{80, 88\}, \{83, 94\}, \{83, 101\}, \{84, 92\}, \{84, 98\}, \{84, 103\}, \{84, 106\}, \{85, 103\}, \{85, 106\}, \{86, 104\}, \{89, 104\}, \{89, 108\}, \{92, 94\}, \{92, 98\}, \{93, 99\}, \{94, 97\}, \{94, 101\}, \{97, 98\}, \{100, 102\}, \{102, 105\}, \{103, 105\}, \{103, 108\}, \{104, 106\}, \{2, 110\}, \{4, 111\}, \{6, 112\}, \{7, 113\}, \{9, 110\}, \{11, 110\}, \{12, 114\}, \{13, 115\}, \{15, 110\}, \{17, 114\}, \{18, 110\}, \{21, 116\}, \{26, 116\}, \{27, 116\}, \{29, 114\}, \{30, 114\}, \{32, 112\}, \{33, 116\}, \{34, 114\}, \{35, 114\}, \{36, 116\}, \{37, 116\}, \{39, 116\}, \{48, 116\}, \{50, 114\}, \{56, 116\}, \{58, 114\}, \{59, 117\}, \{61, 112\}, \{62, 113\}, \{64, 114\}, \{69, 116\}, \{70, 112\}, \{72, 116\}, \{73, 114\}, \{74, 110\}, \{75, 116\}, \{78, 110\}, \{80, 110\}, \{83, 113\}, \{84, 116\}, \{86, 111\}, \{88, 114\}, \{92, 114\}, \{94, 114\}, \{95, 114\}, \{98, 114\}, \{102, 110\}, \{103, 118\}, \{105, 114\}, \{106, 114\}. For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that$

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -972x^8 - 324x^5 - 324x^4 - 27x^2 - 54x - 31.$$

For $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(10 + t)$ and the $\varphi_i(10 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-11 - t)$ and the $\varphi_i(-11 - t)$. Hence, for $x \geq 10$ and for $x \leq -11$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-10 \leq x \leq 9$. Checking the residues of y , we find that for $x \in \{-10, -8, -6, -4, -2, 0, 2, 4, 6, 8\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , for $x \in \{-5, -3, -1, 5, 7, 9\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , for $x = -7$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , for $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , and for $x = -9$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y . Hence (1) has no solution with $-10 \leq x \leq 9$.

For a place v , put $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{109, 111, 114, 115, 116, 117, 118\}$ and $H_i < 0$ for $i \in \{110, 112, 113\}$. The other signs are given in Table 3 for $x \geq 10$ and for $x \leq -11$. In both cases, counting the edges whose two values are negative gives $B_\infty = 1$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13, 17\}$ be the set of primes dividing 2, the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p = (-1)^N$, where $N = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges, then $N = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so B_p is determined by a, b and k .

Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of B_p ; every integer solution lies in one of them. The result is $B_2 = -1, B_3 = -1, B_5 = -1, B_7 = 1, B_{11} = 1, B_{13} = 1, B_{17} = 1$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v = 1$, the product over all primes and $v = \infty$. By Step 3, the factors with $p \notin S$ are 1, so $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 4, $\prod_{p \in S} B_p = -1$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$i = 0, m = 1$, neighbours 83, 86	
W	$x^2 + x$
A	0
B	$4y$
$i = 1, m = 1$, neighbours 12, 93	
W	$x + 1$
A	0
B	$2x^2 + 4x - 4y$
$i = 2, m = 3$, neighbours 110, 13, 63, 79, 102	
W	$6x^3 + 18x^2 + 27x + 3$
A	$-6x^3 + 12x^2y + 6x^2 - 48xy - 39x + 18y^2 - 39y - 45$
B	$18x^3y^2 + 279x^3y - 213x^3 + 12x^2y^3 - 96x^2y - 432x^2 - 12xy^3 - 81xy^2$ $-309xy - 411x + 18y^4 - 3y^3 - 45y^2 - 129y - 102$
$i = 3, m = 1$, neighbours 107	
W	x
A	0
B	2
$i = 4, m = 81$, neighbours 111, 11, 19, 27, 35, 58, 64, 100	
W	$192x^3 - 504x^2 + 801x - 159$
A	$-104976x^9 - 52488x^8y - 104976x^8 - 157464x^7y + 472392x^7 - 78732x^6y^2$ $-603612x^6y - 244944x^6 + 17496x^5y^3 - 104976x^5y^2 - 393660x^5y$ $-227448x^5 + 8748x^4y^4 + 135594x^4y^3 + 244944x^4y^2 + 524880x^4y$ $+568620x^4 + 26244x^3y^4 - 39366x^3y^3 - 852930x^3y^2 - 345546x^3y$ $-405324x^3 + 13122x^2y^5 + 199017x^2y^4 + 674325x^2y^3 - 170586x^2y^2$ $-554040x^2y + 20100x^2 - 2916xy^6 + 17496xy^5 + 212139xy^4 + 836892xy^3$ $+763992xy^2 + 330966xy + 132858x - 1458y^7 - 32319y^6 - 178686y^5 - 421065y^4$ $-254457y^3 + 131541y^2 + 38595y - 48882$
B	$29160x^3y^6 + 173016x^3y^5 + 521802x^3y^4 + 2237409x^3y^3$ $+2638746x^3y^2 - 17370x^3y - 845658x^3 + 1458x^2y^7 + 120771x^2y^6$ $+577044x^2y^5 + 868941x^2y^4 + 1627794x^2y^3 + 1562193x^2y^2$ $+457137x^2y + 518943x^2 - 5832xy^8 - 38394xy^7 - 52407xy^6 + 102978xy^5$ $-277065xy^4 - 49839xy^3 + 1265544xy^2 + 1061919xy - 217944x - 1458y^9 - 30861y^8$ $-149283y^7 - 384291y^6 - 435465y^5 - 426405y^4 - 13692y^3 + 314286y^2 + 13314y$ -23601
$i = 5, m = 17496$, neighbours 29, 30, 44, 52, 60, 62, 68, 82, 90	
W	$1316946x^3 - 2563596x^2 + 2910840x - 1279098$
A	$816293376x^{14} + 816293376x^{13} + 816293376x^{12}y + 204073344x^{12}$ $+816293376x^{11}y + 2516904576x^{11} - 136048896x^{10}y^3 - 3265173504x^{10}y^2$ $-2380855680x^{10}y + 1462525632x^{10} - 136048896x^9y^3 - 2448880128x^9y^2$ $+1564562304x^9y + 2074745664x^9 - 136048896x^8y^4 - 1462525632x^8y^3$ $-8554074336x^8y^2 + 5458961952x^8y + 6587034048x^8 - 136048896x^7y^4$ $-396809280x^7y^3 - 2908045152x^7y^2 - 8066565792x^7y + 969348384x^7$ $+22674816x^6y^6 + 544195584x^6y^5 + 4858079328x^6y^4$ $-3990767616x^6y^3 - 27986391648x^6y^2 - 18933471360x^6y - 2018058624x^6$ $+22674816x^5y^6 + 408146688x^5y^5 + 2386524384x^5y^4$ $+4920435072x^5y^3 - 17762883984x^5y^2 - 15699475728x^5y + 393030144x^5$ $+22674816x^4y^7 + 243754272x^4y^6 + 1754463888x^4y^5$ $+21368179728x^4y^4 + 41554434672x^4y^3 + 26305620912x^4y^2$ $+1632586752x^4y + 679299696x^4 + 22674816x^3y^7 + 62355744x^3y^6$ $-1573065360x^3y^5 + 10725187968x^3y^4 + 30389922144x^3y^3$ $+30012953328x^3y^2 + 10103520096x^3y - 437434992x^3 - 3779136x^2y^9$ $-90699264x^2y^8 - 813459024x^2y^7 - 1779973056x^2y^6$ $-5844433824x^2y^5 + 3276510912x^2y^4 + 39028082256x^2y^3$ $+29096040456x^2y^2 - 7921541448x^2y - 29419490002x^2 - 3779136xy^9$ $-68024448xy^8 - 405312336xy^7 - 1595740176xy^6 - 13919502672xy^5$ $-46038379536xy^4 - 60868968912xy^3 - 32326256952xy^2 - 3026930472xy$ $+1125193777632x - 3149280y^{10} - 30652992y^9 - 206627760y^8 - 2963134224y^7$ $-10524314448y^6 - 25626964032y^5 - 23430078576y^4 - 37320522804y^3$ $+114484758270y^2 - 320676139344y - 1326546938676$
B	$-3779136x^3y^{10} - 59836320x^3y^9 - 537372144x^3y^8 - 1429598160x^3y^7$

$$\begin{aligned}
& -4957923168x^3y^6 - 22965731712x^3y^5 - 74592194976x^3y^4 \\
& + 187134192648x^3y^3 - 575580871764x^3y^2 + 1993066693776x^3y \\
& - 6257782951992x^3 - 3779136x^2y^{11} - 90069408x^2y^{10} - 715621392x^2y^9 \\
& - 863252640x^2y^8 - 10773348624x^2y^7 - 8248224816x^2y^6 \\
& - 32318624160x^2y^5 + 63961595784x^2y^4 - 307652649588x^2y^3 \\
& + 782487956880x^2y^2 - 2259925267392x^2y + 6060263581086x^2 - 6928416xy^{11} \\
& - 78522048xy^{10} - 416264832xy^9 - 3540467232xy^8 - 10481647536xy^7 \\
& - 36379826784xy^6 - 6549847056xy^5 - 204501222108xy^4 + 452285743830xy^3 \\
& - 684409815720xy^2 + 1821067616964xy - 3748988894238x - 3149280y^{12} - 24354432y^{11} \\
& - 161068176y^{10} - 2674800144y^9 - 4799774880y^8 - 16136076096y^7 \\
& - 2196494928y^6 - 77649891876y^5 + 209095318806y^4 - 791658517920y^3 \\
& + 347403173334y^2 - 903446607870y + 154772377464
\end{aligned}$$

$i = 6$, $m = 81$, neighbours 112, 17, 33, 52, 53, 58, 69, 77

$$W \quad 1488x^3 - 3744x^2 + 4374x - 1518$$

$$\begin{aligned}
A \quad & -17496x^9 + 34992x^8y + 17496x^8 - 26244x^7y + 157464x^7 + 122472x^6y^2 \\
& - 139968x^6y - 207036x^6 + 2916x^5y^3 - 39366x^5y^2 + 189540x^5y \\
& - 262440x^5 - 5832x^4y^4 + 145800x^4y^3 - 263898x^4y^2 - 301806x^4y \\
& + 443232x^4 + 4374x^3y^4 - 150903x^3y^3 + 421362x^3y^2 + 272646x^3y \\
& - 44226x^3 - 20412x^2y^5 + 80190x^2y^4 + 486x^2y^3 - 476037x^2y^2 \\
& + 148716x^2y - 710682x^2 - 486xy^6 + 6561xy^5 - 195129xy^4 + 473850xy^3 \\
& + 198288xy^2 - 374949xy + 2141820x + 972y^7 - 20412y^6 + 2349y^5 + 254799y^4 \\
& - 320958y^3 + 81393y^2 + 722046y - 2381484
\end{aligned}$$

$$\begin{aligned}
B \quad & 23328x^3y^6 - 124416x^3y^5 + 400302x^3y^4 + 584199x^3y^3 \\
& + 1078848x^3y^2 + 3696588x^3y + 12304854x^3 - 18954x^2y^7 + 20169x^2y^6 \\
& + 267948x^2y^5 - 495828x^2y^4 - 917199x^2y^3 - 1478385x^2y^2 \\
& - 4324590x^2y - 12575472x^2 - 1458xy^8 + 22113xy^7 - 116802xy^6 + 123390xy^5 \\
& + 248490xy^4 + 243804xy^3 + 1448739xy^2 + 2569473xy + 6778854x + 972y^9 - 18468y^8 \\
& - 33615y^7 + 194373y^6 - 2763y^5 + 254283y^4 + 1269429y^3 + 22785y^2 + 849009y \\
& + 38580
\end{aligned}$$

$i = 7$, $m = 52488$, neighbours 113, 15, 20, 36, 44, 54, 59, 71, 75, 78, 93, 106

$$W \quad 15952866x^3 - 14481900x^2 - 19750428x + 2299482$$

$$\begin{aligned}
A \quad & 88159684608x^{17} + 176319369216x^{16}y + 528958107648x^{16} + 793437161472x^{15}y \\
& + 991796451840x^{15} - 176319369216x^{14}y^2 + 793437161472x^{14}y \\
& - 14693280768x^{14} - 14693280768x^{13}y^3 - 903636767232x^{13}y^2 \\
& - 1366475111424x^{13}y - 2714583621888x^{13} - 29386561536x^{12}y^4 \\
& - 1013836372992x^{12}y^3 - 1461981436416x^{12}y^2 - 3533734024704x^{12}y \\
& - 4430024151552x^{12} - 132239526912x^{11}y^4 - 1664014046976x^{11}y^3 \\
& + 525284787456x^{11}y^2 - 3223338468480x^{11}y - 2978450455680x^{11} \\
& + 29386561536x^{10}y^5 + 749357319168x^{10}y^4 - 9795520512x^{10}y^3 \\
& + 1932166420992x^{10}y^2 - 3208032967680x^{10}y + 730378498176x^{10} \\
& + 2448880128x^9y^6 + 150606127872x^9y^5 + 75303063936x^9y^4 \\
& + 1369536211584x^9y^3 - 989653681728x^9y^2 - 4632056762112x^9y \\
& + 4003612899264x^9 + 4897760256x^8y^7 + 168972728832x^8y^6 \\
& + 1251377745408x^8y^5 + 1115464898304x^8y^4 + 6238522126080x^8y^3 \\
& - 4806233361216x^8y^2 - 3325885323840x^8y + 4323089719296x^8 \\
& + 22039921152x^7y^7 + 277335674496x^7y^6 + 1588098763008x^7y^5 \\
& + 1721409674976x^7y^4 + 8499603759264x^7y^3 - 5190860596320x^7y^2 \\
& + 1746000512928x^7y + 1658504066688x^7 - 4897760256x^6y^8 \\
& - 124892886528x^6y^7 - 764050599936x^6y^6 - 802008241920x^6y^5 \\
& + 4512061635840x^6y^4 + 4412575880640x^6y^3 - 2322303636384x^6y^2 \\
& + 5854660166016x^6y - 1050722629920x^6 - 408146688x^5y^9 \\
& - 25101021312x^5y^8 - 13774950720x^5y^7 + 579007095264x^5y^6 \\
& - 1480807202400x^5y^5 + 6614272170720x^5y^4 - 4237429933152x^5y^3 \\
& + 247540966272x^5y^2 + 5346670594464x^5y - 1456658523360x^5 \\
& - 816293376x^4y^{10} - 28162121472x^4y^9 - 209379250944x^4y^8 \\
& - 733745708352x^4y^7 - 1726307435232x^4y^6 - 1505602113696x^4y^5 \\
& + 5617909577808x^4y^4 - 8111719853712x^4y^3 + 753464295216x^4y^2 \\
& + 1683426523824x^4y - 341244643392x^4 - 3673320192x^3y^{10} \\
& - 46222612416x^3y^9 - 267540153984x^3y^8 - 913814925264x^3y^7 \\
& - 3086481782160x^3y^6 + 3492868337568x^3y^5 + 2698725422448x^3y^4 \\
& - 4862404549152x^3y^3 - 259266680496x^3y^2 - 1485339331248x^3y \\
& + 433171181808x^3 + 816293376x^2y^{11} + 20815481088x^2y^{10}
\end{aligned}$$

$$\begin{aligned}
& +128226084480x^2y^9 + 385392510144x^2y^8 - 419727850272x^2y^7 \\
& -3932008664688x^2y^6 + 3605261731776x^2y^5 - 1323032998320x^2y^4 \\
& -456061408560x^2y^3 - 1070781339024x^2y^2 - 2220726129408x^2y \\
& -42025308644886x^2 + 68024448xy^{12} + 4183503552xy^{11} + 2499898464xy^{10} \\
& -87555967632xy^9 - 57574192176xy^8 - 1716205804704xy^7 - 675414744192xy^6 \\
& +1741298322960xy^5 - 2510089376616xy^4 + 1376736882840xy^3 - 818261833464xy^2 \\
& -1141040673576xy + 77085766367928x + 136048896y^{13} + 3934080576y^{12} \\
& +25783155360y^{11} + 82909204992y^{10} + 343671688512y^9 + 503906914944y^8 \\
& -429631122816y^7 + 899542349424y^6 + 3234599683992y^5 - 5337143052192y^4 \\
& +5542268812740y^3 - 597762599538y^2 - 8708479386672y + 106102868892684 \\
B \quad & -4557638016x^3y^{12} - 55496612160x^3y^{11} - 268942213440x^3y^{10} \\
& -691888627872x^3y^9 - 1684683611424x^3y^8 + 6079871827296x^3y^7 \\
& -11016449991936x^3y^6 + 14211686692368x^3y^5 - 25018499945376x^3y^4 \\
& +28642609680984x^3y^3 - 2599107200892x^3y^2 - 52989111641952x^3y \\
& +216861252781464x^3 + 884317824x^2y^{13} + 18389275776x^2y^{12} \\
& +109538256960x^2y^{11} + 286877677968x^2y^{10} - 600276177648x^2y^9 \\
& -773976115728x^2y^8 + 3532494316464x^2y^7 - 4095955741392x^2y^6 \\
& -5016729740832x^2y^5 + 18653880770136x^2y^4 - 52821978691140x^2y^3 \\
& +56272820470056x^2y^2 - 114825386381400x^2y + 170066400029490x^2 \\
& -68024448xy^{14} + 657569664xy^{13} - 17206406208xy^{12} - 94746403728xy^{11} \\
& -109426037616xy^{10} - 1862016383952xy^9 + 2454486802848xy^8 - 5963043939696xy^7 \\
& +9998244406368xy^6 - 12825139965192xy^5 + 8194340135628xy^4 \\
& +10700069865762xy^3 - 507628591488xy^2 + 14048527817664xy + 18322677391326x \\
& +136048896y^{15} + 3661982784y^{14} + 18595238688y^{13} + 49788857088y^{12} \\
& +273811210272y^{11} + 131096618208y^{10} - 166063820544y^9 + 1798239599280y^8 \\
& -642997451016y^7 - 3167988401952y^6 + 16500020905548y^5 - 35931766950930y^4 \\
& +63054376369752y^3 - 15044847692754y^2 + 20857838133186y + 55695243180504 \\
i = 8, m = 54, \text{neighbours } 16, 26, 38, 50, 57, 103 \\
W \quad & 402x^3 - 324x^2 + 264x - 90 \\
A \quad & 7776x^7 - 7776x^6y + 19440x^6 + 11664x^5y + 17496x^5 - 34992x^4y^2 \\
& +31104x^4y + 5508x^4 - 1296x^3y^3 - 21384x^3y^2 + 28188x^3y - 36612x^3 \\
& +1296x^2y^4 - 59616x^2y^3 - 27864x^2y^2 + 30618x^2y - 51558x^2 \\
& -1944xy^4 - 48600xy^3 - 13608xy^2 - 31914xy + 30834x + 4536y^5 - 31212y^4 \\
& -5652y^3 + 26592y^2 + 2310y + 726 \\
B \quad & -3888x^3y^5 + 47304x^3y^4 + 158652x^3y^3 + 2988x^3y^2 + 14544x^3y \\
& +133578x^3 + 1296x^2y^6 - 36288x^2y^5 + 37152x^2y^4 + 108774x^2y^3 \\
& -4644x^2y^2 - 52794x^2y - 110676x^2 - 11016xy^6 - 6264xy^5 + 44136xy^4 \\
& -45858xy^3 - 36756xy^2 - 14982xy - 1692x + 4536y^7 - 26676y^6 - 16128y^5 \\
& +12948y^4 - 21606y^3 - 29100y^2 - 11862y - 8826 \\
i = 9, m = 54, \text{neighbours } 110, 19, 104, 105, 106, 107 \\
W \quad & 138x^3 - 288x^2 - 219x - 57 \\
A \quad & -972x^6 - 2916x^5 - 3402x^4y + 1944x^4 + 3402x^3y + 17658x^3 + 162x^2y^3 \\
& +1944x^2y^2 + 45846x^2y + 33438x^2 + 486xy^3 + 41796xy^2 + 85617xy + 51318x \\
& +459y^4 + 14706y^3 + 39201y^2 + 38514y + 16809 \\
B \quad & -324x^3y^4 - 1566x^3y^3 - 117378x^3y^2 - 109476x^3y + 30654x^3 \\
& +162x^2y^5 + 2646x^2y^4 - 30762x^2y^3 - 540x^2y^2 + 53910x^2y \\
& +82923x^2 - 432xy^5 + 13734xy^4 + 91986xy^3 + 123084xy^2 + 57774xy + 52977x \\
& +459y^6 + 14247y^5 + 72069y^4 + 74922y^3 + 30786y^2 + 23601y + 20058 \\
i = 10, m = 18, \text{neighbours } 34, 60, 80, 100 \\
W \quad & 24x^3 - 18x^2 - 108x + 54 \\
A \quad & 216x^4 - 756x^3 + 648x^2y + 552x^2 - 1026xy + 756x - 36y^3 + 294y^2 + 42y - 1056 \\
B \quad & 252x^3y^2 - 2664x^3y + 684x^3 + 504x^2y^3 + 216x^2y^2 + 5184x^2y \\
& +1434x^2 - 72xy^4 - 150xy^3 - 2376xy^2 - 504xy - 4080x - 36y^5 + 366y^4 - 78y^3 \\
& +1626y^2 - 1944y + 282 \\
i = 11, m = 2125764, \text{neighbours } 110, 4, 16, 18, 26, 27, 31, 35, 40, 60, 61, 74, 76, 85, 97, 104, 106 \\
W \quad & 3306561175950x^3 + 5857590426660x^2 + 4953847237410x + 1597776265980 \\
A \quad & 771220920950784x^{25} - 385610460475392x^{24}y + 771220920950784x^{24} \\
& -771220920950784x^{23}y^2 + 578415690713088x^{23}y - 2506467993090048x^{23} \\
& +385610460475392x^{22}y^3 - 1349636611663872x^{22}y^2 - 3856104604753920x^{22}y \\
& -5334278036576256x^{22} - 1478173431822336x^{21}y^3 - 2699273223327744x^{21}y^2 \\
& -12982218836004864x^{21}y - 7342665851552256x^{21} + 1028294561267712x^{20}y^4 \\
& +6619646238160896x^{20}y^3 + 3084883683803136x^{20}y^2 - 570382139453184x^{20}y
\end{aligned}$$

$$\begin{aligned}
& +1879850994817536x^{20} + 128536820158464x^{19}y^5 + 3566896759397376x^{19}y^4 \\
& + 14942405343421440x^{19}y^3 + 23393701268840448x^{19}y^2 \\
& + 2432960990619264x^{19}y + 19924546049771904x^{19} - 64268410079232x^{18}y^6 \\
& - 2377931172931584x^{18}y^5 + 10025871972360192x^{18}y^4 \\
& + 37326557003933952x^{18}y^3 + 67943759780638080x^{18}y^2 \\
& + 59645101329157248x^{18}y + 2355035518408576x^{18} + 246362238637056x^{17}y^6 \\
& + 4546990013105664x^{17}y^5 + 14722821608984064x^{17}y^4 \\
& + 42643429012780416x^{17}y^3 + 64062215596894464x^{17}y^2 \\
& + 77116066931633472x^{17}y + 26137828486702656x^{17} - 171382426877952x^{16}y^7 \\
& - 7610450893549056x^{16}y^6 - 38279871753442560x^{16}y^5 \\
& - 86644528188484608x^{16}y^4 - 87809393121170688x^{16}y^3 \\
& - 8144012589727680x^{16}y^2 + 32871952830317184x^{16}y + 2481474722503680x^{16} \\
& - 21422803359744x^{15}y^8 - 594482793232896x^{15}y^7 \\
& - 10550730654673920x^{15}y^6 - 62960280149077632x^{15}y^5 \\
& - 152366341583154240x^{15}y^4 - 209766734797773312x^{15}y^3 \\
& - 144941666562490464x^{15}y^2 - 75450332393312544x^{15}y - 33057617126101632x^{15} \\
& + 10711401679872x^{14}y^9 + 396321862155264x^{14}y^8 \\
& + 4485399453446400x^{14}y^7 - 13507077518318592x^{14}y^6 \\
& - 127159735579995456x^{14}y^5 - 355972681489791168x^{14}y^4 \\
& - 458361296491175136x^{14}y^3 - 337634538659648640x^{14}y^2 \\
& - 164453823647587296x^{14}y - 36147744933643872x^{14} - 41060373106176x^{13}y^9 \\
& - 757831668850944x^{13}y^8 - 4117195020700800x^{13}y^7 \\
& - 21278868899670720x^{13}y^6 - 81322969941403200x^{13}y^5 \\
& - 179681085329432832x^{13}y^4 - 252698200096351536x^{13}y^3 \\
& - 275046981815900304x^{13}y^2 - 200863774768186368x^{13}y - 52091517564665568x^{13} \\
& + 28563737812992x^{12}y^{10} + 1268408482258176x^{12}y^9 \\
& + 19173490006970880x^{12}y^8 + 115324306186341888x^{12}y^7 \\
& + 341194294484592768x^{12}y^6 + 492621882129896976x^{12}y^5 \\
& + 315000844813125360x^{12}y^4 - 77321722124650320x^{12}y^3 \\
& - 307602223045505856x^{12}y^2 - 198103450274520048x^{12}y - 34543526570248320x^{12} \\
& + 3570467226624x^{11}y^{11} + 99080465538816x^{11}y^{10} \\
& + 1760240342725632x^{11}y^9 + 22062140147511360x^{11}y^8 \\
& + 107711512173675360x^{11}y^7 + 260013473769649536x^{11}y^6 \\
& + 409265224265904120x^{11}y^5 + 463040821503885744x^{11}y^4 \\
& + 332471540770941240x^{11}y^3 + 98746403698924992x^{11}y^2 \\
& + 5829977903205888x^{11}y + 1936141642187280x^{11} - 1785233613312x^{10}y^{12} \\
& - 66053643692544x^{10}y^{11} - 749351809187712x^{10}y^{10} \\
& - 7166894725374912x^{10}y^9 - 5143146462851040x^{10}y^8 \\
& + 146962941532612560x^{10}y^7 + 697389544396268568x^{10}y^6 \\
& + 1565571849288775488x^{10}y^5 + 2013774841086994584x^{10}y^4 \\
& + 1474664118145059672x^{10}y^3 + 586657733083988328x^{10}y^2 \\
& + 124957315186672608x^{10}y + 10296124108030944x^{10} + 6843395517696x^9y^{12} \\
& + 126305278141824x^9y^{11} + 688281942665664x^9y^{10} \\
& + 2573488638323136x^9y^9 + 29598727000309632x^9y^8 \\
& + 162466105490540424x^9y^7 + 499169365878094776x^9y^6 \\
& + 960041723619677544x^9y^5 + 1192285804851433296x^9y^4 \\
& + 969814303175693448x^9y^3 + 559470187232658216x^9y^2 \\
& + 222964678660145112x^9y + 41004119651173200x^9 - 4760622968832x^8y^{13} \\
& - 211401413709696x^8y^{12} - 3200923868668416x^8y^{11} \\
& - 28530301875109344x^8y^{10} - 158785260616918368x^8y^9 \\
& - 557647789146677424x^8y^8 - 1226913562834718040x^8y^7 \\
& - 1580347834286942088x^8y^6 - 873041362109502084x^8y^5 \\
& + 423514701720155808x^8y^4 + 1052359332161732820x^8y^3 \\
& + 773419879354617144x^8y^2 + 272793245254286472x^8y + 37434207210689448x^8 \\
& - 595077871104x^7y^{14} - 16513410923136x^7y^{13} - 293670929389824x^7y^{12} \\
& - 3683122906097376x^7y^{11} - 27873577655795664x^7y^{10} \\
& - 129999843899484480x^7y^9 - 399124897433319636x^7y^8 \\
& - 816613476455416680x^7y^7 - 1118056192577546940x^7y^6 \\
& - 1065122429068523196x^7y^5 - 737055925893836064x^7y^4 \\
& - 301939000218857304x^7y^3 - 5272059855724812x^7y^2 + 40833347798985912x^7y \\
& + 10324924430622552x^7 + 297538935552x^6y^{15} + 11008940615424x^6y^{14} \\
& + 125189507133504x^6y^{13} + 1204648367860512x^6y^{12} \\
& + 8509520575869840x^6y^{11} + 27020951941696200x^6y^{10}
\end{aligned}$$

$$\begin{aligned}
& -14107599422134020x^6y^9 - 431382709754792064x^6y^8 \\
& -1651469894129372940x^6y^7 - 3367712280434307516x^6y^6 \\
& -4273335084246245244x^6y^5 - 3466254130764086376x^6y^4 \\
& -1718314197953323788x^6y^3 - 471759533646996024x^6y^2 \\
& -57723753467482596x^6y - 446195793105864x^6 - 1140565919616x^5y^{15} \\
& -21050879690304x^5y^{14} - 115060785869088x^5y^{13} - 415872649712160x^5y^{12} \\
& -796784474496960x^5y^{11} - 2208989495134332x^5y^{10} \\
& -33345912208786092x^5y^9 - 183861054866807280x^5y^8 \\
& -499644307754923788x^5y^7 - 888531882212350944x^5y^6 \\
& -1213700637422036610x^5y^5 - 1254300577548696648x^5y^4 \\
& -924913507471317270x^5y^3 - 468560038068530388x^5y^2 \\
& -146151103447595016x^5y - 20917817282273364x^5 + 793437161472x^4y^{16} \\
& +35233568951616x^4y^{15} + 534379928251392x^4y^{14} \\
& +4793829553785168x^4y^{13} + 28844236445755824x^4y^{12} \\
& +129502733822274888x^4y^{11} + 447827706868616532x^4y^{10} \\
& +1153240641776764260x^4y^9 + 2130136956881814024x^4y^8 \\
& +2703813889088787228x^4y^7 + 2072251010343172230x^4y^6 \\
& +449244980898931980x^4y^5 - 842306295947111442x^4y^4 \\
& -1021405945772259594x^4y^3 - 554913149340030174x^4y^2 \\
& -157039631192234556x^4y - 18823224964757184x^4 + 99179645184x^3y^{17} \\
& +2752235153856x^3y^{16} + 48994744720896x^3y^{15} + 614870409046032x^3y^{14} \\
& +4676428748162520x^3y^{13} + 24050692033450560x^3y^{12} \\
& +90205888709094174x^3y^{11} + 246807758768829048x^3y^{10} \\
& +494074254648553542x^3y^9 + 768470119252737300x^3y^8 \\
& +990471201023595744x^3y^7 + 1070431238856993570x^3y^6 \\
& +964547454125545686x^3y^5 + 672716286749739132x^3y^4 \\
& +296430521228298900x^3y^3 + 53417678302658322x^3y^2 - 7551862670492658x^3y \\
& -3291324393384792x^3 - 49589822592x^2y^{18} - 1834823435904x^2y^{17} \\
& -20914507678176x^2y^{16} - 202469046915312x^2y^{15} \\
& -1437444690654744x^2y^{14} - 7055768581992108x^2y^{13} \\
& -22862278908602082x^2y^{12} - 34777166436587268x^2y^{11} \\
& +55215192112350318x^2y^{10} + 469128733379174466x^2y^9 \\
& +1414256060348822988x^2y^8 + 267937334589018868x^2y^7 \\
& +3511304622129487590x^2y^6 + 3275750432249750997x^2y^5 \\
& +2146354407823124304x^2y^4 + 944949841593900795x^2y^3 \\
& +262603084265921592x^2y^2 + 41786806066262334x^2y \\
& -182224465364903673880732x^2 + 190094319936xy^{18} + 3508479948384xy^{17} \\
& +19234652437872xy^{16} + 67138421061744xy^{15} + 66419368634160xy^{14} \\
& -1402746536818926xy^{13} - 7012877259654918xy^{12} - 12253421750001210xy^{11} \\
& -898652301313968xy^{10} + 52147135822253292xy^9 + 160425810111510873xy^8 \\
& +294603721701189300xy^7 + 453873051950797683xy^6 + 614856258168297507xy^5 \\
& +633198827621900628xy^4 + 454400247596547780xy^3 + 215489214320008671xy^2 \\
& +60618658540935780xy - 6456160355501049443689338x - 66119763456y^{19} \\
& -3500214977952y^{18} - 54006071792832y^{17} - 455204151711720y^{16} \\
& -2693592510178320y^{15} - 12723701855076036y^{14} - 49859339759648100y^{13} \\
& -160841201525631450y^{12} - 424956959166074769y^{11} - 861257888964513216y^{10} \\
& -1544263833431273436y^9 - 586752334362420408y^8 - 10706731069043075634y^7 \\
& +74874107889340024365y^6 - 620965529546568284133y^5 \\
& +5097592300779775507239y^4 - 41841325490798821653090y^3 \\
& +343465934905057722645813y^2 - 1604590910524659522099063y \\
& -4741951252379674389423762 \\
B \quad & 198359290368x^3y^{19} + 7116139541952x^3y^{18} + 105618057150528x^3y^{17} \\
& +1050691942390896x^3y^{16} + 6650644813304904x^3y^{15} \\
& +29117989523415816x^3y^{14} + 95372932524033258x^3y^{13} \\
& +236675128738851936x^3y^{12} + 425184774326683470x^3y^{11} \\
& +611374953731030190x^3y^{10} + 7830662834865960x^3y^9 \\
& +3311585909693911962x^3y^8 - 28166335200780783678x^3y^7 \\
& +226362562477400340252x^3y^6 - 1863201365118562059558x^3y^5 \\
& +15291591535702983424671x^3y^4 - 125524979886481694498592x^3y^3 \\
& +1030397326315789045998843x^3y^2 - 4813772860064960811324828x^3y \\
& +19310027977811301169333410x^3 - 49589822592x^2y^{20} - 2380311484416x^2y^{19} \\
& -35403000845472x^2y^{18} - 323020528141392x^2y^{17} \\
& -2022383738863800x^2y^{16} - 8953595504861724x^2y^{15}
\end{aligned}$$

$$\begin{aligned}
& -29702282236269318x^2y^{14} - 77157278646381864x^2y^{13} \\
& -139759182630266022x^2y^{12} - 140741177585599026x^2y^{11} \\
& +368592794558722260x^2y^{10} - 205803107780411502x^2y^9 \\
& +12775770965908806192x^2y^8 - 82501283851423187292x^2y^7 \\
& +707918173722954069696x^2y^6 - 5780310186658620318309x^2y^5 \\
& +47472340024966508142120x^2y^4 - 389673076588226821399833x^2y^3 \\
& +1376489252311588324254591x^2y^2 - 4841241112573558697418939x^2y \\
& +14079612910082525634921972x^2 + 322333846848xy^{20} + 9599763156768xy^{19} \\
& +94831581989232xy^{18} + 605869626663072xy^{17} + 2869222749220800xy^{16} \\
& +10864189897049826xy^{15} + 37098931511724750xy^{14} + 104938352902979316xy^{13} \\
& +246181918672226538xy^{12} + 384711386742821574xy^{11} + 1159276429404981072xy^{10} \\
& -3137670504223860762xy^9 + 33203342022869589696xy^8 \\
& -265893569589201909786xy^7 + 2187343222535520761535xy^6 \\
& -17952138754304731606143xy^5 + 147366326418322197981555xy^4 \\
& -1209682862540850396028089xy^3 + 1044131591100280814066313xy^2 \\
& -3109677518163266746252476xy + 3410614865409267791786931x - 66119763456y^{21} \\
& -3367975451040y^{20} - 47336240654208y^{19} - 364098005144712y^{18} \\
& -2022902786659680y^{17} - 9187106505261228y^{16} - 34344715565659140y^{15} \\
& -104098570615288998y^{14} - 262424200596863085y^{13} - 507922366010254884y^{12} \\
& -1092363994287716031y^{11} + 194956987539049461y^{10} - 13848981199570461816y^9 \\
& +99999542193769820169y^8 - 832478853300972137505y^7 \\
& +6826875110930678056164y^6 - 56040799858429293549090y^5 \\
& +460024472761546023956283y^4 - 2561383435813008654510888y^3 \\
& +682440274644649232281233y^2 - 2626005517100288439389028y \\
& -1094531128125341342031681
\end{aligned}$$

$i = 12, m = 4$, neighbours 114, 1, 18, 28, 74
 W $24x^3 + 12x^2 + 2x + 3$
 A $-96x^2 + 64xy - 32x + 64y^2 + 128y + 24$
 B $-96x^3y - 192x^3 - 48x^2y^2 - 24x^2y - 52x^2 + 32xy^3 + 64xy^2 + 344xy + 244x$
 $+32y^4 + 96y^3 + 212y^2 + 326y + 129$

$i = 13, m = 4$, neighbours 115, 2, 49
 W $48x^3 + x^2 + 4x + 11$
 A $-384x^2 - 16x + 16y - 48$
 B $-96x^3y - 97x^3 - 192x^2y^2 - 98x^2y - 135x^2 - 4xy^2 - 12xy - 399x + 8y^3$
 $-20y^2 + 110y - 73$

$i = 14, m = 6$, neighbours 18, 39, 42
 W $15x^2 + 9x$
 A $-48x - 48y - 24$
 B $-225x^3 - 162x^2y - 135x^2 - 36xy^2 + 72xy + 12x - 24y^3 + 36y^2 + 48y + 12$

$i = 15, m = 663552$, neighbours 110, 7, 16, 17, 19, 42, 64, 74, 75
 W $4240950x^3 + 3056994x^2 + 721866x + 115326$
 A $2348273369088x^9 - 4696546738176x^8y + 5870683422720x^8$
 $-2348273369088x^7y^2 - 29353417113600x^7y - 11154298503168x^7$
 $+4696546738176x^6y^3 + 14089640214528x^6y^2 - 29353417113600x^6y$
 $-27102988468224x^6 + 16046534688768x^5y^3 + 68686996045824x^5y^2$
 $+46476243763200x^5y - 1663360303104x^5 - 3913788948480x^4y^4$
 $-6849130659840x^4y^3 + 45204262354944x^4y^2 + 62229244280832x^4y$
 $+13062270615552x^4 + 391378894848x^3y^5 - 3326720606208x^3y^4$
 $-17318516097024x^3y^3 + 34490265108480x^3y^2 + 88696242044928x^3y$
 $+38811740405760x^3 - 782757789696x^2y^6 - 4696546738176x^2y^5$
 $-25831007059968x^2y^4 - 98448099508224x^2y^3 - 117584896720896x^2y^2$
 $-30405247893504x^2y + 7373931229722x^2 - 2674422448128xy^6$
 $-13796106043392xy^5 - 35012103634944xy^4 - 99516237742080xy^3$
 $-148104296792064xy^2 - 87334569639936xy - 18843307312356x + 717527973888y^7$
 $+3296144130048y^6 + 7848960196608y^5 + 25619849109504y^4 - 6749522106624y^3$
 $-80282132519016y^2 - 63609897003312y - 14015429659902$
 B $391378894848x^3y^7 - 1161906094080x^3y^6 - 9851868020736x^3y^5$
 $-7284398211072x^3y^4 - 83891267606016x^3y^3 - 113013876167280x^3y^2$
 $+29777681189088x^3y + 50175231622344x^3 - 391378894848x^2y^8$
 $-1394287312896x^2y^7 - 10053290557440x^2y^6 - 54833561763840x^2y^5$
 $-104573286945792x^2y^4 - 204108236080416x^2y^3 - 282173349723984x^2y^2$
 $-137817108972144x^2y - 5642716433118x^2 - 1516593217536xy^8$

$$\begin{aligned}
& -6874101448704xy^7 - 15941295341568xy^6 - 48071149443072xy^5 \\
& -48080658391488xy^4 - 35499582261798xy^3 - 96692096695248xy^2 - 86308901369274xy \\
& -18690107562000x + 358763986944y^9 + 1468690071552y^8 + 4429501562880y^7 \\
& +20967733702656y^6 + 24736518511488y^5 + 34232978953740y^4 + 46372470580194y^3 \\
& -12948204007716y^2 - 42576247408698y - 14002129573626
\end{aligned}$$

$i = 16, m = 24$, neighbours 8, 11, 15, 63, 68

W $288x^3 + 144x + 27$

A $-768x^3 - 2304x^2y - 12672x^2 - 1728xy^2 + 4800xy + 768x - 384y^3 + 4416y^2 - 7488y - 6840$

B $-2880x^3y^2 + 13824x^3y - 29208x^3 - 1632x^2y^3 - 2016x^2y^2 - 33720x^2y - 13356x^2 - 1056xy^4 + 2064xy^3 - 20616xy^2 - 1224xy - 3606x - 192y^5 + 1728y^4 - 7392y^3 - 588y^2 + 198y - 3807$

$i = 17, m = 2$, neighbours 114, 6, 15, 64, 101

W $6x^3 - 3x^2 - 24x + 12$

A $-4x^2 + 6xy + 6x + 4y^2 + 8y + 4$

B $-36x^3y + 147x^3 - 12x^2y^2 + 18x^2y + 34x^2 + 4xy^3 - 12xy^2 + 138xy - 222x + 2y^4 + 26y^3 - 34y^2 - 68y + 122$

$i = 18, m = 3888$, neighbours 110, 11, 12, 14, 35, 42, 57, 58, 60, 91

W $15864x^3 - 8010x^2 - 7191x + 2637$

A $-161243136x^{10}y - 80621568x^{10} + 241864704x^9y^2 - 120932352x^9y - 120932352x^9 + 161243136x^8y^3 + 362797056x^8y^2 + 262020096x^8y + 60466176x^8 + 967458816x^7y^3 + 705438720x^7y^2 + 440059392x^7y + 164602368x^7 + 510603264x^6y^4 + 1928199168x^6y^3 + 1719926784x^6y^2 + 650011392x^6y + 104136192x^6 - 40310784x^5y^5 + 826371072x^5y^4 + 739031040x^5y^3 + 503884800x^5y^2 + 173000448x^5y - 26873856x^4y^6 + 181398528x^4y^5 + 584506368x^4y^4 - 238505472x^4y^3 - 456015744x^4y^2 - 122052096x^4y - 7278336x^4 - 161243136x^3y^6 - 419904000x^3y^5 - 1755758592x^3y^4 - 1950314112x^3y^3 - 646372224x^3y^2 + 78102144x^3y + 55987200x^3 - 85100544x^2y^7 - 563231232x^2y^6 - 2405769984x^2y^5 - 5653587456x^2y^4 - 6034300416x^2y^3 - 3240259200x^2y^2 - 839808000x^2y - 119766624x^2 + 6718464xy^8 - 137728512xy^7 - 651131136xy^6 - 2933449344xy^5 - 5617475712xy^4 - 5654707200xy^3 - 3347334720xy^2 - 1096882560xy - 107595504x + 4478976y^9 - 42550272y^8 - 237012480y^7 - 1049822208y^6 - 2598324480y^5 - 3363544224y^4 - 2373312672y^3 - 835330296y^2 - 120567360y - 22907928$

B $-36951552x^3y^8 + 21275136x^3y^7 - 22021632x^3y^6 + 449001792x^3y^5 + 170981280x^3y^4 + 480324384x^3y^3 + 1522568088x^3y^2 + 1078988544x^3y + 85264290x^3 - 43670016x^2y^9 - 212751360x^2y^8 - 1035016704x^2y^7 - 2597557248x^2y^6 - 5299540992x^2y^5 - 9000846576x^2y^4 - 9322961832x^2y^3 - 5256191628x^2y^2 - 1325298402x^2y - 5321667x^2 + 1119744xy^{10} - 47029248xy^9 - 191382912xy^8 - 1147799808xy^7 - 2945258496xy^6 - 6432968592xy^5 - 9897131808xy^4 - 9291932688xy^3 - 5255170488xy^2 - 1766096316xy - 257452800x + 2239488y^{11} - 22394880y^{10} - 105069312y^9 - 551878272y^8 - 1676966976y^7 - 3315807792y^6 - 4912771896y^5 - 5091704880y^4 - 3353260584y^3 - 1216493124y^2 - 172033146y - 15954159$

$i = 19, m = 3$, neighbours 4, 9, 15, 40, 79, 81

W $48x^5 - 48x^4 + 12x^3 + 12x^2 - 12x - 12$

A $-384x^6 + 816x^5 + 96x^4y - 528x^4 + 284x^3y^2 - 408x^3y + 508x^3 + 40x^2y^3 - 12x^2y^2 - 152x^2y + 180x^2 - 142xy^3 - 38xy - 144x - 20y^4 + 30y^3 - 8y^2 + 306y - 216$

B $-888x^5y^2 + 1224x^5y - 1656x^5 - 444x^3y^3 + 612x^3y^2 - 828x^3y - 72x^3 - 24x^2y^4 + 156x^2y^3 + 518x^2y^2 - 588x^2y - 62x^2 - 142xy^5 - 9xy^4 - 164xy^3 - 265xy^2 - 136xy + 160x - 20y^6 + 12y^5 + 35y^4 + 334y^3 - 383y^2 + 280y + 270$

$i = 20, m = 6$, neighbours 7, 68, 87, 96, 98

W $6x^5 - 36x^4 - 30x^2 - 12x - 36$

A $-6x^6 + 66x^5 - 6x^4y - 192x^4 + 11x^3y^2 - 67x^3y - 213x^3 + x^2y^3 + 6x^2y^2 - 113x^2y - 376x^2 - 11xy^3 + 12xy^2 + 134xy - 387x - y^4 + 7y^3 + 29y^2 + 119y - 586$

B $-66x^5y^2 + 654x^5y + 1464x^5 - 66x^3y^3 + 720x^3y^2 + 810x^3y + 1560x^3 + 12x^2y^4 + 366x^2y^3 - 359x^2y^2 - 845x^2y - 1191x^2 - 11xy^5 + xy^4 + 123xy^3 + 536xy^2 + 1323xy - 467x - y^6 + 6y^5 + 322y^4 + 1023y^3 - 319y^2 - 1107y$

$i = 21$, $m = 18$, neighbours 116, 27, 48, 64, 95, 97, 107

$$W \quad 72x^5 + 36x^4 + 36x^3 + 36x^2 - 36x$$

$$\begin{aligned} A \quad & 216x^8 - 432x^7 - 1836x^6y - 3780x^6 + 2808x^5y + 4500x^5 - 36x^4y^3 \\ & + 6264x^4y^2 + 18972x^4y + 10620x^4 + 468x^3y^3 - 10092x^3y^2 - 37824x^3y \\ & - 16908x^3 + 306x^2y^4 - 10278x^2y^3 - 46134x^2y^2 - 51156x^2y - 14916x^2 \\ & - 468xy^4 + 7248xy^3 + 39096xy^2 + 42924xy + 11544x - 270y^5 + 5400y^4 + 33372y^3 \\ & + 55422y^2 + 34284y + 6564 \\ B \quad & -2376x^5y^3 + 19080x^5y^2 + 77580x^5y - 9684x^5 - 2376x^3y^4 \\ & + 16704x^3y^3 + 96660x^3y^2 + 67896x^3y - 21492x^3 + 36x^2y^6 \\ & - 4644x^2y^5 + 9576x^2y^4 + 92844x^2y^3 + 104628x^2y^2 + 23496x^2y \\ & - 8472x^2 - 468xy^6 + 7716xy^5 + 25458xy^4 - 1458xy^3 + 51246xy^2 + 96564xy \\ & + 38844x - 270y^7 + 5670y^6 + 22248y^5 + 12384y^4 + 19200y^3 + 65790y^2 + 60204y \\ & + 16932 \end{aligned}$$

$i = 22$, $m = 18$, neighbours 40, 43, 48, 49, 85

$$W \quad 36x^5 - 126x^4 + 126x^3 - 18x^2 - 18x$$

$$\begin{aligned} A \quad & 432x^5 + 216x^4y - 2430x^4 + 72x^3y^2 - 2256x^3y + 4619x^3 - 594x^2y^2 \\ & + 1836x^2y - 1098x^2 - 72xy^3 + 1914xy^2 - 203xy - 1523x + 306y^3 + 474y^2 - 875y \\ B \quad & -432x^5y^2 + 6408x^5y - 5034x^5 - 432x^4y^2 + 6408x^4y - 5034x^4 \\ & - 432x^3y^3 + 6156x^3y^2 - 1296x^3y + 3510x^3 - 216x^2y^4 - 918x^2y^3 \\ & - 3852x^2y^2 + 5334x^2y + 5533x^2 - 72xy^5 + 1608xy^4 - 101xy^3 - 2370xy^2 \\ & + 3997xy + 1364x + 306y^5 + 1122y^4 - 1541y^3 - 516y^2 + 571y - 227 \end{aligned}$$

$i = 23$, $m = 72$, neighbours 31, 55, 100, 104

$$W \quad 72x^5 - 216x^4 - 72x^3 - 216x^2 - 144x$$

$$\begin{aligned} A \quad & -864x^6 + 5184x^5 - 4320x^4 - 720x^3y^2 + 2172x^3y - 15005x^3 + 144x^2y^3 \\ & + 6192x^2y - 9360x^2 - 864xy^3 + 300xy^2 - 14729xy - 6739x + 144y^4 - 1584y^3 \\ & + 10092y^2 - 8813y - 10368 \\ B \quad & 4320x^5y^2 - 13032x^5y + 64974x^5 + 4320x^4y^2 - 13032x^4y + 64974x^4 \\ & - 4320x^3y^3 + 15552x^3y^2 - 72576x^3y - 35424x^3 - 864x^2y^4 \\ & + 5184x^2y^3 - 15696x^2y^2 - 4764x^2y + 40925x^2 + 720xy^5 + 1284xy^4 \\ & + 5213xy^3 + 8772xy^2 - 30091xy - 22838x - 144y^6 + 1584y^5 - 10236y^4 - 10483y^3 \\ & - 70716y^2 - 94591y - 32659 \end{aligned}$$

$i = 24$, $m = 18$, neighbours 36, 43, 74, 80, 100, 103, 106

$$W \quad 18x^5 + 234x^4 - 576x^3 + 54x^2 + 198x - 216$$

$$\begin{aligned} A \quad & 864x^{10} - 864x^9 + 6048x^8y - 2808x^8 - 7776x^7y + 4824x^7 - 144x^6y^3 \\ & + 15552x^6y^2 - 3168x^6y - 2214x^6 + 144x^5y^3 - 24624x^5y^2 + 15552x^5y \\ & - 2304x^5 - 1008x^4y^4 + 17532x^4y^3 + 16542x^4y^2 - 28080x^4y - 2118x^4 \\ & - 24x^3y^5 + 4224x^3y^4 - 36150x^3y^3 - 30108x^3y^2 - 10977x^3y \\ & - 8757x^3 + 24x^2y^6 - 2592x^2y^5 + 6114x^2y^4 + 28557x^2y^3 \\ & - 10812x^2y^2 - 40791x^2y - 55191x^2 - 24xy^6 + 4104xy^5 - 22374xy^4 \\ & - 30642xy^3 + 19599xy^2 + 36816xy + 17469x + 24y^7 - 1752y^6 - 42y^5 + 15369y^4 \\ & + 7161y^3 - 38625y^2 - 38418y + 40731 \\ B \quad & 144x^5y^5 - 17568x^5y^4 + 8748x^5y^3 + 134964x^5y^2 + 289062x^5y \\ & + 287748x^5 - 144x^3y^6 + 17712x^3y^5 - 26316x^3y^4 - 126216x^3y^3 \\ & - 154098x^3y^2 + 1314x^3y + 158196x^3 + 864x^2y^7 - 7020x^2y^6 \\ & - 32754x^2y^5 - 20082x^2y^4 + 48060x^2y^3 + 132456x^2y^2 + 124587x^2y \\ & + 70725x^2 + 24xy^8 - 4080xy^7 + 18294xy^6 + 45312xy^5 + 32361xy^4 + 12132xy^3 \\ & - 4047xy^2 - 14061xy - 27336x - 24y^9 + 1728y^8 + 1746y^7 - 14487y^6 - 30222y^5 \\ & - 15348y^4 + 8964y^3 + 10065y^2 + 89700y + 87387 \end{aligned}$$

$i = 25$, $m = 5184$, neighbours 30, 37, 64, 85, 90, 104, 108

$$W \quad 15552x^5 - 46656x^4 + 145152x^3 - 57024x^2 - 5184x - 20736$$

$$\begin{aligned} A \quad & 8957952x^{10} + 35831808x^9 + 58226688x^8y + 40310784x^8 + 188116992x^7y \\ & + 79128576x^7 - 1492992x^6y^3 + 107495424x^6y^2 + 65691648x^6y \\ & + 19408896x^6 - 5971968x^5y^3 + 255301632x^5y^2 + 163482624x^5y \\ & + 153031680x^5 - 9704448x^4y^4 - 38071296x^4y^3 - 453123072x^4y^2 \\ & + 68677632x^4y - 947054592x^4 + 746496x^3y^5 - 45225216x^3y^4 \\ & - 235452096x^3y^3 + 742242096x^3y^2 - 257820588x^3y + 1389527331x^3 \\ & + 248832x^2y^6 - 17915904x^2y^5 - 261522432x^2y^4 - 1296414720x^2y^3 \\ & + 4727808x^2y^2 + 1096229376x^2y - 3979196928x^2 + 995328xy^6 - 42985728xy^5 \\ & - 365902272xy^4 + 312815088xy^3 - 973686492xy^2 - 1659753777xy + 4234449117x \\ & + 248832y^7 - 10450944y^6 - 137168640y^5 - 696672576y^4 - 1177200432y^3 \end{aligned}$$

$$\begin{aligned}
& +829760940y^2 + 2729513181y - 859963392 \\
B \quad & -4478976x^5y^5 + 83234304x^5y^4 + 663230592x^5y^3 - 5098425120x^5y^2 \\
& -2360236536x^5y + 4490623278x^5 + 4478976x^4y^5 - 83234304x^4y^4 \\
& -663230592x^4y^3 + 5098425120x^4y^2 + 2360236536x^4y - 4490623278x^4 \\
& +4478976x^3y^6 - 85100544x^3y^5 - 628549632x^3y^4 + 5374771200x^3y^3 \\
& +235892736x^3y^2 - 5474055168x^3y - 6255138816x^3 + 8211456x^2y^7 \\
& +91819008x^2y^6 + 808455168x^2y^5 + 497601792x^2y^4 - 123861312x^2y^3 \\
& +2978293968x^2y^2 + 1925990316x^2y + 8688168669x^2 - 746496xy^8 \\
& +32534784xy^7 + 227738304xy^6 - 1182239280xy^5 + 2237336172xy^4 \\
& +4340350893xy^3 - 4980590676xy^2 - 2843577123xy - 777475584x - 248832y^9 \\
& +10450944y^8 + 136919808y^7 + 1136856384y^6 + 1969792560y^5 - 1500855660y^4 \\
& -257556697y^3 + 2927329668y^2 - 2054338185y - 2514522333 \\
i = 26, m = 9, \text{ neighbours } 116, 8, 11, 39, 57, 75, 76, 81 \\
W \quad & 144x^5 - 216x^4 - 108x^3 + 144x^2 - 36 \\
A \quad & 3456x^8 + 3456x^7y + 15552x^7 + 864x^6y^2 + 16416x^6y + 18144x^6 \\
& +5184x^5y^2 + 25920x^5y + 11952x^5 - 144x^4y^3 + 7992x^4y^2 + 5976x^4y \\
& -25200x^4 - 1728x^3y^4 - 10800x^3y^3 - 14304x^3y^2 - 24984x^3y - 4680x^3 \\
& -144x^2y^5 - 3168x^2y^4 - 8676x^2y^3 - 21024x^2y^2 - 32652x^2y \\
& +3624x^2 - 864xy^5 - 6264xy^4 - 11688xy^3 - 16776xy^2 - 4932xy - 1296x - 72y^6 \\
& -1440y^5 - 4500y^4 - 9738y^3 - 8406y^2 + 3108y \\
B \quad & 3456x^5y^4 + 23328x^5y^3 + 31032x^5y^2 - 12744x^5y - 94176x^5 \\
& -1728x^3y^5 - 11664x^3y^4 - 15516x^3y^3 + 6372x^3y^2 + 47088x^3y \\
& +54576x^3 + 288x^2y^6 - 324x^2y^5 + 4806x^2y^4 + 36324x^2y^3 \\
& +42558x^2y^2 + 45828x^2y + 8460x^2 + 864xy^7 + 6264xy^6 + 13596xy^5 \\
& +29880xy^4 + 31920xy^3 + 32370xy^2 + 14238xy - 5952x + 72y^8 + 1440y^7 + 4734y^6 \\
& +11898y^5 + 14346y^4 + 8214y^3 + 14742y^2 + 6270y + 132 \\
i = 27, m = 9, \text{ neighbours } 116, 4, 11, 21, 47, 63, 71, 77, 92 \\
W \quad & 396x^5 + 108x^4 + 576x^3 + 144x^2 + 90x + 162 \\
A \quad & -864x^{10} + 2160x^9 + 216x^8y^2 + 4536x^8y + 864x^8 - 648x^7y^2 - 12960x^7y \\
& -6768x^7 - 828x^6y^3 - 8532x^6y^2 + 9216x^6y - 21384x^6 + 2880x^5y^3 \\
& +26856x^5y^2 + 9468x^5y - 8820x^5 - 36x^4y^5 + 702x^4y^4 + 2898x^4y^3 \\
& -31086x^4y^2 - 23634x^4y - 88008x^4 + 960x^3y^5 + 4086x^3y^4 \\
& -1896x^3y^3 + 61518x^3y^2 + 128892x^3y + 123336x^3 + 138x^2y^6 \\
& +693x^2y^5 + 4137x^2y^4 + 35343x^2y^3 + 19575x^2y^2 + 3744x^2y \\
& -59874x^2 - 480xy^6 - 2202xy^5 - 1041xy^4 - 25245xy^3 - 32031xy^2 - 9021xy \\
& +68400x - 60y^7 - 381y^6 - 1533y^5 - 9978y^4 + 6561y^3 + 8349y^2 + 4896y - 16602 \\
B \quad & -2556x^5y^5 - 19872x^5y^4 - 49158x^5y^3 - 153558x^5y^2 - 332496x^5y \\
& -468360x^5 - 1278x^3y^6 - 8658x^3y^5 - 14643x^3y^4 - 52200x^3y^3 \\
& -89469x^3y^2 - 67932x^3y - 96594x^3 + 18x^2y^8 - 171x^2y^7 - 774x^2y^6 \\
& +4668x^2y^5 + 8991x^2y^4 + 48678x^2y^3 + 24285x^2y^2 + 3093x^2y \\
& -49830x^2 - 480xy^8 - 2682xy^7 - 3291xy^6 - 28035xy^5 - 60306xy^4 - 108183xy^3 \\
& -103134xy^2 - 145881xy - 82254x - 60y^9 - 441y^8 - 1710y^7 - 11319y^6 - 7806y^5 \\
& -15180y^4 - 14997y^3 - 12639y^2 + 3360y - 8994 \\
i = 28, m = 1, \text{ neighbours } 12, 76, 78 \\
W \quad & 4x^2 - 1 \\
A \quad & 0 \\
B \quad & -4x^2y - 10x^2 + 2y^2 + 7y + 5 \\
i = 29, m = 10368, \text{ neighbours } 114, 5, 31, 39, 50, 55, 65, 78, 83 \\
W \quad & 746496x^5 - 839808x^4 + 723168x^3 - 46656x^2 - 124416x + 85536 \\
A \quad & 429981696x^{11} - 429981696x^{10}y - 1719926784x^{10} - 644972544x^9y \\
& +1612431360x^9 + 1504935936x^8y^2 + 5267275776x^8y + 573308928x^8 \\
& -71663616x^7y^3 - 1182449664x^7y^2 - 6288482304x^7y - 322486272x^7 \\
& +71663616x^6y^4 - 680804352x^6y^3 + 1522851840x^6y^2 + 9549176832x^6y \\
& -92428148736x^6 + 107495424x^5y^4 - 644972544x^5y^3 - 9710419968x^5y^2 \\
& -19259596800x^5y + 202372079616x^5 - 250822656x^4y^5 - 2329067520x^4y^4 \\
& -9364045824x^4y^3 - 9531260928x^4y^2 + 8244301824x^4y - 289150746624x^4 \\
& +47775744x^3y^6 - 194088960x^3y^5 + 2530372608x^3y^4 \\
& +23163542784x^3y^3 + 83945972160x^3y^2 - 61461343152x^3y + 309092381052x^3 \\
& -11943936x^2y^7 + 113467392x^2y^6 + 1507921920x^2y^5 \\
& +3436867584x^2y^4 + 5159780352x^2y^3 - 43655086080x^2y^2 \\
& -23986409472x^2y - 98982508032x^2 - 17915904xy^7 + 108988416xy^6 \\
& -523915776xy^5 - 6843325824xy^4 - 41599307808xy^3 + 17720488296xy^2
\end{aligned}$$

$$\begin{aligned}
& -19753438674xy + 7563437985x + 5971968y^8 - 14929920y^7 - 114213888y^6 \\
& + 157946112y^5 - 3832080192y^4 + 14189891472y^3 + 51888025836y^2 - 29199997023y \\
& - 7316407296 \\
B \quad & -107495424x^5y^6 + 1173491712x^5y^5 + 7485115392x^5y^4 \\
& + 6132402432x^5y^3 - 121159312704x^5y^2 + 252088230672x^5y \\
& - 223594181748x^5 - 53747712x^4y^6 + 586745856x^4y^5 + 3742557696x^4y^4 \\
& + 3066201216x^4y^3 - 60579656352x^4y^2 + 126044115336x^4y - 111797090874x^4 \\
& - 53747712x^3y^7 + 555393024x^3y^6 + 4084826112x^3y^5 \\
& + 5249359872x^3y^4 - 58791038976x^3y^3 + 90705982464x^3y^2 \\
& - 38271356928x^3y - 167539829760x^3 + 107495424x^2y^8 + 1155575808x^2y^7 \\
& + 5213528064x^2y^6 + 15235983360x^2y^5 + 60293859840x^2y^4 \\
& + 142091895168x^2y^3 + 11513452320x^2y^2 + 225027903960x^2y \\
& + 94732338498x^2 - 23887872xy^9 + 123918336xy^8 - 427617792xy^7 \\
& - 7883630208xy^6 - 45584409312xy^5 - 32423709096xy^4 - 195381345630xy^3 \\
& - 113567758200xy^2 - 143060723706xy - 136314951453x + 5971968y^{10} - 14929920y^9 \\
& - 108241920y^8 - 495984384y^7 - 8476031808y^6 + 4559035536y^5 + 50148809388y^4 \\
& + 19856166897y^3 + 120678874236y^2 + 63966271629y - 26196843615 \\
i = 30, m = 144, \text{ neighbours } 114, 5, 25, 33, 69, 96, 106 \\
W \quad & 1296x^5 - 11664x^4 + 7776x^3 - 5616x^2 + 2160x - 2592 \\
A \quad & 10368x^7 - 10368x^6y - 279936x^6 + 10368x^5y + 4955904x^5 - 31104x^4y^2 \\
& - 31104x^4y - 25994304x^4 + 769680x^3y^2 - 887172x^3y + 30025001x^3 \\
& + 1728x^2y^4 + 57024x^2y^3 - 91584x^2y^2 + 627264x^2y - 33791040x^2 \\
& - 1728xy^4 - 754128xy^3 - 182460xy^2 - 1141481xy + 21513600x - 1728y^5 - 13824y^4 \\
& + 1100016y^3 + 3054804y^2 + 8408467y - 14787223 \\
B \quad & -10368x^5y^3 - 4618080x^5y^2 + 5260824x^5y + 16634634x^5 - 10368x^4y^3 \\
& - 4618080x^4y^2 + 5260824x^4y + 16634634x^4 - 10368x^3y^4 - 4613760x^3y^3 \\
& + 7185024x^3y^2 + 14442624x^3y + 37274688x^3 + 41472x^2y^5 + 777600x^2y^4 \\
& + 8773056x^2y^3 - 603792x^2y^2 - 458940x^2y - 18025769x^2 - 738576xy^5 \\
& - 2131644xy^4 - 9346745xy^3 - 1037532xy^2 - 2290913xy + 12539822x - 1728y^7 \\
& - 15552y^6 + 833904y^5 + 3997572y^4 + 14611975y^3 + 3481908y^2 + 14447515y \\
& + 8068759 \\
i = 31, m = 432, \text{ neighbours } 11, 23, 29, 59, 85, 107 \\
W \quad & 4320x^5 - 6480x^4 - 7344x^3 - 1728x^2 + 2592x + 864 \\
A \quad & -248832x^7 - 124416x^6y - 2737152x^6 + 311040x^5y + 10077696x^5 + 746496x^4y \\
& + 3929472x^4 + 31104x^3y^3 + 1312416x^3y^2 + 4814136x^3y - 18806862x^3 \\
& + 20736x^2y^4 + 1047168x^2y^3 + 1264896x^2y^2 + 1078272x^2y - 18175104x^2 \\
& - 51840xy^4 - 2110752xy^3 - 803448xy^2 - 4362018xy + 14824860x - 20736y^5 \\
& - 974592y^4 + 325728y^3 - 4063320y^2 + 5304342y + 11393742 \\
B \quad & 62208x^5y^3 + 1270080x^5y^2 - 14141520x^5y + 36325332x^5 + 62208x^4y^3 \\
& + 1270080x^4y^2 - 14141520x^4y + 36325332x^4 + 62208x^3y^4 \\
& + 1368576x^3y^3 - 12130560x^3y^2 + 13934592x^3y - 34038144x^3 \\
& + 124416x^2y^5 + 5038848x^2y^4 + 1264896x^2y^3 + 4410720x^2y^2 \\
& - 5384376x^2y - 6781362x^2 - 31104xy^6 - 1125792xy^5 - 1009080xy^4 \\
& + 4306350xy^3 + 3675528xy^2 - 7348002xy + 18557340x - 20736y^7 - 953856y^6 \\
& + 512352y^5 - 7437240y^4 + 12092718y^3 - 2570328y^2 + 4557846y + 12140238 \\
i = 32, m = 36, \text{ neighbours } 112, 46, 57, 64, 91 \\
W \quad & 72x^5 - 360x^4 + 1008x^3 - 576x^2 + 72 \\
A \quad & -864x^6 + 8640x^5 - 46656x^4 + 1152x^3y^2 - 5808x^3y + 124466x^3 \\
& + 144x^2y^3 - 288x^2y - 237600x^2 - 1440xy^3 - 2256xy^2 + 9934xy + 219256x \\
& - 144y^4 - 1440y^3 + 29184y^2 + 83708y - 15552 \\
B \quad & -6912x^5y^2 + 34848x^5y + 49812x^5 - 13824x^4y^2 + 69696x^4y + 99624x^4 \\
& - 6912x^3y^3 + 20736x^3y^2 + 120960x^3y + 114048x^3 + 864x^2y^4 \\
& + 27648x^2y^3 - 49536x^2y^2 - 38256x^2y + 12910x^2 - 1152xy^5 + 624xy^4 \\
& - 49730xy^3 + 42096xy^2 - 26402xy + 11946x - 144y^6 - 1440y^5 + 29040y^4 + 87308y^3 \\
& + 40704y^2 + 167324y + 109628 \\
i = 33, m = 20736, \text{ neighbours } 116, 6, 30, 44, 57, 69, 72, 92 \\
W \quad & 1907712x^5 - 1451520x^4 + 2363904x^3 - 331776x^2 + 829440x + 414720 \\
A \quad & -573308928x^9 - 573308928x^8y - 1719926784x^8 - 1719926784x^7y + 2149908480x^7 \\
& + 1146617856x^6y^2 - 1863254016x^6y - 597865660416x^6 + 95551488x^5y^3 \\
& + 5016453120x^5y^2 - 6401949696x^5y + 925416161280x^5 + 95551488x^4y^4 \\
& + 1289945088x^4y^3 + 238878720x^4y^2 + 382205952x^4y - 1861749080064x^4 \\
& - 15925248x^3y^5 - 1537449984x^3y^4 + 4429513728x^3y^3
\end{aligned}$$

$$\begin{aligned}
& -120473465472x^3y^2 - 92747857008x^3y + 1015261382470x^3 - 191102976x^2y^5 \\
& -1552711680x^2y^4 + 106746937344x^2y^3 - 14189395968x^2y^2 \\
& +102885064704x^2y - 1673902817280x^2 - 15925248xy^6 - 804888576xy^5 \\
& +1597473792xy^4 - 154228245120xy^3 + 211229134224xy^2 + 66065676550xy \\
& +2130945696488x - 318504960y^6 - 4197629952y^5 + 67830220800y^4 \\
& -292783613184y^3 + 141003720480y^2 - 1182427869556y - 165112971264 \\
B \quad & 95551488x^5y^5 + 10944626688x^5y^4 - 27007064064x^5y^3 \\
& +701915016960x^5y^2 + 588019133088x^5y + 2619621987420x^5 \\
& +191102976x^4y^5 + 21889253376x^4y^4 - 54014128128x^4y^3 \\
& +1403830033920x^4y^2 + 1176038266176x^4y + 5239243974840x^4 \\
& -95551488x^3y^6 - 10749542400x^3y^5 + 49352343552x^3y^4 \\
& -757054439424x^3y^3 + 845057359872x^3y^2 - 1419082924032x^3y \\
& +982006530048x^3 - 95551488x^2y^7 + 238878720x^2y^6 - 22820880384x^2y^5 \\
& -256212688896x^2y^4 + 798609079296x^2y^3 - 446863494528x^2y^2 \\
& +1152724401264x^2y + 926320965818x^2 + 15925248xy^8 + 1441898496xy^7 \\
& +6813711360xy^6 + 25503912576xy^5 + 350656944240xy^4 - 290781995206xy^3 \\
& +785445262320xy^2 + 291163975178xy + 1085464062510x + 318504960y^8 + 4197629952y^7 \\
& -65791789056y^6 + 284400297216y^5 - 169377812256y^4 + 1370402836084y^3 \\
& +310873120224y^2 + 1084422318676y + 1316582158708 \\
i = 34, m = 5184, \text{ neighbours } 114, 10, 37, 62, 64, 70, 91, 96, 99 \\
W \quad & 559872x^5 + 746496x^4 - 326592x^3 + 93312x^2 + 839808x + 419904 \\
A \quad & 107495424x^{11} - 53747712x^{10}y + 161243136x^9y - 644972544x^9 \\
& -107495424x^8y^2 + 214990848x^8y + 143327232x^8 - 17915904x^7y^3 \\
& -644972544x^7y^2 - 250822656x^7y - 313528320x^7 + 8957952x^6y^4 \\
& +282175488x^6y^3 - 421023744x^6y^2 + 6122760192x^6y - 52000911360x^6 \\
& -26873856x^5y^4 - 456855552x^5y^3 - 1997623296x^5y^2 - 4416270336x^5y \\
& -140816019456x^5 + 17915904x^4y^5 + 528519168x^4y^4 - 473278464x^4y^3 \\
& +7372394496x^4y^2 - 461334528x^4y - 28404172800x^4 + 2985984x^3y^6 \\
& +17169408x^3y^5 + 1209634560x^3y^4 + 891186624x^3y^3 \\
& -7347529872x^3y^2 + 5785521948x^3y + 85652175255x^3 - 1492992x^2y^7 \\
& -47029248x^2y^6 - 213497856x^2y^5 - 1252620288x^2y^4 \\
& +14315553792x^2y^3 + 11475883008x^2y^2 + 22088070144x^2y - 165991348224x^2 \\
& +4478976xy^7 + 76142592xy^6 + 1258903296xy^5 + 3722770368xy^4 \\
& +15288727536xy^3 - 6818030532xy^2 + 9097908063xy - 258089984814x - 1492992y^8 \\
& -51508224y^7 - 344881152y^6 - 1362044160y^5 + 5548264128y^4 + 27760306608y^3 \\
& +6050297772y^2 - 10000765317y - 86371299735 \\
B \quad & 541956096x^5y^5 + 1709102592x^5y^4 - 14730574464x^5y^3 \\
& -50215181472x^5y^2 - 77599326888x^5y - 79613622666x^5 + 541956096x^4y^5 \\
& +1709102592x^4y^4 - 14730574464x^4y^3 - 50215181472x^4y^2 \\
& -77599326888x^4y - 79613622666x^4 - 541956096x^3y^6 - 851005440x^3y^5 \\
& +17436653568x^3y^4 + 26891771904x^3y^3 - 1908043776x^3y^2 \\
& -43251978240x^3y - 116622084096x^3 - 8957952x^2y^8 - 219469824x^2y^7 \\
& -1948354560x^2y^6 - 22501628928x^2y^5 - 26983155456x^2y^4 \\
& -40761537984x^2y^3 + 32340215952x^2y^2 + 28463714532x^2y + 182510089833x^2 \\
& -2985984xy^9 - 26127360xy^8 - 888641280xy^7 - 2181131712xy^6 - 15508995696xy^5 \\
& -33719894748xy^4 - 125225392263xy^3 + 15558577668xy^2 + 18621762609xy \\
& +272803040466x + 1492992y^{10} + 53001216y^9 + 399375360y^8 + 2459393280y^7 \\
& -757563840y^6 - 28907105904y^5 - 46722051036y^4 - 65676291783y^3 \\
& -3602251404y^2 - 5779023027y + 88658124393 \\
i = 35, m = 4478976, \text{ neighbours } 114, 4, 11, 18, 57, 64, 69, 88, 100, 104, 105, 106 \\
W \quad & 779341824x^5 - 4340127744x^4 + 9002741760x^3 - 7121571840x^2 + 3359232000x \\
& -550914048 \\
A \quad & 160489808068608x^{15} + 160489808068608x^{14}y + 1203673560514560x^{14} \\
& -40122452017152x^{13}y^2 + 2688204285149184x^{13}y + 2688204285149184x^{13} \\
& -40122452017152x^{12}y^3 + 1283918464548864x^{12}y^2 + 13240409165660160x^{12}y \\
& +314292540801024x^{12} - 708829985636352x^{11}y^3 + 15577541995659264x^{11}y^2 \\
& +23869515412537344x^{11}y - 5483401775677440x^{11} - 448034047524864x^{10}y^4 \\
& +2397316508024832x^{10}y^3 + 58919820787187712x^{10}y^2 \\
& +7479493763530752x^{10}y - 6105299781943296x^{10} + 6687075336192x^9y^5 \\
& -4781258865377280x^9y^4 + 40008771736436736x^9y^3 \\
& +88105561091997696x^9y^2 - 18259059205472256x^9y - 480354911649792x^9 \\
& +6687075336192x^8y^6 - 1979374299512832x^8y^5 - 8298660492214272x^8y^4
\end{aligned}$$

$$\begin{aligned}
& +127417762480914432x^8y^3 + 39167872012910592x^8y^2 \\
& -18554962289098752x^8y + 8123682020917248x^8 + 118138330939392x^7y^6 \\
& -16358258041159680x^7y^5 + 34982877365010432x^7y^4 \\
& +149037077042823168x^7y^3 - 14538816293437440x^7y^2 - 5935893873426432x^7y \\
& +3014756464066560x^7 + 74672341254144x^6y^7 - 3990512206872576x^6y^6 \\
& -46007078313000960x^6y^5 + 113052810146217984x^6y^4 \\
& +45878352112779264x^6y^3 - 33664966267502592x^6y^2 + 20267410831441920x^6y \\
& -108918154315431936x^6 - 1114512556032x^5y^8 + 796876477562880x^5y^7 \\
& -29981502269816832x^5y^6 - 40962237228122112x^5y^5 \\
& +56401579666833408x^5y^4 - 58296808268365824x^5y^3 \\
& -22692868781506560x^5y^2 + 19917546764894208x^5y + 1132747095961239552x^5 \\
& -1114512556032x^4y^9 + 329895716585472x^4y^8 - 1627188331806720x^4y^7 \\
& -83437982507335680x^4y^6 - 28097976050122752x^4y^5 \\
& -92414266633617408x^4y^4 - 144664565657370624x^4y^3 \\
& +3992926984077312x^4y^2 + 20963516798730240x^4y - 5470677014321037312x^4 \\
& +185752092672x^3y^{10} - 144221020618752x^3y^9 + 5939393494450176x^3y^8 \\
& -10661376050675712x^3y^7 - 76925837260541952x^3y^6 \\
& -16059867463406592x^3y^5 - 59675242929437952x^3y^4 \\
& -44265131581370688x^3y^3 - 562096165201031376x^3y^2 \\
& +1902973289433254316x^3y + 10066306671494207139x^3 - 12445390209024x^2y^{10} \\
& +665085367812096x^2y^9 + 8779015403864064x^2y^8 \\
& -58805861878333440x^2y^7 - 87039529832153088x^2y^6 \\
& -166257689174212608x^2y^5 - 245586371011411968x^2y^4 \\
& -132690211796680704x^2y^3 - 90028048813129728x^2y^2 \\
& -23581398437462016x^2y - 24859627903627296768x^2 + 185752092672xy^{11} \\
& -132890142965760xy^{10} + 5049176590319616xy^9 + 1769896999305216xy^8 \\
& -37452867435417600xy^7 - 56886691415970816xy^6 - 172444688258284800xy^5 \\
& -150270971711844672xy^4 - 206611271287746768xy^3 + 232873577251981740xy^2 \\
& -994244490494830941xy + 19820720855100948480x - 11145125560320y^{11} \\
& +250037796077568y^{10} + 12868147782549504y^9 + 16570501198626816y^8 \\
& +16893602138419200y^7 + 147989463462927360y^6 + 467614094196895488y^5 \\
& +651322092689445312y^4 - 1388454895281414672y^3 + 4497505061656008924y^2 \\
& -9220894559752658865y - 15122115624561109341 \\
B \quad & -1114512556032x^5y^{10} + 747187792773120x^5y^9 - 19278102925541376x^5y^8 \\
& -86760093542547456x^5y^7 - 141137702337060864x^5y^6 \\
& -651750661380704256x^5y^5 - 991776493254578688x^5y^4 \\
& -594084569813112960x^5y^3 + 3139640523457832160x^5y^2 \\
& -11580288857741638152x^5y + 28917847836998531118x^5 - 1114512556032x^4y^{10} \\
& +747187792773120x^4y^9 - 19278102925541376x^4y^8 \\
& -86760093542547456x^4y^7 - 141137702337060864x^4y^6 \\
& -651750661380704256x^4y^5 - 991776493254578688x^4y^4 \\
& -594084569813112960x^4y^3 + 3139640523457832160x^4y^2 \\
& -11580288857741638152x^4y + 28917847836998531118x^4 + 1114512556032x^3y^{11} \\
& -746723412541440x^3y^{10} + 18966774678552576x^3y^9 \\
& +94792636428189696x^3y^8 + 177287741313122304x^3y^7 \\
& +710558037354479616x^3y^6 + 1263339268829872128x^3y^5 \\
& +1007324775335854080x^3y^4 - 2892105286035701760x^3y^3 \\
& +10272105306300874752x^3y^2 - 24092727479606181888x^3y \\
& +43653706849308377088x^3 + 1114512556032x^2y^{12} - 260795938111488x^2y^{11} \\
& -1364906376953856x^2y^{10} + 68578635498651648x^2y^9 \\
& +153270396903555072x^2y^8 + 473428165053726720x^2y^7 \\
& +985588760077848576x^2y^6 + 632357541011248128x^2y^5 \\
& -367847965646537472x^2y^4 + 9008244481988227392x^2y^3 \\
& -7568789321870036784x^2y^2 + 17493458703488005716x^2y \\
& -15097774534461134499x^2 - 185752092672xy^{13} + 144221020618752xy^{12} \\
& -5454766284668928xy^{11} - 8767100933554176xy^{10} + 52247091566333952xy^9 \\
& +73743887855195136xy^8 + 294545518088097024xy^7 + 3564449571104064xy^6 \\
& -77648521924962864xy^5 + 559424780953035156xy^4 - 7251785789782885203xy^3 \\
& +5268680775874615284xy^2 - 13921499303392979451xy + 13819855896047178426x \\
& +11145125560320y^{13} - 261182921637888y^{12} - 12595819735351296y^{11} \\
& -361818740146176y^{10} - 4613558918197248y^9 - 122741733408500736y^8 \\
& -390854669154792192y^7 - 627326912590342848y^6 + 1062307081775497680y^5 \\
& -6928510594124618604y^4 + 16827919054072187757y^3 - 4777178613441566940y^2
\end{aligned}$$

$$+9532585828247904177y + 14819572275125774685$$

$i = 36, m = 54, \text{neighbours } 116, 7, 24, 56, 61, 85, 87$

$W \quad 1080x^5 - 1944x^4 + 270x^3 + 324x^2 - 540x - 162$

$A \quad -3888x^7 - 3888x^6y - 198288x^6 - 15552x^5y + 718308x^5 - 19440x^4y^2$
 $-14580x^4y - 710856x^4 + 5832x^3y^3 - 211788x^3y^2 + 230832x^3y - 8664x^3$
 $+648x^2y^4 + 2916x^2y^3 + 5508x^2y^2 + 105300x^2y + 290952x^2 + 2592xy^4$
 $-113670xy^3 + 76428xy^2 - 192612xy - 168906x + 324y^5 + 7290y^4 - 69876y^3$
 $+140328y^2 + 93090y - 49572$

$B \quad -15552x^5y^3 + 585792x^5y^2 - 478656x^5y + 249552x^5 - 7776x^4y^3$
 $+292896x^4y^2 - 239328x^4y + 124776x^4 + 7776x^3y^4 - 297432x^3y^3$
 $+410184x^3y^2 - 264384x^3y - 440640x^3 + 8748x^2y^5 - 38880x^2y^4$
 $+454572x^2y^3 + 274536x^2y^2 - 453996x^2y + 31548x^2 - 2916xy^6$
 $+106380xy^5 + 71856xy^4 + 235938xy^3 + 35568xy^2 - 103344xy + 157482x - 324y^7$
 $-7290y^6 + 92880y^5 - 101286y^4 - 106320y^3 - 125856y^2 - 122610y - 20190$

$i = 37, m = 18, \text{neighbours } 116, 25, 34, 52, 58, 73, 79, 81, 82$

$W \quad 144x^5 + 72x^4 - 252x^3 - 126x^2 + 162x + 108$

$A \quad 3456x^{12} + 15552x^{11} + 19008x^{10}y + 10368x^{10} + 74304x^9y - 59328x^9$
 $-576x^8y^3 + 44928x^8y^2 + 37440x^8y - 131040x^8 - 2592x^7y^3$
 $+149472x^7y^2 - 229536x^7y - 38880x^7 - 3168x^6y^4 + 57888x^6y^3$
 $+47808x^6y^2 - 416448x^6y + 194784x^6 - 12384x^5y^4 + 174144x^5y^3$
 $-376272x^5y^2 - 84768x^5y + 297840x^5 + 96x^4y^6 - 7488x^4y^5$
 $+42456x^4y^4 + 41232x^4y^3 - 535872x^4y^2 + 488448x^4y + 122496x^4$
 $+4128x^3y^6 - 59496x^3y^5 + 222444x^3y^4 - 338592x^3y^3$
 $-155448x^3y^2 + 963936x^3y + 1216x^3 + 528x^2y^7 - 9648x^2y^6$
 $+17508x^2y^5 + 59760x^2y^4 - 385392x^2y^3 + 516408x^2y^2 + 123648x^2y$
 $-246048x^2 + 2064xy^7 - 29040xy^6 + 106740xy^5 - 162144xy^4 - 19152xy^3$
 $+482376xy^2 - 318232xy - 71684x + 240y^8 - 3528y^7 + 5208y^6 + 23784y^5$
 $-122418y^4 + 204600y^3 - 28416y^2 - 155820y + 72462$

$B \quad -11088x^5y^6 + 103752x^5y^5 - 230544x^5y^4 + 27576x^5y^3$
 $+334728x^5y^2 - 1118592x^5y - 449664x^5 + 5544x^3y^7 - 57420x^3y^6$
 $+167148x^3y^5 - 129060x^3y^4 - 153576x^3y^3 + 726660x^3y^2$
 $-334464x^3y - 382596x^3 - 48x^2y^9 + 3024x^2y^8 - 10692x^2y^7$
 $-10956x^2y^6 + 183252x^2y^5 - 309000x^2y^4 + 259320x^2y^3$
 $+220380x^2y^2 - 729516x^2y - 171188x^2 - 2064xy^9 + 26976xy^8 - 82068xy^7$
 $+115116xy^6 + 29976xy^5 - 403932xy^4 + 271150xy^3 - 410526xy^2 - 541070xy + 35770x$
 $-240y^{10} + 3288y^9 - 2160y^8 - 22008y^7 + 96138y^6 - 138102y^5 + 1908y^4$
 $+65178y^3 - 193836y^2 + 768y + 84126$

$i = 38, m = 3, \text{neighbours } 8, 60, 75, 87$

$W \quad 12x^4 + 12x^3$

$A \quad -24x^4 - 72x^3 - 6x^2y - 24x^2 - 27xy + 30x + 3y^2 - 6y + 18$

$B \quad 36x^5 - 18x^4 - 18x^3y + 30x^3 + 12x^2y^3 - 45x^2y^2 + 30x^2y + 15x^2$
 $+30xy^3 - 9xy^2 + 117xy - 9x - 3y^4 + 24y^3 + 6y^2 - 33y - 39$

$i = 39, m = 216, \text{neighbours } 116, 14, 26, 29, 56, 77, 92$

$W \quad 10368x^5 - 2592x^4 - 14256x^3 + 3888x^2 + 3888x - 1296$

$A \quad 248832x^7 - 497664x^6y - 17915904x^6 - 995328x^5y + 8522496x^5$
 $+995328x^4y^2 + 1119744x^4y + 48418560x^4 - 248832x^3y^3 + 8131968x^3y^2$
 $-20009952x^3y + 21226488x^3 + 82944x^2y^4 + 3234816x^2y^3$
 $+1140480x^2y^2 + 2706048x^2y - 42882048x^2 + 165888xy^4 - 1733184xy^3$
 $-3004272xy^2 + 5322156xy - 24883002x - 41472y^5 - 1907712y^4 + 1198368y^3$
 $-5862168y^2 + 8567430y + 17494206$

$B \quad 622080x^5y^3 - 26262144x^5y^2 + 57728160x^5y - 131548392x^5$
 $-311040x^4y^3 + 13131072x^4y^2 - 28864080x^4y + 65774196x^4$
 $+311040x^3y^4 - 12845952x^3y^3 + 16827264x^3y^2 - 39315456x^3y$
 $+26645760x^3 - 373248x^2y^5 + 3359232x^2y^4 - 12825216x^2y^3$
 $-4750272x^2y^2 - 4219920x^2y - 22671228x^2 + 124416xy^6 - 3723840xy^5$
 $+2560752xy^4 - 9028668xy^3 - 2380464xy^2 - 5626308xy + 1707330x - 41472y^7$
 $-1866240y^6 + 2649888y^5 - 9341496y^4 + 16546398y^3 - 3995928y^2 + 6887814y$
 $+19173822$

$i = 40, m = 6, \text{neighbours } 11, 19, 22, 65, 77, 78$

$W \quad 24x^4 + 72x^3 + 96x^2 + 96x$

$A \quad -96x^7 - 48x^6y + 384x^6 + 576x^5y - 576x^5 + 168x^4y^2 - 1488x^4y + 112x^4$

$$\begin{aligned}
& +192x^3y^3 - 1944x^3y^2 - 40x^3y + 192x^3 + 8x^2y^4 - 292x^2y^3 \\
& +1904x^2y^2 - 200x^2y - 2368x^2 - 96xy^4 + 1032xy^3 - 1276xy^2 - 1144xy + 672x \\
& -4y^5 + 118y^4 - 788y^3 + 680y^2 + 1960y - 2096 \\
B \quad & -528x^5y^3 + 2520x^5y^2 + 4656x^5y - 720x^5 - 264x^3y^4 + 1788x^3y^3 \\
& -192x^3y^2 - 5016x^3y - 9416x^3 - 72x^2y^5 + 840x^2y^4 - 824x^2y^3 \\
& -1488x^2y^2 - 2004x^2y - 4808x^2 - 96xy^6 + 840xy^5 - 20xy^4 - 130xy^3 \\
& +820xy^2 - 100xy + 712x - 4y^7 + 110y^6 - 608y^5 - 422y^4 + 734y^3 - 736y^2 + 592y \\
& +1048 \\
i = 41, m = 3, \text{ neighbours } 63, 78, 80 \\
W \quad & 12x^2 - 6x \\
A \quad & 12x^2 - 6y - 12 \\
B \quad & -108x^3 + 54x^2 - 18xy - 42x - 6y^3 - 9y^2 - 33y - 24 \\
i = 42, m = 1119744, \text{ neighbours } 14, 15, 18, 43, 47, 54, 55, 60, 75, 95, 98 \\
W \quad & 436700160x^5 + 243544320x^4 - 558472320x^3 - 311568768x^2 + 145286784x + 51228288 \\
A \quad & 13374150672384x^{15} + 13374150672384x^{14}y - 40122452017152x^{14} \\
& -53496602689536x^{13}y^2 - 160489808068608x^{13}y - 40122452017152x^{13} \\
& -53496602689536x^{12}y^3 + 46809527353344x^{12}y^2 + 267483013447680x^{12}y \\
& +251879837663232x^{12} + 425743796404224x^{11}y^3 + 755639512989696x^{11}y^2 \\
& +507103212994560x^{11}y - 158539411095552x^{11} + 238505686990848x^{10}y^4 \\
& -167176883404800x^{10}y^3 - 1292277308719104x^{10}y^2 - 1489267402997760x^{10}y \\
& -378655640911872x^{10} + 8916100448256x^9y^5 - 855945643032576x^9y^4 \\
& -2078565916999680x^9y^3 - 1004454441123840x^9y^2 + 344941636091904x^9y \\
& +522659950755840x^9 + 8916100448256x^8y^6 - 275284601339904x^8y^5 \\
& -186123596857344x^8y^4 + 1843310891630592x^8y^3 + 2686253888176128x^8y^2 \\
& +1649060640718848x^8y + 37939864928256x^8 - 70957299400704x^7y^6 \\
& -5572562780160x^7y^5 + 1228285712793600x^7y^4 + 1780758874423296x^7y^3 \\
& -244798039130112x^7y^2 - 1251574381412352x^7y - 381488360325120x^7 \\
& -39750947831808x^6y^7 - 92504542150656x^6y^6 + 957366285631488x^6y^5 \\
& +1586415747465216x^6y^4 - 697383012925440x^6y^3 - 956762591330304x^6y^2 \\
& -153454447558656x^6y - 31629626410401792x^6 - 1486016741376x^5y^8 \\
& +142657607172096x^5y^7 + 1216118950723584x^5y^6 + 1467023589900288x^5y^5 \\
& -592479518588928x^5y^4 - 1399084762005504x^5y^3 - 476546993750016x^5y^2 \\
& +282699205705728x^5y - 35387398984826880x^5 - 1486016741376x^4y^9 \\
& +45880766889984x^4y^8 + 280299907842048x^4y^7 - 353393356308480x^4y^6 \\
& -2488636880584704x^4y^5 - 3198000903487488x^4y^4 - 458923763957760x^4y^3 \\
& +723159985618944x^4y^2 + 92934093864960x^4y + 71356265871998976x^4 \\
& -495338913792x^3y^{10} + 43940689477632x^3y^9 + 249606094454784x^3y^8 \\
& +48536620498944x^3y^7 + 77963110612992x^3y^6 - 1059374794825728x^3y^5 \\
& +1041218936985600x^3y^4 + 1574933234010624x^3y^3 \\
& -25835537330211456x^3y^2 + 59896010880290976x^3y - 60429196034789288x^3 \\
& +6625157971968x^2y^{10} + 15417423691776x^2y^9 - 137642300669952x^2y^8 \\
& -578339140534272x^2y^7 - 790851144056832x^2y^6 - 1831585290780672x^2y^5 \\
& -2684469894119424x^2y^4 + 3785011054903296x^2y^3 - 42099937837056x^2y^2 \\
& +5524651639701504x^2y - 36584951272169472x^2 + 247669456896xy^{11} \\
& -23796906983424xy^{10} - 200853049835520xy^9 - 547846701907968xy^8 \\
& -1025610460250112xy^7 - 781218420387840xy^6 + 1231987385981952xy^5 \\
& +3110098123397376xy^4 + 8589820579319232xy^3 - 19289778421872xy^2 \\
& +8552933539199836xy + 64809438499740606x - 2817240072192y^{11} - 27640943345664y^{10} \\
& -92495942516736y^9 - 56610601795584y^8 + 48323928858624y^7 \\
& +1713464902121472y^6 + 4334010042106368y^5 - 587490743117952y^4 \\
& +148034236297248y^3 - 5983619372794248y^2 + 16031611347963842y + 28032900622980074 \\
B \quad & 1486016741376x^5y^{10} - 96343418732544x^5y^9 - 746032001974272x^5y^8 \\
& -1215959857496064x^5y^7 - 3924806632243200x^5y^6 \\
& +2075070410661888x^5y^5 + 1799528262727680x^5y^4 \\
& -2720906126286336x^5y^3 + 75818659505512320x^5y^2 - 180624107092881888x^5y \\
& +469192502011981560x^5 + 743008370688x^4y^{10} - 48171709366272x^4y^9 \\
& -373016000987136x^4y^8 - 607979928748032x^4y^7 - 1962403316121600x^4y^6 \\
& +1037535205330944x^4y^5 + 899764131363840x^4y^4 - 1360453063143168x^4y^3 \\
& +37909329752756160x^4y^2 - 90312053546440944x^4y + 234596251005990780x^4 \\
& +743008370688x^3y^{11} - 46995279446016x^3y^{10} - 449287874150400x^3y^9 \\
& -1198588596977664x^3y^8 - 2925038203305984x^3y^7 \\
& -2069603378528256x^3y^6 + 2542528206471168x^3y^5 + 64173478182912x^3y^4
\end{aligned}$$

$$\begin{aligned}
& +35755279069446144x^3y^3 - 30288948104577024x^3y^2 + 91602166224125952x^3y \\
& -92292377464848384x^3 + 743008370688x^2y^{12} - 13745654857728x^2y^{11} \\
& -103448436277248x^2y^{10} - 240879185952768x^2y^9 - 271976752152576x^2y^8 \\
& +825025085964288x^2y^7 + 338989124763648x^2y^6 - 2722218376384512x^2y^5 \\
& +5007885562598400x^2y^4 - 1562420477797632x^2y^3 \\
& -12367906190080704x^2y^2 + 3502437476956080x^2y + 1624164629563348x^2 \\
& +247669456896xy^{13} - 21227336368128xy^{12} - 154554340737024xy^{11} \\
& -349169359454208xy^{10} - 1052581277122560xy^9 - 1216742096719872xy^8 \\
& -1663563008535552xy^7 + 865723841637120xy^6 + 13129129568395584xy^5 \\
& -11567737628979024xy^4 + 40533013542966676xy^3 + 6906770136874896xy^2 \\
& -1048870759154852xy + 77080718048025534x - 2817240072192y^{13} - 24823703273472y^{12} \\
& -70489479315456y^{11} - 56701757915136y^{10} - 175601007157248y^9 \\
& +1619592008914944y^8 + 2956896685361664y^7 - 1388897037081216y^6 \\
& +7591929177359520y^5 - 10089825189276840y^4 + 25556597367866378y^3 \\
& -3336713002123656y^2 + 13412289163055042y + 30657238114391018
\end{aligned}$$

$i = 43, m = 18$, neighbours 22, 24, 42, 55, 77, 80

$$W \quad 72x^5 - 180x^4 + 396x^3 - 180x^2 + 18x + 18$$

$$\begin{aligned}
A \quad & 864x^8 - 3456x^7 + 5400x^6 - 864x^5y - 1656x^5 - 144x^4y^3 - 1296x^4y^2 \\
& +2880x^4y - 13824x^4 + 3168x^3y^2 - 8844x^3y + 30226x^3 - 900x^2y^3 \\
& -5454x^2y^2 + 6804x^2y - 45336x^2 + 144xy^4 + 972xy^3 + 5808xy^2 - 5477xy \\
& +25514x + 36y^5 + 792y^4 + 1476y^3 + 2460y^2 + 12595y - 324
\end{aligned}$$

$$\begin{aligned}
B \quad & 1728x^5y^3 + 1512x^5y^2 + 10764x^5y + 8790x^5 + 1728x^4y^3 \\
& +1512x^4y^2 + 10764x^4y + 8790x^4 + 864x^3y^4 + 1764x^3y^3 \\
& +6264x^3y^2 + 10674x^3y + 27468x^3 + 72x^2y^6 + 540x^2y^5 - 558x^2y^4 \\
& +4104x^2y^3 - 8802x^2y^2 + 1434x^2y - 3089x^2 + 72xy^6 - 612xy^5 \\
& +2640xy^4 - 7877xy^3 + 8256xy^2 - 5405xy - 1366x + 36y^7 + 792y^6 + 1836y^5 \\
& +3612y^4 + 13243y^3 + 1992y^2 + 16351y + 13567
\end{aligned}$$

$i = 44, m = 12$, neighbours 5, 7, 33, 47, 87

$$W \quad 24x^5 + 24x^4 + 72x^3$$

$$\begin{aligned}
A \quad & -96x^6 - 192x^5 - 696x^4 - 216x^3y - 1041x^3 + 16x^2y^3 + 24x^2y^2 \\
& +64x^2y - 912x^2 + 32xy^3 - 120xy^2 + 185xy + 627x - 16y^4 + 88y^3 - 256y^2 - 914y \\
& +1098
\end{aligned}$$

$$\begin{aligned}
B \quad & 1296x^5y + 3606x^5 - 2592x^4y - 7212x^4 + 1296x^3y^2 + 4848x^3y + 5496x^3 \\
& +96x^2y^4 + 456x^2y^3 - 1200x^2y^2 - 3744x^2y - 3591x^2 + 24xy^4 \\
& +257xy^3 + 960xy^2 + 3713xy + 1623x - 16y^6 + 72y^5 + 88y^4 - 34y^3 + 464y^2 - 1858y \\
& -1098
\end{aligned}$$

$i = 45, m = 6$, neighbours 47, 66, 82, 100

$$W \quad 54x^3 - 54x$$

$$A \quad 12x^4 + 66x^3 + 6x^2y - 402x^2 - 36xy - 540x - 18y^2 + 396y + 864$$

$$\begin{aligned}
B \quad & -756x^5 + 756x^4 - 756x^3y + 816x^3 - 12x^2y^3 - 108x^2y^2 - 660x^2y \\
& +1338x^2 - 54xy^3 - 432xy^2 - 180xy + 270x - 18y^4 + 360y^3 - 126y^2 + 522y + 504
\end{aligned}$$

$i = 46, m = 216$, neighbours 32, 56, 66, 74, 80, 101

$$W \quad 1944x^5 - 3888x^4 - 1944x^3 + 3888x^2$$

$$\begin{aligned}
A \quad & 62208x^8 - 311040x^7 + 93312x^6y - 334368x^6 - 590976x^5y + 2928960x^5 \\
& -10368x^4y^3 - 139968x^4y^2 + 806112x^4y - 1684800x^4 - 62208x^3y^3 \\
& +901584x^3y^2 - 2029716x^3y + 1558473x^3 - 15552x^2y^4 - 92016x^2y^3 \\
& +738720x^2y^2 - 475632x^2y + 3649968x^2 + 98496xy^4 + 82080xy^3 + 9252xy^2 \\
& +638787xy - 17011179x + 25920y^5 + 221616y^4 - 637200y^3 + 655524y^2 - 2045841y \\
& -14527794
\end{aligned}$$

$$\begin{aligned}
B \quad & 684288x^5y^3 - 5316192x^5y^2 + 14728824x^5y - 38098710x^5 + 684288x^4y^3 \\
& -5316192x^4y^2 + 14728824x^4y - 38098710x^4 + 684288x^3y^4 \\
& -3548448x^3y^3 + 995328x^3y^2 - 49248x^3y - 36288x^3 + 10368x^2y^6 \\
& -15552x^2y^5 - 476928x^2y^4 + 510624x^2y^3 - 222480x^2y^2 \\
& +2480724x^2y - 3497289x^2 + 72576xy^6 - 232848xy^5 + 1316916xy^4 - 5377353xy^3 \\
& +801972xy^2 + 120207xy - 12884034x + 25920y^7 + 169776y^6 - 1044144y^5 \\
& +2618100y^4 - 7790937y^3 + 2005524y^2 - 5028117y - 7263897
\end{aligned}$$

$i = 47, m = 144$, neighbours 27, 42, 44, 45, 100

$$W \quad 864x^4 - 2592x^2 + 1728x$$

$$\begin{aligned}
A \quad & 13824x^6 + 6912x^5 - 13824x^4y - 186624x^4 + 5760x^3y^2 + 218640x^3y \\
& +35050x^3 - 2304x^2y^3 - 38016x^2y^2 + 104832x^2y + 819072x^2 - 1152xy^3 \\
& -194448xy^2 - 444442xy - 1114302x + 2304y^4 + 24192y^3 - 506400y^2 + 53836y + 426068
\end{aligned}$$

B	$-34560x^5y^2 - 793440x^5y - 244860x^5 - 69120x^4y^2 - 1586880x^4y$ $-489720x^4 - 34560x^3y^3 - 829440x^3y^2 - 1071360x^3y + 3082752x^3$ $+317952x^2y^3 - 67968x^2y^2 - 1780752x^2y - 2551018x^2 - 5760xy^5$ $-253200xy^4 + 466070xy^3 + 389616xy^2 + 88934xy - 60222x + 2304y^6 + 26496y^5$ $-519072y^4 - 500948y^3 - 453408y^2 - 802292y - 384596$
$i = 48, m = 3, \text{neighbours } 116, 21, 22, 97, 98$	
W	$6x^3 - 12x^2 - 18x$
A	$3x^4 - 27x^3 - 18x^2y + 3x^2 + 51xy + 93x + 15y^2 + 51y + 36$
B	$90x^5 + 180x^4 + 90x^3y + 141x^3 - 3x^2y^3 + 144x^2y^2 - 21x^2y + 180x^2$ $+21xy^3 - 36xy^2 - 126xy + 93x + 15y^4 - 36y^3 - 21y^2 + 87y + 36$
$i = 49, m = 1, \text{neighbours } 13, 22$	
W	$2x^2 - 3x - 1$
A	0
B	$-4x^2 + 5x + 2y + 1$
$i = 50, m = 2592, \text{neighbours } 114, 8, 29, 58, 64, 70, 76, 106$	
W	$69984x^5 + 69984x^4 + 147744x^3 + 318816x^2 + 186624x + 139968$
A	$-53747712x^9 - 26873856x^8y - 376233984x^8 - 282175488x^7y - 927148032x^7$ $-40310784x^6y^2 - 900274176x^6y - 1559803392x^6 + 8957952x^5y^3$ $-67184640x^5y^2 - 510603264x^5y - 875639808x^5 + 4478976x^4y^4$ $+123171840x^4y^3 + 501645312x^4y^2 + 1941636096x^4y - 2446640640x^4$ $-1492992x^3y^5 + 24758784x^3y^4 + 287203968x^3y^3 + 2045742048x^3y^2$ $+2106099720x^3y - 6822847074x^3 + 6718464x^2y^5 + 200434176x^2y^4$ $+973990656x^2y^3 + 1525091328x^2y^2 - 122985216x^2y - 16030814976x^2$ $-1492992xy^6 + 12068352xy^5 + 68978304xy^4 - 18703008xy^3 - 2110886424xy^2$ $-2109796698xy - 12887940450x + 3732480y^6 + 59968512y^5 + 126665856y^4$ $-1195089120y^3 - 591958728y^2 - 7297033902y - 19956703548$
B	$8957952x^5y^5 + 133622784x^5y^4 + 1562105088x^5y^3 - 523858752x^5y^2$ $+4750786512x^5y - 19783275444x^5 - 8957952x^4y^5 - 133622784x^4y^4$ $-1562105088x^4y^3 + 523858752x^4y^2 - 4750786512x^4y + 19783275444x^4$ $-8957952x^3y^6 - 147806208x^3y^5 - 1773674496x^3y^4 - 1949474304x^3y^3$ $-3921343488x^3y^2 + 12261196800x^3y - 42065422848x^3 - 4478976x^2y^7$ $-127650816x^2y^6 - 806962176x^2y^5 - 1891372032x^2y^4$ $+4124587392x^2y^3 - 4882053600x^2y^2 + 19154479608x^2y - 42936336798x^2$ $+1492992xy^8 - 11321856xy^7 + 7661952xy^6 + 135394848xy^5 + 1484369784xy^4$ $+3220847490xy^3 - 1182310920xy^2 + 9033035106xy - 9790481664x - 3732480y^8$ $-52503552y^7 + 14919552y^6 + 1323258336y^5 - 1898550648y^4 + 12122247678y^3$ $-2903851800y^2 + 13535453094y + 182831262$
$i = 51, m = 6, \text{neighbours } 78, 89, 96, 104$	
W	$18x^4 + 72x^3 - 18x^2 - 72x$
A	$-42x^4 - 276x^3 + 18x^2y - 756x^2 - 72xy - 636x + 60y^2 - 474y + 414$
B	$-1008x^5 + 2016x^4 + 1008x^3y + 3930x^3 + 42x^2y^3 - 720x^2y^2 - 606x^2y$ $+1818x^2 + 192xy^3 + 288xy^2 - 1020xy - 1122x - 60y^4 + 702y^3 + 516y^2 + 66y - 162$
$i = 52, m = 1728, \text{neighbours } 5, 6, 37, 53, 58, 85, 91$	
W	$25920x^5 + 93312x^4 + 112320x^3 + 115776x^2 + 58752x + 8640$
A	$-3981312x^8 + 11943936x^7 + 3981312x^6y^2 - 21897216x^6y - 113965056x^6$ $+3981312x^5y^2 + 73654272x^5y - 811524096x^5 + 22560768x^4y^3$ $-40808448x^4y^2 - 31186944x^4y - 2432913408x^4 + 663552x^3y^4$ $-19270656x^3y^3 + 276094080x^3y^2 - 76710576x^3y - 4046103358x^3$ $-663552x^2y^5 + 46448640x^2y^4 - 17998848x^2y^3 - 104675328x^2y^2$ $-11529216x^2y - 6047142912x^2 - 663552xy^5 + 1299456xy^4 + 323455104xy^3$ $-356235312xy^2 + 162852050xy - 6082883142x - 663552y^6 + 25878528y^5$ $+110785536y^4 - 388650240y^3 + 225386592y^2 - 1760680996y - 2402227588$
B	$-3981312x^5y^4 + 187287552x^5y^3 - 734890752x^5y^2 + 195506208x^5y$ $-2656457868x^5 + 7962624x^4y^4 - 374575104x^4y^3 + 1469781504x^4y^2$ $-391012416x^4y + 5312915736x^4 + 3981312x^3y^5 - 191102976x^3y^4$ $+914374656x^3y^3 - 899776512x^3y^2 + 2843817984x^3y - 5612156928x^3$ $-18579456x^2y^6 + 87588864x^2y^5 - 54245376x^2y^4 + 1084105728x^2y^3$ $-692307072x^2y^2 + 2901451440x^2y - 1327174850x^2 - 663552xy^7$ $+52448256xy^6 - 225332352xy^5 - 270299472xy^4 + 1352744638xy^3 - 651149520xy^2$ $+1195330702xy + 2814720194x + 663552y^8 - 26542080y^7 - 71635968y^6 + 435071232y^5$ $-853855584y^4 + 2889415300y^3 - 288092256y^2 + 1835330596y + 2327577988$

$i = 53, m = 9$, neighbours 6, 52, 58, 75

$W \quad 72x^5 - 252x^4 + 180x^3 - 18x^2 - 18x$

$A \quad -864x^6 + 6048x^5 - 14904x^4 - 1728x^3y^2 - 288x^3y + 11004x^3 + 144x^2y^3$
 $+ 90x^2y - 7452x^2 - 1008xy^3 + 720xy^2 - 1212xy + 6816x + 72y^4 - 972y^3 + 1413y^2$
 $- 3084y$

$B \quad 5184x^5y^2 + 864x^5y + 13914x^5 + 5184x^4y^2 + 864x^4y + 13914x^4$
 $- 2592x^3y^3 + 2592x^3y^2 - 6453x^3y - 3159x^3 - 216x^2y^4 + 5184x^2y^3$
 $- 1539x^2y^2 + 2223x^2y + 3084x^2 + 864xy^5 + 1224xy^4 - 822xy^3 + 1683xy^2$
 $- 1641xy + 2361x - 72y^6 + 972y^5 - 1485y^4 + 3903y^3 - 441y^2 + 1914y + 3246$

$i = 54, m = 18$, neighbours 7, 42, 80

$W \quad 144x^5 - 216x^4 - 216x^3 + 198x^2 + 72x - 54$

$A \quad -3456x^6 + 10368x^5 + 2592x^4 - 2880x^3y^2 - 1968x^3y - 21116x^3$
 $+ 576x^2y^3 + 576x^2y + 1872x^2 - 1728xy^3 - 120xy^2 - 1810xy + 8745x + 288y^4$
 $- 1440y^3 - 132y^2 + 1909y + 347$

$B \quad 8640x^5y^2 + 5904x^5y - 10092x^5 + 4320x^4y^2 + 2952x^4y - 5046x^4$
 $- 4320x^3y^3 + 3888x^3y^2 + 9720x^3y + 25164x^3 - 864x^2y^4$
 $- 1296x^2y^3 + 4464x^2y^2 + 2172x^2y + 2242x^2 + 1440xy^5 + 2712xy^4$
 $+ 6958xy^3 + 1740xy^2 + 5590xy - 1299x - 288y^6 + 1152y^5 + 996y^4 + 239y^3$
 $+ 1788y^2 + 1631y + 1967$

$i = 55, m = 72$, neighbours 23, 29, 42, 43, 81, 104

$W \quad 1296x^4 - 2376x^3 + 2376x^2 - 1080x + 216$

$A \quad 13824x^8 - 13824x^7 + 27648x^6y - 17280x^6 - 62208x^5y + 1152x^5$
 $- 2304x^4y^3 - 13824x^4y^2 - 16128x^4y - 290304x^4 + 18432x^3y^3$
 $+ 38592x^3y^2 + 161040x^3y + 191236x^3 - 4608x^2y^4 - 21312x^2y^3$
 $+ 48096x^2y^2 - 58752x^2y - 2057952x^2 + 10368xy^4 + 33696xy^3 - 34248xy^2$
 $- 49594xy + 949201x - 1728y^5 - 1440y^4 + 52272y^3 + 261588y^2 - 805259y - 968207$

$B \quad -48384x^5y^3 - 167616x^5y^2 - 106416x^5y + 2811444x^5 - 24192x^4y^3$
 $- 83808x^4y^2 - 53208x^4y + 1405722x^4 + 24192x^3y^4 + 93888x^3y^3$
 $+ 88128x^3y^2 - 1383552x^3y + 3649536x^3 + 1152x^2y^6 + 12096x^2y^5$
 $+ 66240x^2y^4 + 181440x^2y^3 - 518400x^2y^2 + 1312248x^2y - 1520066x^2$
 $- 8640xy^6 - 25344xy^5 - 66696xy^4 - 150050xy^3 + 205032xy^2 - 649550xy + 1407117x$
 $+ 1728y^7 - 288y^6 - 60624y^5 - 197796y^4 + 983327y^3 - 308244y^2 + 851915y$
 $+ 921551$

$i = 56, m = 36$, neighbours 116, 36, 39, 46, 81, 108

$W \quad 432x^4 - 216x^3 - 432x^2 + 216x$

$A \quad 3456x^6 + 6912x^5 + 5184x^4y - 54432x^4 - 1728x^3y^2 - 15408x^3y - 53580x^3$
 $- 576x^2y^3 - 6048x^2y^2 - 36432x^2y + 97632x^2 - 1152xy^3 - 12456xy^2$
 $- 402xy + 52497x - 288y^4 - 3168y^3 + 4860y^2 + 39423y - 40827$

$B \quad 5184x^5y^2 + 25488x^5y + 146052x^5 - 2592x^4y^2 - 12744x^4y - 73026x^4$
 $- 2592x^3y^3 - 10368x^3y^2 - 61344x^3y - 94176x^3 - 1728x^2y^4$
 $+ 25056x^2y^3 + 14688x^2y^2 + 56520x^2y + 13398x^2 + 864xy^5 + 9864xy^4$
 $+ 18006xy^3 + 34920xy^2 + 71970xy - 8421x + 288y^6 + 3456y^5 + 4068y^4 - 31899y^3$
 $- 18684y^2 - 62967y - 40827$

$i = 57, m = 144$, neighbours 8, 18, 26, 32, 33, 35

$W \quad 1152x^5 + 432x^4 + 1224x^3 + 1008x^2 + 504x$

$A \quad -193536x^6 + 41472x^5y - 304128x^5 + 13824x^4y^2 - 172800x^4y - 259200x^4$
 $- 13824x^3y^3 - 123264x^3y^2 + 608736x^3y - 627784x^3 + 52992x^2y^3$
 $- 169344x^2y^2 - 144000x^2y - 457056x^2 - 6912xy^4 - 55872xy^3 + 283920xy^2$
 $+ 117076xy - 233842x + 26496y^4 + 288y^3 - 161400y^2 - 318662y - 183470$

$B \quad 41472x^5y^3 + 349056x^5y^2 - 1141920x^5y - 269736x^5 - 20736x^4y^3$
 $- 174528x^4y^2 + 570960x^4y + 134868x^4 - 20736x^3y^4 - 196992x^3y^3$
 $+ 381888x^3y^2 + 753408x^3y - 235008x^3 - 6912x^2y^5 + 6912x^2y^4$
 $+ 438912x^2y^3 - 178560x^2y^2 + 44688x^2y - 163900x^2 + 6912xy^6$
 $+ 75456xy^5 - 420144xy^4 + 64772xy^3 - 283920xy^2 - 117076xy + 233842x - 26496y^6$
 $- 15264y^5 + 129432y^4 + 203918y^3 + 161400y^2 + 318662y + 183470$

$i = 58, m = 52488$, neighbours 114, 4, 6, 18, 37, 50, 52, 53, 62, 63, 73, 80, 86, 100

$W \quad 16376256x^5 + 43460064x^4 + 11967264x^3 - 10077696x^2 + 6141096x - 1732104$

$A \quad 352638738432x^{19} + 528958107648x^{18}y - 1234235584512x^{18} + 1057916215296x^{17}y$
 $+ 88159684608x^{17} - 176319369216x^{16}y^3 + 3790866438144x^{16}y^2$
 $- 9036367672320x^{16}y + 3335374734336x^{16} + 205705930752x^{15}y^3$
 $- 4055345491968x^{15}y^2 + 8536796126208x^{15}y - 2101139149824x^{15}$

$$\begin{aligned}
& -1322395269120x^{14}y^4 + 11181586664448x^{14}y^3 - 19637569746432x^{14}y^2 \\
& + 9242073603072x^{14}y - 977103171072x^{14} + 1895433219072x^{13}y^4 \\
& - 25639774940160x^{13}y^3 + 35998537881600x^{13}y^2 - 14818173654528x^{13}y \\
& - 2972940475392x^{13} + 29386561536x^{12}y^6 - 3937799245824x^{12}y^5 \\
& + 16280155090944x^{12}y^4 - 3815355239424x^{12}y^3 - 10924454251008x^{12}y^2 \\
& + 5775683781888x^{12}y + 1370148431616x^{12} - 34284321792x^{11}y^6 \\
& + 7508266472448x^{11}y^5 - 53449257673728x^{11}y^4 + 48034783710720x^{11}y^3 \\
& - 9529817018112x^{11}y^2 - 13291296894720x^{11}y + 13593733590528x^{11} \\
& + 220399211520x^{10}y^7 - 5918943269376x^{10}y^6 + 8838008381952x^{10}y^5 \\
& + 47633167369728x^{10}y^4 - 42620309747712x^{10}y^3 + 8752909797504x^{10}y^2 \\
& + 26162610847488x^{10}y - 16395048383616x^{10} - 315905536512x^9y^7 \\
& + 15927516352512x^9y^6 - 53133352137216x^9y^5 + 4435534131840x^9y^4 \\
& + 29280035250432x^9y^3 - 17704791105408x^9y^2 - 2570711914368x^9y \\
& - 7160321420928x^9 - 4897760256x^8y^9 + 656299874304x^8y^8 \\
& - 4426350831360x^8y^7 - 9016164411264x^8y^6 + 77665927149504x^8y^5 \\
& - 25866602462016x^8y^4 - 31296994145856x^8y^3 + 44077087313856x^8y^2 \\
& - 44256773893248x^8y + 25546309348608x^8 + 5714053632x^7y^9 \\
& - 1251377745408x^7y^8 + 18943924450176x^7y^7 - 20827725488640x^7y^6 \\
& - 47462664090816x^7y^5 + 31547392138944x^7y^4 + 9343804165056x^7y^3 \\
& - 59606762682240x^7y^2 + 51465868843392x^7y - 10723884166080x^7 \\
& - 36733201920x^6y^{10} + 986490544896x^6y^9 - 484878265344x^6y^8 \\
& - 22036553941824x^6y^7 + 48401656564896x^6y^6 + 14996941903872x^6y^5 \\
& - 66029767111296x^6y^4 + 26437871776320x^6y^3 + 45003631313952x^6y^2 \\
& - 8717843194560x^6y - 57119176643328x^6 + 52650922752x^5y^{10} \\
& - 2653769765376x^5y^9 + 11881660271040x^5y^8 + 7661627590464x^5y^7 \\
& - 43380278880768x^5y^6 - 149840852832x^5y^5 + 77103041848416x^5y^4 \\
& - 266813481824x^5y^3 - 16959464234784x^5y^2 - 30910819354560x^5y \\
& - 224824507972416x^5 + 816293376x^4y^{12} - 109383312384x^4y^{11} \\
& + 738541431936x^4y^{10} + 2701795025664x^4y^9 - 20059695458496x^4y^8 \\
& + 7449544367712x^4y^7 + 28921325330016x^4y^6 - 37773261717696x^4y^5 \\
& - 61007726188800x^4y^4 - 12248329051872x^4y^3 - 15528893087232x^4y^2 \\
& + 27834821840448x^4y - 381011559962976x^4 - 28162121472x^3y^{12} \\
& + 1043268284160x^3y^{11} - 3571755912000x^3y^{10} - 6812265514752x^3y^9 \\
& + 5831233092000x^3y^8 - 3770807274144x^3y^7 - 25111345060080x^3y^6 \\
& + 17811380298816x^3y^5 - 44910067949568x^3y^4 - 116341358158944x^3y^3 \\
& - 98444960959392x^3y^2 - 241245753038664x^3y - 381605015084924x^3 \\
& + 6122200320x^2y^{13} - 164415090816x^2y^{12} + 86935244544x^2y^{11} \\
& + 3830779783008x^2y^{10} - 9212474465136x^2y^9 - 6351884868672x^2y^8 \\
& + 14607136307664x^2y^7 + 3397104086544x^2y^6 - 23172749290656x^2y^5 \\
& - 5093667831888x^2y^4 - 4221540555840x^2y^3 - 26066809749888x^2y^2 \\
& + 1999805397120x^2y + 137299191022944x^2 - 8775153792xy^{13} + 458031283200xy^{12} \\
& - 2457732754080xy^{11} - 2099233835424xy^{10} + 11588036100480xy^9 \\
& + 8876719926576xy^8 - 20430450931680xy^7 + 4414484713680xy^6 \\
& + 31233734393904xy^5 + 45532964626128xy^4 + 112761038897640xy^3 \\
& + 84003517234572xy^2 + 181553855675306xy + 324505916991164x + 2857026816y^{14} \\
& - 44658050112y^{13} - 580713375168y^{12} + 2142334566576y^{11} + 4618864113264y^{10} \\
& - 713257017552y^9 - 5769719115552y^8 + 7882952736960y^7 + 17658396738648y^6 \\
& + 30123240096240y^5 + 58396795178976y^4 + 49375440536904y^3 \\
& + 117525724819068y^2 + 153116033683006y - 3000184266816 \\
B \quad & 81629337600x^5y^{12} - 2504115979776x^5y^{11} + 1258815085056x^5y^{10} \\
& + 26864290586880x^5y^9 + 22106375998848x^5y^8 - 21614455103616x^5y^7 \\
& + 998304403968x^5y^6 + 78818140599552x^5y^5 + 251243940908736x^5y^4 \\
& + 461944930254624x^5y^3 + 405134530393104x^5y^2 + 713113949096280x^5y \\
& + 803060609643348x^5 - 81629337600x^4y^{12} + 2504115979776x^4y^{11} \\
& - 1258815085056x^4y^{10} - 26864290586880x^4y^9 - 22106375998848x^4y^8 \\
& + 21614455103616x^4y^7 - 998304403968x^4y^6 - 78818140599552x^4y^5 \\
& - 251243940908736x^4y^4 - 461944930254624x^4y^3 - 405134530393104x^4y^2 \\
& - 713113949096280x^4y - 803060609643348x^4 - 40814668800x^3y^{13} \\
& + 1286070213888x^3y^{12} - 1672789200768x^3y^{11} - 12907639008000x^3y^{10} \\
& + 140266411776x^3y^9 + 20018217551328x^3y^8 - 9505175161824x^3y^7 \\
& - 38993110131456x^3y^6 - 92781078537888x^3y^5 - 126287489748672x^3y^4 \\
& - 10090210923792x^3y^3 - 187750920217680x^3y^2 - 104399492698224x^3y \\
& + 18156351562992x^3 - 408146688x^2y^{15} + 46120575744x^2y^{14}
\end{aligned}$$

$$\begin{aligned}
& -154075374720x^2y^{13} - 2078521020864x^2y^{12} + 4696903901520x^2y^{11} \\
& + 8811732049344x^2y^{10} + 6114849270624x^2y^9 + 3891734517312x^2y^8 \\
& + 21841599897936x^2y^7 + 61888987683216x^2y^6 + 149735494023408x^2y^5 \\
& + 227499699067776x^2y^4 + 375818225336784x^2y^3 + 266756744587032x^2y^2 \\
& + 348313880877780x^2y + 321537809835838x^2 + 14489207424xy^{15} \\
& - 547347383424xy^{14} + 1307938184352xy^{13} + 5895702842688xy^{12} \\
& - 1119815523648xy^{11} - 5360437427040xy^{10} + 7851170308176xy^9 \\
& + 2196430977744xy^8 + 14374805708016xy^7 + 27982143037728xy^6 \\
& - 8630023090320xy^5 - 8803192876188xy^4 + 13879925692246xy^3 \\
& - 40670524193964xy^2 - 50256033527978xy - 116966612656944x - 2857026816y^{16} \\
& + 44658050112y^{15} + 577856348352y^{14} - 2100533543280y^{13} - 4059612451440y^{12} \\
& - 980603700768y^{11} - 412931634048y^{10} - 10482694504416y^9 - 12530875756392y^8 \\
& - 36435078697104y^7 - 84020794358904y^6 - 92412984343824y^5 \\
& - 203797688136780y^4 - 262342904194918y^3 - 165223376358276y^2 - 270906237555898y \\
& - 150115849416190
\end{aligned}$$

$i = 59, m = 18, \text{neighbours } 117, 7, 31, 63, 71, 75, 103$

$W \quad 720x^5 + 2520x^4 + 2520x^3 + 720x^2$

$A \quad -2160x^7 - 2160x^6y - 88560x^6 - 1080x^5y - 590760x^5 - 3240x^4y^2$
 $+ 14040x^4y - 1654200x^4 + 360x^3y^3 - 178020x^3y^2 + 136740x^3y$
 $- 2373850x^3 + 360x^2y^4 + 18540x^2y^3 + 8460x^2y^2 - 5760x^2y$
 $- 1558080x^2 + 180xy^4 + 103050xy^3 + 94440xy^2 + 5075xy - 464575x - 180y^5$
 $- 7830y^4 - 77130y^3 + 233520y^2 - 137315y - 11065$

$B \quad 577800x^5y^2 - 378900x^5y + 212250x^5 - 577800x^4y^2 + 378900x^4y$
 $- 212250x^4 + 288900x^3y^3 - 237600x^3y^2 + 137700x^3y + 857340x^3$
 $+ 2160x^2y^5 + 25380x^2y^4 + 453780x^2y^3 + 38250x^2y^2 + 752250x^2y$
 $+ 1056245x^2 - 180xy^6 + 86850xy^5 + 31710xy^4 + 509765xy^3 + 160140xy^2$
 $+ 723335xy + 459990x - 180y^7 - 8010y^6 - 82080y^5 + 159630y^4 - 36725y^3$
 $+ 115080y^2 + 188605y + 17545$

$i = 60, m = 432, \text{neighbours } 5, 10, 11, 18, 38, 42, 70, 78$

$W \quad 15552x^5 + 41472x^4 - 44064x^3 - 36288x^2 + 23328x$

$A \quad 1990656x^8 - 995328x^7 - 2488320x^6y^2 - 5971968x^6y - 50761728x^6$
 $- 1741824x^5y^2 - 10450944x^5y - 221045760x^5 + 497664x^4y^4$
 $+ 1658880x^4y^3 + 7589376x^4y^2 + 21067776x^4y - 34504704x^4$
 $+ 3068928x^3y^4 + 15683328x^3y^3 - 12886848x^3y^2 + 323310288x^3y$
 $+ 147744028x^3 + 663552x^2y^5 + 3608064x^2y^4 + 17832960x^2y^3$
 $+ 13893120x^2y^2 + 4105728x^2y + 85238784x^2 - 953856xy^5 - 8982144xy^4$
 $- 2906784xy^3 - 68154648xy^2 - 131666834xy + 236765341x - 456192y^6 - 2363904y^5$
 $- 5576256y^4 - 2601360y^3 + 30527508y^2 - 61134161y - 222417778$

$B \quad -7713792x^5y^4 - 17065728x^5y^3 + 156171456x^5y^2 - 784675440x^5y$
 $+ 2057280684x^5 + 3856896x^4y^4 + 8532864x^4y^3 - 78085728x^4y^2$
 $+ 392337720x^4y - 1028640342x^4 - 3856896x^3y^5 - 15925248x^3y^4$
 $+ 61731072x^3y^3 - 242673408x^3y^2 + 276659712x^3y - 678896640x^3$
 $- 248832x^2y^7 - 580608x^2y^6 - 4105728x^2y^5 - 38651904x^2y^4$
 $- 214413696x^2y^3 - 89035488x^2y^2 - 360617064x^2y + 276773746x^2$
 $- 1410048xy^7 - 10226304xy^6 - 13494240xy^5 - 163296744xy^4 - 45356846xy^3$
 $- 157412184xy^2 - 26748098xy - 51416647x - 456192y^8 - 1824768y^7 - 6675264y^6$
 $- 12786192y^5 - 5961228y^4 - 139503881y^3 + 12852516y^2 - 89157509y - 111208889$

$i = 61, m = 3, \text{neighbours } 112, 11, 36, 63, 78$

$W \quad 12x^3 - 6x^2 - 6x$

$A \quad 144x^3 + 24x^2y - 96x^2 - 84xy - 132x - 12y^2 - 12y + 24$

$B \quad -288x^5 - 144x^4 - 144x^3y - 180x^3 - 72x^2y - 72xy^3 - 72xy^2 - 156xy - 24x$
 $- 12y^4 - 6y^3 - 66y^2 - 36y + 12$

$i = 62, m = 1, \text{neighbours } 113, 5, 34, 58, 71$

$W \quad 12x^4 - 18x^3 + 12x^2 - 6x$

$A \quad -24x^4 + 56x^3 + 2x^2y - 96x^2 - 6xy + 68x + 5y^2 + 30y - 35$

$B \quad 42x^5 + 42x^4 + 21x^3y + 114x^3 + 12x^2y^3 - 3x^2y^2 - 15x^2y - 22x^2$
 $- 16xy^3 - 6xy^2 + 4xy + 25x + 5y^4 + 32y^3 - 4y^2 + 34y + 35$

$i = 63, m = 81, \text{neighbours } 2, 16, 27, 41, 58, 59, 61, 71$

$W \quad 2592x^5 - 9072x^4 - 648x^3 + 972x^2 + 1944x + 324$

$A \quad 279936x^{10} - 419904x^9y + 139968x^8y^2 - 419904x^8y - 629856x^8$
 $+ 489888x^7y^2 + 944784x^7y - 116640x^7 - 186624x^6y^3 - 314928x^6y^2$

$$\begin{aligned}
& +933120x^6y - 851472x^6 + 69984x^5y^4 + 69984x^5y^3 - 915624x^5y^2 \\
& -542376x^5y + 7978176x^5 - 23328x^4y^5 + 34992x^4y^4 + 554040x^4y^3 \\
& +262440x^4y^2 - 489888x^4y - 13015080x^4 + 23328x^3y^6 - 77760x^3y^5 \\
& -607176x^3y^4 + 593352x^3y^3 + 2889792x^3y^2 - 5486532x^3y + 10363552x^3 \\
& +31104x^2y^6 + 122472x^2y^5 - 241056x^2y^4 - 433512x^2y^3 \\
& +189540x^2y^2 + 234252x^2y - 6804x^2 - 11664xy^7 - 21384xy^6 + 186948xy^5 \\
& +172800xy^4 - 2108448xy^3 - 1133610xy^2 + 1263430xy - 12993828x - 21384y^7 \\
& -55404y^6 + 153900y^5 - 90288y^4 - 1059714y^3 + 1476732y^2 - 2043878y - 4197140 \\
B \quad & -69984x^5y^6 - 11664x^5y^5 + 614304x^5y^4 - 1610928x^5y^3 \\
& -5974992x^5y^2 + 16077600x^5y - 38952192x^5 - 69984x^4y^6 - 11664x^4y^5 \\
& +614304x^4y^4 - 1610928x^4y^3 - 5974992x^4y^2 + 16077600x^4y \\
& -38952192x^4 - 34992x^3y^7 - 81648x^3y^6 + 294516x^3y^5 - 139968x^3y^4 \\
& -4732668x^3y^3 + 1565892x^3y^2 - 2058696x^3y + 523908x^3 + 11664x^2y^8 \\
& -23328x^2y^7 - 259524x^2y^6 + 88452x^2y^5 + 503334x^2y^4 \\
& +823554x^2y^3 - 377640x^2y^2 + 1987050x^2y - 1177136x^2 - 11664xy^9 \\
& +33048xy^8 + 210276xy^7 - 346248xy^6 - 715815xy^5 + 1396992xy^4 - 6908741xy^3 \\
& +264630xy^2 - 1729640xy - 7746928x - 21384y^9 - 34020y^8 + 153414y^7 - 366417y^6 \\
& -601362y^5 + 2240121y^4 - 5388074y^3 + 1555464y^2 - 2148854y - 4092164
\end{aligned}$$

$i = 64$, $m = 3439853568$, neighbours 114, 4, 15, 17, 21, 25, 32, 34, 35, 50, 70, 93, 101

$$\begin{aligned}
W \quad & 8338205048832x^5 + 1976938883186688x^4 - 1090199671013376x^3 \\
& -2603274300555264x^2 + 3579628577882112x - 1058043919859712 \\
A \quad & -11832592569282330624x^{17} + 5916296284641165312x^{16}y + 5916296284641165312x^{16} \\
& +47330370277129322496x^{15}y^2 + 35497777707846991872x^{15}y \\
& +130158518262105636864x^{15} - 23665185138564661248x^{14}y^3 \\
& +5916296284641165312x^{14}y^2 - 35497777707846991872x^{14}y \\
& +7888395046188220416x^{14} + 120298024454370361344x^{13}y^3 \\
& -17748888539234959360x^{13}y^2 - 176502839158461431808x^{13}y \\
& -495982838529084358656x^{13} - 71981604796467511296x^{12}y^4 \\
& +229749505720231919616x^{12}y^3 - 215944814389402533888x^{12}y^2 \\
& -451610616394275618816x^{12}y - 525564319952290185216x^{12} \\
& -7888395046188220416x^{11}y^5 - 266233332808852439040x^{11}y^4 \\
& -1268059503674756431872x^{11}y^3 - 440764073205766815744x^{11}y^2 \\
& +226791357577911336960x^{11}y + 793441068395765170176x^{11} \\
& +3944197523094110208x^{10}y^6 + 70009506034920456192x^{10}y^5 \\
& +1057044936189221535744x^{10}y^4 + 17091522600074477568x^{10}y^3 \\
& +850960615607554277376x^{10}y^2 + 2063308329268606402560x^{10}y \\
& +1986396477568271253504x^{10} - 20049670742395060224x^9y^6 \\
& -940691109257945284608x^9y^5 - 2404317073452784680960x^9y^4 \\
& -869038187588402282496x^9y^3 + 1077094606931616595968x^9y^2 \\
& +1465269379829461942272x^9y + 324081563147566055424x^9 \\
& +11996934132744585216x^8y^7 + 364016563068893921280x^8y^6 \\
& +2279746168348395700224x^8y^5 + 2678767484434749849600x^8y^4 \\
& +4627365402406702546944x^8y^3 + 2060185839562823565312x^8y^2 \\
& -1898309399552502792192x^8y - 2641900193698050736128x^8 \\
& +1314732507698036736x^7y^8 + 44372222134808739840x^7y^7 \\
& -426959381874937430016x^7y^6 - 2034219872535787339776x^7y^5 \\
& -1863304646535042564096x^7y^4 + 1321141828673064665088x^7y^3 \\
& -638631315614321344512x^7y^2 - 3815216786036817395712x^7y \\
& -2202697365345175339008x^7 - 657366253849018368x^6y^9 \\
& -11668251005820076032x^6y^8 + 36155143961696010240x^6y^7 \\
& +2041122218201202032640x^6y^6 + 2714265262142596841472x^6y^5 \\
& +1611286858965675147264x^6y^4 - 1721231364921923469312x^6y^3 \\
& -1913565774693915426816x^6y^2 + 376506521892025270272x^6y \\
& -11585895367220314276626432x^6 + 3341611790399176704x^5y^9 \\
& +156781851542990880768x^5y^8 + 899030522920263745536x^5y^7 \\
& +1581376694415544811520x^5y^6 + 483164196579028500480x^5y^5 \\
& -263439526229994110976x^5y^4 - 4255871298200276041728x^5y^3 \\
& -4435195334198172844032x^5y^2 - 159329145776655826944x^5y \\
& -5494705997091557234297536512x^5 - 1999489022124097536x^4y^{10} \\
& -60669427178148986880x^4y^9 - 879145193741330939904x^4y^8 \\
& -1812194420298281385984x^4y^7 - 4862045155030802104320x^4y^6 \\
& -9066231031522199076864x^4y^5 - 10110073862113286160384x^4y^4
\end{aligned}$$

$$\begin{aligned}
& -10532112127171214966784x^4y^3 - 11794547497340818685952x^4y^2 \\
& -8156331140196278992896x^4y - 648351124708799030768934322176x^4 \\
& +776057383016202240x^3y^{11} + 10796708131070017536x^3y^{10} \\
& -401606446287610183680x^3y^9 - 2604252018335507546112x^3y^8 \\
& +105532296289736384839680x^3y^7 - 232355523726571888705536x^3y^6 \\
& -12325476830123925199060992x^3y^5 + 133248334119327164321685504x^3y^4 \\
& +701548927306940695834063872x^3y^3 - 25657954599635761881712390272x^3y^2 \\
& +106936989199197293709178248528x^3y + 189613269727599864139492573518x^3 \\
& +109561042308169728x^2y^{12} + 1944708500970012672x^2y^{11} \\
& -5916296284641165312x^2y^{10} - 882988960308975894528x^2y^9 \\
& -4535251956086108848128x^2y^8 - 10928796141021661495296x^2y^7 \\
& -17092316917631211798528x^2y^6 - 19355510238504214855680x^2y^5 \\
& -8975541878751611584512x^2y^4 + 1945579757768707814719488x^2y^3 \\
& +26080472011378551422976x^2y^2 + 1946991360497979991523328x^2y \\
& +1508384002713805364276729217024x^2 - 556935298399862784xy^{12} \\
& -28162133336914919424xy^{11} - 187204267858785730560xy^{10} \\
& +992516548747811880960xy^9 + 10958918615397920931840xy^8 \\
& -191237845679904224968704xy^7 + 514890795684642918432768xy^6 \\
& +25198492637327996382289920xy^5 - 272025196128381685404939264xy^4 \\
& -516514976698833196884051840xy^3 + 52385016717493054349009716656xy^2 \\
& -217413891867618529101625028910xy + 564133149979739689883737028652x \\
& -109561042308169728y^{13} + 2045139456419168256y^{12} + 132204418559096389632y^{11} \\
& +1651312050196426260480y^{10} - 14048268526135988453376y^9 \\
& -30331113938017113341952y^8 + 2498303926884058474217472y^7 \\
& -15607992632530688649068544y^6 - 235216931150534524343304192y^5 \\
& +4122141627579210211847399424y^4 - 1157978331412135538249634048y^3 \\
& +44008849160302259562433771680y^2 - 501464139349832459446260664356y \\
& -1136006768771632358482625362632 \\
B \quad & -5971076805795250176x^5y^{11} - 109152470921228845056x^5y^{10} \\
& +2835940693346749513728x^5y^9 + 22027437714685172908032x^5y^8 \\
& -601378894467760442572800x^5y^7 + 1448040626347883545755648x^5y^6 \\
& +74010982346760739077881856x^5y^5 - 799418108952243166869798912x^5y^4 \\
& -4209199519108050294043686912x^5y^3 + 153947791613619445178843308800x^5y^2 \\
& -641621918265482821669758611424x^5y + 3216261478485666397529674381356x^5 \\
& +11942153611590500352x^4y^{11} + 218304941842457690112x^4y^{10} \\
& -5671881386693499027456x^4y^9 - 44054875429370345816064x^4y^8 \\
& +1202757788935520885145600x^4y^7 - 2896081252695767091511296x^4y^6 \\
& -148021964693521478155763712x^4y^5 + 1598836217904486333739597824x^4y^4 \\
& +8418399038216100588087373824x^4y^3 - 307895583227238890357686617600x^4y^2 \\
& +1283243836530965643339517222848x^4y - 6432522956971332795059348762712x^4 \\
& -5971076805795250176x^3y^{12} - 132787983277501710336x^3y^{11} \\
& +2403878829283552002048x^3y^{10} + 33253036292516056399872x^3y^9 \\
& -514186953513798299811840x^3y^8 - 932417497587001539428352x^3y^7 \\
& +79742809826054444779831296x^3y^6 - 506457970496315241353183232x^3y^5 \\
& -7373562867044012829569974272x^3y^4 + 137286376850483412764920381440x^3y^3 \\
& -32245243128239184503503847424x^3y^2 + 676508052018130228420213211136x^3y \\
& -3245331237982557621801027895296x^3 + 2656855275973115904x^2y^{13} \\
& +72282897662814978048x^2y^{12} + 594907329646502608896x^2y^{11} \\
& -17660861501892596858880x^2y^{10} - 42415944498168201216000x^2y^9 \\
& +2292158407044940292947968x^2y^8 - 13772209472562478398308352x^2y^7 \\
& -219429159503730130552356864x^2y^6 + 3739365163664686964440498176x^2y^5 \\
& +1497101425693469906224619520x^2y^4 + 88965851646708943749754850304x^2y^3 \\
& +493547951188471685044589523072x^2y^2 - 1668852451843000911938941689168x^2y \\
& +6111754063108677180331742248626x^2 - 776057383016202240xy^{14} \\
& -22081495488811499520xy^{13} + 111424895647563644928xy^{12} \\
& +3539222747755994677248xy^{11} - 31043097746269909549056xy^{10} \\
& -122486878566815551193088xy^9 + 5707960192386275703717888xy^8 \\
& -27616837786551136205070336xy^7 - 597005253655303959038260224xy^6 \\
& +9671855877619641381668098176xy^5 + 11773202979171342487854304944xy^4 \\
& -100263083219656676610049236174xy^3 - 456357441667547188690730651088xy^2 \\
& +1148837551468320360757866388482xy - 4081615977789946323404990758578x \\
& -109561042308169728y^{15} + 2264261541035507712y^{14} + 127566334434717204480y^{13} \\
& +1398114959741102850048y^{12} - 17278213330928230465536y^{11}
\end{aligned}$$

$$\begin{aligned}
& +2033078089316007149568y^{10} + 2472043427582121127968768y^9 \\
& -20603558425700653405175808y^8 - 191531476135096484850450432y^7 \\
& +4492117041757046247896377344y^6 - 10393026973358202555713134848y^5 \\
& +68681820703891798791965762208y^4 - 635863627258835629722364180068y^3 \\
& +178571352599177921707424045472y^2 - 814597995955850486624226853572y \\
& -8274916209763844680432319844
\end{aligned}$$

$i = 65, m = 6, \text{ neighbours } 29, 40, 72, 81$

$$\begin{aligned}
W & 18x^4 + 18x^3 - 36x^2 \\
A & 42x^4 - 120x^3 - 6x^2 - 48xy + 252x - 42y^2 + 246y - 324 \\
B & 72x^5 + 144x^4 + 72x^3y + 318x^3 - 42x^2y^3 - 36x^2y^2 - 78x^2y + 210x^2 \\
& + 36xy^3 + 66xy + 54x - 42y^4 + 126y^3 + 102y^2 - 24y + 162
\end{aligned}$$

$i = 66, m = 3, \text{ neighbours } 45, 46, 82$

$$\begin{aligned}
W & 9x^2 \\
A & 3x^2 - 6x - 3y - 6 \\
B & 54x^3 - 9x^2y - 9x^2 - 3x - 3y^3 - 9y^2 - 39y - 39
\end{aligned}$$

$i = 67, m = 3, \text{ neighbours } 69, 70, 72$

$$\begin{aligned}
W & 12x^2 + 12x \\
A & 3x^2 - 6x + 3y - 6 \\
B & 99x^3 + 27x^2y + 108x^2 + 45xy + 15x - 3y^3 + 9y^2 - 3y - 39
\end{aligned}$$

$i = 68, m = 6, \text{ neighbours } 5, 16, 20, 69, 70$

$$\begin{aligned}
W & 48x^4 - 96x^3 + 120x^2 - 72x \\
A & -372x^4 + 48x^3y + 1305x^3 + 12x^2y^2 - 54x^2y - 3504x^2 + 24xy^2 - 171xy \\
& + 4398x + 12y^3 + 438y^2 - 810y - 3276 \\
B & -432x^5y + 1494x^5 - 864x^4y + 2988x^4 + 432x^3y^2 - 1512x^3y + 6600x^3 \\
& + 372x^2y^3 - 432x^2y^2 + 1152x^2y - 2637x^2 - 48xy^4 - 849xy^3 + 192xy^2 \\
& - 1203xy + 1293x - 12y^5 - 486y^4 + 1662y^3 - 102y^2 + 1080y + 1494
\end{aligned}$$

$i = 69, m = 20736, \text{ neighbours } 116, 6, 30, 33, 35, 67, 68, 77$

$$\begin{aligned}
W & 995328x^5 - 497664x^4 - 995328x^3 + 497664x^2 \\
A & -286654464x^9 - 286654464x^8y + 3153199104x^8 + 1863254016x^7y \\
& - 15479341056x^7 - 716636160x^6y^2 - 4013162496x^6y - 119487135744x^6 \\
& + 47775744x^5y^3 + 4729798656x^5y^2 + 238878720x^5y + 74625712128x^5 \\
& + 47775744x^4y^4 - 668860416x^4y^3 - 15001583616x^4y^2 + 18704203776x^4y \\
& + 414550130688x^4 - 7962624x^3y^5 - 819818496x^3y^4 - 10336826880x^3y^3 \\
& - 55070170560x^3y^2 - 394304465688x^3y - 1791711573311x^3 + 119439360x^2y^5 \\
& + 1528823808x^2y^4 + 15582855168x^2y^3 - 43714805760x^2y^2 \\
& + 110553071616x^2y - 14308835328x^2 - 7962624xy^6 - 772042752xy^5 \\
& - 2742474240xy^4 + 29755745856xy^3 + 213789181800xy^2 + 583861472593xy \\
& + 8336686628730x + 87588864y^6 + 3709919232y^5 + 39856942080y^4 + 83324822400y^3 \\
& + 318031126320y^2 + 1403203730974y - 5507164722428 \\
B & 47775744x^5y^5 + 3055656960x^5y^4 + 80940155904x^5y^3 \\
& + 519326315136x^5y^2 + 2048930284176x^5y + 8214189621114x^5 \\
& - 95551488x^4y^5 - 6111313920x^4y^4 - 161880311808x^4y^3 \\
& - 1038652630272x^4y^2 - 4097860568352x^4y - 16428379242228x^4 \\
& - 47775744x^3y^6 - 2866544640x^3y^5 - 68844847104x^3y^4 \\
& - 198938198016x^3y^3 + 6736379904x^3y^2 - 103840579584x^3y \\
& - 205961232384x^3 - 47775744x^2y^7 + 334430208x^2y^6 + 7795408896x^2y^5 \\
& + 61041143808x^2y^4 + 68623234560x^2y^3 + 110705024448x^2y^2 \\
& + 412148706072x^2y + 1486854550847x^2 + 7962624xy^8 + 963145728xy^7 \\
& + 12446922240xy^6 + 95559781824xy^5 + 321599418648xy^4 + 1465649405183xy^3 \\
& + 266877930840xy^2 + 1053373975079xy - 1352580822335x - 87588864y^8 - 3885096960y^7 \\
& - 48574688256y^6 - 180831853440y^5 - 722401731120y^4 - 2988388225534y^3 \\
& - 857208263088y^2 - 3421891969198y - 2753582361214
\end{aligned}$$

$i = 70, m = 41472, \text{ neighbours } 112, 34, 50, 60, 64, 67, 68, 104$

$$\begin{aligned}
W & 995328x^5 + 3981312x^4 + 4976640x^3 + 1990656x^2 \\
A & -1146617856x^{10} + 573308928x^9 - 5159780352x^8y + 8599633920x^8 \\
& - 2293235712x^7y - 955514880x^7 + 191102976x^6y^3 - 6306398208x^6y^2 \\
& + 40896036864x^6y - 165590728704x^6 - 95551488x^5y^3 - 21212430336x^5y^2 \\
& - 18728091648x^5y - 1351671349248x^5 + 859963392x^4y^4 + 859963392x^4y^3 \\
& + 71376961536x^4y^2 + 24461180928x^4y - 4286184947712x^4 + 79626240x^3y^5 \\
& + 3627638784x^3y^4 - 19946234880x^3y^3 + 277235005824x^3y^2 \\
& + 141350828784x^3y - 7055804477386x^3 - 31850496x^2y^6 + 1051066368x^2y^5
\end{aligned}$$

$$\begin{aligned}
& +2325086208x^2y^4 + 64752058368x^2y^3 - 9682550784x^2y^2 \\
& -39160184832x^2y - 6548860108800x^2 + 15925248xy^6 + 3372834816xy^5 \\
& -17964868608xy^4 + 220556771712xy^3 - 505867827216xy^2 - 12223194538xy \\
& -3121579663468x - 31850496y^7 + 159252480y^6 - 7828586496y^5 - 22266427392y^4 \\
& -924063720192y^3 - 2099598580704y^2 - 990727015468y - 884090421464 \\
B \quad & -477757440x^5y^5 - 19472596992x^5y^4 - 99708807168x^5y^3 \\
& -1823363225856x^5y^2 - 957989183904x^5y - 792623898948x^5 \\
& +955514880x^4y^5 + 38945193984x^4y^4 + 199417614336x^4y^3 \\
& +3646726451712x^4y^2 + 1915978367808x^4y + 1585247797896x^4 \\
& +477757440x^3y^6 + 19492503552x^3y^5 + 100520165376x^3y^4 \\
& +1827517759488x^3y^3 + 1033962651648x^3y^2 + 832540114944x^3y \\
& +2679582228480x^3 - 668860416x^2y^7 - 3726508032x^2y^6 \\
& -81107288064x^2y^5 - 532791779328x^2y^4 + 1221155841024x^2y^3 \\
& -545734687104x^2y^2 + 1592749425936x^2y + 3466189877194x^2 - 79626240xy^8 \\
& -2767675392xy^7 + 6903456768xy^6 - 190486862208xy^5 - 625951618800xy^4 \\
& +871759761994xy^3 - 601168150896xy^2 + 345333279418xy + 1615827488822x \\
& +31850496y^9 - 222953472y^8 + 8306343936y^7 - 1353369600y^6 + 893161550592y^5 \\
& +295212543456y^4 + 1497985633132y^3 + 1297045055712y^2 + 606310798924y \\
& +331969896556 \\
i = 71, m = 27, \text{ neighbours } 7, 27, 59, 62, 63, 75 \\
W \quad & 864x^5 - 2160x^4 + 2160x^3 - 1296x^2 + 432x \\
A \quad & -3888x^7 - 3888x^6y - 132192x^6 + 5832x^5y + 624024x^5 + 5832x^4y^2 \\
& +17496x^4y - 1391256x^4 - 648x^3y^3 - 194616x^3y^2 + 69624x^3y \\
& +1493364x^3 + 648x^2y^4 + 20088x^2y^3 - 9072x^2y^2 + 19764x^2y \\
& -1911276x^2 - 972xy^4 - 110268xy^3 + 63432xy^2 - 140250xy + 1333722x + 324y^5 \\
& +8100y^4 - 115938y^3 + 5382y^2 - 433992y - 563088 \\
B \quad & 3888x^5y^3 + 589680x^5y^2 - 169992x^5y + 1402452x^5 + 3888x^4y^3 \\
& +589680x^4y^2 - 169992x^4y + 1402452x^4 - 1944x^3y^4 - 294516x^3y^3 \\
& +134136x^3y^2 - 715392x^3y + 830736x^3 - 3888x^2y^5 - 39852x^2y^4 \\
& +449550x^2y^3 - 23706x^2y^2 + 599094x^2y + 3540x^2 + 324xy^6 + 94392xy^5 \\
& +84501xy^4 - 114774xy^3 + 320832xy^2 - 270366xy + 365019x - 324y^7 - 7776y^6 \\
& +122661y^5 - 137196y^4 + 681942y^3 - 6840y^2 + 431805y + 564546 \\
i = 72, m = 18, \text{ neighbours } 116, 33, 65, 67, 74, 75 \\
W \quad & 108x^4 + 216x^3 - 108x^2 - 216x \\
A \quad & 432x^6 - 1944x^5 + 864x^4y + 216x^4 + 576x^3y^2 - 3588x^3y - 3248x^3 \\
& -72x^2y^3 + 108x^2y^2 + 5220x^2y - 3204x^2 + 324xy^3 - 1812xy^2 - 2530xy \\
& +8065x - 36y^4 + 108y^3 + 2316y^2 - 7757y + 5962 \\
B \quad & -1728x^5y^2 + 720x^5y - 14772x^5 - 864x^4y^2 + 360x^4y - 7386x^4 \\
& +864x^3y^3 - 2592x^3y^2 + 8316x^3y + 2268x^3 - 324x^2y^4 - 1836x^2y^3 \\
& +7812x^2y^2 + 25608x^2y + 18202x^2 - 288xy^5 + 1200xy^4 + 994xy^3 + 876xy^2 \\
& +1354xy + 339x + 36y^6 - 36y^5 - 3000y^4 + 2333y^3 + 168y^2 + 3185y + 2981 \\
i = 73, m = 1, \text{ neighbours } 114, 37, 58, 102 \\
W \quad & 3x^2 + 3x - 6 \\
A \quad & 2x^2 + x + y - 2 \\
B \quad & -3x^3 + 15x^2y + 6x^2 + 3xy - 4x - y^3 + 6y^2 + 11y + 17 \\
i = 74, m = 864, \text{ neighbours } 110, 11, 12, 15, 24, 46, 72, 108 \\
W \quad & 15552x^5 - 7776x^4 + 75168x^3 + 51840x^2 + 54432x + 23328 \\
A \quad & 995328x^6y - 39813120x^6 + 2488320x^5y^2 + 3732480x^5y + 41554944x^5 \\
& +995328x^4y^3 + 6718464x^4y^2 - 3359232x^4y - 402983424x^4 \\
& -829440x^3y^4 - 6566400x^3y^3 - 91638144x^3y^2 + 266547360x^3y \\
& -450002680x^3 + 2322432x^2y^4 - 3068928x^2y^3 - 45868032x^2y^2 \\
& -5101056x^2y - 1085322240x^2 - 414720xy^5 - 3863808xy^4 - 51905088xy^3 \\
& +20304816xy^2 - 44375300xy - 666416442x + 1119744y^5 + 1040256y^4 - 5066784y^3 \\
& +24697752y^2 - 228008594y - 515688580 \\
B \quad & 2488320x^5y^4 + 39854592x^5y^3 + 229751424x^5y^2 - 878521824x^5y \\
& +1122824424x^5 - 1244160x^4y^4 - 19927296x^4y^3 - 114875712x^4y^2 \\
& +439260912x^4y - 561412212x^4 - 1244160x^3y^5 - 22519296x^3y^4 \\
& -156390912x^3y^3 + 199936512x^3y^2 + 353714688x^3y - 2443903488x^3 \\
& -497664x^2y^6 - 5474304x^2y^5 - 62664192x^2y^4 + 313452288x^2y^3 \\
& -39094848x^2y^2 + 530858928x^2y - 1491690628x^2 + 414720xy^7 + 4154112xy^6 \\
& +27882432xy^5 - 58036560xy^4 + 346655420xy^3 - 312032112xy^2 + 614224724xy \\
& -662652482x - 1119744y^7 - 4772736y^6 + 7520544y^5 - 1588056y^4 + 265071074y^3
\end{aligned}$$

$$\begin{aligned}
& -107241144y^2 + 390919274y - 8281534 \\
i = 75, m = 13122, \text{ neighbours } 116, 7, 15, 26, 38, 42, 53, 59, 71, 72, 78, 85, 104 \\
W \quad & 16218792x^5 + 5589972x^4 - 16691184x^3 - 16848648x^2 - 6495390x - 905418 \\
A \quad & 14693280768x^{18} + 44079842304x^{17}y + 22039921152x^{17} + 29386561536x^{16}y^2 \\
& + 80813044224x^{16}y - 40406522112x^{16} + 176319369216x^{15}y^2 - 29386561536x^{15}y \\
& - 96118545024x^{15} + 144483927552x^{14}y^3 + 84486364416x^{14}y^2 \\
& - 434064002688x^{14}y - 96118545024x^{14} - 7346640384x^{13}y^4 \\
& + 69793083648x^{13}y^3 - 763438379904x^{13}y^2 - 550691918784x^{13}y \\
& + 93057444864x^{13} - 4897760256x^{12}y^5 + 206930370816x^{12}y^4 \\
& - 154279448064x^{12}y^3 - 903636767232x^{12}y^2 + 294783945408x^{12}y \\
& + 289988221824x^{12} - 29386561536x^{11}y^5 - 355087618560x^{11}y^4 \\
& - 1692278205120x^{11}y^3 - 479368285056x^{11}y^2 + 940880152512x^{11}y \\
& + 151881586272x^{11} - 24080654592x^{10}y^6 + 22652141184x^{10}y^5 \\
& - 427125508992x^{10}y^4 - 412126118208x^{10}y^3 + 1983592903680x^{10}y^2 \\
& + 1124648198784x^{10}y - 65150415072x^{10} + 1224440064x^9y^7 \\
& - 11632180608x^9y^6 - 155401851456x^9y^5 - 570691106496x^9y^4 \\
& + 1664473212000x^9y^3 + 2579589104832x^9y^2 + 82930305168x^9y \\
& - 230058683136x^8y^8 + 816293376x^8y^7 - 34488395136x^8y^6 \\
& - 120403272960x^8y^5 - 516866762016x^8y^4 - 351133697520x^8y^3 \\
& + 1658044901664x^8y^2 - 194915552688x^8y - 1349154386352x^8 \\
& - 255933482544x^8 + 4897760256x^7y^8 + 59181269760x^7y^7 \\
& + 402993836064x^7y^6 + 805987672128x^7y^5 + 2514808572696x^7y^4 \\
& + 2031448102848x^7y^3 - 1312416932904x^7y^2 - 960743291328x^7y \\
& - 28230145920x^7 + 4013442432x^6y^9 - 3775356864x^6y^8 - 7652750400x^6y^7 \\
& - 225526554288x^6y^6 - 67739595624x^6y^5 + 1567959274872x^6y^4 \\
& - 179384720904x^6y^3 - 1246084593048x^6y^2 + 81880177752x^6y \\
& - 43681675369392x^6 - 204073344x^5y^{10} + 1938696768x^5y^9 \\
& + 25560186336x^5y^8 + 235755730656x^5y^7 + 235283811048x^5y^6 \\
& + 687465700308x^5y^5 - 303814190880x^5y^4 - 2616874296804x^5y^3 \\
& - 1120792062888x^5y^2 + 325142689680x^5y - 30105887478552x^5 \\
& - 136048896x^4y^{11} + 5748065856x^4y^{10} + 19931163264x^4y^9 \\
& + 113898435120x^4y^8 + 197645033664x^4y^7 + 515161899288x^4y^6 \\
& + 1434148808364x^4y^5 - 520457177412x^4y^4 - 1730331506484x^4y^3 \\
& - 156074301468x^4y^2 + 205173072576x^4y + 85028130960336x^4 \\
& + 68024448x^3y^{12} + 306110016x^3y^{11} - 4070129472x^3y^{10} \\
& - 63624588912x^3y^9 - 64684321632x^3y^8 - 163283213340x^3y^7 \\
& + 1196565113124x^3y^6 + 2060908797852x^3y^5 + 1516183030620x^3y^4 \\
& - 3667519146960x^3y^3 - 11012029913520x^3y^2 + 1452804252768x^3y \\
& + 140025520715322x^3 - 668907072x^2y^{12} + 629226144x^2y^{11} \\
& + 640563552x^2y^{10} + 62479038312x^2y^9 + 149094709668x^2y^8 \\
& + 206420187456x^2y^7 + 459238008564x^2y^6 - 844475692230x^2y^5 \\
& - 1324674796716x^2y^4 + 7613647089318x^2y^3 + 550379431476x^2y^2 \\
& + 7354783241100x^2y + 32405911454844x^2 + 34012224xy^{13} - 362797056xy^{12} \\
& - 4858079328xy^{11} - 43919229024xy^{10} - 68704535016xy^9 - 224541052722xy^8 \\
& - 93396893508xy^7 - 693740477412xy^6 - 2119669514316xy^5 - 1814299761690xy^4 \\
& + 7736183013114xy^3 + 7067933034408xy^2 + 4256401111611xy - 29961015738873x \\
& - 334453536y^{13} - 1295298864y^{12} - 3252418920y^{11} + 26684243100y^{10} \\
& + 63926289936y^9 - 47601931446y^8 - 1039275618228y^7 - 3675619876446y^6 \\
& - 6400856454552y^5 - 472357691478y^4 + 12087111001680y^3 + 6966250090182y^2 \\
& - 28357274411241y - 25626233845185 \\
B \quad & -204073344x^5y^{12} - 3367210176x^5y^{11} - 17380246464x^5y^{10} \\
& - 15078752640x^5y^9 - 237715212672x^5y^8 - 919397968848x^5y^7 \\
& - 4733315981856x^5y^6 - 8371785016656x^5y^5 - 10109456308008x^5y^4 \\
& + 9427897757880x^5y^3 + 34723049284152x^5y^2 - 3988204580412x^5y \\
& - 31690833155190x^5 + 204073344x^4y^{12} + 3367210176x^4y^{11} \\
& + 17380246464x^4y^{10} + 15078752640x^4y^9 + 237715212672x^4y^8 \\
& + 919397968848x^4y^7 + 4733315981856x^4y^6 + 8371785016656x^4y^5 \\
& + 10109456308008x^4y^4 - 9427897757880x^4y^3 - 34723049284152x^4y^2 \\
& + 3988204580412x^4y + 31690833155190x^4 + 102036672x^3y^{13} \\
& + 1700611200x^3y^{12} + 8970724080x^3y^{11} + 8987730192x^3y^{10} \\
& + 120114169056x^3y^9 + 479508585480x^3y^8 + 2443274488332x^3y^7 \\
& + 4580335506816x^3y^6 + 5752376905392x^3y^5 - 3871494186606x^3y^4
\end{aligned}$$

$$\begin{aligned}
& -18147182788566x^3y^3 - 899485150140x^3y^2 + 16177766959296x^3y \\
& -183592930793796x^3 + 68024448x^2y^{14} - 2074745664x^2y^{13} \\
& -9531925776x^2y^{12} - 69495476688x^2y^{11} - 209731419180x^2y^{10} \\
& -669674945196x^2y^9 - 1515046095828x^2y^8 - 2201808378204x^2y^7 \\
& -1997755548264x^2y^6 + 1255964569668x^2y^5 + 4764474004158x^2y^4 \\
& -35348928412734x^2y^3 + 1248063252246x^2y^2 + 7831325335050x^2y \\
& -113182548212913x^2 - 34012224xy^{15} - 272097792xy^{14} + 3174474240xy^{13} \\
& +36155938896xy^{12} + 94425807024xy^{11} + 253806052050xy^{10} - 123529935006xy^9 \\
& -607108983840xy^8 - 2453420642292xy^7 - 3717161732142xy^6 + 1055505385674xy^5 \\
& +6925008061872xy^4 - 21417452670057xy^3 - 4208527915578xy^2 + 1495517349477xy \\
& -5787740642904x + 334453536y^{15} + 960845328y^{14} + 2626027128y^{13} \\
& -27680517828y^{12} - 34453044468y^{11} + 44848201014y^{10} + 892108176066y^9 \\
& +2814022174338y^8 + 4781156445504y^7 + 459819048456y^6 - 2527619764626y^5 \\
& +2353311611226y^4 + 14353394617941y^3 - 7786031844906y^2 + 29177056165965y \\
& +24806452090461
\end{aligned}$$

$i = 76, m = 54, \text{ neighbours } 11, 26, 28, 50, 77, 94$

$W \quad 1296x^5 + 1944x^4 - 2592x^3 - 1944x^2 + 1296x$

$A \quad -7776x^7 - 3888x^6y - 295488x^6 + 7776x^5y - 874800x^5 + 5832x^4y^2$
 $+5832x^4y + 460080x^4 + 3888x^3y^3 - 149256x^3y^2 + 445248x^3y + 885288x^3$
 $+648x^2y^4 + 56052x^2y^3 + 12312x^2y^2 + 73872x^2y - 244296x^2$
 $-1296xy^4 + 124848xy^3 + 87228xy^2 - 64356xy + 1476552x - 324y^5 - 27702y^4$
 $-99414y^3 - 91872y^2 - 502392y - 999744$

$B \quad -7776x^5y^3 + 500256x^5y^2 - 1219104x^5y + 4972392x^5 + 7776x^4y^3$
 $-500256x^4y^2 + 1219104x^4y - 4972392x^4 - 3888x^3y^4 + 239112x^3y^3$
 $+99144x^3y^2 + 759132x^3y - 1859112x^3 - 1944x^2y^5 + 71928x^2y^4$
 $-445824x^2y^3 + 436212x^2y^2 - 933804x^2y + 647940x^2 - 1944xy^6$
 $+74628xy^5 - 192816xy^4 + 321942xy^3 - 94158xy^2 + 37848xy - 261468x - 324y^7$
 $-27054y^6 - 51462y^5 - 28638y^4 - 543072y^3 - 46566y^2 - 524460y - 499872$

$i = 77, m = 162, \text{ neighbours } 6, 27, 39, 40, 43, 69, 76$

$W \quad 3888x^4 - 5832x^3 + 1944x^2$

$A \quad -139968x^9 - 139968x^8y - 69984x^8 - 139968x^7y + 979776x^7 + 209952x^6y^2$
 $+629856x^6y - 221616x^6 + 23328x^5y^3 + 769824x^5y^2 - 1667952x^5y$
 $-839808x^5 + 23328x^4y^4 + 11664x^4y^3 - 1131408x^4y^2 + 180792x^4y$
 $-1446336x^4 + 7776x^3y^5 + 274752x^3y^4 - 1100736x^3y^3 + 365760x^3y^2$
 $+3180948x^3y + 5727170x^3 - 34992x^2y^5 - 279936x^2y^4 + 1012824x^2y^3$
 $+177876x^2y^2 - 2781864x^2y - 7136424x^2 - 3888xy^6 - 131544xy^5$
 $+576612xy^4 - 207180xy^3 - 1975386xy^2 + 2455199xy + 3940488x + 7776y^6 - 45036y^5$
 $+25650y^4 + 357606y^3 - 43734y^2 - 797521y - 557782$

$B \quad -23328x^5y^5 - 754272x^5y^4 + 292896x^5y^3 + 2174472x^5y^2$
 $-2247012x^5y - 974382x^5 - 23328x^4y^5 - 754272x^4y^4 + 292896x^4y^3$
 $+2174472x^4y^2 - 2247012x^4y - 974382x^4 - 11664x^3y^6 - 367416x^3y^5$
 $+460728x^3y^4 + 965196x^3y^3 - 2029536x^3y^2 + 449064x^3y + 3283416x^3$
 $-11664x^2y^7 - 52488x^2y^6 + 357696x^2y^5 - 496044x^2y^4$
 $+1221048x^2y^3 + 2286972x^2y^2 + 1544226x^2y - 1563049x^2 - 3888xy^8$
 $-154872xy^7 + 321948xy^6 + 360144xy^5 - 1577298xy^4 - 389071xy^3 - 732972xy^2$
 $-3423049xy - 992558x + 7776y^8 - 29484y^7 - 51786y^6 + 321750y^5 + 606624y^4$
 $+274103y^3 + 724200y^2 + 1027049y + 278891$

$i = 78, m = 8748, \text{ neighbours } 110, 7, 28, 29, 40, 41, 51, 60, 61, 75, 81$

$W \quad 5458752x^5 - 4723920x^4 - 10497600x^3 + 997272x^2 + 7033392x + 2361960$

$A \quad 3265173504x^{15} + 3265173504x^{14}y + 17958454272x^{14} + 8162933760x^{13}y$
 $+18774747648x^{13} - 3265173504x^{12}y^2 - 32651735040x^{12}y - 50746238208x^{12}$
 $-544195584x^{11}y^3 - 25305094656x^{11}y^2 - 105165796608x^{11}y$
 $-101492476416x^{11} - 544195584x^{10}y^4 - 6258249216x^{10}y^3$
 $-18094503168x^{10}y^2 + 9183300480x^{10}y + 9795520512x^{10} - 1360488960x^9y^4$
 $+8707129344x^9y^3 + 118770686208x^9y^2 + 241758888192x^9y$
 $+119201507712x^9 + 544195584x^8y^5 + 10339716096x^8y^4$
 $+73171631232x^8y^3 + 160809795072x^8y^2 + 142443194112x^8y$
 $+86901232320x^8 + 90699264x^7y^6 + 4217515776x^7y^5 + 29749358592x^7y^4$
 $+22878889344x^7y^3 - 118090441728x^7y^2 - 126672859584x^7y$
 $+24976309824x^7 + 90699264x^6y^7 + 1043041536x^6y^6 + 2698303104x^6y^5$
 $-33536052864x^6y^4 - 170344555200x^6y^3 - 254116662912x^6y^2$
 $-161461696032x^6y - 5046397454016x^6 + 226748160x^5y^7 - 1451188224x^5y^6$

$$\begin{aligned}
& -32005502784x^5y^5 - 115125709536x^5y^4 - 121468989312x^5y^3 \\
& -36891925632x^5y^2 - 122674533696x^5y + 8457855643872x^5 - 90699264x^4y^8 \\
& -1723286016x^4y^7 - 13933674432x^4y^6 - 37889617536x^4y^5 \\
& -144551952x^4y^4 + 102546855360x^4y^3 + 61763364432x^4y^2 \\
& -53896148064x^4y + 15363994267008x^4 + 105815808x^3y^9 - 549234432x^3y^8 \\
& -7719935040x^3y^7 - 11440214496x^3y^6 + 45542916144x^3y^5 \\
& +136337681088x^3y^4 + 62592634656x^3y^3 + 737399822688x^3y^2 \\
& -276658069464x^3y - 21518143075164x^3 - 15116544x^2y^{10} - 173840256x^2y^9 \\
& -464833728x^2y^8 + 5974814016x^2y^7 + 40329994608x^2y^6 \\
& +77685336792x^2y^5 + 45079423776x^2y^4 + 854825604120x^2y^3 \\
& +130613553648x^2y^2 + 1000917711504x^2y - 30140028350640x^2 - 37791360xy^{10} \\
& +191476224xy^9 + 5247540288xy^8 + 22628171664xy^7 + 29142655632xy^6 \\
& +1039270176xy^5 + 8865198576xy^4 - 1339495361640xy^3 - 200467525356xy^2 \\
& -1240746959586xy + 24036551104776x + 7558272y^{11} + 60466176y^{10} + 711107424y^9 \\
& +3893349888y^8 + 6147254592y^7 - 3434738904y^6 - 7489691568y^5 \\
& -432849353904y^4 - 874078479648y^3 - 2040998396652y^2 + 5605486726686y \\
& +20015754801708 \\
B \quad & -362797056x^5y^9 - 461054592x^5y^8 + 8194426560x^5y^7 \\
& +25562495808x^5y^6 + 19257777216x^5y^5 - 39623284512x^5y^4 \\
& +50364351072x^5y^3 - 2095381650384x^5y^2 + 1192972503384x^5y \\
& +12267467503380x^5 - 362797056x^4y^9 - 461054592x^4y^8 \\
& +8194426560x^4y^7 + 25562495808x^4y^6 + 19257777216x^4y^5 \\
& -39623284512x^4y^4 + 50364351072x^4y^3 - 2095381650384x^4y^2 \\
& +1192972503384x^4y + 12267467503380x^4 - 181398528x^3y^{10} \\
& -804955968x^3y^9 + 3367210176x^3y^8 + 25755756624x^3y^7 \\
& +50102840304x^3y^6 + 10679838336x^3y^5 - 37554691608x^3y^4 \\
& -967947269328x^3y^3 - 2721201361416x^3y^2 + 8022606882048x^3y \\
& -28770258147552x^3 + 22674816x^2y^{11} + 317447424x^2y^{10} \\
& +3427676352x^2y^9 + 15755847840x^2y^8 + 30331398024x^2y^7 \\
& +30002000832x^2y^6 - 22282532352x^2y^5 + 1283576615928x^2y^4 \\
& -7047304344216x^2y^3 + 1347746813280x^2y^2 - 9181461083004x^2y \\
& +10985374704318x^2 - 52907904xy^{12} + 206592768xy^{11} + 3548188800xy^{10} \\
& +10335062160xy^9 + 11147599728xy^8 + 11450486592xy^7 + 28946912976xy^6 \\
& -471008125152xy^5 + 774239969412xy^4 + 406351997286xy^3 + 2230288028604xy^2 \\
& -5132042167386xy + 25843887480036x + 7558272y^{13} + 45349632y^{12} + 627966432y^{11} \\
& +2705441472y^{10} + 1431417744y^9 - 2075754600y^8 + 6778949616y^7 \\
& -440557109784y^6 - 215370360y^5 - 2487335011740y^4 + 9259760075910y^3 \\
& -142006666476y^2 + 6414417183498y + 12797304921654 \\
i = 79, m = 3, \text{ neighbours } 2, 19, 37, 82 \\
W \quad & 24x^3 + 12x^2 - 24x - 12 \\
A \quad & -24x^3 - 12x^2y - 72x^2 + 120x - 6y^2 - 36y + 96 \\
B \quad & 144x^5 + 144x^4 - 72x^3y - 324x^3 - 54x^2y + 12x^2 + 12xy^3 + 18xy^2 + 72xy + 276x \\
& +6y^4 + 57y^3 + 33y^2 + 90y + 120 \\
i = 80, m = 648, \text{ neighbours } 110, 10, 24, 41, 43, 46, 54, 58, 88 \\
W \quad & 77760x^5 - 171072x^4 + 171072x^3 - 104976x^2 + 34992x \\
A \quad & -2239488x^{11} + 1119744x^{10}y + 7838208x^{10} - 7278336x^9y + 2239488x^8y^2 \\
& +12317184x^8y - 26500608x^8 + 373248x^7y^3 - 1959552x^7y^2 + 17542656x^7y \\
& +22581504x^7 - 186624x^6y^4 - 2985984x^6y^3 - 10264320x^6y^2 \\
& -54587520x^6y - 982202112x^6 + 1213056x^5y^4 + 5738688x^5y^3 \\
& +9237888x^5y^2 - 8118144x^5y + 4389334272x^5 - 373248x^4y^5 \\
& -5412096x^4y^4 - 7216128x^4y^3 + 4059072x^4y^2 + 79501824x^4y \\
& -9306658944x^4 - 342144x^3y^6 + 295488x^3y^5 + 8245152x^3y^4 \\
& +19387296x^3y^3 - 1208976336x^3y^2 + 1374142008x^3y + 9489938012x^3 \\
& +31104x^2y^7 + 497664x^2y^6 + 2371680x^2y^5 + 15513120x^2y^4 \\
& +191126304x^2y^3 + 19346688x^2y^2 + 119843712x^2y - 12362118912x^2 \\
& -202176xy^7 - 839808xy^6 - 1806624xy^5 - 2304720xy^4 - 745840296xy^3 \\
& +482515548xy^2 - 1316920658xy + 7499661696x + 15552y^8 + 178848y^7 + 2083968y^6 \\
& +14157504y^5 + 130663584y^4 - 478942920y^3 - 194745684y^2 - 658787546y \\
& -6122175652 \\
B \quad & 839808x^5y^6 + 93312x^5y^5 - 24992064x^5y^4 - 76404384x^5y^3 \\
& +3623429808x^5y^2 - 4137029352x^5y + 9559491564x^5 + 839808x^4y^6 \\
& +93312x^4y^5 - 24992064x^4y^4 - 76404384x^4y^3 + 3623429808x^4y^2
\end{aligned}$$

$$\begin{aligned}
& -4137029352x^4y + 9559491564x^4 - 419904x^3y^7 - 396576x^3y^6 \\
& + 12457152x^3y^5 + 48615552x^3y^4 - 1779879744x^3y^3 + 558752256x^3y^2 \\
& - 3055983552x^3y + 3863972160x^3 + 139968x^2y^8 + 1329696x^2y^7 \\
& + 209952x^2y^6 - 5139936x^2y^5 - 319000032x^2y^4 + 2513105136x^2y^3 \\
& + 120548808x^2y^2 + 2256238308x^2y + 1123547090x^2 + 171072xy^9 + 202176xy^8 \\
& - 2019168xy^7 - 17220816xy^6 + 547049448xy^5 - 216342636xy^4 - 243886198xy^3 \\
& + 1047920628xy^2 - 2094929926xy + 2372344804x - 15552y^{10} - 147744y^9 - 1804032y^8 \\
& - 11583648y^7 - 113538240y^6 + 687258504y^5 - 1316614236y^4 + 3647853626y^3 \\
& - 504881844y^2 + 1859937686y + 3061087826 \\
i = 81, m = 162, \text{ neighbours } 19, 26, 37, 55, 56, 65, 78 \\
W \quad & 3888x^5 - 5832x^4 - 1944x^3 + 5832x^2 - 1944x \\
A \quad & 279936x^{10} - 839808x^9 + 629856x^8 - 699840x^7y + 723168x^7 - 46656x^6y^3 \\
& - 349920x^6y^2 + 1353024x^6y - 4070736x^6 + 139968x^5y^3 + 419904x^5y^2 \\
& + 104976x^5y + 8048160x^5 - 174960x^4y^3 + 594864x^4y^2 - 1265544x^4y \\
& - 2814912x^4 - 62208x^3y^5 - 292896x^3y^4 - 1436400x^3y^3 \\
& - 7527528x^3y^2 - 21929664x^3y - 77695120x^3 + 7776x^2y^6 + 58320x^2y^5 \\
& - 95256x^2y^4 + 441288x^2y^3 - 175932x^2y^2 + 487944x^2y + 12739032x^2 \\
& - 23328xy^6 - 33696xy^5 + 215568xy^4 - 373860xy^3 + 3812232xy^2 + 11644696xy \\
& + 134653592x + 3888y^7 + 46656y^6 + 127008y^5 + 401490y^4 + 1970154y^3 \\
& + 4062300y^2 + 11940284y - 69862744 \\
B \quad & 186624x^5y^5 + 1228608x^5y^4 + 5440608x^5y^3 + 21690288x^5y^2 \\
& + 66150576x^5y + 198962328x^5 - 186624x^4y^5 - 1228608x^4y^4 \\
& - 5440608x^4y^3 - 21690288x^4y^2 - 66150576x^4y - 198962328x^4 \\
& - 93312x^3y^6 - 349920x^3y^5 - 979776x^3y^4 - 3137616x^3y^3 \\
& - 2347380x^3y^2 - 5767848x^3y - 8232840x^3 - 11664x^2y^7 + 134136x^2y^6 \\
& + 235224x^2y^5 + 1516968x^2y^4 + 5242860x^2y^3 + 5487120x^2y^2 \\
& + 16492596x^2y + 38118560x^2 + 31104xy^8 + 204768xy^7 + 692928xy^6 \\
& + 3580704xy^5 + 11159124xy^4 + 35639654xy^3 + 10312134xy^2 + 31424396xy - 2535948x \\
& - 3888y^9 - 54432y^8 - 239760y^7 - 957798y^6 - 4033170y^5 - 12525918y^4 \\
& - 39494984y^3 - 15264654y^2 - 46426916y - 34931372 \\
i = 82, m = 18, \text{ neighbours } 5, 37, 45, 66, 79 \\
W \quad & 216x^4 + 540x^3 + 324x^2 \\
A \quad & 216x^5 - 216x^4y - 8640x^4 - 72x^3y^2 - 1644x^3y - 25802x^3 + 216x^2y^2 \\
& + 2160x^2y - 66996x^2 - 36xy^3 - 444xy^2 - 1777xy - 58140x + 198y^3 + 4566y^2 \\
& - 12149y - 40978 \\
B \quad & 216x^5y^2 + 5580x^5y - 40854x^5 - 216x^4y^2 - 5580x^4y + 40854x^4 \\
& - 108x^3y^3 - 2916x^3y^2 + 17172x^3y - 2160x^3 + 108x^2y^4 + 4428x^2y^3 \\
& - 6012x^2y^2 + 2388x^2y + 27913x^2 + 36xy^5 + 768xy^4 + 9913xy^3 + 1902xy^2 \\
& + 2587xy + 29070x - 198y^5 - 4494y^4 + 19967y^3 - 2424y^2 + 17615y + 17897 \\
i = 83, m = 2, \text{ neighbours } 113, 0, 29, 94, 101 \\
W \quad & 6x^3 - 6x \\
A \quad & 3x^3 + 4x^2y - 6x^2 - 2xy - 9x - 8y^2 + 18y \\
B \quad & -18x^5 + 18x^4 - 36x^3y + 36x^3 + 36x^2y^2 - 60x^2y + 33x^2 - 3xy^3 - 12xy^2 \\
& + 41xy - 30x - 4y^4 - 15y^3 + 44y^2 - 19y + 9 \\
i = 84, m = 3, \text{ neighbours } 116, 92, 98, 103, 106 \\
W \quad & 12x^3 - 12x \\
A \quad & 3x^4 - 9x^3 - 27x^2 - 18xy + 9x - 12y^2 - 42y + 24 \\
B \quad & 108x^5 - 216x^3y - 111x^3 - 3x^2y^3 - 72x^2y^2 - 75x^2y + 96x^2 + 9xy^3 \\
& + 216xy^2 - 201xy + 87x + 6y^4 + 168y^3 - 138y^2 - 6y + 60 \\
i = 85, m = 576, \text{ neighbours } 11, 22, 25, 31, 36, 52, 75, 103, 106 \\
W \quad & 24192x^5 - 918432x^4 - 9792x^3 + 145728x^2 - 161856x - 61344 \\
A \quad & -442368x^{13} - 221184x^{12}y - 2433024x^{12} - 3207168x^{11}y - 3317760x^{11} \\
& - 1105920x^{10}y^2 - 7852032x^{10}y + 5271552x^{10} + 73728x^9y^3 \\
& - 5253120x^9y^2 + 8239104x^9y + 19169280x^9 + 36864x^8y^4 - 976896x^8y^3 \\
& + 8828928x^8y^2 + 49747968x^8y + 13455360x^8 + 534528x^7y^4 \\
& + 6082560x^7y^3 + 63590400x^7y^2 + 44587008x^7y - 16472064x^7 \\
& + 184320x^6y^5 + 2414592x^6y^4 + 46295040x^6y^3 + 67009536x^6y^2 \\
& - 44792832x^6y - 131475456x^6 - 12288x^5y^6 + 875520x^5y^5 \\
& + 17464320x^5y^4 + 53403648x^5y^3 - 71033856x^5y^2 - 108355584x^5y \\
& + 7390402560x^5 - 6144x^4y^7 + 162816x^4y^6 + 1729536x^4y^5 \\
& + 18760704x^4y^4 - 83251200x^4y^3 - 194543616x^4y^2 - 55400448x^4y
\end{aligned}$$

$$\begin{aligned}
& -140494371328x^4 + 1024x^3y^8 - 98048x^3y^7 - 1226624x^3y^6 \\
& -4002368x^3y^5 - 76135136x^3y^4 - 262218896x^3y^3 - 64509112x^3y^2 \\
& +111122268x^3y - 4254663598x^3 - 30720x^2y^8 - 402432x^2y^7 \\
& -6638592x^2y^6 - 47413248x^2y^5 - 188030976x^2y^4 - 162636288x^2y^3 \\
& +44686848x^2y^2 + 90291712x^2y + 44696586240x^2 + 2048xy^9 - 145920xy^8 \\
& -2907648xy^7 - 20130816xy^6 - 99001344xy^5 - 161826816xy^4 - 1224707584xy^3 \\
& +127018496xy^2 - 1191612416xy - 26967839662x - 36864y^9 - 577536y^8 - 6497280y^7 \\
& -35592192y^6 - 36523008y^5 - 1108008960y^4 + 24583475712y^3 + 979410432y^2 \\
& +22830299648y + 4276951552 \\
B \quad & -6144x^5y^8 + 53760x^5y^7 + 1277184x^5y^6 + 3831168x^5y^5 \\
& +12028224x^5y^4 + 179559264x^5y^3 - 356796336x^5y^2 - 426106920x^5y \\
& +719406612x^5 + 12288x^3y^9 - 101376x^3y^8 - 2608128x^3y^7 \\
& -8939520x^3y^6 - 27887616x^3y^5 - 371146752x^3y^4 + 534033408x^3y^3 \\
& +1209010176x^3y^2 - 1012706304x^3y + 2696192512x^3 + 6144x^2y^{10} \\
& +58368x^2y^9 + 1740800x^2y^8 + 11869952x^2y^7 + 49629056x^2y^6 \\
& -269344192x^2y^5 + 6519271136x^2y^4 - 6181141872x^2y^3 \\
& -5389154120x^2y^2 + 3444575908x^2y - 18006996562x^2 - 1024xy^{11} + 73472xy^{10} \\
& +1416064xy^9 + 9447744xy^8 + 40288608xy^7 + 53142096xy^6 + 643072472xy^5 \\
& -227909612xy^4 + 1976773366xy^3 + 760602460xy^2 - 337045934xy + 2833099182x \\
& +18432y^{11} + 279552y^{10} + 3127296y^9 + 15876096y^8 + 10874880y^7 \\
& +565630464y^6 - 12538495488y^5 + 6355789824y^4 - 26327792128y^3 \\
& -1755359232y^2 - 11027175424y - 8040037888 \\
i = 86, m = 1, \text{ neighbours } 111, 0, 58, 104 \\
W \quad & 3x^2 + 3x \\
A \quad & 0 \\
B \quad & -12x^2y - 9x^2 - 12xy - 9x - 6y^2 - 6y \\
i = 87, m = 24, \text{ neighbours } 20, 36, 38, 44 \\
W \quad & 54x^4 + 36x^3 + 18x^2 + 12x + 6 \\
A \quad & -102x^4 + 12x^3 - 384x^2y + 132x^2 + 150x - 360y^2 + 60y + 60 \\
B \quad & 648x^5 + 648x^4 - 1296x^3y + 1362x^3 + 102x^2y^3 + 432x^2y^2 - 1698x^2y \\
& +1056x^2 + 90xy^3 - 432xy^2 + 414xy + 258x + 180y^4 + 348y^3 - 252y^2 + 312y + 96 \\
i = 88, m = 2, \text{ neighbours } 114, 35, 80 \\
W \quad & 3x^2 - 3x + 3 \\
A \quad & 0 \\
B \quad & 15x^2 - 15x + 6y + 15 \\
i = 89, m = 6, \text{ neighbours } 51, 104, 108 \\
W \quad & 6x^2 + 24x \\
A \quad & 6x^2 + 12y - 12 \\
B \quad & 180x^3 + 108x^2y + 288x^2 + 252xy + 174x - 6y^3 + 72y^2 - 6y - 78 \\
i = 90, m = 2, \text{ neighbours } 5, 25 \\
W \quad & x^2 + 2x - 2 \\
A \quad & 0 \\
B \quad & -x^2 - 2y - 2 \\
i = 91, m = 12, \text{ neighbours } 18, 32, 34, 52 \\
W \quad & 6x^4 - 12x^3 + 12x^2 - 6x \\
A \quad & -6x^4 + 12x^3 - 48x^2y - 24x^2 + 270x - 72y^2 + 180y + 108 \\
B \quad & -216x^5 + 216x^4 + 432x^3y + 42x^3 + 6x^2y^3 + 144x^2y^2 + 222x^2y \\
& -912x^2 - 18xy^3 + 432xy^2 - 414xy - 558x + 36y^4 + 36y^3 + 36y^2 - 216y - 108 \\
i = 92, m = 2304, \text{ neighbours } 114, 27, 33, 39, 84, 94, 98 \\
W \quad & 20736x^5 - 13824x^4 - 13824x^3 + 13824x^2 - 6912x \\
A \quad & 10616832x^8 - 26542080x^7 - 21233664x^6y - 66355200x^6 + 55738368x^5y \\
& +85819392x^5 - 1769472x^4y^3 - 50429952x^4y^2 + 884736x^4y + 98205696x^4 \\
& +1105920x^3y^3 + 95740416x^3y^2 - 144855456x^3y - 66028430x^3 \\
& +3538944x^2y^4 + 114573312x^2y^3 - 190218240x^2y^2 + 201719808x^2y \\
& -13860864x^2 - 9289728xy^4 - 216916992xy^3 + 135200064xy^2 + 29588476xy + 6687402x \\
& -12386304y^4 + 282986496y^3 - 443550336y^2 + 209917192y - 28670180 \\
B \quad & 19906560x^5y^3 - 224087040x^5y^2 + 359524800x^5y - 163867308x^5 \\
& -39813120x^4y^3 + 448174080x^4y^2 - 719049600x^4y + 327734616x^4 \\
& +39813120x^3y^4 - 429096960x^3y^3 + 504299520x^3y^2 + 16809984x^3y \\
& -413171712x^3 + 1769472x^2y^6 + 50429952x^2y^5 + 75202560x^2y^4 \\
& -14376960x^2y^3 - 72737280x^2y^2 - 207269472x^2y + 319947662x^2
\end{aligned}$$

$$\begin{aligned}
& -4644864xy^6 - 116974080xy^5 + 238416288xy^4 - 430718578xy^3 + 328184736xy^2 \\
& -21319186xy - 99058574x - 6193152y^6 + 74695680y^5 - 63218496y^4 + 192724196y^3 \\
& -109731648y^2 - 92056156y + 28670180
\end{aligned}$$

$i = 93, m = 12, \text{ neighbours } 1, 7, 64, 99$

$$\begin{aligned}
W & 6x^4 + 24x^3 + 12x^2 - 24x - 18 \\
A & -6x^4 - 36x^3 - 120x^2 + 24xy - 132x + 24y^2 + 168y + 78 \\
B & -72x^5 + 144x^4 - 144x^3y + 294x^3 + 6x^2y^3 + 6x^2y + 642x^2 + 24xy^3 \\
& -144xy^2 - 396xy - 402x + 12y^4 + 222y^3 + 588y^2 + 594y - 174
\end{aligned}$$

$i = 94, m = 144, \text{ neighbours } 114, 76, 83, 92, 97, 101$

$$\begin{aligned}
W & 864x^3 + 864x^2 - 1728x \\
A & 20736x^6 + 41472x^5 - 93312x^4y - 124416x^4 + 1728x^3y^2 + 94680x^3y \\
& -149253x^3 - 3456x^2y^3 + 41472x^2y^2 + 639360x^2y - 114048x^2 - 6912xy^3 \\
& -436464xy^2 + 565986xy + 255099x + 6912y^4 + 120960y^3 - 491472y^2 - 163002y + 526602 \\
B & -10368x^5y^2 - 879120x^5y - 1240290x^5 + 10368x^4y^2 + 879120x^4y \\
& +1240290x^4 - 20736x^3y^3 - 1741824x^3y^2 - 1088640x^3y + 2778624x^3 \\
& +51840x^2y^4 + 1430784x^2y^3 - 323136x^2y^2 - 3619800x^2y - 1924347x^2 \\
& -1728xy^5 - 187992xy^4 + 1498821xy^3 + 2113704xy^2 + 505221xy - 508032x + 3456y^6 \\
& +63936y^5 - 676008y^4 - 1440069y^3 - 366696y^2 + 52563y - 263301
\end{aligned}$$

$i = 95, m = 2, \text{ neighbours } 114, 21, 42$

$$\begin{aligned}
W & 3x^2 - 3x \\
A & 0 \\
B & 15x^2 - 9x - 6y - 6
\end{aligned}$$

$i = 96, m = 24, \text{ neighbours } 20, 30, 34, 51$

$$\begin{aligned}
W & 18x^4 - 18x^2 \\
A & 42x^4 + 12x^3 - 180x^2 - 192xy + 318x - 168y^2 + 588y - 420 \\
B & -72x^5 - 72x^4 - 144x^3y + 426x^3 - 42x^2y^3 - 144x^2y^2 + 750x^2y \\
& -744x^2 - 54xy^3 - 432xy^2 + 726xy - 276x - 84y^4 - 78y^3 + 780y^2 - 738y + 498
\end{aligned}$$

$i = 97, m = 144, \text{ neighbours } 11, 21, 48, 94, 98$

$$\begin{aligned}
W & 72x^4 - 504x^2 - 432x \\
A & -6912x^5 - 3456x^4y + 12960x^4 - 576x^3y^2 + 80232x^3y - 21985x^3 \\
& +20736x^2y^2 - 58752x^2y + 13248x^2 + 1152xy^3 - 138000xy^2 - 75766xy + 63939x \\
& -26496y^3 - 45264y^2 - 9934y + 8834 \\
B & 3456x^5y^2 - 170352x^5y + 214854x^5 + 3456x^4y^2 - 170352x^4y + 214854x^4 \\
& +6912x^3y^3 - 331776x^3y^2 - 10368x^3y + 151200x^3 + 3456x^2y^4 \\
& -109728x^2y^3 + 24768x^2y^2 - 138120x^2y - 48863x^2 + 576xy^5 - 62952xy^4 \\
& +196225xy^3 - 54888xy^2 - 332879xy + 75650x - 13248y^5 + 32088y^4 + 53377y^3 \\
& -105576y^2 - 67175y + 45889
\end{aligned}$$

$i = 98, m = 2304, \text{ neighbours } 114, 20, 42, 48, 84, 92, 97$

$$\begin{aligned}
W & 6912x^5 - 13824x^4 - 27648x^3 + 13824x^2 + 20736x \\
A & 5308416x^8 - 7962624x^7 - 23887872x^6y - 39813120x^6 + 34504704x^5y \\
& +49545216x^5 - 884736x^4y^3 + 15925248x^4y^2 + 68124672x^4y + 101302272x^4 \\
& +2433024x^3y^3 + 3257856x^3y^2 - 77286432x^3y - 130869338x^3 \\
& +3981312x^2y^4 + 78299136x^2y^3 - 6635520x^2y^2 - 58834944x^2y \\
& -123715584x^2 - 5750784xy^4 - 135023616xy^3 + 51902016xy^2 + 63616276xy \\
& +239253774x - 4423680y^5 - 116342784y^4 - 96196608y^3 + 347069568y^2 + 51431720y \\
& -93049012 \\
B & -6635520x^5y^3 - 67322880x^5y^2 + 272615616x^5y + 36729372x^5 \\
& -13271040x^4y^3 - 134645760x^4y^2 + 545231232x^4y + 73458744x^4 \\
& -13271040x^3y^4 - 141557760x^3y^3 + 475103232x^3y^2 + 357433344x^3y \\
& -425115648x^3 + 884736x^2y^6 + 10616832x^2y^5 - 282230784x^2y^4 \\
& -367386624x^2y^3 - 26260992x^2y^2 + 457280544x^2y + 141928538x^2 \\
& -663552xy^6 - 8566272xy^5 + 192080928xy^4 - 264990118xy^3 - 321328608xy^2 \\
& +324242618xy + 278771982x - 2211840y^7 - 59277312y^6 - 16247808y^5 + 168728640y^4 \\
& -107666252y^3 - 70472640y^2 + 227715316y + 124899508
\end{aligned}$$

$i = 99, m = 1, \text{ neighbours } 34, 93$

$$\begin{aligned}
W & x^2 - 1 \\
A & 0 \\
B & x - y
\end{aligned}$$

$i = 100, m = 15925248, \text{ neighbours } 4, 10, 23, 24, 35, 45, 47, 58, 102$

$$W \quad 13532479488x^5 + 49268736000x^4 + 25966116864x^3 - 23720656896x^2 + 31591710720x$$

$$\begin{aligned}
& +4263985152 \\
A \quad & 338151365148672x^{12} - 338151365148672x^{11}y - 1521681143169024x^{11} \\
& + 84537841287168x^{10}y^2 + 2451597397327872x^{10}y + 2028908190892032x^{10} \\
& - 1437143301881856x^9y^2 - 5579497524953088x^9y + 2310700995182592x^9 \\
& + 239523883646976x^8y^3 + 4184623143714816x^8y^2 - 1690756825743360x^8y \\
& - 12821572595220480x^8 + 56358560858112x^7y^4 - 929916254158848x^7y^3 \\
& + 464958127079424x^7y^2 + 30419533223165952x^7y + 20542695432781824x^7 \\
& - 14089640214528x^6y^5 - 281792804290560x^6y^4 - 1958459989819392x^6y^3 \\
& - 29813678693941248x^6y^2 - 60684080403972096x^6y - 30532201324675596288x^6 \\
& + 239523883646976x^5y^5 + 2747479841832960x^5y^4 \\
& + 19004576376029184x^5y^3 + 76851942550142976x^5y^2 + 51042069950496768x^5y \\
& - 222257875529553149952x^5 - 39920647274496x^4y^6 - 1204664238342144x^4y^5 \\
& - 8263573985820672x^4y^4 - 51366131675430912x^4y^3 \\
& - 90060980251262976x^4y^2 + 5971659177590784x^4y - 521662824899810426880x^4 \\
& - 17612050268160x^3y^7 + 43858897403904x^3y^6 + 2190593029570560x^3y^5 \\
& + 262679629529088x^3y^4 + 663917130251919360x^3y^3 \\
& + 1845500086776254976x^3y^2 + 12598317614094538080x^3y \\
& - 327933649418411995358x^3 + 2348273369088x^2y^8 + 46965467381760x^2y^7 \\
& - 178468776050688x^2y^6 - 136199855407104x^2y^5 - 29196082797871104x^2y^4 \\
& + 5021591655816364032x^2y^3 + 25902629397725184x^2y^2 \\
& + 5150075476311539712x^2y + 176815334022614876160x^2 - 39920647274496xy^8 \\
& - 441817849921536xy^7 - 3067610451148800xy^6 - 3721629194256384xy^5 \\
& + 69414329159958528xy^4 + 35913595586353136640xy^3 - 3590133749120191296xy^2 \\
& + 12222470301676379684xy - 255072453077002136934x + 4696546738176y^9 \\
& + 72796474441728y^8 + 487755947704320y^7 - 4034886063095808y^6 \\
& + 168559410707693568y^5 + 3407198415310651392y^4 + 106512002958536607744y^3 \\
& + 122509418246411385216y^2 - 52046666757843982520y - 92028274427106746812 \\
B \quad & 49313740750848x^5y^7 + 666762869735424x^5y^6 + 2073253254266880x^5y^5 \\
& + 104335747715432448x^5y^4 - 3525843088063217664x^5y^3 \\
& - 10827812601644313600x^5y^2 - 75953502939943338048x^5y \\
& + 81325493818377020724x^5 + 98627481501696x^4y^7 + 1333525739470848x^4y^6 \\
& + 4146506508533760x^4y^5 + 208671495430864896x^4y^4 \\
& - 7051686176126435328x^4y^3 - 21655625203288627200x^4y^2 \\
& - 151907005879886676096x^4y + 162650987636754041448x^4 - 98627481501696x^3y^8 \\
& - 1282157259522048x^3y^7 - 3451961852559360x^3y^6 \\
& - 206511856624336896x^3y^5 + 7160369246663344128x^3y^4 \\
& + 17982871986556108800x^3y^3 + 140628034419840516096x^3y^2 \\
& - 241769219865861685248x^3y + 521922063418993410048x^3 + 11741366845440x^2y^9 \\
& + 373375465684992x^2y^8 + 825418089234432x^2y^7 + 63289073866899456x^2y^6 \\
& - 1838817463913938944x^2y^5 + 7604493232331685888x^2y^4 \\
& - 31569271715244957696x^2y^3 - 129871063448620651008x^2y^2 \\
& + 164371650063107697312x^2y - 340578838963024924450x^2 + 17612050268160xy^{10} \\
& + 176878799290368xy^9 + 1234363679244288xy^8 + 2907073578663936xy^7 \\
& - 120636768853057536xy^6 - 19543305585042805248xy^5 - 41047968232650257760xy^4 \\
& - 42073245758417862946xy^3 + 68600916877686355104xy^2 - 163100192856561028930xy \\
& + 225380966732650992282x - 2348273369088y^{11} - 35224100536320y^{10} \\
& - 228614196953088y^9 + 2346617143296000y^8 - 84465222517260288y^7 \\
& - 1658071350195830784y^6 - 52510749747635856384y^5 - 36791017365254859456y^4 \\
& - 10541519465945793092y^3 - 63223784747756834496y^2 - 3618468916309838084y \\
& + 73846705050630283708
\end{aligned}$$

$i = 101, m = 72$, neighbours 17, 46, 64, 83, 94

$$W \quad 216x^5 - 1080x^3 + 864x$$

$$\begin{aligned}
A \quad & -7776x^6 + 864x^5 - 1728x^4y + 78624x^4 + 576x^3y^2 - 18696x^3y - 44165x^3 \\
& + 1296x^2y^3 + 10368x^2y^2 + 2160x^2y - 261504x^2 - 144xy^3 + 33456xy^2 \\
& + 51998xy + 154671x + 2592y^4 - 36000y^3 - 113712y^2 - 154906y + 302026 \\
B \quad & -3456x^5y^2 + 49968x^5y + 236478x^5 - 3456x^4y^2 + 49968x^4y + 236478x^4 \\
& + 6912x^3y^3 - 108864x^3y^2 - 343872x^3y - 287712x^3 - 13824x^2y^4 \\
& + 16416x^2y^3 + 301824x^2y^2 + 346152x^2y + 318917x^2 - 576xy^5 - 8952xy^4 \\
& + 41429xy^3 + 162696xy^2 + 232325xy + 194314x - 1296y^6 + 16704y^5 + 51528y^4 \\
& + 62741y^3 - 75624y^2 + 141629y + 151013
\end{aligned}$$

$i = 102, m = 12$, neighbours 110, 2, 73, 100, 105

$$W \quad 18x^3 + 36x^2 - 18x - 36$$

$$\begin{aligned}
A & -180x^3 + 48x^2y - 102x^2 - 408xy - 78x + 96y^2 - 468y - 564 \\
B & 504x^5 + 504x^4 - 1008x^3y + 1548x^3 + 720x^2y^2 - 1800x^2y + 2304x^2 \\
& + 180xy^3 - 144xy^2 + 1284xy - 798x - 48y^4 + 570y^3 - 912y^2 + 1026y - 366 \\
i = 103, m = 20736, \text{ neighbours } 118, 8, 24, 59, 84, 85, 105, 108 \\
W & 51840x^5 - 155520x^4 - 51840x^3 + 155520x^2 \\
A & -2866544640x^9 - 1433272320x^8y - 12899450880x^8 - 31531991040x^7y \\
& - 5733089280x^7 - 12182814720x^6y^2 - 94595973120x^6y + 29411942400x^6 \\
& + 477757440x^5y^3 - 128994508800x^5y^2 - 5016453120x^5y + 22514319360x^5 \\
& + 238878720x^4y^4 - 37265080320x^4y^3 - 252017049600x^4y^2 \\
& + 181786705920x^4y - 5762949120x^4 - 39813120x^3y^5 + 11119472640x^3y^4 \\
& - 247320000000x^3y^3 + 62177883840x^3y^2 + 141017346600x^3y \\
& + 12215946395x^3 + 2030469120x^2y^5 - 41564897280x^2y^4 \\
& - 298921881600x^2y^3 + 406213263360x^2y^2 - 36314542080x^2y \\
& - 26839019520x^2 - 79626240xy^6 + 21522309120xy^5 - 203362237440xy^4 \\
& + 125468254080xy^3 + 237156122640xy^2 - 126781140290xy - 82059679185x \\
& + 3185049600y^6 - 28768296960y^5 - 103419417600y^4 + 339554862720y^3 \\
& - 253957002960y^2 - 28613719270y + 72018524470 \\
B & 238878720x^5y^5 - 35184844800x^5y^4 + 27715322880x^5y^3 \\
& + 23232493440x^5y^2 - 181065723120x^5y - 155022060450x^5 - 238878720x^4y^5 \\
& + 35184844800x^4y^4 - 27715322880x^4y^3 - 23232493440x^4y^2 \\
& + 181065723120x^4y + 155022060450x^4 - 477757440x^3y^6 + 70946979840x^3y^5 \\
& - 140460687360x^3y^4 + 20513710080x^3y^3 + 418276638720x^3y^2 \\
& - 127531376640x^3y - 33532600320x^3 - 238878720x^2y^7 + 19110297600x^2y^6 \\
& + 110640660480x^2y^5 - 258060349440x^2y^4 + 517969589760x^2y^3 \\
& - 86244914880x^2y^2 - 159231849000x^2y - 15978286235x^2 + 39813120xy^8 \\
& - 9925079040xy^7 + 85400040960xy^6 + 30495447360xy^5 - 133531581480xy^4 \\
& + 241906478245xy^3 - 196914967080xy^2 - 11285918875xy + 23733596160x \\
& - 1592524800y^8 + 12791623680y^7 + 97307343360y^6 - 214471874880y^5 \\
& + 245189135400y^4 - 60663124645y^3 - 105072781080y^2 + 79210749635y + 31709445275 \\
i = 104, m = 10319560704, \text{ neighbours } 9, 11, 23, 25, 35, 51, 55, 70, 75, 86, 89, 106 \\
W & 29828690214912x^5 - 153539583934464x^4 + 40045055311872x^3 + 80848598335488x^2 \\
& + 26965012119552x + 30587177926656 \\
A & 283982221662775934976x^{18} + 283982221662775934976x^{17}y \\
& + 2271857773302207479808x^{17} + 70995555415693983744x^{16}y^2 \\
& + 5466657767008436748288x^{16}y + 7028559986153704390656x^{16} \\
& + 3869257770155322114048x^{15}y^2 + 30137613273962096099328x^{15}y \\
& + 9253087389178782547968x^{15} + 804616294711198482432x^{14}y^3 \\
& + 41070928807978969595904x^{14}y^2 + 74521668001340118269952x^{14}y \\
& - 816448887280480813056x^{14} - 47330370277129322496x^{13}y^4 \\
& + 22464176992782504689664x^{13}y^3 + 169939694480032832421888x^{13}y^2 \\
& + 82561914652167461928960x^{13}y - 19511945146746563198976x^{13} \\
& - 11832592569282330624x^{12}y^5 + 3579359252207905013760x^{12}y^4 \\
& + 166135515969008563126272x^{12}y^3 + 343523827471404622675968x^{12}y^2 \\
& - 3750931844462498807808x^{12}y - 27349065625134560182272x^{12} \\
& - 644876295025887019008x^{11}y^5 + 72095986524637240492032x^{11}y^4 \\
& + 539392676468258135605248x^{11}y^3 + 331939719346077220995072x^{11}y^2 \\
& - 128726774561222474858496x^{11}y - 19758457491939945086976x^{11} \\
& - 134102715785199747072x^{10}y^6 + 6839238505045187100672x^{10}y^5 \\
& + 407232477963182237810688x^{10}y^4 + 911591805981937713020928x^{10}y^3 \\
& + 19448837986377057435648x^{10}y^2 - 182305739764313641451520x^{10}y \\
& - 13437880961181633478656x^{10} + 7888395046188220416x^9y^7 \\
& - 3744029498797084114944x^9y^6 + 136839988866227059556352x^9y^5 \\
& + 1069949364435404690817024x^9y^4 + 796347284604031680380928x^9y^3 \\
& - 361630652250548909113344x^9y^2 - 165221448193031503085568x^9y \\
& - 9157769282370674884608x^9 + 1972098761547055104x^8y^8 \\
& - 596559875367984168960x^8y^7 - 992951726438942244864x^8y^6 \\
& + 630560830115816946008064x^8y^5 + 1547180501890318876016640x^8y^4 \\
& + 134997062573561336561664x^8y^3 - 551876061628850994413568x^8y^2 \\
& - 129836408797719617863680x^8y + 6857316077026035105792x^8 \\
& + 107479382504314503168x^7y^8 - 12015997754106206748672x^7y^7 \\
& + 144942356628043135451136x^7y^6 + 1381357563575015521124352x^7y^5 \\
& + 1244973129522705831297024x^7y^4 - 592148783463093793456128x^7y^3
\end{aligned}$$

$$\begin{aligned}
& -545420725016053634039808x^7y^2 - 56331193366393544245248x^7y \\
& + 25340483036498884558848x^7 + 22350452630866624512x^6y^9 \\
& - 1139873084174197850112x^6y^8 - 33027724009009305354240x^6y^7 \\
& + 603978910908924190261248x^6y^6 + 1745943405819742692311040x^6y^5 \\
& + 289975922823807822200832x^6y^4 - 971706142139232357974016x^6y^3 \\
& - 384350873399205606457344x^6y^2 + 45926235958907819261952x^6y \\
& - 148264208372315258741587968x^6 - 1314732507698036736x^5y^{10} \\
& + 624004916466180685824x^5y^9 - 22808308226672465805312x^5y^8 \\
& + 46760926759732610334720x^5y^7 + 1152428450939587176431616x^5y^6 \\
& + 1280735168464600128552960x^5y^5 - 658962339113363784597504x^5y^4 \\
& - 915377030432642668953600x^5y^3 - 111666476858205825662976x^5y^2 \\
& + 98890837617255924105216x^5y + 1526645082680259343338700800x^5 \\
& - 328683126924509184x^4y^{11} + 99426645894664028160x^4y^{10} \\
& + 165163271279565864960x^4y^9 - 74253462569769718775808x^4y^8 \\
& + 302180749034332159475712x^4y^7 + 1297931705323726986805248x^4y^6 \\
& + 289544690561282866151424x^4y^5 - 1074899441887930107297792x^4y^4 \\
& - 521788216458357384413184x^4y^3 + 108908934984351501778944x^4y^2 \\
& + 81552063134952484503552x^4y - 4327225730615763721997254656x^4 \\
& + 109561042308169728x^3y^{12} - 53511439080681897984x^3y^{11} \\
& + 2818674272520282046464x^3y^{10} - 24616332654157190135808x^3y^9 \\
& - 100059016971125979611136x^3y^8 + 589557419914662537854976x^3y^7 \\
& + 838128134591455189008384x^3y^6 - 587926459704742700580864x^3y^5 \\
& - 675874973451452628049920x^3y^4 - 4865322963164727995255808x^3y^3 \\
& + 1112995978824050737840512x^3y^2 - 45690764450965733815832880x^3y \\
& + 1415392631066798098264039494x^3 - 3725075438477770752x^2y^{12} \\
& + 189978847362366308352x^2y^{11} + 5501114714814356127744x^2y^{10} \\
& - 8242378568077436944384x^2y^9 - 11604979503887108014080x^2y^8 \\
& + 600561592438290068275200x^2y^7 + 132860923541418374332416x^2y^6 \\
& - 787235832252296784248832x^2y^5 - 335273114058303284969472x^2y^4 \\
& + 24919061892271930236469248x^2y^3 + 109891752314376938323968x^2y^2 \\
& + 24709745265299739918729216x^2y + 3372713463677956561318182912x^2 \\
& + 219122084616339456xy^{13} - 104064730019043213312xy^{12} \\
& + 3822424228771437477888xy^{11} - 8389845253363020595200xy^{10} \\
& - 130586618232497932075008xy^9 + 115490175723506057084928xy^8 \\
& + 341855337393952587251712xy^7 - 349185828502584962187264xy^6 \\
& - 466876156803309928857600xy^5 + 14048024964870931531776xy^4 \\
& - 251419474397769990527437056xy^3 - 536853219652902570434784xy^2 \\
& - 227790113296413964364222628xy + 1137174570448745843223633990x \\
& - 6354540453873844224y^{13} + 220308995908011294720y^{12} \\
& + 4622915245903504736256y^{11} - 37934806772956457336832y^{10} \\
& - 40841168387630885240832y^9 + 103789935436622129528832y^8 \\
& - 1133562584342134259712y^7 - 281527558380194286403584y^6 \\
& + 1859035694083517711204352y^5 - 7016907193414168696399872y^4 \\
& + 791571967857922009275733248y^3 + 334569016710797102307450720y^2 \\
& + 59906817784761956887978452y + 1461539259286750686393515892 \\
B \quad & -657366253849018368x^5y^{12} + 213589251979776884736x^5y^{11} \\
& - 4896047881015485530112x^5y^{10} + 2625809461764824236032x^5y^9 \\
& + 87079901356426939858944x^5y^8 - 123029226584562481496064x^5y^7 \\
& + 25677425148307449053184x^5y^6 + 235940876605962736238592x^5y^5 \\
& - 1160705692945868911165440x^5y^4 + 28837708084706778235717632x^5y^3 \\
& - 5660351054558083513518336x^5y^2 + 274358505353047879292020512x^5y \\
& - 870291693608313347628515748x^5 + 657366253849018368x^4y^{12} \\
& - 213589251979776884736x^4y^{11} + 4896047881015485530112x^4y^{10} \\
& - 2625809461764824236032x^4y^9 - 87079901356426939858944x^4y^8 \\
& + 123029226584562481496064x^4y^7 - 25677425148307449053184x^4y^6 \\
& - 235940876605962736238592x^4y^5 + 1160705692945868911165440x^4y^4 \\
& - 28837708084706778235717632x^4y^3 + 5660351054558083513518336x^4y^2 \\
& - 274358505353047879292020512x^4y + 870291693608313347628515748x^4 \\
& + 1314732507698036736x^3y^{13} - 426137674057626157056x^3y^{12} \\
& + 9453912779729657659392x^3y^{11} + 2500456888078203617280x^3y^{10} \\
& - 178317334360648184758272x^3y^9 + 108181942688115641548800x^3y^8 \\
& + 143441425128942364262400x^3y^7 - 512537676363412266811392x^3y^6 \\
& + 1947838331265630156619776x^3y^5 - 55837632155582597362089984x^3y^4
\end{aligned}$$

$$\begin{aligned}
& -34339002358336231846182912x^3y^3 - 539754788203045459687636992x^3y^2 \\
& + 1306182420407634219711332352x^3y - 3171710803890171869097099264x^3 \\
& + 328683126924509184x^2y^{14} - 63928868186817036288x^2y^{13} \\
& - 634796679133535404032x^2y^{12} + 27651554532860557787136x^2y^{11} \\
& - 43589679727180637011968x^2y^{10} - 109670281340556819824640x^2y^9 \\
& + 207522951368475030847488x^2y^8 - 145617500775508806205440x^2y^7 \\
& + 349996810396465422139392x^2y^6 - 15131152956221602422128640x^2y^5 \\
& + 154796250229607264663347200x^2y^4 - 336204339735221387015500800x^2y^3 \\
& - 931500647561830569001255296x^2y^2 + 1515336719910207722338341168x^2y \\
& - 3729720810493541699839533126x^2 - 109561042308169728xy^{15} \\
& + 50553290938361315328xy^{14} - 1905209082275916939264xy^{13} \\
& + 7252706075423514034176xy^{12} + 45543581421847732813824xy^{11} \\
& - 97954455066616572936192xy^{10} + 53085384432203881512960xy^9 \\
& + 153342641110295856218112xy^8 - 204458704102124936257536xy^7 \\
& + 670411259033796434119680xy^6 + 123206946572704396748070528xy^5 \\
& + 76971010393505836013395248xy^4 + 47877495069833553131757114xy^3 \\
& - 556883545327525938677198544xy^2 + 821299063760420600460402618xy \\
& - 1524861860162005038840938496x + 3177270226936922112y^{15} \\
& - 113331768180942569472y^{14} - 2194948584543872876544y^{13} \\
& + 21055374743295032819712y^{12} - 2630429337891184312320y^{11} \\
& - 29839765955681338785792y^{10} + 75758161483064371838976y^9 \\
& + 35874087511255050682368y^8 - 1025665560961531345870848y^7 \\
& + 4663309478174800233182208y^6 - 401239751368317443239059072y^5 \\
& + 236540355375905382954517200y^4 - 658767034867215340049148474y^3 \\
& - 635072235113854570611396528y^2 + 437834317866075041013681942y \\
& - 1198557356401831362654429114
\end{aligned}$$

$i = 105$, $m = 4$, neighbours 114, 9, 35, 102, 103

$$W \quad 6x^3 - 18x^2 - 6x + 18$$

$$A \quad -28x^3 - 8x^2y + 2x^2 - 72xy + 144x - 16y^2 + 124y - 58$$

$$B \quad 72x^5 - 144x^4 - 144x^3y - 240x^3 - 144x^2y^2 - 24x^2y - 20x^2 + 28xy^3 + 84xy \\ + 18x + 8y^4 - 154y^3 - 232y^2 - 170y + 74$$

$i = 106$, $m = 191102976$, neighbours 114, 7, 9, 11, 24, 30, 35, 50, 84, 85, 104

$$W \quad 308440203264x^5 - 1117952409600x^4 + 580188635136x^3 + 585921724416x^2 \\ + 202951360512x - 96315899904$$

$$\begin{aligned}
A \quad & -73040694872113152x^{13} - 146081389744226304x^{12}y - 255642432052396032x^{12} \\
& - 91300868590141440x^{11}y^2 - 894748512183386112x^{11}y - 474764516668735488x^{11} \\
& - 18260173718028288x^{10}y^3 - 1259951986543951872x^{10}y^2 \\
& - 2410342930779734016x^{10}y - 681713152139722752x^{10} \\
& - 718233499575779328x^9y^3 - 4072018739120308224x^9y^2 \\
& - 3487693180143403008x^9y - 419983995514650624x^9 - 121734491453521920x^8y^4 \\
& - 3974631145957490688x^8y^3 - 8104473768518221824x^8y^2 \\
& - 4312444359741014016x^8y - 206948635470987264x^8 + 15216811431690240x^7y^5 \\
& - 1987315572978745344x^7y^4 - 8174471101103996928x^7y^3 \\
& - 8993135556128931840x^7y^2 - 2930757881743540224x^7y + 229266625570799616x^7 \\
& + 3043362286338048x^6y^6 - 228252171475353600x^6y^5 \\
& - 4653300935810875392x^6y^4 - 9560215395483254784x^6y^3 \\
& - 9099653236150763520x^6y^2 - 1934563960015552512x^6y \\
& - 15855052182477814628352x^6 + 119705583262629888x^5y^6 \\
& - 1676892619772264448x^5y^5 - 3843766567644954624x^5y^4 \\
& - 6266282947570040832x^5y^3 - 8308886268750594048x^5y^2 \\
& - 748159895391436800x^5y + 114941397421916285829120x^5 \\
& + 20289081908920320x^4y^7 + 114633312785399808x^4y^6 \\
& + 2281507260658089984x^4y^5 + 7557683011072819200x^4y^4 \\
& - 3492258223572910080x^4y^3 - 7259940734059413504x^4y^2 \\
& + 3213790574372978688x^4y - 267952604924764691103744x^4 \\
& - 9468238224162816x^3y^8 + 317753088528089088x^3y^7 \\
& + 2668662908157689856x^3y^6 + 16454883040190005248x^3y^5 \\
& + 26999546153580232704x^3y^4 - 315247631192189558784x^3y^3 \\
& - 917641676845018400256x^3y^2 - 6388403238577219294944x^3y \\
& + 164718375645639738861834x^3 - 507227047723008x^2y^9 \\
& + 38042028579225600x^2y^8 + 720262407766671360x^2y^7 \\
& + 11215804479251152896x^2y^6 + 32300725626048872448x^2y^5
\end{aligned}$$

	$+13356809847689969664x^2y^4 + 262701510875571343360x^2y^3$ $+813507646706417664x^2y^2 + 2647995625668379410432x^2y$ $+124860950380028872359936x^2 - 19950930543771648xy^9 + 293635146890870784xy^8$ $+4753327841473462272xy^7 + 24682626363099709440xy^6 + 28004422183380320256xy^5$ $-34440136365780615168xy^4 - 18556496929018266577920xy^3$ $+1862722011309953721408xy^2 - 6101382906196440525132xy + 14281370880701876569314x$ $-1014454095446016y^{10} + 56640353662402560y^9 + 461844316592013312y^8$ $+4263741463600300032y^7 + 659390801834409984y^6 + 76292250637816037376y^5$ $+1765689344852298006528y^4 + 54755601212355940767744y^3$ $+61205532101481341904768y^2 - 13667088109594073167464y - 40136433929654062961172$
B	$41592617913286656x^5y^8 + 80797041810210816x^5y^7$ $-3254202744617041920x^5y^6 - 17964549886300913664x^5y^5$ $-76104463114162667520x^5y^4 + 1823214040983715921920x^5y^3$ $+5461246543401539970048x^5y^2 + 38353790424914201086272x^5y$ $-36646580867377564153404x^5 - 83185235826573312x^4y^8$ $-161594083620421632x^4y^7 + 6508405489234083840x^4y^6$ $+35929099772601827328x^4y^5 + 152208926228325335040x^4y^4$ $-3646428081967431843840x^4y^3 - 10922493086803079940096x^4y^2$ $-76707580849828402172544x^4y + 73293161734755128306808x^4$ $-83185235826573312x^3y^9 - 121734491453521920x^3y^8$ $+6585835987635535872x^3y^7 + 32810488809010495488x^3y^6$ $+134992899253953626112x^3y^5 - 3719361525785171066880x^3y^4$ $-9175246297527018848256x^3y^3 - 71473886245735259701248x^3y^2$ $+110048877558631237681152x^3y - 209308846447429611945984x^3$ $-14202357336244224x^2y^{10} - 121734491453521920x^2y^9$ $-1126720348675375104x^2y^8 + 5021318815804293120x^2y^7$ $+45283060502432317440x^2y^6 - 971643048549106581504x^2y^5$ $+3812798984297037299712x^2y^4 - 16523533199240137285632x^2y^3$ $-59009972030733641955840x^2y^2 + 66375657885956928952032x^2y$ $-104783958670745652373770x^2 + 9468238224162816xy^{11} - 122977902202454016xy^{10}$ $-2088564241245143040xy^9 - 7000230751604047872xy^8 - 326924302499315712xy^7$ $+62680220528169443328xy^6 + 10082000425774386267648xy^5$ $+20942661297339477374688xy^4 + 19007404783742568840438xy^3$ $-23252568203472661016352xy^2 + 14856294371056379297814xy + 3229709698849109911818x$ $+507227047723008y^{12} - 28573790355062784y^{11} - 216128036070752256y^{10}$ $-1613375494315573248y^9 + 3285459195742126080y^8 - 35249302141460545536y^7$ $-865819049768897200128y^6 - 26996952471259560053760y^5$ $-18040400036694475369920y^4 - 12403066568979059177964y^3$ $-31698436009821257087424y^2 - 7916717159763760733100y + 30857928378659785036308$
$i = 107, m = 24, \text{neighbours } 3, 9, 21, 31$	
W	$6x^4 + 12x^3$
A	$-6x^4 - 120x^3 + 168x^2 + 288xy + 480x + 24y^2 + 240y + 216$
B	$576x^5 + 1152x^4 + 1152x^3y + 1158x^3 + 6x^2y^3 + 432x^2y^2 + 294x^2y$ $-6x^2 + 132xy^3 - 864xy^2 - 1584xy - 624x + 12y^4 - 180y^3 - 852y^2 - 744y - 180$
$i = 108, m = 288, \text{neighbours } 25, 56, 74, 89, 103$	
W	$432x^4 + 864x^2$
A	$55296x^6 + 165888x^5 + 331776x^4y - 44928x^4 - 4608x^3y^2 + 917088x^3y$ $-301730x^3 - 9216x^2y^3 + 732672x^2y^2 - 64512x^2y - 506880x^2 - 27648xy^3$ $+1336512xy^2 - 1451908xy - 585780x - 18432y^4 + 617472y^3 - 1120128y^2 - 267256y$ $+788344$
B	$27648x^5y^2 - 691776x^5y - 650292x^5 - 55296x^4y^2 + 1383552x^4y$ $+1300584x^4 - 55296x^3y^3 + 1492992x^3y^2 - 1437696x^3y - 590976x^3$ $-221184x^2y^4 + 957312x^2y^3 - 133632x^2y^2 - 1867488x^2y + 477986x^2$ $+4608xy^5 - 364128xy^4 + 529826xy^3 - 907872xy^2 - 499966xy + 827230x + 9216y^6$ $-299520y^5 + 435648y^4 + 103868y^3 - 536640y^2 + 986492y + 394172$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 83\}, k_{ij} = 2 \cdot 3$	
α	$5x^3 + 6$
β	$60x^5 - 60x^4 + 5x^3y^2 + 65x^3 + 22x^2 - 12x + 6y^2 + 18$

γ	$-30x^5 + 30x^4 - 30x^3 - 11x^2 + 6x - 6$
$\{i, j\} = \{0, 86\}, k_{ij} = 2 \cdot 3$	
α	$5x^3 + 6$
β	$-60x^5 + 60x^4 + 5x^3y^2 - 55x^3 - 22x^2 + 12x + 6y^2 - 6$
γ	$-30x^5 + 30x^4 - 30x^3 - 11x^2 + 6x - 6$
$\{i, j\} = \{1, 12\}, k_{ij} = 1$	
α	0
β	-2
γ	1
$\{i, j\} = \{1, 93\}, k_{ij} = 2^4 \cdot 3$	
α	$23x^3 + 35$
β	$276x^5 - 552x^4 + 23x^3y^2 - 23x^3y + 874x^3 - 638x^2 + 402x + 35y^2 - 35y - 96$
γ	$-138x^5 + 276x^4 - 414x^3 + 319x^2 - 201x + 83$
$\{i, j\} = \{2, 13\}, k_{ij} = 2^8$	
α	108
β	$-324x^3 - 162x^2y - 81xy^2 - 54y^3 - 45y^2 - 30y - 128$
γ	$27y^3 + 90y^2 + 60y + 148$
$\{i, j\} = \{2, 63\}, k_{ij} = 3^3$	
α	$2464x^4 - 12342y - 5446$
β	$6160x^7 - 792x^6y + 2640x^5y + 11352x^5 + 2860x^4y^2 - 4708x^4y - 39688x^4$ $+ 28240x^3 - 12654x^2y - 8712xy^2 + 3372xy - 37988x - 9680y^3 + 4148y^2 - 10332y$ $- 17237$
γ	$-13552x^6 + 3872x^5y - 13552x^5 - 396x^4y^2 - 5544x^4y - 31680x^4$ $- 5324x^2y^2 - 2662x^2y - 28974x^2 - 5324xy^2 + 24112xy - 28974x - 2662y^3$ $- 6932y^2 + 4922y - 5473$
$\{i, j\} = \{2, 79\}, k_{ij} = 2^4 \cdot 3^2$	
α	$-12x^4 + 11y - 2$
β	$36x^7 + 18x^6y - 24x^5y - 36x^5 - 3x^4y^2 + 6x^4y + 42x^4 + 48x^3 + 24x^2y$ $+ 22xy^2 - 4xy + 73x + 11y^3 - 12y^2 + 35y - 72$
γ	$9x^4y^2 + 24x^4y + 36x^4 - 10y^2 + 4y - 73$
$\{i, j\} = \{2, 102\}, k_{ij} = 2^3 \cdot 3^3$	
α	$-1364x^4 - 10261y - 19212$
β	$5084x^7 + 2790x^6y + 4650x^5y - 16740x^5 + 4216x^4y^2 - 8804x^4y + 20398x^4$ $+ 6138x^3 - 28545x^2y - 9610xy^2 - 11615xy - 21573x + 651y^3 - 117718y^2 - 128603y$ $- 18969$
γ	$-1984x^6 - 496x^5y - 1984x^5 + 2790x^4y^2 - 5828x^4y + 35464x^4$ $- 2728x^2y^2 + 24056x^2y + 33432x^2 - 2728xy^2 + 87284xy + 33432x + 5456y^3$ $- 54709y^2 - 4462y + 9714$
$\{i, j\} = \{3, 107\}, k_{ij} = 1$	
α	$6x^4 - 3y - 4$
β	$36x^7 - 36x^6y - 36x^6 - 6x^5y + 12x^5 + 78x^4y^2 + 150x^4y + 42x^4 - 12x^3$ $+ 12x^2y + 12x^2 + 3xy^2 - 2xy + x - 3y^3 - 31y^2 - 46y - 31$
γ	$-36x^4y^2 - 78x^4y - 24x^4 + 15y^2 + 22y + 13$
$\{i, j\} = \{4, 11\}, k_{ij} = 3 \cdot 5^2 \cdot 7$	
α	-128
β	$384x^3 + 192x^2y + 192x^2 + 96xy^2 + 192xy + 96x + 24y^3 + 20y^2 + 186y - 143$
γ	$-24y^3 - 124y^2 + 42y - 255$
$\{i, j\} = \{4, 19\}, k_{ij} = 2$	
α	$48x^4 - 4y + 12$
β	$-144x^7 - 72x^6y - 72x^6 + 168x^5y + 144x^5 + 84x^4y^2 + 156x^4y + 48x^4$ $- 36x^3 - 18x^2y - 18x^2 - 8xy^2 + 42xy + 32x - 4y^3 + 17y^2 + 39y + 10$
γ	$-36x^4y^2 - 240x^4y - 180x^4 - y^2 - 60y - 41$
$\{i, j\} = \{4, 27\}, k_{ij} = 2^2 \cdot 3 \cdot 7$	
α	$24x^4 + 18y + 10$
β	$-72x^7 - 36x^6y - 36x^6 + 84x^5y + 36x^5 + 42x^4y^2 + 60x^4y + 6x^4 - 12x^3$ $- 6x^2y - 6x^2 + 36xy^2 + 26xy - 48x + 18y^3 + 31y^2 - 20y - 29$
γ	$-18x^4y^2 - 120x^4y - 54x^4 - 39y^2 - 32y + 45$
$\{i, j\} = \{4, 35\}, k_{ij} = 2^4 \cdot 3$	
α	$46x^4 - 48$

$$\begin{aligned}
\beta & -92x^7 + 46x^4y^2 + 138x^4y + 23x^4 - 88x^3 + 39x - 184y^3 - 255y^2 - 573y - 262 \\
\gamma & -92x^6 - 92x^5y - 184x^5 - 92x^4 + 184x^2y^2 + 207x^2y + 533x^2 + 184xy^2 \\
& + 119xy + 445x - 184y^3 - 207y^2 - 621y - 166 \\
\{i, j\} & = \{4, 58\}, k_{ij} = 3 \cdot 5 \\
\alpha & -48x^4 - 234y - 100 \\
\beta & 864x^7 - 1008x^6y - 1008x^6 - 672x^5y - 1008x^5 - 552x^4y^2 - 504x^4y \\
& + 1176x^4 - 8748x^3 + 27636x^2y + 27636x^2 - 1008xy^2 + 23800xy + 36882x - 324y^3 \\
& + 14285y^2 + 18387y - 115 \\
\gamma & -720x^6 + 1440x^5y + 2160x^5 - 504x^4y^2 - 1416x^4y - 1656x^4 + 180x^2y^2 \\
& + 1890x^2y + 9660x^2 - 180xy^2 - 33900xy - 41670x - 90y^3 + 14061y^2 + 32906y + 18606 \\
\{i, j\} & = \{4, 64\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \\
\alpha & -12x^4 + 17y + 43 \\
\beta & 36x^7 + 18x^6y + 18x^6 - 24x^5y + 12x^5 - 30x^4y^2 - 90x^4y + 18x^4 - 120x^3 \\
& - 60x^2y - 60x^2 + 34xy^2 + 18xy + 237x + 17y^3 + 154y^2 + 275y + 631 \\
\gamma & 18x^4y^2 + 84x^4y - 6x^4 - 128y^2 - 156y - 534 \\
\{i, j\} & = \{4, 100\}, k_{ij} = 2^2 \\
\alpha & y + 3 \\
\beta & 24x^7 + 18x^6y + 18x^6 + 12x^5y + 36x^4y^2 + 204x^4y - 96x^4 - 732x^3 \\
& - 306x^2y - 306x^2 + 2xy^2 - 210xy + 2881x - 215y^3 - 1458y^2 - 4281y - 1809 \\
\gamma & -48x^6 - 12x^5y - 108x^5 + 18x^4y^2 + 84x^4y - 150x^4 + 54x^2y^2 + 318x^2y \\
& + 1500x^2 + 108xy^2 + 516xy + 2880x - 108y^3 - 784y^2 - 2478y - 1808 \\
\{i, j\} & = \{5, 29\}, k_{ij} = 2^4 \cdot 3 \\
\alpha & 6x^4 - 2y - 12 \\
\beta & -36x^7 - 36x^6y - 72x^6 + 42x^5y + 54x^5 + 24x^4y^2 + 45x^4y + 3x^4 + 72x^3 \\
& + 72x^2y + 144x^2 - 2xy^2 - 78xy - 124x - 2y^3 - 45y^2 - 95y - 30 \\
\gamma & -18x^4y^2 - 93x^4y - 99x^4 + 37y^2 + 183y + 206 \\
\{i, j\} & = \{5, 30\}, k_{ij} = 2 \cdot 3 \cdot 11^2 \\
\alpha & -14x^4 - 49y - 159 \\
\beta & 84x^7 + 84x^6y + 168x^6 - 182x^5y - 392x^5 - 98x^4y^2 - 238x^4y - 42x^4 \\
& + 156x^3 + 156x^2y + 312x^2 - 49xy^2 - 373xy - 609x - 49y^3 - 266y^2 - 309y + 174 \\
\gamma & 84x^4y^2 + 518x^4y + 728x^4 + 205y^2 + 997y + 1233 \\
\{i, j\} & = \{5, 44\}, k_{ij} = 2^4 \cdot 3^3 \\
\alpha & x^4 - 8y - 22 \\
\beta & -6x^7 - 6x^6y - 12x^6 - 5x^5y - 29x^5 + 7x^4y^2 - x^4y - 69x^4 - 12x^3 \\
& - 12x^2y - 24x^2 - 8xy^2 + 6xy - 82x - 8y^3 - 2y^2 + 46y - 210 \\
\gamma & -6x^4y^2 - 19x^4y + 5x^4 - 4y^2 - 54y + 34 \\
\{i, j\} & = \{5, 52\}, k_{ij} = 2 \cdot 3^5 \cdot 7 \\
\alpha & 26x^4 + 676y + 497 \\
\beta & -156x^7 - 156x^6y - 312x^6 - 442x^5y + 442x^5 - 130x^4y^2 - 78x^4y + 2990x^4 \\
& + 4038x^3 + 4038x^2y + 8076x^2 + 676xy^2 + 7879xy + 8761x + 676y^3 + 2507y^2 + 7297y \\
& + 2243 \\
\gamma & -156x^4y^2 - 182x^4y - 1066x^4 + 3362y^2 + 8273y + 7391 \\
\{i, j\} & = \{5, 60\}, k_{ij} = 2 \cdot 3^4 \\
\alpha & 12x^4 - 28y - 74 \\
\beta & -72x^7 - 72x^6y - 144x^6 + 12x^5y - 48x^5 + 48x^4y^2 + 126x^4y + 12x^4 \\
& + 216x^3 + 216x^2y + 432x^2 - 28xy^2 + 38xy - 96x - 28y^3 - 140y^2 - 405y - 598 \\
\gamma & -36x^4y^2 - 150x^4y - 120x^4 + 122y^2 + 413y + 480 \\
\{i, j\} & = \{5, 62\}, k_{ij} = 2^6 \cdot 3 \\
\alpha & 688x^4 + 776y + 2305 \\
\beta & -5152x^7 - 4896x^6y - 9792x^6 + 8160x^5y + 16320x^5 + 3136x^4y^2 + 5536x^4y \\
& - 4208x^4 - 56950x^3 - 47430x^2y - 94860x^2 + 136xy^2 + 80206xy + 125526x + 648y^3 \\
& + 25464y^2 + 52924y - 76019 \\
\gamma & 512x^6 - 128x^5y + 256x^5 - 2448x^4y^2 - 13744x^4y - 17952x^4 + 256x^2y^2 \\
& - 1344x^2y + 25984x^2 + 256xy^2 - 6104xy + 16464x + 128y^3 - 24327y^2 - 127403y \\
& - 154151 \\
\{i, j\} & = \{5, 68\}, k_{ij} = 2 \cdot 3^3 \\
\alpha & -10x^4 + 25y + 44 \\
\beta & 60x^7 + 60x^6y + 120x^6 - 70x^5y - 150x^5 + 50x^4y^2 + 330x^4y + 490x^4 \\
& - 54x^3 - 54x^2y - 108x^2 + 25xy^2 + 23xy + 255x + 25y^3 - 85y^2 - 202y - 476 \\
\gamma & 60x^4y^2 + 310x^4y + 390x^4 - 79y^2 - 239y - 471
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{5, 82\}, k_{ij} = 3^3 \cdot 11 \\
&\alpha \quad -1360x^4 - 4600y - 9233 \\
&\beta \quad 21920x^7 + 28800x^6y + 57600x^5y + 38400x^4y + 48000x^5 + 13040x^4y^2 \\
&\quad + 41120x^4y + 16400x^4 - 332134x^3 - 443160x^2y - 886320x^2 + 4000xy^2 - 612080xy \\
&\quad - 892776x - 2880y^3 - 222753y^2 - 725510y - 555723 \\
&\gamma \quad -6880x^6 - 3440x^5y + 14400x^4y^2 + 45280x^4y + 40480x^4 - 3440x^2y^2 \\
&\quad - 1720x^2y + 170366x^2 + 3440xy^2 + 57233xy - 59340x + 1720y^3 - 222720y^2 - 720976y \\
&\quad - 550958 \\
&\{i, j\} = \{5, 90\}, k_{ij} = 3 \cdot 11 \\
&\alpha \quad -12x^4 - 128y + 63 \\
&\beta \quad 72x^7 + 72x^6y + 144x^6 + 132x^5y - 72x^5 + 132x^4y^2 + 192x^4y - 132x^4 \\
&\quad - 1350x^3 - 1350x^2y - 2700x^2 - 128xy^2 - 1869xy - 1774x - 128y^3 - 2125y^2 - 5384y \\
&\quad - 3611 \\
&\gamma \quad 72x^4y^2 + 156x^4y + 360x^4 - 1222y^2 - 3531y - 3626 \\
&\{i, j\} = \{6, 17\}, k_{ij} = 3^3 \cdot 11 \\
&\alpha \quad 3 \\
&\beta \quad 18x^3 - 18x^2y + 36x^2 + 18xy^2 - 72xy + 72x - 12y^3 + 61y^2 - 72y - 45 \\
&\gamma \quad 6y^3 - 47y^2 + 144y - 192 \\
&\{i, j\} = \{6, 33\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 11 \\
&\alpha \quad -534x^4 - 7921y + 8554 \\
&\beta \quad -3204x^7 + 3204x^6y - 6408x^6 - 9078x^5y + 42186x^5 + 2670x^4y^2 - 16554x^4y \\
&\quad - 26166x^4 - 1542x^3 + 1542x^2y - 3084x^2 + 7921xy^2 - 5259xy + 19413x - 7921y^3 \\
&\quad + 2175y^2 - 14998y - 3782 \\
&\gamma \quad 3204x^4y^2 - 21894x^4y + 55002x^4 + 9463y^2 - 11427y + 25581 \\
&\{i, j\} = \{6, 52\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5^2 \\
&\alpha \quad -21x^4 - 49y + 272 \\
&\beta \quad -126x^7 + 126x^6y - 252x^6 + 147x^5y - 399x^5 + 105x^4y^2 - 21x^4y - 609x^4 \\
&\quad + 498x^3 - 498x^2y + 996x^2 + 49xy^2 - 525xy + 1416x - 49y^3 - 275y^2 + 419y + 1952 \\
&\gamma \quad 126x^4y^2 - 357x^4y + 105x^4 - 449y^2 + 1467y - 576 \\
&\{i, j\} = \{6, 53\}, k_{ij} = 1 \\
&\alpha \quad 166x^4 + 62y + 83 \\
&\beta \quad 1028x^7 - 1044x^6y + 2088x^6 + 1392x^5y - 4872x^5 - 356x^4y^2 + 2152x^4y \\
&\quad - 594x^4 + 2676x^3 + 348x^2y - 696x^2 + 58xy^2 - 464xy - 1580x + 38y^3 + 646y^2 \\
&\quad + 666y - 41 \\
&\gamma \quad 16x^6 - 8x^5y + 32x^5 - 522x^4y^2 + 2768x^4y - 4492x^4 + 48x^2y^2 \\
&\quad - 472x^2y + 1600x^2 + 48xy^2 + 1040xy - 1424x - 24y^3 + 487y^2 - 688y - 1518 \\
&\{i, j\} = \{6, 58\}, k_{ij} = 3^2 \cdot 5 \\
&\alpha \quad -80x^4 - 72y + 207 \\
&\beta \quad -224x^7 + 96x^6y - 192x^6 + 64x^5y - 96x^5 - 32x^4y^2 - 96x^4y + 16x^4 \\
&\quad + 410x^3 + 414x^2y - 828x^2 - 24xy^2 + 192xy - 936x - 104y^3 + 114y^2 - 504y - 81 \\
&\gamma \quad 128x^6 - 64x^5y + 48x^4y^2 - 160x^4y + 144x^4 + 64x^2y^2 + 288x^2y \\
&\quad - 392x^2 - 64xy^2 + 124xy - 432x - 32y^3 + 115y^2 - 660y + 792 \\
&\{i, j\} = \{6, 69\}, k_{ij} = 2^2 \cdot 3^2 \cdot 11 \\
&\alpha \quad -125x^4 + 100y - 164 \\
&\beta \quad -875x^7 + 900x^6y - 1800x^6 + 1050x^5y - 150x^5 + 775x^4y^2 - 2550x^4y \\
&\quad + 4775x^4 - 1834x^3 + 5754x^2y - 11508x^2 - 625xy^2 + 4563xy - 2394x - 75y^3 \\
&\quad + 4565y^2 - 11546y + 8532 \\
&\gamma \quad -125x^6 + 25x^5y + 200x^5 + 900x^4y^2 - 2475x^4y + 2800x^4 + 175x^2y^2 \\
&\quad + 1650x^2y - 4675x^2 - 350xy^2 + 620xy + 1510x - 175y^3 + 4529y^2 - 11718y + 8252 \\
&\{i, j\} = \{6, 77\}, k_{ij} = 3^3 \cdot 11 \\
&\alpha \quad 1904x^4 + 204y - 381 \\
&\beta \quad -23392x^7 + 14688x^6y - 29376x^6 - 4896x^5y + 26928x^5 - 5440x^4y^2 \\
&\quad + 53040x^4y - 64736x^4 + 6706x^3 + 44910x^2y - 89820x^2 - 3468xy^2 - 24660xy \\
&\quad + 72456x - 884y^3 - 12534y^2 + 68232y - 74093 \\
&\gamma \quad -17408x^6 - 4352x^5y - 8704x^5 + 7344x^4y^2 - 44880x^4y + 51544x^4 \\
&\quad + 2176x^2y^2 - 16320x^2y + 16736x^2 + 2176xy^2 + 9488xy - 34880x + 1088y^3 \\
&\quad + 12561y^2 - 67974y + 74432 \\
&\{i, j\} = \{7, 15\}, k_{ij} = 2 \cdot 3^5 \\
&\alpha \quad 1 \\
&\beta \quad 6x^3 - 6x^2y - 12x^2 + 6xy^2 + 24xy + 24x - 12y^3 - 91y^2 - 271y - 438
\end{aligned}$$

$$\begin{aligned}
&\gamma \quad 6y^3 + 55y^2 + 199y + 391 \\
&\{i, j\} = \{7, 20\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5^2 \\
&\alpha \quad -39x^4 - 169y - 88 \\
&\beta \quad -234x^7 + 234x^6y + 468x^6 + 273x^5y - 117x^5 - 273x^4y^2 - 429x^4y + 195x^4 \\
&\quad -138x^3 + 138x^2y + 276x^2 + 169xy^2 - 281xy + 412x - 169y^3 - 57y^2 - 19y - 912 \\
&\gamma \quad 234x^4y^2 + 1209x^4y + 819x^4 + 307y^2 + 271y + 964 \\
&\{i, j\} = \{7, 36\}, k_{ij} = 2^3 \cdot 5^2 \\
&\alpha \quad 6x^4 - 2y - 76 \\
&\beta \quad 36x^7 - 36x^6y - 72x^6 + 30x^5y + 30x^5 - 12x^4y^2 - 33x^4y + 3x^4 - 456x^3 \\
&\quad + 456x^2y + 912x^2 + 2xy^2 - 390xy - 340x - 2y^3 + 157y^2 + 401y - 138 \\
&\gamma \quad -18x^4y^2 - 57x^4y - 57x^4 + 229y^2 + 717y + 742 \\
&\{i, j\} = \{7, 44\}, k_{ij} = 2 \cdot 3^3 \\
&\alpha \quad x + 1 \\
&\beta \quad 6x^4 - 6x^3y - 6x^3 - 25x^2y - 83x^2 + 7xy^2 + 4xy - 64x + 7y^2 + 23y - 47 \\
&\gamma \quad -6xy^2 - 43xy - 95x - 6y^2 - 43y - 41 \\
&\{i, j\} = \{7, 54\}, k_{ij} = 3^2 \\
&\alpha \quad 560x^4 + 2268y + 2488 \\
&\beta \quad 9184x^7 - 12096x^6y - 24192x^6 + 10080x^5y + 4032x^5 - 5488x^4y^2 - 2856x^4y \\
&\quad + 15400x^4 - 67808x^3 + 116064x^2y + 232128x^2 - 1176xy^2 - 92184xy - 63460x \\
&\quad + 1904y^3 + 58126y^2 + 41882y - 97159 \\
&\gamma \quad 2912x^6 - 1456x^5y - 1456x^5 - 6048x^4y^2 - 21336x^4y - 21448x^4 \\
&\quad + 728x^2y^2 - 2184x^2y - 36972x^2 + 364xy^2 + 23036xy + 29770x - 364y^3 \\
&\quad + 59082y^2 + 214948y + 196797 \\
&\{i, j\} = \{7, 59\}, k_{ij} = 2 \cdot 3^2 \cdot 5^2 \\
&\alpha \quad -16x^4 - 8y - 79 \\
&\beta \quad -96x^7 + 96x^6y + 192x^6 + 160x^5y + 320x^5 - 64x^4y^2 - 96x^4y + 48x^4 \\
&\quad - 234x^3 + 234x^2y + 468x^2 + 8xy^2 + 410xy + 710x - 8y^3 - 184y^2 - 192y + 147 \\
&\gamma \quad 48x^4y^2 + 272x^4y + 352x^4 + 121y^2 + 673y + 823 \\
&\{i, j\} = \{7, 71\}, k_{ij} = 2^7 \\
&\alpha \quad 2288x^4 + 11132y + 19103 \\
&\beta \quad 21472x^7 - 25344x^6y - 50688x^6 + 33792x^5y + 29568x^5 - 10384x^4y^2 \\
&\quad - 23760x^4y + 18832x^4 - 32694x^3 + 122256x^2y + 244512x^2 - 3872xy^2 - 152624xy \\
&\quad - 223614x + 9680y^3 + 64501y^2 + 174279y + 111359 \\
&\gamma \quad 3872x^6 - 1936x^5y - 12672x^4y^2 - 37664x^4y - 37840x^4 + 2904x^2y^2 \\
&\quad - 11132x^2y - 37048x^2 + 2904xy^2 + 33649xy + 52514x - 1452y^3 + 66210y^2 + 225939y \\
&\quad + 203743 \\
&\{i, j\} = \{7, 75\}, k_{ij} = 2^2 \cdot 5 \cdot 13 \\
&\alpha \quad 4752x^4 - 2916y - 7603 \\
&\beta \quad 38880x^7 - 44064x^6y - 88128x^6 - 29376x^5y - 51408x^5 - 17280x^4y^2 \\
&\quad - 108432x^4y - 114912x^4 - 277434x^3 + 100878x^2y + 201756x^2 + 16524xy^2 \\
&\quad + 30838xy + 345826x + 1620y^3 + 75398y^2 + 295990y + 226639 \\
&\gamma \quad 5184x^6 - 2592x^5y - 10368x^5 - 22032x^4y^2 - 102816x^4y - 106056x^4 \\
&\quad - 9072x^2y^2 - 55080x^2y - 180168x^2 + 9072xy^2 - 33198xy + 3612x + 4536y^3 \\
&\quad + 81705y^2 + 367997y + 461141 \\
&\{i, j\} = \{7, 78\}, k_{ij} = 3^2 \cdot 5 \cdot 13 \\
&\alpha \quad -22736x^4 - 2900y - 5865 \\
&\beta \quad -199520x^7 + 183744x^6y + 367488x^6 - 61248x^5y - 275616x^5 - 114608x^4y^2 \\
&\quad - 165184x^4y + 321552x^4 - 649370x^3 - 798996x^2y - 1597992x^2 - 74008xy^2 \\
&\quad + 497288xy + 2093572x - 28536y^3 + 131763y^2 + 1173926y + 1643087 \\
&\gamma \quad -31552x^6 - 7888x^5y - 47328x^5 + 91872x^4y^2 + 313200x^4y + 150800x^4 \\
&\quad + 51272x^2y^2 + 404260x^2y + 176630x^2 + 51272xy^2 - 319923xy - 1271736x \\
&\quad + 25636y^3 - 131828y^2 - 1176696y - 1646312 \\
&\{i, j\} = \{7, 93\}, k_{ij} = 2^3 \cdot 3^5 \\
&\alpha \quad 1446x^4 - 58081y - 59041 \\
&\beta \quad 8676x^7 - 8676x^6y - 17352x^6 - 36150x^5y - 182196x^5 + 18798x^4y^2 \\
&\quad + 112788x^4y - 67962x^4 + 24x^3 - 24x^2y - 48x^2 + 58081xy^2 + 58945xy + 57577x \\
&\quad - 58081y^3 - 58993y^2 - 115850y - 59233 \\
&\gamma \quad -8676x^4y^2 - 70854x^4y - 216900x^4 + 58057y^2 + 58849y + 57481 \\
&\{i, j\} = \{7, 106\}, k_{ij} = 2 \cdot 3^2 \\
&\alpha \quad 2x^4 - 5y - 19
\end{aligned}$$

β	$12x^7 - 12x^6y - 24x^6 - 2x^5y - 28x^5 - 22x^4y^2 - 68x^4y - 94x^4 + 5xy^2$
	$-xy - x - 5y^3 + y^2 - 4y - 19$
γ	$-12x^4y^2 - 50x^4y - 76x^4 + 5y^2 - y - 1$
$\{i, j\}$	$\{8, 16\}, k_{ij} = 3^2 \cdot 5$
α	24
β	$72x^3 - 36x^2y + 36x^2 + 18xy^2 - 36xy + 18x - 6y^3 + 23y^2 - 9y + 13$
γ	$3y^3 - 4y^2 + 18y - 8$
$\{i, j\}$	$\{8, 26\}, k_{ij} = 2$
α	$48x^4 + 4y + 12$
β	$144x^7 - 72x^6y + 72x^6 - 24x^5y - 144x^5 + 12x^4y^2 + 60x^4y - 48x^4$
	$+36x^3 - 18x^2y + 18x^2 - 8xy^2 - 6xy - 32x + 4y^3 - y^2 + 15y - 10$
γ	$-36x^4y^2 + 48x^4y - 180x^4 - 17y^2 + 12y - 41$
$\{i, j\}$	$\{8, 38\}, k_{ij} = 2^5$
α	$-240x^4 + 25y - 23$
β	$-720x^7 + 360x^6y - 360x^6 + 300x^5y + 480x^5 - 60x^4y^2 - 120x^4y + 450x^4$
	$+36x^3 - 18x^2y + 18x^2 - 50xy^2 + 5xy + 71x + 25y^3 - 57y^2 - 9y + 55$
γ	$180x^4y^2 - 60x^4y + 660x^4 - 59y^2 + 23y + 62$
$\{i, j\}$	$\{8, 50\}, k_{ij} = 2 \cdot 3^4 \cdot 7$
α	$-4x^4 + 3y + 49$
β	$-12x^7 + 6x^6y - 6x^6 + 2x^5y - 18x^5 + 2x^4y^2 + 6x^4y - 26x^4 + 102x^3$
	$-51x^2y + 51x^2 - 6xy^2 - 47xy + 216x + 3y^3 - 11y^2 - 126y + 323$
γ	$6x^4y^2 - 8x^4y - 30x^4 - 63y^2 + 8y + 381$
$\{i, j\}$	$\{8, 57\}, k_{ij} = 2^2 \cdot 3 \cdot 5$
α	$-48x^4 + y - 65$
β	$-144x^7 + 72x^6y - 72x^6 + 60x^5y + 24x^5 - 12x^4y^2 + 12x^4y + 18x^4$
	$-180x^3 + 90x^2y - 90x^2 - 2xy^2 + 83xy + 45x + y^3 - 21y^2 + 16y + 35$
γ	$36x^4y^2 - 12x^4y + 60x^4 + 43y^2 - 7y + 90$
$\{i, j\}$	$\{8, 103\}, k_{ij} = 2^4 \cdot 3^3 \cdot 5$
α	$3808x^4 + 51527y - 50807$
β	$12376x^7 - 6426x^6y + 6426x^6 + 10710x^5y - 100674x^5 - 9044x^4y^2$
	$+70210x^4y - 148274x^4 - 33791x^3 + 136107x^2y - 136107x^2 - 84966xy^2$
	$+202269xy - 128007x + 33439y^3 + 218194y^2 - 453539y + 149899$
γ	$1904x^6 - 476x^5y - 1428x^5 - 6426x^4y^2 + 30940x^4y - 202538x^4$
	$+4522x^2y^2 - 64498x^2y - 23350x^2 - 4522xy^2 + 302921xy - 215073x - 9044y^3$
	$+108737y^2 - 252893y + 48466$
$\{i, j\}$	$\{9, 19\}, k_{ij} = 2 \cdot 5$
α	$48x^4 + 4y + 4$
β	$144x^7 - 72x^6y - 72x^6 - 168x^5y - 144x^5 + 84x^4y^2 + 156x^4y + 96x^4$
	$-12x^3 + 6x^2y + 6x^2 - 8xy^2 - 2xy + 16x + 4y^3 + 5y^2 - 5y - 6$
γ	$-36x^4y^2 - 240x^4y - 180x^4 - 5y^2 + 4y + 19$
$\{i, j\}$	$\{9, 104\}, k_{ij} = 2^3 \cdot 5 \cdot 7$
α	$1504x^4 + 20257y + 19977$
β	$7144x^7 - 4230x^6y - 4230x^6 + 7050x^5y - 18330x^5 - 6956x^4y^2 - 4042x^4y$
	$-19082x^4 - 227293x^3 + 242205x^2y + 242205x^2 - 22090xy^2 - 249985xy + 4885x$
	$+1833y^3 + 583958y^2 + 1595315y + 1020129$
γ	$5264x^6 - 1316x^5y - 6580x^5 - 4230x^4y^2 - 3572x^4y - 44838x^4$
	$+4606x^2y^2 - 49350x^2y - 505358x^2 - 4606xy^2 + 306467xy + 762475x - 9212y^3$
	$+292119y^2 + 787389y + 500216$
$\{i, j\}$	$\{9, 105\}, k_{ij} = 2^6 \cdot 5 \cdot 7$
α	$-330x^4 - 3267y + 3877$
β	$-2706x^7 + 1782x^6y + 1782x^6 - 1188x^5y + 7128x^5 + 3234x^4y^2 + 10659x^4y$
	$-10428x^4 + 174424x^3 - 109026x^2y - 109026x^2 + 6534xy^2 + 23778xy - 976493x$
	$-15279y^3 - 265424y^2 - 687880y + 413143$
γ	$-3432x^6 + 858x^5y + 7722x^5 + 1782x^4y^2 + 6336x^4y - 2838x^4$
	$+3003x^2y^2 + 32175x^2y + 312650x^2 - 6006xy^2 - 107978xy - 668928x - 6006y^3$
	$-145293y^2 - 547665y - 411506$
$\{i, j\}$	$\{9, 106\}, k_{ij} = 2^2$
α	$y + 7$
β	$24x^7 - 18x^6y - 18x^6 + 12x^5y - 36x^4y^2 - 204x^4y + 96x^4 - 1284x^3$
	$+726x^2y + 726x^2 - 2xy^2 - 470xy + 5497x + 217y^3 + 2338y^2 + 7911y + 3395$

$$\begin{aligned}
&\gamma \quad 48x^6 - 12x^5y - 108x^5 - 18x^4y^2 - 84x^4y + 150x^4 - 54x^2y^2 - 330x^2y \\
&\quad - 2664x^2 + 108xy^2 + 828xy + 5496x + 108y^3 + 1220y^2 + 4514y + 3392 \\
&\{i, j\} = \{9, 107\}, k_{ij} = 2 \cdot 5^4 \\
&\alpha \quad -252x^4 + 1323y + 73 \\
&\beta \quad -756x^7 + 378x^6y + 378x^6 + 1260x^5y + 4788x^5 - 1008x^4y^2 - 6300x^4y \\
&\quad - 12474x^4 - 3624x^3 + 1812x^2y + 1812x^2 - 2646xy^2 - 1814xy - 2017x + 1323y^3 \\
&\quad + 5710y^2 + 2581y + 3267 \\
&\gamma \quad 378x^4y^2 + 3276x^4y + 9954x^4 - 3480y^2 - 4y - 2222 \\
&\{i, j\} = \{10, 34\}, k_{ij} = 3^2 \cdot 17^2 \\
&\alpha \quad 352982448x^4 - 1035182592x^3 - 86265216x^2y^2 - 654177888x^2y \\
&\quad + 1474571760x^2 + 86987062xy^2 - 389666856xy + 692881711x - 86626139y^3 \\
&\quad - 566468980y^2 + 2276130425y - 2292727428 \\
&\beta \quad -352982448x^7 + 529473672x^6y + 1058947344x^6 - 176491224x^5y + 1058947344x^5 \\
&\quad + 882456120x^4y^2 + 3000350808x^4y - 4059298152x^4 - 29956156776x^3 \\
&\quad - 22485412404x^2y - 42334044612x^2 + 2412046728xy^2 - 5309538448xy + 94125775506x \\
&\quad - 360923y^5 - 1617836220y^3 - 20222787847y^2 - 45993263385y + 67164555386 \\
&\gamma \quad -1411929792x^6 - 1411929792x^5y + 1975306176x^5 + 1047064968x^4y^2 \\
&\quad + 6395944464x^4y - 8284054176x^4 - 4331076x^3y^2 + 51972912x^3y \\
&\quad - 582529722x^3 + 2165538x^2y^3 + 2810866356x^2y^2 + 2663610018x^2y \\
&\quad + 57365698062x^2 + 1041679206xy^3 + 2840827584xy^2 + 20002141164xy + 81099554844x \\
&\quad + 86265216y^5 + 652012350y^4 - 3155328156y^3 - 23667965896y^2 - 111816466420y \\
&\quad - 136621840801 \\
&\{i, j\} = \{10, 60\}, k_{ij} = 2^2 \cdot 3^2 \cdot 17 \\
&\alpha \quad 3408x^4 - 5041y - 10150 \\
&\beta \quad -10224x^7 - 5112x^6y - 10224x^6 + 9372x^5y - 12780x^5 + 5964x^4y^2 \\
&\quad + 12780x^4y - 15762x^4 + 15540x^3 + 7770x^2y + 15540x^2 - 10082xy^2 + 6061xy \\
&\quad - 36452x - 5041y^3 - 14035y^2 - 14384y - 57373 \\
&\gamma \quad -2556x^4y^2 - 19596x^4y + 2556x^4 + 13967y^2 + 9479y + 51992 \\
&\{i, j\} = \{10, 80\}, k_{ij} = 2 \cdot 3^3 \\
&\alpha \quad 4x^4 - 9y - 24 \\
&\beta \quad -12x^7 - 6x^6y - 12x^6 + 8x^5y - 32x^5 + x^4y^2 - 28x^4y - 14x^4 + 36x^3 \\
&\quad + 18x^2y + 36x^2 - 18xy^2 + 24xy - 51x - 9y^3 + 21y^2 + 15y + 48 \\
&\gamma \quad -3x^4y^2 - 20x^4y + 20x^4 + 27y^2 + 12y + 87 \\
&\{i, j\} = \{10, 100\}, k_{ij} = 2^6 \cdot 3^2 \\
&\alpha \quad -110x^4 - 1089y + 182 \\
&\beta \quad -2266x^7 - 1782x^6y - 3564x^6 - 1188x^5y - 1188x^5 - 3674x^4y^2 - 27951x^4y \\
&\quad + 8393x^4 + 251588x^3 + 126738x^2y + 253476x^2 - 2178xy^2 + 98748xy - 1247633x \\
&\quad + 32659y^3 + 404732y^2 + 1678714y + 742054 \\
&\gamma \quad 5192x^6 + 1298x^5y + 12980x^5 - 1782x^4y^2 - 12540x^4y + 16016x^4 \\
&\quad - 8437x^2y^2 - 60357x^2y - 503388x^2 - 16874xy^2 - 122602xy - 1010552x + 16874y^3 \\
&\quad + 226497y^2 + 1131385y + 1484502 \\
&\{i, j\} = \{11, 16\}, k_{ij} = 3^3 \\
&\alpha \quad -24 \\
&\beta \quad -72x^3 + 36x^2y + 72x^2 - 18xy^2 - 72xy - 72x + 6y^3 + 25y^2 + 65y + 3 \\
&\gamma \quad -3y^3 - 29y^2 - 43y - 57 \\
&\{i, j\} = \{11, 18\}, k_{ij} = 3 \cdot 5 \cdot 17 \\
&\alpha \quad 8 \\
&\beta \quad 24x^3 - 12x^2y - 24x^2 + 6xy^2 + 24xy + 24x - 6y^3 - 53y^2 - 143y - 227 \\
&\gamma \quad 3y^3 + 35y^2 + 107y + 207 \\
&\{i, j\} = \{11, 26\}, k_{ij} = 2^2 \cdot 5^2 \cdot 13 \\
&\alpha \quad -1128x^4 + 4418y + 30852 \\
&\beta \quad -3384x^7 + 1692x^6y + 3384x^6 + 564x^5y - 10152x^5 - 282x^4y^2 + 4512x^4y \\
&\quad + 9588x^4 + 82404x^3 - 41202x^2y - 82404x^2 - 8836xy^2 + 32514xy + 650x + 4418y^3 \\
&\quad - 7421y^2 - 30630y + 14776 \\
&\gamma \quad 846x^4y^2 + 3948x^4y - 6768x^4 - 29437y^2 - 49890y - 81754 \\
&\{i, j\} = \{11, 27\}, k_{ij} = 2 \cdot 3^4 \cdot 7 \\
&\alpha \quad 80x^4 + 100y + 152 \\
&\beta \quad 240x^7 - 120x^6y - 240x^6 - 280x^5y + 200x^5 + 140x^4y^2 + 180x^4y - 160x^4 \\
&\quad + 636x^3 - 318x^2y - 636x^2 - 200xy^2 - 222xy + 590x + 100y^3 + 311y^2 - 23y - 514 \\
&\gamma \quad -60x^4y^2 - 520x^4y - 40x^4 - 359y^2 - 858y - 46
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{11, 31\}, k_{ij} = 2 \cdot 3 \cdot 13 \\
&\alpha \quad -60x^4 - 25y - 46 \\
&\beta \quad -180x^7 + 90x^6y + 180x^6 + 210x^5y - 60x^5 - 150x^4y^2 - 570x^4y - 90x^4 \\
&\quad -198x^3 + 99x^2y + 198x^2 + 50xy^2 + 91xy + 39x - 25y^3 - 195y^2 - 412y - 299 \\
&\gamma \quad 90x^4y^2 + 780x^4y + 240x^4 + 199y^2 + 578y + 474 \\
&\{i, j\} = \{11, 35\}, k_{ij} = 2 \cdot 3^4 \\
&\alpha \quad 126x^4 + 2072y + 2840 \\
&\beta \quad 700x^7 - 672x^6y - 1344x^6 + 1120x^5y - 448x^5 - 546x^4y^2 - 672x^4y \\
&\quad + 2625x^4 - 28632x^3 + 68688x^2y + 137376x^2 - 1568xy^2 - 101488xy - 138523x \\
&\quad + 1428y^3 + 69561y^2 + 177999y + 108936 \\
&\gamma \quad 644x^6 - 644x^5y - 644x^5 - 672x^4y^2 - 1736x^4y - 4872x^4 + 644x^2y^2 \\
&\quad - 4669x^2y - 64308x^2 + 644xy^2 + 104075xy + 153180x - 644y^3 + 70221y^2 + 244828y \\
&\quad + 215194 \\
&\{i, j\} = \{11, 40\}, k_{ij} = 2^8 \cdot 5 \cdot 7 \\
&\alpha \quad 120x^4 + 250y + 316 \\
&\beta \quad 360x^7 - 180x^6y - 360x^6 - 420x^5y + 480x^5 + 210x^4y^2 + 180x^4y - 420x^4 \\
&\quad + 1548x^3 - 774x^2y - 1548x^2 - 500xy^2 - 406xy + 3014x + 250y^3 + 703y^2 - 976y - 2856 \\
&\gamma \quad -90x^4y^2 - 780x^4y + 120x^4 - 887y^2 - 1954y + 1466 \\
&\{i, j\} = \{11, 60\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5 \cdot 7 \cdot 17 \\
&\alpha \quad -5232x^4 + 11881y + 16622 \\
&\beta \quad -15696x^7 + 7848x^6y + 15696x^6 + 22236x^5y + 48396x^5 - 9156x^4y^2 \\
&\quad - 27468x^4y - 18966x^4 + 2124x^3 - 1062x^2y - 2124x^2 - 23762xy^2 - 11075xy \\
&\quad - 57888x + 11881y^3 + 17153y^2 + 39360y + 36193 \\
&\gamma \quad 3924x^4y^2 + 37932x^4y + 64092x^4 - 24293y^2 - 13199y - 60012 \\
&\{i, j\} = \{11, 61\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5 \\
&\alpha \quad 112x^4 - 147y - 314 \\
&\beta \quad 336x^7 - 168x^6y - 336x^6 - 308x^5y + 420x^5 + 196x^4y^2 + 420x^4y - 406x^4 \\
&\quad - 564x^3 + 282x^2y + 564x^2 + 294xy^2 - 113xy + 744x - 147y^3 - 455y^2 - 558y - 1555 \\
&\gamma \quad -84x^4y^2 - 644x^4y + 84x^4 + 435y^2 + 451y + 1308 \\
&\{i, j\} = \{11, 74\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5^2 \cdot 13 \\
&\alpha \quad 272x^4 - 867y - 3034 \\
&\beta \quad 816x^7 - 408x^6y - 816x^6 - 340x^5y + 2924x^5 + 68x^4y^2 - 1700x^4y \\
&\quad - 1734x^4 - 5124x^3 + 2562x^2y + 5124x^2 + 1734xy^2 - 3883xy + 2192x - 867y^3 \\
&\quad + 1715y^2 + 2974y - 51 \\
&\gamma \quad -204x^4y^2 - 1156x^4y + 2108x^4 + 3015y^2 + 1241y + 7316 \\
&\{i, j\} = \{11, 76\}, k_{ij} = 2 \cdot 3^3 \cdot 5 \cdot 7 \\
&\alpha \quad -76x^4 + 361y + 512 \\
&\beta \quad -228x^7 + 114x^6y + 228x^6 + 380x^5y + 1368x^5 - 133x^4y^2 - 456x^4y \\
&\quad + 190x^4 - 288x^3 + 144x^2y + 288x^2 - 722xy^2 - 736xy - 1521x + 361y^3 + 440y^2 \\
&\quad + 1229y + 544 \\
&\gamma \quad 57x^4y^2 + 608x^4y + 1596x^4 - 650y^2 - 448y - 1233 \\
&\{i, j\} = \{11, 85\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 13 \\
&\alpha \quad -68x^4 - 1445y + 15364 \\
&\beta \quad -204x^7 + 102x^6y + 204x^6 - 272x^5y + 1428x^5 + 136x^4y^2 - 442x^4y \\
&\quad - 1462x^4 + 49152x^3 - 24576x^2y - 49152x^2 + 2890xy^2 + 32896xy + 25941x - 1445y^3 \\
&\quad - 19338y^2 - 46589y - 18259 \\
&\gamma \quad 102x^4y^2 - 136x^4y + 3264x^4 - 18796y^2 - 32512y - 46422 \\
&\{i, j\} = \{11, 97\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 7 \\
&\alpha \quad 44x^4 - 363y + 26 \\
&\beta \quad 132x^7 - 66x^6y - 132x^6 - 286x^5y - 1496x^5 + 176x^4y^2 + 1452x^4y \\
&\quad + 3146x^4 + 210x^3 - 105x^2y - 210x^2 + 726xy^2 + 1085xy + 1063x - 363y^3 - 1942y^2 \\
&\quad - 2673y - 1903 \\
&\gamma \quad -66x^4y^2 - 836x^4y - 3256x^4 + 1347y^2 + 1750y + 1706 \\
&\{i, j\} = \{11, 104\}, k_{ij} = 2^4 \cdot 5 \cdot 13 \\
&\alpha \quad 196x^4 + 3577y + 1102 \\
&\beta \quad 1372x^7 - 882x^6y - 1764x^6 + 1470x^5y - 2352x^5 - 1568x^4y^2 - 4704x^4y \\
&\quad - 4410x^4 - 129128x^3 + 112005x^2y + 224010x^2 - 4802xy^2 - 142967xy + 94541x \\
&\quad + 1225y^3 + 243046y^2 + 943607y + 703221 \\
&\gamma \quad 1568x^6 - 392x^5y - 2352x^5 - 882x^4y^2 - 3332x^4y - 9016x^4 + 588x^2y^2 \\
&\quad - 10780x^2y - 255852x^2 - 588xy^2 + 105662xy + 445616x - 1176y^3 + 123373y^2
\end{aligned}$$

	$+589688y + 703190$
$\{i, j\} = \{11, 106\}, k_{ij} = 2^3 \cdot 3 \cdot 5$	
α	$6x^4 + 63y + 14$
β	$-162x^7 + 126x^6y + 252x^6 - 84x^5y - 84x^5 + 258x^4y^2 + 1947x^4y - 567x^4$ $+17820x^3 - 9030x^2y - 18060x^2 - 126xy^2 + 6860xy - 88213x - 2277y^3 - 28756y^2$ $-118772y - 52566$
γ	$-360x^6 + 90x^5y + 900x^5 + 126x^4y^2 + 876x^4y - 1104x^4 + 585x^2y^2$ $+4275x^2y + 35520x^2 - 1170xy^2 - 8790xy - 71520x - 1170y^3 - 16077y^2 - 80135y$ -105026
$\{i, j\} = \{12, 18\}, k_{ij} = 3$	
α	8
β	$-24x^3 + 4y^2 - 2y + 1$
γ	$24x^3 + 4$
$\{i, j\} = \{12, 28\}, k_{ij} = 5$	
α	$-64x^3 + 12$
β	$-96x^5 - 32x^3y^2 + 16x^3y - 64x^3 + 2x^2 - 12x + 6y^2 - 3y + 8$
γ	$192x^5 + 48x^3 - 4x^2 + 24x - 1$
$\{i, j\} = \{12, 74\}, k_{ij} = 3^2 \cdot 5^3$	
α	$608x^3 - 212$
β	$-912x^5 + 456x^4 + 304x^3y^2 - 152x^3y + 836x^3 - 404x^2 - 197x - 106y^2 + 53y$ $+269$
γ	$-1824x^5 + 912x^4 + 912x^3 - 808x^2 - 394x + 803$
$\{i, j\} = \{13, 49\}, k_{ij} = 2^3 \cdot 5^2$	
α	$125460x^4 - 1223399y + 439341$
β	$-755589x^7 - 1511424x^6y + 755712x^6 + 2298624x^5y + 999744x^5$ $-1448694x^4y^2 + 1139349x^4y - 4602783x^4 + 235920x^3 - 1753248x^2y$ $+876624x^2 - 336200xy^2 + 2192848xy - 2583928x - 603028y^3 - 1385752y^2 + 1274331y$ -1637925
γ	$2829x^6 + 246x^5y + 5535x^5 - 3022848x^4y^2 + 721887x^4y - 1741557x^4$ $-17343x^2y^2 - 666537x^2y - 136200x^2 - 34686xy^2 + 892014xy - 1384944x$ $+17343y^3 - 2590474y^2 + 2567339y - 1198784$
$\{i, j\} = \{14, 18\}, k_{ij} = 3^2$	
α	24
β	$-144x^3 - 288x^2y - 144x^2 - 576xy^2 - 576xy - 144x - 384y^3 - 568y^2 - 292y - 46$
γ	$384y^3 + 580y^2 + 286y + 61$
$\{i, j\} = \{14, 39\}, k_{ij} = 2$	
α	$2822656x^4 + 795064y + 176781$
β	$-19415296x^7 - 38449152x^6y - 19224576x^6 - 8811264x^5y + 300384x^5$ $+20635904x^4y^2 + 24211904x^4y + 5547568x^4 - 9981454x^3 - 6739068x^2y$ $-3369534x^2 - 310814xy^2 - 1449196xy + 4374680x + 737252y^3 + 5660815y^2 + 6865010y$ $+1323166$
γ	$1239680x^6 - 190720x^5y - 715200x^5 - 38449152x^4y^2 - 33328320x^4y$ $-8785040x^4 - 1358880x^2y^2 - 7199680x^2y + 1430320x^2 + 679440xy^2$ $-3012080xy - 4021120x - 679440y^3 - 11202661y^2 - 7393110y - 1146387$
$\{i, j\} = \{14, 42\}, k_{ij} = 2^3 \cdot 7^2$	
α	$38518656x^4 - 61382476y - 38518607$
β	$-262149696x^7 - 519524352x^6y - 259762176x^6 + 400466688x^5y + 399113760x^5$ $+279021504x^4y^2 + 489680352x^4y + 48765096x^4 - 41877486x^3 - 171005652x^2y$ $-85502826x^2 - 23366194xy^2 + 131989496xy + 97857328x - 33899468y^3 + 83656253y^2$ $+150131248y + 55439788$
γ	$15518880x^6 - 2387520x^5y + 6565680x^5 - 519524352x^4y^2 - 713192016x^4y$ $-318395688x^4 + 12832920x^2y^2 - 57300480x^2y - 17580060x^2 + 6416460xy^2$ $+14975100xy + 13022640x + 6416460y^3 - 178348105y^2 - 266953120y - 93958787$
$\{i, j\} = \{15, 16\}, k_{ij} = 3^2$	
α	24
β	$144x^3 - 288x^2y - 144x^2 + 576xy^2 + 576xy + 144x - 384y^3 - 568y^2 - 292y - 30$
γ	$384y^3 + 580y^2 + 286y + 45$
$\{i, j\} = \{15, 17\}, k_{ij} = 3^2 \cdot 5$	
α	64
β	$384x^3 - 768x^2y - 384x^2 + 1536xy^2 + 1536xy + 384x - 1536y^3 - 2672y^2 - 1452y - 207$

$$\begin{aligned}
&\gamma \quad 1536y^3 + 1936y^2 + 852y + 113 \\
&\{i, j\} = \{15, 19\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \\
&\alpha \quad 85376x^4 + 18860y + 5481 \\
&\beta \quad 416576x^7 - 847872x^6y - 423936x^6 - 229632x^5y - 11040x^5 + 466624x^4y^2 \\
&\quad + 255392x^4y + 96416x^4 + 48166x^3 - 84852x^2y - 42426x^2 - 5290xy^2 - 18812xy \\
&\quad + 60x + 10580y^3 + 37174y^2 + 34682y + 5421 \\
&\gamma \quad -47840x^6 - 7360x^5y - 3680x^5 - 847872x^4y^2 - 986608x^4y - 217488x^4 \\
&\quad - 4600x^2y^2 + 16100x^2y + 13160x^2 + 5740xy + 2870x - 2300y^3 - 79447y^2 - 87678y \\
&\quad - 21183 \\
&\{i, j\} = \{15, 42\}, k_{ij} = 2^4 \cdot 17 \\
&\alpha \quad -328704x^4 - 91592y - 19629 \\
&\beta \quad -1972224x^7 + 3944448x^6y + 1972224x^6 - 903936x^5y + 30816x^5 \\
&\quad - 2136576x^4y^2 - 2650176x^4y - 1361040x^4 - 86958x^3 + 173916x^2y + 86958x^2 \\
&\quad + 22898xy^2 - 64332xy + 272x - 45796y^3 - 79599y^2 - 169554y - 63516 \\
&\gamma \quad 3944448x^4y^2 + 3492480x^4y + 1001520x^4 + 185365y^2 + 141750y + 43615 \\
&\{i, j\} = \{15, 64\}, k_{ij} = 2 \cdot 3^5 \cdot 5 \\
&\alpha \quad -63104x^4 + 2916588y + 2194841 \\
&\beta \quad -378624x^7 + 757248x^6y + 378624x^6 + 1908896x^5y + 3884840x^5 \\
&\quad - 788800x^4y^2 - 2831792x^4y + 4578984x^4 + 30x^3 - 60x^2y - 30x^2 \\
&\quad - 729147xy^2 - 731147xy - 770x + 1458294y^3 + 732907y^2 + 2186741y + 2194011 \\
&\gamma \quad 1514496x^4y^2 + 3423392x^4y + 4263464x^4 - 729267y^2 - 731267y - 800 \\
&\{i, j\} = \{15, 74\}, k_{ij} = 2 \cdot 3^3 \cdot 13 \cdot 17 \\
&\alpha \quad 35840x^4 + 29400y - 1479 \\
&\beta \quad 215040x^7 - 430080x^6y - 215040x^6 - 116480x^5y + 101920x^5 - 197120x^4y^2 \\
&\quad - 575680x^4y - 122640x^4 + 36486x^3 - 72972x^2y - 36486x^2 - 7350xy^2 - 11442xy \\
&\quad + 20118x + 14700y^3 - 46413y^2 - 78087y - 29781 \\
&\gamma \quad -430080x^4y^2 - 488320x^4y - 56560x^4 - 76647y^2 - 78693y - 8184 \\
&\{i, j\} = \{15, 75\}, k_{ij} = 2 \\
&\alpha \quad -33792x^4 - 2112x - 968y - 3169 \\
&\beta \quad -202752x^7 + 405504x^6y + 202752x^6 + 312576x^5y + 210144x^5 + 185856x^4y^2 \\
&\quad + 390720x^4y + 156288x^4 + 25344x^3y - 2646x^3 + 50172x^2y + 28452x^2 \\
&\quad + 11858xy^2 + 48038xy + 26344x - 484y^3 + 14036y^2 + 29099y + 12380 \\
&\gamma \quad 405504x^4y^2 + 561792x^4y + 206448x^4 + 25344xy^2 + 35112xy + 12903x + 30757y^2 \\
&\quad + 42445y + 15551 \\
&\{i, j\} = \{16, 63\}, k_{ij} = 2 \cdot 3^4 \\
&\alpha \quad -2368x^4 + 10952y + 5004 \\
&\beta \quad 7104x^7 + 7104x^6y + 3552x^6 - 11840x^5y - 23680x^5 - 4736x^4y^2 - 10656x^4y \\
&\quad + 14800x^4 + 6300x^3 + 6300x^2y + 3150x^2 + 5476xy^2 - 4210xy - 2796x + 5476y^3 \\
&\quad - 648y^2 + 133y + 471 \\
&\gamma \quad 7104x^4y^2 + 18944x^4y + 25456x^4 + 824y^2 + 10510y + 4371 \\
&\{i, j\} = \{16, 68\}, k_{ij} = 3^2 \cdot 5^2 \\
&\alpha \quad -128x^4 - 240y - 32 \\
&\beta \quad 384x^7 + 384x^6y + 192x^6 - 448x^5y + 320x^4y^2 + 1440x^4y + 256x^4 + 96x^3 \\
&\quad + 96x^2y + 48x^2 - 120xy^2 - 112xy - 90x - 120y^3 + 260y^2 + 390y + 199 \\
&\gamma \quad 768x^4y^2 + 1664x^4y + 192x^4 + 432y^2 + 416y + 228 \\
&\{i, j\} = \{17, 64\}, k_{ij} = 3^2 \cdot 5^2 \cdot 17 \\
&\alpha \quad -13616980x^4 - 173612179y - 129352633 \\
&\beta \quad 82052555x^7 + 164077056x^6y + 164077056x^6 + 242697312x^5y - 11109384x^5 \\
&\quad - 170885546x^4y^2 - 362446890x^4y - 681349656x^4 + 90454471x^3 \\
&\quad - 303391836x^2y - 303391836x^2 - 38419953xy^2 - 371526606xy - 876720312x \\
&\quad - 88229830y^3 + 230741170y^2 + 608825653y + 569272661 \\
&\gamma \quad -350675x^6 + 28054x^5y + 729404x^5 + 328154112x^4y^2 + 413204129x^4y \\
&\quad + 337103338x^4 + 2847481x^2y^2 + 131208558x^2y + 10785671x^2 - 5694962xy^2 \\
&\quad + 221883662xy + 462729436x + 2847481y^3 - 420070275y^2 - 1012605275y - 633950890 \\
&\{i, j\} = \{17, 101\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5 \cdot 11 \cdot 17 \\
&\alpha \quad 24x^4 - 28y - 44 \\
&\beta \quad -144x^7 - 288x^6y - 288x^6 + 6x^5y + 564x^5 - 564x^4y^2 - 18x^4y - 36x^4 \\
&\quad - 2112x^3 - 4224x^2y - 4224x^2 - 7xy^2 + 3xy + 8421x - 14y^3 - 8449y^2 - 23y + 17 \\
&\gamma \quad -576x^4y^2 - 1158x^4y - 1140x^4 - 8441y^2 - 16899y - 16869 \\
&\{i, j\} = \{18, 35\}, k_{ij} = 7 \cdot 17
\end{aligned}$$

$$\begin{aligned}
\alpha & 301760x^4 + 1812400y + 1437424 \\
\beta & 923680x^7 - 927360x^6y - 463680x^6 + 1700160x^5y - 1854720x^5 - 776480x^4y^2 \\
& + 266800x^4y + 3183200x^4 + 1446592x^3 + 1586256x^2y + 793128x^2 - 740600xy^2 \\
& - 1272376xy - 1222198x + 851000y^3 + 2017468y^2 + 4360210y + 2818885 \\
\gamma & 36800x^6 - 7360x^5y + 33120x^5 - 1854720x^4y^2 + 1501440x^4y - 4176800x^4 \\
& + 110400x^2y^2 - 1867600x^2y + 1326880x^2 + 110400xy^2 + 4198096xy + 4359728x \\
& - 110400y^3 + 3558912y^2 + 5366576y + 1381580 \\
\{i, j\} &= \{18, 42\}, k_{ij} = 7 \\
\alpha & -384x^4 + 64y + 32 \\
\beta & -1152x^7 + 1152x^6y + 576x^6 + 192x^5y + 96x^5 - 768x^4y^2 - 864x^4y \\
& - 432x^4 - 72x^3 + 72x^2y + 36x^2 - 32xy^2 + 16xy + 2x + 32y^3 - 20y^2 - 22y + 5 \\
\gamma & 1152x^4y^2 + 1344x^4y + 384x^4 + 40y^2 + 88y + 20 \\
\{i, j\} &= \{18, 57\}, k_{ij} = 5 \cdot 17 \\
\alpha & 384x^4 - 32y + 32 \\
\beta & 1152x^7 - 1152x^6y - 576x^6 - 192x^5y + 480x^5 - 384x^4y^2 - 1056x^4y \\
& + 48x^4 + 360x^3 - 360x^2y - 180x^2 + 16xy^2 - 32xy + 114x - 16y^3 - 148y^2 - 310y - 29 \\
\gamma & -1152x^4y^2 - 1344x^4y + 192x^4 - 344y^2 - 392y + 24 \\
\{i, j\} &= \{18, 58\}, k_{ij} = 5 \\
\alpha & 2112x^4 - 968y - 604 \\
\beta & 6336x^7 - 6336x^6y - 3168x^6 - 4224x^5y - 4224x^5 - 2112x^4y^2 - 4224x^4y \\
& - 2640x^4 + 564x^3 - 564x^2y - 282x^2 + 484xy^2 - 222xy - 332x - 484y^3 - 100y^2 \\
& - 827y - 609 \\
\gamma & -6336x^4y^2 - 10560x^4y - 5808x^4 - 80y^2 - 786y - 473 \\
\{i, j\} &= \{18, 60\}, k_{ij} = 3 \cdot 5 \cdot 17 \\
\alpha & -6144x^4 + 16384y + 14728 \\
\beta & -18432x^7 + 18432x^6y + 9216x^6 + 21504x^5y + 41472x^5 - 12288x^4y^2 \\
& - 32256x^4y - 768x^4 - 4584x^3 + 4584x^2y + 2292x^2 - 8192xy^2 - 2780xy + 2122x \\
& + 8192y^3 + 5072y^2 + 8362y + 7937 \\
\gamma & 18432x^4y^2 + 39936x^4y + 46080x^4 - 3608y^2 + 1804y + 3268 \\
\{i, j\} &= \{18, 91\}, k_{ij} = 3 \cdot 5^2 \cdot 13 \\
\alpha & -4224x^4 - 1936y + 1632 \\
\beta & -12672x^7 + 12672x^6y + 6336x^6 - 10560x^5y + 2112x^5 + 23232x^4y^2 \\
& + 5280x^4y + 2112x^4 + 7008x^3 - 7008x^2y - 3504x^2 + 968xy^2 + 4256xy + 1274x \\
& - 968y^3 - 10780y^2 - 7122y - 299 \\
\gamma & 25344x^4y^2 + 4224x^4y + 10560x^4 - 12080y^2 - 5504y - 956 \\
\{i, j\} &= \{19, 40\}, k_{ij} = 2 \\
\alpha & 0 \\
\beta & -1 \\
\gamma & 1 \\
\{i, j\} &= \{19, 79\}, k_{ij} = 2^2 \cdot 3^4 \\
\alpha & 352x^4 - 352y + 370 \\
\beta & 352x^7 - 1408x^6 + 352x^5y - 352x^5 + 176x^4y^2 - 880x^4y - 176x^4 - 70x^3 \\
& - 704x^2y^2 + 880x^2y - 1040x^2 - 70xy - 92x - 352y^3 + 405y^2 - 471y + 139 \\
\gamma & -352x^7 - 176x^4y^2 + 176x^4y + 176x^4 + 70x^3 + 162x + 35y^2 - 49y + 23 \\
\{i, j\} &= \{19, 81\}, k_{ij} = 2^5 \cdot 5 \\
\alpha & 96x^4 - 96y + 106 \\
\beta & -96x^7 - 384x^6 - 96x^5y + 96x^5 + 48x^4y^2 - 240x^4y + 48x^4 + 62x^3 \\
& - 192x^2y^2 + 336x^2y - 352x^2 + 62xy - 88x - 96y^3 + 137y^2 - 167y + 1 \\
\gamma & 96x^7 - 48x^4y^2 + 48x^4y - 48x^4 - 62x^3 + 26x + 31y^2 - 9y - 27 \\
\{i, j\} &= \{20, 68\}, k_{ij} = 3^4 \cdot 5^2 \\
\alpha & -16x^4 + 112y - 4 \\
\beta & -8x^7 + 112x^6 - 16x^5y - 8x^5 - 8x^4y^2 + 116x^4y - 100x^4 + 536x^3 \\
& + 112x^2y^2 - 852x^2y - 908x^2 + 1072xy + 194x + 112y^3 - 428y^2 - 722y + 1155 \\
\gamma & 8x^7 + 8x^4y^2 - 4x^4y - 12x^4 - 536x^3 + 342x - 536y^2 + 666y + 437 \\
\{i, j\} &= \{20, 87\}, k_{ij} = 2 \cdot 3^4 \cdot 5^3 \\
\alpha & 801x^4 - 6764y + 12892 \\
\beta & 89x^7 - 4895x^6 + 267x^5y - 178x^5 + 267x^4y^2 - 5340x^4y + 4272x^4 \\
& - 5470x^3 - 6764x^2y^2 + 48594x^2y - 76332x^2 - 16410xy + 15202x - 6764y^3 \\
& + 38948y^2 - 78026y + 39714 \\
\gamma & -89x^7 - 267x^4y^2 + 445x^4y + 623x^4 + 5470x^3 - 4262x + 16410y^2 - 46900y
\end{aligned}$$

	+32356
$\{i, j\} = \{20, 96\}, k_{ij} = 2^4 \cdot 3^3$	
α	$13x^4 - 130y + 274$
β	$13x^7 - 91x^6 - 13x^5y + 13x^5 - 13x^4y^2 - 65x^4y + 65x^4 - 987x^3$ $- 130x^2y^2 + 1001x^2y - 787x^2 + 987xy - 1029x - 130y^3 + 2118y^2 - 3590y + 1500$
γ	$-13x^7 + 13x^4y^2 - 26x^4y + 26x^4 + 987x^3 + 42x - 987y^2 + 1802y - 671$
$\{i, j\} = \{20, 98\}, k_{ij} = 2^{10} \cdot 3^2$	
α	$98x^4 - 1078y + 2634$
β	$196x^7 - 980x^6 - 196x^5y - 98x^4y^2 - 882x^4y + 931x^4 - 15802x^3$ $- 1078x^2y^2 + 8379x^2y + 14722x^2 + 15802xy - 10652x - 1078y^3 + 17358y^2 + 4307y$ $+ 5265$
γ	$-196x^7 + 98x^4y^2 - 98x^4y + 49x^4 + 15802x^3 + 10652x - 7901y^2 + 2036y + 1317$
$\{i, j\} = \{21, 27\}, k_{ij} = 2 \cdot 3^3$	
α	$x^3 - 10$
β	$-6x^5 + 2x^3y^2 + 2x^3 + 59x^2 - 5x - 20y^2 + y - 23$
γ	$-x^3y^2 - 3x^3y - 4x^3 + 5x + 10y^2 + 29y + 41$
$\{i, j\} = \{21, 48\}, k_{ij} = 2 \cdot 3$	
α	$2y - 4$
β	$-x^7 + 2x^6 + 2x^4y + 2x^4 + 5x^3 + 2x^2y^2 - 8x^2y + 16x^2 + 13x + 2y^3$ $- 6y^2 + 7y - 8$
γ	$x^7 - 5x^3 - 13x + y - 2$
$\{i, j\} = \{21, 64\}, k_{ij} = 3^3 \cdot 5$	
α	$-64y + 144$
β	$-32x^7 - 64x^6 + 32x^5 - 80x^4y - 80x^4 + 184x^3 - 64x^2y^2 + 272x^2y$ $- 544x^2 + 294x - 64y^3 + 208y^2 - 212y + 815$
γ	$32x^7 + 16x^4y + 16x^4 - 184x^3 - 478x - 60y - 403$
$\{i, j\} = \{21, 95\}, k_{ij} = 2^3$	
α	$-3x^4 + 12y + 4$
β	$3x^7 + 21x^6 - 3x^5y - 3x^5 + 3x^4y^2 + 27x^4y + 21x^4 - 13x^3 + 12x^2y^2$ $- 33x^2y - 13x^2 + 13xy - 39x + 12y^3 - 34y^2 - 112y - 126$
γ	$-3x^7 - 3x^4y^2 - 6x^4y + 13x^3 + 52x + 13y^2 + 66y + 61$
$\{i, j\} = \{21, 97\}, k_{ij} = 2 \cdot 3^2$	
α	$-7x^4 + 28y + 10$
β	$-7x^7 + 49x^6 + 7x^5y + 7x^5 + 7x^4y^2 + 63x^4y + 63x^4 + 45x^3 + 28x^2y^2$ $- 91x^2y - 63x^2 - 45xy + 37x + 28y^3 - 108y^2 - 354y - 282$
γ	$7x^7 - 7x^4y^2 - 14x^4y - 14x^4 - 45x^3 - 82x + 45y^2 + 200y + 137$
$\{i, j\} = \{21, 107\}, k_{ij} = 2^2 \cdot 5^3$	
α	$-42x^4 + 294y - 206$
β	$-84x^7 + 420x^6 + 84x^5y + 84x^5 + 42x^4y^2 + 504x^4y + 483x^4 + 470x^3$ $+ 294x^2y^2 - 1029x^2y + 695x^2 - 470xy + 552x + 294y^3 - 970y^2 - 1462y - 1992$
γ	$84x^7 - 42x^4y^2 - 84x^4y - 63x^4 - 470x^3 - 1022x + 235y^2 + 1128y + 643$
$\{i, j\} = \{22, 40\}, k_{ij} = 2^7 \cdot 3 \cdot 5$	
α	$-10x^4 + 32$
β	$40x^7 + 100x^6 + 20x^5y + 140x^5 - 20x^4y^2 - 140x^4y - 10x^4 + 52x^3$ $- 30x^2y^2 - 185x^2y - 280x^2 - 30xy^2 - 159xy + 846x + 30y^3 + 159y^2 - 309y - 1042$
γ	$-20x^7 + 10x^4y^2 + 130x^4y + 75x^4 - 26x^3 - 537x - 30y^3 - 187y^2 - 65y + 944$
$\{i, j\} = \{22, 43\}, k_{ij} = 2^3$	
α	$-7x^3 + 8$
β	$42x^5 - 126x^4 - 14x^3y^2 - 42x^3y - 56x^3 + x^2 - 10x + 16y^2 - y + 8$
γ	$84x^4 + 7x^3y^2 + 42x^3y + 49x^3 + 9x - 8y^2 + y$
$\{i, j\} = \{22, 48\}, k_{ij} = 2^5 \cdot 3$	
α	$-y + 8$
β	$x^7 - x^6 - 2x^5 - 4x^4y - 4x^4 - 51x^3 - x^2y^2 + 7x^2y + 5x^2 - xy^2 + 7xy$ $+ 36x - 3y^3 + 19y^2 + 81y + 164$
γ	$-x^7 + 3x^4y + 2x^4 + 51x^3 + 20x + 2y^3 - 13y^2 - 69y - 88$
$\{i, j\} = \{22, 49\}, k_{ij} = 2^7 \cdot 3^2 \cdot 5^2$	
α	$164x^4 - 25215y + 33838$
β	$41x^7 - 1025x^6 + 82x^5y - 1066x^5 + 82x^4y^2 - 2952x^4y - 2296x^4$ $- 177291x^3 - 5043x^2y^2 + 149609x^2y - 24671x^2 - 5043xy^2 - 204973xy + 214530x$

$$\begin{aligned}
& -15129y^3 + 84159y^2 - 157633y + 85672 \\
\gamma & -41x^7 - 82x^4y^2 + 2009x^4y + 1230x^4 + 177291x^3 - 61910x + 10086y^3 \\
& + 60407y^2 - 72011y + 5038 \\
\{i, j\} &= \{22, 85\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
\alpha & 72x^4 - 302y - 59 \\
\beta & 19x^7 + 550x^6 + 30x^5y + 560x^5 + 24x^4y^2 - 515x^4y + 81x^4 + 32055x^3 \\
& - 2870x^2y^2 - 88540x^2y + 33325x^2 - 2870xy^2 + 61949xy + 31490x + 2870y^3 \\
& - 170597y^2 - 18336y + 33831 \\
\gamma & -19x^7 - 963x^6 + 27x^5y + 9x^5 - 24x^4y^2 + 3021x^4y + 1442x^4 - 32055x^3 \\
& - 3663x^2y^2 + 84456x^2y - 2430x^2 - 54324xy + 33890x + 1586y^3 - 86062y^2 + 34014y \\
& + 89 \\
\{i, j\} &= \{23, 31\}, k_{ij} = 3 \cdot 13 \\
\alpha & -128x^3 + 328 \\
\beta & 768x^5 + 64x^3y^2 - 416x^3y + 272x^3 - 304x^2 + 312x - 164y^2 + 234y + 31 \\
\gamma & 768x^4 - 64x^3y^2 - 352x^3y + 496x^3 + 8x + 164y^2 + 70y + 289 \\
\{i, j\} &= \{23, 55\}, k_{ij} = 2 \cdot 3 \cdot 13 \\
\alpha & 2x^4 - 6y + 13 \\
\beta & 4x^7 - 10x^6 - 2x^5y - 8x^5 - 4x^4y^2 - 24x^4y + 12x^4 + 34x^3 + 6x^2y^2 \\
& + 16x^2y - 53x^2 + 6xy^2 - xy + 78x + 18y^3 + 8y^2 - 22y - 109 \\
\gamma & -2x^7 + 2x^4y^2 + 16x^4y - 10x^4 - 17x^3 - 57x - 12y^3 - 9y^2 + 37y + 65 \\
\{i, j\} &= \{23, 100\}, k_{ij} = 2^4 \cdot 7 \\
\alpha & -3x^4 + 21y + 17 \\
\beta & 3x^7 + 15x^6 + 3x^5y + 18x^5 - 3x^4y^2 - 42x^4y + 39x^4 + 175x^3 \\
& - 21x^2y^2 - 150x^2y - 253x^2 - 21xy^2 + 25xy - 218x + 63y^3 + 233y^2 + 473y - 33 \\
\gamma & -3x^7 + 3x^4y^2 + 30x^4y - 21x^4 - 175x^3 + 140x - 42y^3 - 104y^2 - 195y + 95 \\
\{i, j\} &= \{23, 104\}, k_{ij} = 2^4 \cdot 3 \cdot 13 \cdot 17 \\
\alpha & 98x^4 - 588y - 21080 \\
\beta & 196x^7 - 196x^6 + 196x^5y + 196x^5 + 98x^4y^2 + 294x^4y + 147x^4 - 53822x^3 \\
& - 294x^2y^2 + 2989x^2y + 18640x^2 - 294xy^2 - 50833xy - 64101x - 27058y^2 - 91984y \\
& - 55042 \\
\gamma & -196x^7 - 98x^4y^2 + 98x^4y + 49x^4 + 53822x^3 - 24903x + 294y^3 + 23775y^2 \\
& + 22511y + 15844 \\
\{i, j\} &= \{24, 36\}, k_{ij} = 2 \cdot 3^4 \\
\alpha & 2x^4 - 20y + 31 \\
\beta & -4x^7 - 10x^6 + 2x^5y - 4x^5 - 4x^4y^2 + 22x^4y - 22x^4 + 330x^3 \\
& - 20x^2y^2 + 156x^2y - 375x^2 - 165xy + 94x + 20y^3 + 154y^2 - 183y - 131 \\
\gamma & 2x^7 + 2x^4y^2 - 6x^4y + 6x^4 - 165x^3 + 118x - 165y^2 + 357y - 63 \\
\{i, j\} &= \{24, 43\}, k_{ij} = 2^7 \cdot 3^2 \cdot 5^2 \cdot 17 \\
\alpha & -18594x^4 + 146686y + 79794 \\
\beta & 8264x^7 + 103300x^6 - 12396x^5y + 8264x^5 + 12396x^4y^2 - 123960x^4y \\
& + 130158x^4 - 667660x^3 + 146686x^2y^2 - 1129069x^2y + 337648x^2 + 1001490xy \\
& - 361260x - 146686y^3 + 274265y^2 - 403853y - 409806 \\
\gamma & -4132x^7 - 6198x^4y^2 + 10330x^4y - 13429x^4 + 333830x^3 - 153200x + 500745y^2 \\
& - 531432y + 220527 \\
\{i, j\} &= \{24, 74\}, k_{ij} = 2 \cdot 3^4 \cdot 5^3 \\
\alpha & 11x^4 - 154y + 559 \\
\beta & 22x^7 - 55x^6 - 11x^5y + 44x^5 - 22x^4y^2 + 165x^4y - 121x^4 - 2050x^3 \\
& - 154x^2y^2 + 1276x^2y - 4577x^2 + 1025xy + 400x + 154y^3 + 620y^2 - 205y - 11009 \\
\gamma & -11x^7 + 11x^4y^2 - 55x^4y + 33x^4 + 1025x^3 - 2250x - 1025y^2 + 3029y + 4341 \\
\{i, j\} &= \{24, 80\}, k_{ij} = 2^3 \cdot 3^6 \cdot 17 \\
\alpha & 22x^4 - 242y - 298 \\
\beta & -88x^7 - 44x^6 + 44x^5y - 176x^5 - 44x^4y^2 + 264x^4y - 242x^4 + 11356x^3 \\
& - 242x^2y^2 + 2057x^2y - 10030x^2 - 5678xy - 22432x + 242y^3 + 3379y^2 + 6511y \\
& - 32164 \\
\gamma & 44x^7 + 22x^4y^2 - 110x^4y + 99x^4 - 5678x^3 + 22572x - 2839y^2 + 2788y + 33639 \\
\{i, j\} &= \{24, 100\}, k_{ij} = 2^7 \cdot 3^4 \\
\alpha & -22x^4 - 22y + 1318 \\
\beta & 44x^7 + 44x^6 + 44x^5y + 88x^5 - 22x^4y^2 - 66x^4y + 99x^4 - 2394x^3 \\
& - 22x^2y^2 + 77x^2y - 2944x^2 - 2394xy - 5796x + 22y^3 + 1098y^2 + 3725y - 4989 \\
\gamma & -44x^7 + 22x^4y^2 + 22x^4y - 55x^4 + 2394x^3 + 1008x - 1197y^2 - 704y + 4061
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{24, 103\}, k_{ij} = 2^3 \cdot 3^3 \\
&\alpha \quad -5x^4 + 40y + 32 \\
&\beta \quad -5x^7 + 25x^6 - 5x^5y + 5x^5 - 5x^4y^2 - 15x^4y + 15x^4 + 417x^3 \\
&\quad + 40x^2y^2 - 305x^2y + 265x^2 + 417xy - 753x - 40y^3 + 762y^2 - 1028y + 42 \\
&\gamma \quad 5x^7 + 5x^4y^2 - 10x^4y + 10x^4 - 417x^3 + 336x - 417y^2 + 458y - 113 \\
&\{i, j\} = \{24, 106\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5 \\
&\alpha \quad -2x^4 - 2y + 122 \\
&\beta \quad -4x^7 + 4x^6 - 4x^5y - 8x^5 - 2x^4y^2 - 6x^4y + 7x^4 + 226x^3 - 2x^2y^2 \\
&\quad + 5x^2y - 268x^2 + 226xy + 572x + 2y^3 + 106y^2 + 325y - 313 \\
&\gamma \quad 4x^7 + 2x^4y^2 + 2x^4y - 3x^4 - 226x^3 - 120x - 113y^2 - 52y + 285 \\
&\{i, j\} = \{25, 30\}, k_{ij} = 3 \cdot 11^2 \\
&\alpha \quad 4x^4 - 16y + 159 \\
&\beta \quad -2x^7 - 20x^6 + 4x^5y + 18x^5 - 2x^4y^2 - x^4y - 11x^4 - 111x^3 + 16x^2y^2 \\
&\quad + 79x^2y - 729x^2 - 16xy^2 + 143xy + 329x - 103y^2 + 281y - 244 \\
&\gamma \quad 2x^7 + 2x^4y^2 + 17x^4y - 7x^4 + 111x^3 + 289x - 16y^3 + 40y^2 + 305y + 204 \\
&\{i, j\} = \{25, 37\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 11 \\
&\alpha \quad -34x^4 + 224 \\
&\beta \quad 136x^7 + 68x^6 - 68x^5y - 136x^5 + 68x^4y^2 + 136x^4y + 34x^4 - 284x^3 \\
&\quad - 102x^2y^2 - 119x^2y - 975x^2 + 102xy^2 + 261xy - 245x - 102y^3 - 261y^2 - 436y \\
&\quad + 457 \\
&\gamma \quad -68x^7 - 34x^4y^2 - 68x^4y + 51x^4 + 142x^3 + 681x + 102y^3 + 139y^2 + 575y - 106 \\
&\{i, j\} = \{25, 64\}, k_{ij} = 2 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 11 \cdot 17 \\
&\alpha \quad -4316x^4 + 609635y + 1295256 \\
&\beta \quad -1079x^7 + 24817x^6 + 2158x^5y - 22659x^5 + 2158x^4y^2 - 76609x^4y \\
&\quad + 41002x^4 + 4810296x^3 + 121927x^2y^2 - 3613571x^2y - 2993610x^2 \\
&\quad - 121927xy^2 - 6007021xy - 9385429x - 365781y^3 + 1463975y^2 + 6467021y + 3515432 \\
&\gamma \quad 1079x^7 - 2158x^4y^2 + 49634x^4y - 18343x^4 - 4810296x^3 + 2758447x \\
&\quad + 243854y^3 + 2271523y^2 + 2533610y + 353103 \\
&\{i, j\} = \{25, 85\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad -49x^4 - 567y - 271 \\
&\beta \quad 91x^7 + 665x^6 + 133x^5y - 532x^5 - 49x^4y^2 + 658x^4y + 91x^4 - 46877x^3 \\
&\quad + 1995x^2y^2 - 35378x^2y - 19437x^2 - 1995xy^2 - 95817xy - 47558x + 1995y^3 \\
&\quad - 72819y^2 - 76429y - 19847 \\
&\gamma \quad -91x^7 - 462x^6 - 42x^5y - 42x^5 + 49x^4y^2 - 532x^4y + 903x^4 + 46877x^3 \\
&\quad - 1638x^2y^2 + 37464x^2y + 67086x^2 + 84318xy + 20118x - 714y^3 + 36902y^2 + 20047y \\
&\quad + 319 \\
&\{i, j\} = \{25, 90\}, k_{ij} = 3 \cdot 5 \cdot 11^2 \\
&\alpha \quad -64x^4 + 2496y + 15385 \\
&\beta \quad 64x^7 + 320x^6 + 64x^5y - 192x^5 - 64x^4y^2 + 192x^4y + 64x^4 - 55961x^3 \\
&\quad + 832x^2y^2 - 14080x^2y - 36541x^2 - 832xy^2 - 41881xy - 42864x + 832y^3 \\
&\quad + 41881y^2 + 105402y + 49649 \\
&\gamma \quad -64x^7 + 64x^4y^2 - 576x^4y + 128x^4 + 55961x^3 - 32517x - 1664y^3 - 26969y^2 \\
&\quad - 26980y - 6785 \\
&\{i, j\} = \{25, 104\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 17 \\
&\alpha \quad -29x^3 - 226 \\
&\beta \quad 174x^5 - 522x^4 - 29x^3y^2 - 290x^3y + 551x^3 + 515x^2 + 21x - 226y^2 - 578y - 636 \\
&\gamma \quad 348x^4 + 29x^3y^2 + 116x^3y - 377x^3 + 247x + 226y^2 + 63y + 368 \\
&\{i, j\} = \{25, 108\}, k_{ij} = 3^6 \\
&\alpha \quad 10x^4 - 10y + 3 \\
&\beta \quad 10x^7 - 50x^6 + 10x^5y + 30x^5 + 10x^4y^2 + 110x^4y - 50x^4 - 117x^3 \\
&\quad + 10x^2y^2 - 80x^2y - 105x^2 - 10xy^2 - 37xy + 572x - 30y^3 + 143y^2 + 380y + 257 \\
&\gamma \quad -10x^7 - 10x^4y^2 - 70x^4y + 20x^4 + 117x^3 - 233x + 20y^3 - 53y^2 - 238y - 363 \\
&\{i, j\} = \{26, 39\}, k_{ij} = 2 \cdot 3 \\
&\alpha \quad -9536x^4 + 894y + 499 \\
&\beta \quad -4768x^7 + 26224x^6 + 4768x^5y + 2384x^5 + 4768x^4y^2 - 10728x^4y - 9536x^4 \\
&\quad - 2358x^3 + 1788x^2y^2 + 10430x^2y + 6860x^2 + 2358xy + 187x - 894y^3 - 2857y^2 \\
&\quad - 3341y - 989 \\
&\gamma \quad 4768x^7 - 4768x^4y^2 - 2384x^4y + 9536x^4 + 2358x^3 + 992x - 2358y^2 - 89y + 493 \\
&\{i, j\} = \{26, 57\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad 2y + 1
\end{aligned}$$

$$\begin{aligned}
&\beta \quad 32x^7 + 16x^6 + 16x^5 + 8x^4y + 14x^3 + 4x^2y^2 + 2x^2y + 4x^2 + 19x - 2y^3 \\
&\quad -y^2 + 7y - 5 \\
&\gamma \quad -32x^7 - 16x^4y - 14x^3 - 12x - 9y + 11 \\
&\{i, j\} = \{26, 75\}, k_{ij} = 3 \\
&\alpha \quad -8y + 8 \\
&\beta \quad -64x^7 - 64x^6 - 32x^5 + 16x^4y + 8x^3 - 16x^2y^2 + 16x^2y - 16x^2 + 2x + 8y^3 \\
&\quad -8y^2 + 6y + 3 \\
&\gamma \quad 64x^7 + 16x^4y - 8x^3 + 2x + 2y - 5 \\
&\{i, j\} = \{26, 76\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \\
&\alpha \quad -224x^4 - 56y + 82 \\
&\beta \quad -224x^7 + 448x^6 + 224x^5y + 224x^5 + 112x^4y^2 - 112x^4y - 112x^4 - 58x^3 \\
&\quad -112x^2y^2 + 280x^2y - 80x^2 + 58xy + 36x + 56y^3 - 111y^2 + 19y - 3 \\
&\gamma \quad 224x^7 - 112x^4y^2 - 112x^4y + 112x^4 + 58x^3 + 22x - 29y^2 + 21y - 19 \\
&\{i, j\} = \{26, 81\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad 4y + 2 \\
&\beta \quad 32x^7 + 32x^6 - 32x^5 - 32x^4y + 14x^3 + 8x^2y^2 + 4x^2y + 8x^2 - 10x - 4y^3 \\
&\quad -2y^2 - 9y + 9 \\
&\gamma \quad -32x^7 + 16x^4y - 14x^3 - 4x + 5y - 5 \\
&\{i, j\} = \{27, 47\}, k_{ij} = 2^2 \cdot 3^3 \cdot 7^2 \cdot 17 \\
&\alpha \quad 3320x^4 - 10375y + 40363 \\
&\beta \quad 26560x^7 - 63080x^6 - 13280x^5y + 13280x^5 - 3320x^4y^2 - 24900x^4y \\
&\quad + 27390x^4 - 108696x^3 - 20750x^2y^2 + 66400x^2y + 76798x^2 + 54348xy - 95013x \\
&\quad - 10375y^3 + 57162y^2 - 9215y + 36261 \\
&\gamma \quad -53120x^7 + 6640x^4y^2 - 13280x^4y + 8300x^4 + 217392x^3 + 81330x - 27174y^2 \\
&\quad + 28828y + 13340 \\
&\{i, j\} = \{27, 63\}, k_{ij} = 2^3 \cdot 3^4 \\
&\alpha \quad -4y + 30 \\
&\beta \quad 32x^7 - 32x^6 - 32x^5 - 32x^4y + 32x^4 - 202x^3 - 8x^2y^2 + 44x^2y + 88x^2 \\
&\quad -96x - 4y^3 + 26y^2 + 125y + 436 \\
&\gamma \quad -32x^7 + 16x^4y - 16x^4 + 202x^3 + 298x - 103y - 182 \\
&\{i, j\} = \{27, 71\}, k_{ij} = 2^4 \cdot 3^2 \\
&\alpha \quad -32x^4 + 24y + 94 \\
&\beta \quad -32x^7 + 128x^6 - 32x^5y - 16x^4y^2 + 80x^4y - 48x^4 + 194x^3 + 48x^2y^2 \\
&\quad -104x^2y - 428x^2 + 194xy + 164x + 24y^3 + 21y^2 - 107y + 3 \\
&\gamma \quad 32x^7 + 16x^4y^2 - 16x^4y - 16x^4 - 194x^3 - 164x - 97y^2 - 55y + 47 \\
&\{i, j\} = \{27, 77\}, k_{ij} = 3^5 \\
&\alpha \quad -8y + 48 \\
&\beta \quad 64x^7 - 64x^6 + 32x^5 - 16x^4y + 16x^4 - 296x^3 - 16x^2y^2 + 64x^2y + 128x^2 \\
&\quad -354x - 8y^3 + 40y^2 - 38y + 89 \\
&\gamma \quad -64x^7 - 16x^4y + 16x^4 + 296x^3 + 206x + 70y + 53 \\
&\{i, j\} = \{27, 92\}, k_{ij} = 2^3 \cdot 3^6 \cdot 17 \\
&\alpha \quad 58392x^4 - 59203y + 81045 \\
&\beta \quad -51904x^7 - 278984x^6 + 77856x^5y - 25952x^5 - 19464x^4y^2 - 113540x^4y \\
&\quad + 147602x^4 + 161528x^3 - 118406x^2y^2 + 288716x^2y - 14966x^2 - 242292xy \\
&\quad + 103461x - 59203y^3 + 264134y^2 - 230383y + 87655 \\
&\gamma \quad 103808x^7 + 38928x^4y^2 - 51904x^4y - 16220x^4 - 323056x^3 - 45394x \\
&\quad - 121146y^2 + 157084y - 69556 \\
&\{i, j\} = \{28, 76\}, k_{ij} = 3 \cdot 5 \\
&\alpha \quad 8y + 16 \\
&\beta \quad 64x^7 + 64x^6 - 32x^5 + 48x^4y + 48x^4 + 40x^3 + 16x^2y^2 + 18x + 8y^3 + 8y^2 \\
&\quad + 14y + 75 \\
&\gamma \quad -64x^7 - 16x^4y - 16x^4 - 40x^3 - 38x - 14y - 37 \\
&\{i, j\} = \{28, 78\}, k_{ij} = 3 \\
&\alpha \quad 8y + 16 \\
&\beta \quad -64x^7 + 64x^6 + 32x^5 + 48x^4y + 48x^4 - 40x^3 + 16x^2y^2 - 10x + 8y^3 + 8y^2 \\
&\quad + 6y + 47 \\
&\gamma \quad 64x^7 - 16x^4y - 16x^4 + 40x^3 + 30x - 6y - 17 \\
&\{i, j\} = \{29, 31\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 13 \\
&\alpha \quad 22x^4 - 33y - 79 \\
&\beta \quad 44x^7 - 88x^6 - 22x^5y - 88x^5 - 22x^4y^2 - 198x^4y - 110x^4 + 348x^3
\end{aligned}$$

$$\begin{aligned}
& +66x^2y^2 + 242x^2y + 118x^2 + 33xy^2 - 53xy - 409x + 99y^3 + 222y^2 - 71y + 150 \\
\gamma & -88x^7 + 44x^4y^2 + 286x^4y + 132x^4 - 696x^3 + 240x - 132y^3 - 169y^2 + 207y \\
& -469 \\
\{i, j\} &= \{29, 39\}, k_{ij} = 3 \cdot 5 \\
\alpha & 16y + 8 \\
\beta & 128x^7 + 128x^6 + 32x^4y + 96x^3 + 32x^2y^2 - 32x^2y + 72x^2 + 16xy^2 - 16xy \\
& + 18x + 4y^2 + 28y + 31 \\
\gamma & -128x^7 + 32x^4y - 96x^3 - 30x + 16y^3 - 12y^2 - 7 \\
\{i, j\} &= \{29, 50\}, k_{ij} = 5^6 \\
\alpha & 2916x^4 - 3402y - 3915 \\
\beta & -1944x^7 - 11664x^6 - 2916x^5y - 9720x^5 + 972x^4y^2 - 1620x^4y - 2268x^4 \\
& - 10620x^3 + 6804x^2y^2 + 21546x^2y - 459x^2 + 3402xy^2 - 5157xy - 17565x \\
& + 1134y^3 + 10035y^2 + 246y - 9763 \\
\gamma & 3888x^7 - 1944x^4y^2 - 11340x^4y - 5184x^4 + 21240x^3 - 7809x + 4536y^3 \\
& + 4878y^2 - 6108y + 9770 \\
\{i, j\} &= \{29, 55\}, k_{ij} = 13 \\
\alpha & -16x + 32 \\
\beta & 48x^3 - 96x^2 - 8xy^2 + 16xy + 2x + 16y^2 - 32y - 3 \\
\gamma & 24x^2 + 8xy^2 + 8xy - 38x - 16y^2 - 16y - 19 \\
\{i, j\} &= \{29, 65\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5 \cdot 13 \cdot 17 \\
\alpha & -1752x^4 + 1606y + 12080 \\
\beta & -10512x^7 + 21024x^6 + 5256x^5y - 10512x^5 + 1752x^4y^2 + 14016x^4y \\
& - 6424x^4 - 10010x^3 + 9636x^2y^2 - 4234x^2y - 22685x^2 + 4818xy^2 + 2888xy \\
& - 52436x + 8030y^3 + 1352y^2 - 17163y - 10886 \\
\gamma & 21024x^7 - 3504x^4y^2 - 1752x^4y + 2336x^4 + 20020x^3 + 42147x - 6424y^3 \\
& - 2120y^2 + 14529y + 53630 \\
\{i, j\} &= \{29, 78\}, k_{ij} = 3^2 \cdot 5 \cdot 13 \cdot 17 \\
\alpha & -2y + 29 \\
\beta & 32x^7 - 16x^6 - 40x^5 - 56x^4y - 40x^4 - 426x^3 - 4x^2y^2 + 44x^2y + 146x^2 \\
& - 2xy^2 + 22xy - 1396x - 6y^3 + 64y^2 + 669y + 6155 \\
\gamma & -32x^7 + 48x^4y + 20x^4 + 426x^3 + 1895x + 4y^3 - 43y^2 - 585y - 4958 \\
\{i, j\} &= \{29, 83\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 11 \\
\alpha & 216x^4 - 810y + 660 \\
\beta & -144x^7 - 864x^6 + 216x^5y - 432x^5 - 72x^4y^2 - 72x^4y - 72x^4 + 1378x^3 \\
& - 540x^2y^2 + 1998x^2y - 21x^2 - 270xy^2 - 1068xy + 163x - 90y^3 + 932y^2 - 456y \\
& + 390 \\
\gamma & 288x^7 + 144x^4y^2 - 504x^4y - 288x^4 - 2756x^3 - 347x - 360y^3 - 136y^2 - 177y \\
& - 270 \\
\{i, j\} &= \{30, 33\}, k_{ij} = 2 \cdot 3 \cdot 5^2 \cdot 13 \\
\alpha & 2848x^4 - 12460y + 9807 \\
\beta & 3026x^7 - 46725x^6 + 3738x^5y - 48594x^5 + 1424x^4y^2 - 234337x^4y \\
& + 250268x^4 + 2114184x^3 - 237363x^2y^2 + 7328349x^2y + 1248009x^2 \\
& - 237363xy^2 - 10932789xy - 11785292x - 41029y^3 + 6624946y^2 + 5878610y - 7857 \\
\gamma & -3026x^7 + 26611x^6 + 2314x^5y + 52065x^5 - 1424x^4y^2 + 169367x^4y \\
& - 148007x^4 - 2114184x^3 - 167765x^2y^2 - 6623825x^2y - 3674151x^2 \\
& - 335530xy^2 + 9241856xy + 3570815x - 28569y^3 + 6656168y^2 - 733876y - 5883814 \\
\{i, j\} &= \{30, 69\}, k_{ij} = 2^2 \cdot 3^5 \cdot 5 \cdot 13 \\
\alpha & -8436x^4 - 4921y + 24954 \\
\beta & 6327x^7 + 69597x^6 + 12654x^5y + 50616x^5 - 4218x^4y^2 + 42180x^4y + 1406x^4 \\
& - 24691x^3 + 14763x^2y^2 + 97717x^2y - 257541x^2 + 14763xy^2 + 48335xy - 117888x \\
& + 4921y^3 + 53954y^2 - 172708y + 193284 \\
\gamma & -6327x^7 + 4218x^4y^2 + 40071x^4y - 20387x^4 + 24691x^3 - 65580x + 9842y^3 \\
& + 43763y^2 - 134215y + 77529 \\
\{i, j\} &= \{30, 96\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
\alpha & -7x^3 + 10 \\
\beta & 42x^5 + 126x^4 + 7x^3y^2 + 91x^3y + 14x^3 - 11x^2 + 9x - 10y^2 - 32y + 22 \\
\gamma & -84x^4 - 7x^3y^2 - 49x^3y - 14x^3 + x + 10y^2 + 21y - 1 \\
\{i, j\} &= \{30, 106\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5^2 \\
\alpha & -3x^4 + 18y + 129 \\
\beta & x^7 + 21x^6 + 3x^5y + 21x^5 - x^4y^2 + 10x^4y + 6x^4 - 6x^3 - 18x^2y^2
\end{aligned}$$

$$\begin{aligned}
& -111x^2y - 789x^2 - 18xy^2 - 129xy - 532x - 6y^3 - 37y^2 - 569y + 294 \\
\gamma & -x^7 + x^4y^2 + 14x^4y - 6x^4 + 6x^3 - 257x - 12y^3 - 74y^2 - 238y + 477 \\
\{i, j\} &= \{31, 59\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5^2 \\
\alpha & -868x^4 + 3038y - 1270 \\
\beta & 6944x^7 + 19096x^6 + 3472x^5y + 29512x^5 - 1736x^4y^2 - 5208x^4y + 9114x^4 \\
& + 16236x^3 + 6076x^2y^2 - 7595x^2y + 30327x^2 + 6076xy^2 + 523xy + 35095x \\
& - 1021y^2 - 9559y + 9632 \\
\gamma & -3472x^7 + 868x^4y^2 + 13888x^4y + 19747x^4 - 8118x^3 + 9793x + 3038y^3 \\
& + 2789y^2 + 16407y + 18102 \\
\{i, j\} &= \{31, 85\}, k_{ij} = 2^3 \cdot 3 \cdot 5^2 \cdot 17 \\
\alpha & -9819x^4 + 1647410y + 1658929 \\
\beta & 1091x^7 + 60005x^6 + 3273x^5y + 62187x^5 - 3273x^4y^2 - 65460x^4y - 72006x^4 \\
& + 6031344x^3 - 329482x^2y^2 - 10529241x^2y - 4178615x^2 - 329482xy^2 \\
& + 7564791xy + 4323178x + 329482y^3 - 7235309y^2 - 3118769y + 896646 \\
\gamma & -1091x^7 + 3273x^4y^2 + 128738x^4y + 194198x^4 - 6031344x^3 + 3560895x \\
& - 658964y^3 - 3952896y^2 - 4024296y - 752083 \\
\{i, j\} &= \{31, 107\}, k_{ij} = 2 \cdot 5 \\
\alpha & 3x^4 - 30y - 40 \\
\beta & 3x^7 - 21x^6 - 3x^5y - 24x^5 - 3x^4y^2 - 75x^4y - 117x^4 + 251x^3 \\
& + 30x^2y^2 + 231x^2y + 13x^2 + 30xy^2 - 20xy - 486x + 90y^3 + 592y^2 + 468y + 96 \\
\gamma & -3x^7 + 3x^4y^2 + 51x^4y + 72x^4 - 251x^3 + 248x - 60y^3 - 301y^2 - 244y - 73 \\
\{i, j\} &= \{32, 46\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 11 \\
\alpha & y + 10 \\
\beta & x^7 + x^6 - x^4y + 69x^3 + x^2y^2 - 2x^2y + 9x^2 + 2xy^2 - 4xy + 369x - 3y^3 \\
& + 2y^2 + 120y + 739 \\
\gamma & -x^7 + 2x^4y - 69x^3 - 489x + 4y^3 - 119y - 490 \\
\{i, j\} &= \{32, 57\}, k_{ij} = 3^4 \cdot 5^2 \cdot 11 \\
\alpha & -487800x^4 + 49322y + 59029 \\
\beta & -1084000x^7 - 325200x^4y^2 - 314360x^4y - 281840x^4 + 146472x^3 + 8679x \\
& + 197288y^3 + 361662y^2 + 128358y - 4034 \\
\gamma & 542000x^7 + 108400x^6 - 813000x^5y - 325200x^5 + 162600x^4y^2 + 509480x^4y \\
& + 249320x^4 - 73236x^3 - 493220x^2y^2 - 424386x^2y + 125243x^2 + 246610xy^2 \\
& + 322047xy - 30343x + 147966y^3 + 154667y^2 - 75631y - 36754 \\
\{i, j\} &= \{32, 64\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \\
\alpha & 32y + 152 \\
\beta & 8x^7 + 32x^6 + 48x^5 + 4x^4y + 48x^4 + 240x^3 + 32x^2y^2 - 40x^2y + 144x^2 \\
& + 64xy^2 - 80xy - 18x + 64y^2 + 48y + 281 \\
\gamma & -8x^7 + 28x^4y + 48x^4 - 240x^3 - 174x + 32y^3 + 24y^2 - 64y + 31 \\
\{i, j\} &= \{32, 91\}, k_{ij} = 1 \\
\alpha & -27 \\
\beta & 9x^3 + 189x^2 + 27xy + 174x - 9y^2 + 165y + 107 \\
\gamma & -9x^3 + 213x + 9y^2 + 78y + 28 \\
\{i, j\} &= \{33, 44\}, k_{ij} = 3^2 \\
\alpha & 16x^4 - 16y + 12 \\
\beta & 16x^7 - 32x^6 - 32x^5y - 48x^5 - 8x^4y^2 - 28x^4y - 44x^4 + 72x^3 \\
& + 16x^2y^2 + 76x^2y + 252x^2 + 32xy^2 + 8xy + 320x - 4y^2 - 106y + 131 \\
\gamma & -16x^7 + 8x^4y^2 - 4x^4y - 52x^4 - 72x^3 + 256x - 16y^3 - 8y^2 - 130y - 3 \\
\{i, j\} &= \{33, 57\}, k_{ij} = 3 \cdot 5^3 \cdot 11^2 \\
\alpha & 1675x^4 + 2613y - 47128 \\
\beta & 16750x^7 + 31825x^6 - 8375x^5y + 72025x^5 - 3350x^4y^2 + 19095x^4y \\
& + 29480x^4 - 121674x^3 + 13065x^2y^2 - 70283x^2y - 80377x^2 + 26130xy^2 \\
& - 79729xy + 15091x + 7839y^3 - 7383y^2 - 56071y - 36483 \\
\gamma & -8375x^7 + 1675x^4y^2 - 33835x^4y + 57285x^4 + 60837x^3 - 118341x - 10452y^3 \\
& + 64963y^2 - 11505y + 92503 \\
\{i, j\} &= \{33, 69\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 13 \\
\alpha & -8y + 134 \\
\beta & -2x^7 - 8x^6 - 12x^5 + x^4y - 12x^4 + 342x^3 - 8x^2y^2 + 86x^2y + 540x^2 \\
& - 16xy^2 + 172xy + 69x - 16y^2 - y - 388 \\
\gamma & 2x^7 + 7x^4y - 12x^4 - 342x^3 + 327x + 8y^3 - 102y^2 - 195y - 128 \\
\{i, j\} &= \{33, 72\}, k_{ij} = 2^2 \cdot 3 \cdot 5^3 \cdot 17
\end{aligned}$$

$$\begin{aligned}
&\alpha \quad -87x^4 + 145y + 2492 \\
&\beta \quad -522x^7 - 261x^6 + 261x^5y + 174x^4y^2 - 261x^4y - 58x^4 + 5092x^3 \\
&\quad -435x^2y^2 + 290x^2y - 7169x^2 - 870xy^2 - 1966xy + 3420x - 725y^3 - 634y^2 \\
&\quad -6319y + 18423 \\
&\gamma \quad 261x^7 - 87x^4y^2 + 522x^4y + 29x^4 - 2546x^3 - 11425x + 580y^3 - 698y^2 + 4778y \\
&\quad -11504 \\
&\{i, j\} = \{33, 92\}, k_{ij} = 2^5 \cdot 3^4 \cdot 13 \cdot 17 \\
&\alpha \quad 29196x^4 - 376304y + 468260 \\
&\beta \quad 12976x^7 - 123272x^6 - 38928x^5y - 233568x^5 - 9732x^4y^2 - 35684x^4y \\
&\quad -226269x^4 + 1639080x^3 + 188152x^2y^2 + 1892063x^2y + 3958431x^2 \\
&\quad +376304xy^2 - 1133114xy + 5407362x - 853006y^2 - 3928855y + 2412992 \\
&\gamma \quad -12976x^7 + 9732x^4y^2 + 81100x^4y - 240867x^4 - 1639080x^3 + 4148580x \\
&\quad -188152y^3 - 286449y^2 - 2295804y + 104572 \\
&\{i, j\} = \{34, 37\}, k_{ij} = 2^3 \cdot 3 \\
&\alpha \quad -2x^4 + 8 \\
&\beta \quad -8x^7 + 4x^6 + 4x^5y + 4x^4y^2 - 4x^4y - 2x^4 + 40x^3 + 4x^2y^2 - x^2y \\
&\quad -5x^2 + 4xy^2 - 21xy - 55x + 4y^3 - 25y^2 + y + 28 \\
&\gamma \quad 4x^7 - 2x^4y^2 + 2x^4y + 3x^4 - 20x^3 + 35x - 4y^3 + 17y^2 - 9y - 44 \\
&\{i, j\} = \{34, 62\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \\
&\alpha \quad -21x^3 + 30 \\
&\beta \quad 126x^5 - 42x^4 + 14x^3y^2 - 28x^3y + 112x^3 - 33x^2 + 32x - 20y^2 - 9y - 41 \\
&\gamma \quad 84x^4 - 7x^3y^2 - 49x^3y + 70x^3 - x + 10y^2 + 21y + 19 \\
&\{i, j\} = \{34, 64\}, k_{ij} = 2 \cdot 3^5 \cdot 5^3 \\
&\alpha \quad 564x^4 - 329y + 5840 \\
&\beta \quad 423x^7 - 2115x^6 - 846x^5y - 2538x^5 - 282x^4y^2 + 282x^4y - 2068x^4 \\
&\quad +6401x^3 + 987x^2y^2 + 3149x^2y - 23639x^2 + 987xy^2 - 9653xy - 22234x - 329y^3 \\
&\quad -4330y^2 + 9722y - 5612 \\
&\gamma \quad -423x^7 + 282x^4y^2 + 987x^4y - 2585x^4 - 6401x^3 - 7806x - 658y^3 + 3155y^2 \\
&\quad +7413y - 24649 \\
&\{i, j\} = \{34, 70\}, k_{ij} = 2^2 \cdot 3^3 \\
&\alpha \quad -3y + 24 \\
&\beta \quad -x^7 - 3x^6 - 6x^5 - 4x^4 + 51x^3 - 3x^2y^2 + 21x^2y + 15x^2 - 3xy^2 + 21xy \\
&\quad +148x + y^3 - 10y^2 + 66y + 54 \\
&\gamma \quad x^7 + 3x^4y - 5x^4 - 51x^3 + 20x + 2y^3 - 17y^2 - 39y + 69 \\
&\{i, j\} = \{34, 91\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 7 \\
&\alpha \quad 66x^4 + 1496y + 700 \\
&\beta \quad -374x^7 + 880x^6 - 880x^5y - 880x^5 + 66x^4y^2 + 4180x^4y + 1485x^4 \\
&\quad +171160x^3 + 1760x^2y^2 - 20460x^2y - 68560x^2 + 1760xy^2 + 320900xy - 68313x \\
&\quad +2024y^3 - 41144y^2 - 499230y - 239228 \\
&\gamma \quad 374x^7 - 2024x^6 + 506x^5y + 3036x^5 - 66x^4y^2 - 2398x^4y - 2497x^4 \\
&\quad -171160x^3 - 1012x^2y^2 + 21505x^2y + 410596x^2 + 1012xy^2 - 191705xy - 68523x \\
&\quad -1760y^3 + 22114y^2 + 217238y - 239718 \\
&\{i, j\} = \{34, 96\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \\
&\alpha \quad -21x^3 + 30 \\
&\beta \quad 126x^5 + 42x^4 + 7x^3y^2 - 77x^3y + 182x^3 - 33x^2 + 31x - 10y^2 + 12y - 22 \\
&\gamma \quad 84x^4 - 7x^3y^2 - 49x^3y + 70x^3 - x + 10y^2 + 21y + 19 \\
&\{i, j\} = \{34, 99\}, k_{ij} = 2^5 \cdot 5 \\
&\alpha \quad 3x^4 - 12y + 25 \\
&\beta \quad -3x^7 - 15x^6 - 3x^5y - 15x^5 + 3x^4y^2 + 36x^4y - 78x^4 - 88x^3 \\
&\quad +12x^2y^2 + 63x^2y - 95x^2 + 12xy^2 - 25xy - 150x - 36y^3 - 41y^2 + 293y - 390 \\
&\gamma \quad 3x^7 - 3x^4y^2 - 24x^4y + 48x^4 + 88x^3 + 55x + 24y^3 + 2y^2 - 160y + 255 \\
&\{i, j\} = \{35, 57\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 17 \\
&\alpha \quad 381x^4 + 762y - 768 \\
&\beta \quad 2286x^7 + 1143x^6 - 1143x^5y - 762x^4y^2 - 1524x^4y - 127x^4 + 218x^3 \\
&\quad +2286x^2y^2 - 2413x^2y + 5189x^2 + 2286xy^2 - 2522xy + 5880x + 762y^3 + 647y^2 \\
&\quad +714y + 3505 \\
&\gamma \quad -1143x^7 + 381x^4y^2 - 381x^4y - 508x^4 - 109x^3 - 400x - 1524y^3 + 883y^2 \\
&\quad -3006y - 1207 \\
&\{i, j\} = \{35, 64\}, k_{ij} = 2^3 \cdot 3^4 \cdot 5 \cdot 7 \\
&\alpha \quad -6324x^4 + 12121y + 41742
\end{aligned}$$

$$\begin{aligned}
&\beta \quad -4743x^7 + 23715x^6 + 9486x^5y + 37944x^5 + 3162x^4y^2 + 6324x^4y + 20026x^4 \\
&\quad -49061x^3 - 36363x^2y^2 - 139655x^2y - 422931x^2 - 36363xy^2 - 41533xy - 204100x \\
&\quad +12121y^3 + 67138y^2 + 293414y + 235064 \\
&\gamma \quad 4743x^7 - 3162x^4y^2 - 20553x^4y - 5797x^4 + 49061x^3 - 71648x + 24242y^3 \\
&\quad +72517y^2 + 227639y + 127063 \\
&\{i, j\} = \{35, 69\}, k_{ij} = 2 \cdot 3^2 \cdot 7 \\
&\alpha \quad -3y + 18 \\
&\beta \quad -x^7 - 3x^6 + 2x^4y + 2x^4 + 39x^3 - 3x^2y^2 + 15x^2y + 9x^2 - 3xy^2 + 15xy \\
&\quad -42x + y^3 - 6y^2 - 38y + 90 \\
&\gamma \quad x^7 + x^4y + x^4 - 39x^3 - 66x + 2y^3 - 9y^2 + 29y - 9 \\
&\{i, j\} = \{35, 88\}, k_{ij} = 3 \cdot 7 \\
&\alpha \quad 5x - 4 \\
&\beta \quad -30x^3 - 66x^2 + 5xy^2 + 60xy + 107x - 4y^2 - 48y + 22 \\
&\gamma \quad 60x^2 - 5xy^2 - 30xy - 83x + 4y^2 + 24y + 3 \\
&\{i, j\} = \{35, 100\}, k_{ij} = 2^2 \cdot 3 \cdot 5^2 \\
&\alpha \quad x^4 - 6y + 41 \\
&\beta \quad -x^7 - 5x^6 - x^5y - 5x^5 + x^4y^2 + 14x^4y + 4x^4 - 76x^3 + 6x^2y^2 \\
&\quad +35x^2y - 171x^2 + 6xy^2 - 41xy - 430x - 18y^3 - 35y^2 + 149y + 648 \\
&\gamma \quad x^7 - x^4y^2 - 10x^4y - 4x^4 + 76x^3 + 259x + 12y^3 - 54y - 389 \\
&\{i, j\} = \{35, 104\}, k_{ij} = 2^3 \cdot 5 \cdot 7 \\
&\alpha \quad -94x^4 - 1316y - 1596 \\
&\beta \quad 1880x^7 - 3948x^6 + 3948x^5y - 94x^4y^2 - 13724x^4y - 11515x^4 - 405635x^3 \\
&\quad -7896x^2y^2 + 77973x^2y - 14469x^2 - 7896xy^2 - 731703xy - 419415x - 14476y^3 \\
&\quad +139498y^2 + 932104y + 777878 \\
&\gamma \quad -1880x^7 + 8272x^6 - 2068x^5y - 10340x^5 + 94x^4y^2 + 7144x^4y + 9259x^4 \\
&\quad +405635x^3 + 7238x^2y^2 - 77550x^2y - 794134x^2 - 7238xy^2 + 481591xy + 793445x \\
&\quad +7896y^3 - 69609y^2 - 465534y - 388001 \\
&\{i, j\} = \{35, 105\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5^3 \cdot 7 \\
&\alpha \quad -201x^4 + 670y + 21871 \\
&\beta \quad -603x^7 - 603x^6 - 603x^5y + 603x^5 - 201x^4y^2 + 938x^4 + 92078x^3 \\
&\quad +402x^2y^2 - 5293x^2y + 144539x^2 + 402xy^2 + 86785xy + 36388x - 134y^3 \\
&\quad +32591y^2 - 44363y + 73994 \\
&\gamma \quad 603x^7 + 201x^4y^2 + 268x^4 - 92078x^3 - 76005x - 268y^3 - 27298y^2 - 8098y \\
&\quad -30135 \\
&\{i, j\} = \{35, 106\}, k_{ij} = 2 \cdot 3 \\
&\alpha \quad -10185021x^4 - 48519440640x^3 - 4043286720x^2y^2 - 22575017520x^2y \\
&\quad +29501349912x^2 + 4033174971xy^2 - 25924199622xy - 29447420584x - 4046657303y^3 \\
&\quad -26598080742y^2 - 41603464237y - 19049525320 \\
&\beta \quad -30555063x^7 - 30555063x^6 - 30555063x^5y + 291086088777x^5 \\
&\quad +24249535299x^4y^2 + 135429735078x^4y + 114094964340x^4 + 60670494x^3y^2 \\
&\quad -121340988x^3y + 3544054260x^3 + 20223498x^2y^3 - 81040536x^2y^2 \\
&\quad -62600061x^2y + 75318983949x^2 + 48478993644xy^3 + 4022916678xy^2 \\
&\quad +79053423300xy + 109902682651x + 4039916137y^5 + 22554794022y^4 + 23068167462y^3 \\
&\quad +26890857213y^2 + 80261770997y + 38301875811 \\
&\gamma \quad 30555063x^7 + 10185021x^4y^2 + 20370042x^4y + 13580028x^4 - 3867630228x^3 \\
&\quad -15321436020x^2 - 3709018017xy + 15400840034x + 3370583y^5 + 13580028y^3 \\
&\quad -139170669y^2 - 19405956281y - 19252350485 \\
&\{i, j\} = \{36, 56\}, k_{ij} = 3 \cdot 5 \cdot 7 \\
&\alpha \quad -16y + 56 \\
&\beta \quad -128x^7 - 128x^6 + 32x^4y + 240x^3 - 32x^2y^2 + 64x^2y + 72x^2 - 16xy^2 + 32xy \\
&\quad -42x - 4y^2 - 60y + 49 \\
&\gamma \quad 128x^7 + 32x^4y - 240x^3 - 42x + 16y^3 - 36y^2 + 40y - 49 \\
&\{i, j\} = \{36, 61\}, k_{ij} = 2^2 \cdot 3^4 \\
&\alpha \quad 304x^3 - 250 \\
&\beta \quad -912x^5 - 152x^3y^2 + 380x^3y - 190x^3 + 28x^2 - 95x + 125y^2 - 132y + 199 \\
&\gamma \quad -456x^4 + 152x^3y^2 + 76x^3y - 38x^3 - 81x - 125y^2 + 118y - 287 \\
&\{i, j\} = \{36, 85\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5^3 \\
&\alpha \quad -78552x^4 + 1298290y + 1378342 \\
&\beta \quad -17456x^7 + 226928x^6 - 26184x^5y + 96008x^5 - 26184x^4y^2 + 96008x^4y \\
&\quad -78552x^4 + 3215874x^3 + 370940x^2y^2 - 3846866x^2y - 2614369x^2
\end{aligned}$$

$$\begin{aligned}
& +185470xy^2 + 2900378xy + 1804420x + 185470y^3 + 2900378y^2 + 3519788y + 1586881 \\
\gamma \quad & 34912x^7 + 52368x^4y^2 - 445128x^4y + 253112x^4 - 6431748x^3 + 208539x \\
& - 741880y^3 - 1768420y^2 - 1524829y - 1369342 \\
\{i, j\} = \{36, 87\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5^4 \\
\alpha \quad & 3204x^4 - 22250y + 42736 \\
\beta \quad & -712x^7 - 9256x^6 - 1068x^5y - 3916x^5 + 1068x^4y^2 + 12104x^4y - 7120x^4 \\
& - 32538x^3 + 8900x^2y^2 + 50374x^2y - 108379x^2 + 4450xy^2 - 23620xy - 48029x \\
& - 13350y^3 - 22304y^2 + 49528y + 14745 \\
\gamma \quad & 1424x^7 - 2136x^4y^2 - 16020x^4y + 10324x^4 + 65076x^3 + 52755x + 17800y^3 \\
& - 1316y^2 - 14297y - 24764 \\
\{i, j\} = \{37, 52\}, k_{ij} = 2^2 \cdot 3^4 \cdot 13^2 \\
\alpha \quad & -1832x^4 - 5267y + 28683 \\
\beta \quad & -14656x^7 - 23816x^6 - 7328x^5y - 21984x^5 - 1832x^4y^2 + 8244x^4y - 6870x^4 \\
& + 42600x^3 - 10534x^2y^2 + 25648x^2y + 1670x^2 + 21300xy + 65881x + 5267y^3 \\
& - 12766y^2 + 20827y - 5493 \\
\gamma \quad & 29312x^7 + 3664x^4y^2 + 7328x^4y - 10076x^4 - 85200x^3 - 3962x - 10650y^2 \\
& - 17676y + 20580 \\
\{i, j\} = \{37, 58\}, k_{ij} = 3^3 \\
\alpha \quad & -4y + 20 \\
\beta \quad & -32x^7 - 32x^6 - 16x^5 + 8x^4y - 8x^4 + 112x^3 - 8x^2y^2 + 24x^2y + 48x^2 \\
& + 30x + 4y^3 - 16y^2 + 14y - 7 \\
\gamma \quad & 32x^7 + 8x^4y - 8x^4 - 112x^3 + 26x - 26y + 5 \\
\{i, j\} = \{37, 73\}, k_{ij} = 2 \cdot 3^3 \\
\alpha \quad & -2y + 7 \\
\beta \quad & -32x^7 - 16x^6 + 16x^5 + 24x^4y - 24x^4 + 58x^3 - 4x^2y^2 + 6x^2y + 12x^2 + 5x \\
& + 2y^3 - 5y^2 - 34y + 93 \\
\gamma \quad & 32x^7 - 16x^4y + 16x^4 - 58x^3 - 34x + 31y - 70 \\
\{i, j\} = \{37, 79\}, k_{ij} = 3^3 \\
\alpha \quad & -4y + 20 \\
\beta \quad & 32x^7 - 32x^6 - 16x^5 + 24x^4y - 24x^4 - 112x^3 - 8x^2y^2 + 24x^2y + 48x^2 \\
& + 82x + 4y^3 - 16y^2 - 38y + 3 \\
\gamma \quad & -32x^7 - 8x^4y + 8x^4 + 112x^3 - 26x + 26y - 5 \\
\{i, j\} = \{37, 81\}, k_{ij} = 3^2 \cdot 5 \\
\alpha \quad & -4y + 20 \\
\beta \quad & -32x^7 - 32x^6 + 16x^5 + 24x^4y - 24x^4 + 112x^3 - 8x^2y^2 + 24x^2y + 48x^2 \\
& - 78x + 4y^3 - 16y^2 - 42y + 29 \\
\gamma \quad & 32x^7 - 8x^4y + 8x^4 - 112x^3 + 22x + 30y - 27 \\
\{i, j\} = \{37, 82\}, k_{ij} = 3^3 \cdot 11 \\
\alpha \quad & 4y - 4 \\
\beta \quad & 32x^7 + 32x^6 + 48x^5 + 8x^4y - 8x^4 + 32x^3 + 8x^2y^2 + 8x^2y + 16x^2 + 114x \\
& - 4y^3 + 22y - 51 \\
\gamma \quad & -32x^7 - 24x^4y + 24x^4 - 32x^3 - 66x - 26y + 125 \\
\{i, j\} = \{38, 60\}, k_{ij} = 2^3 \cdot 3 \\
\alpha \quad & -176x^3 - 170 \\
\beta \quad & 528x^5 + 88x^3y^2 - 220x^3y + 110x^3 + 268x^2 + 55x + 85y^2 - 152y + 109 \\
\gamma \quad & -264x^4 - 88x^3y^2 - 44x^3y + 22x^3 - 79x - 85y^2 + 18y - 97 \\
\{i, j\} = \{38, 75\}, k_{ij} = 2^2 \cdot 3^2 \\
\alpha \quad & -2y + 7 \\
\beta \quad & -32x^7 - 16x^6 - 8x^5 + 8x^4y + 8x^4 + 54x^3 - 4x^2y^2 + 4x^2y + 8x^2 \\
& + 2xy^2 - 2xy + 72x + 6y^3 - 8y^2 + 15y - 16 \\
\gamma \quad & 32x^7 - 4x^4 - 54x^3 - 49x - 4y^3 + 5y^2 - 18y + 29 \\
\{i, j\} = \{38, 87\}, k_{ij} = 2^3 \\
\alpha \quad & 27 \\
\beta \quad & -18x^3 - 54x^2 - 27xy + 93x + 9y^2 + 60y - 104 \\
\gamma \quad & 36x^3 - 96x - 18y^2 - 39y + 19 \\
\{i, j\} = \{39, 56\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5 \\
\alpha \quad & 176x^3 + 266 \\
\beta \quad & -528x^5 + 88x^3y^2 - 132x^3y + 22x^3 - 556x^2 - 143x + 133y^2 - 139y + 135 \\
\gamma \quad & 264x^4 - 88x^3y^2 - 132x^3y - 418x^3 + 135x - 133y^2 - 139y - 409
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{39, 77\}, k_{ij} = 2 \cdot 3^3 \\
&\alpha \quad 6y - 9 \\
&\beta \quad -32x^7 + 48x^6 - 72x^5 - 8x^4y + 16x^4 - 30x^3 + 12x^2y^2 - 24x^2y + 36x^2 \\
&\quad - 6xy^2 + 12xy - 66x + 2y^3 - 19y + 15 \\
&\gamma \quad 32x^7 + 32x^4y + 44x^4 + 30x^3 + 3x + 4y^3 - 3y^2 + 19y + 39 \\
&\{i, j\} = \{39, 92\}, k_{ij} = 2^2 \cdot 3^3 \\
&\alpha \quad 24x^4 - 6y - 6 \\
&\beta \quad -16x^7 - 96x^6 + 24x^5y + 24x^5 - 8x^4y^2 - 96x^4y - 80x^4 + 38x^3 \\
&\quad + 12x^2y^2 - 78x^2y + 3x^2 - 6xy^2 - 18xy + 60x + 10y^3 - 32y^2 - 48y - 51 \\
&\gamma \quad 32x^7 + 16x^4y^2 + 72x^4y + 40x^4 - 76x^3 - 9x - 8y^3 + 4y^2 + 39y + 63 \\
&\{i, j\} = \{40, 65\}, k_{ij} = 2^2 \cdot 3 \cdot 5^2 \cdot 11 \cdot 13 \cdot 17 \\
&\alpha \quad 4376x^4 - 16957y + 100214 \\
&\beta \quad 35008x^7 - 83144x^6 - 17504x^5y + 35008x^5 - 4376x^4y^2 - 24068x^4y \\
&\quad + 64546x^4 - 375432x^3 - 33914x^2y^2 + 158630x^2y + 346292x^2 + 187716xy \\
&\quad - 704831x - 16957y^3 + 160158y^2 - 86611y + 310469 \\
&\gamma \quad -70016x^7 + 8752x^4y^2 - 35008x^4y + 37196x^4 + 750864x^3 + 658798x - 93858y^2 \\
&\quad + 202254y + 4074 \\
&\{i, j\} = \{40, 77\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 11 \\
&\alpha \quad -4y + 38 \\
&\beta \quad 32x^7 - 32x^6 - 16x^4y + 32x^4 - 346x^3 - 8x^2y^2 + 44x^2y + 200x^2 - 346x \\
&\quad - 4y^3 + 30y^2 + 58y + 127 \\
&\gamma \quad -32x^7 + 346x^3 + 346x - 2y + 19 \\
&\{i, j\} = \{40, 78\}, k_{ij} = 2 \cdot 3 \cdot 5^2 \cdot 13 \cdot 17 \\
&\alpha \quad -4y + 70 \\
&\beta \quad 32x^7 - 32x^6 - 64x^5 - 48x^4y + 96x^4 - 826x^3 - 8x^2y^2 + 108x^2y + 456x^2 \\
&\quad - 2422x - 4y^3 + 62y^2 + 948y + 10079 \\
&\gamma \quad -32x^7 + 32x^4y - 64x^4 + 826x^3 + 4074x - 828y - 6461 \\
&\{i, j\} = \{41, 63\}, k_{ij} = 3^3 \\
&\alpha \quad -2y - 12 \\
&\beta \quad 16x^7 - 16x^6 + 8x^5 - 4x^4y - 8x^4 - 20x^3 - 4x^2y^2 - 8x^2y + 44x^2 + 72x \\
&\quad - 2y^3 - 8y^2 + 10y - 1 \\
&\gamma \quad -16x^7 - 4x^4y - 8x^4 + 20x^3 - 82x + 4y - 37 \\
&\{i, j\} = \{41, 78\}, k_{ij} = 2 \cdot 3^4 \\
&\alpha \quad -4y - 26 \\
&\beta \quad 32x^7 - 32x^6 - 16x^4y - 32x^4 - 58x^3 - 8x^2y^2 - 20x^2y + 104x^2 + 198x \\
&\quad - 4y^3 - 18y^2 + 34y - 81 \\
&\gamma \quad -32x^7 + 58x^3 - 198x - 2y - 13 \\
&\{i, j\} = \{41, 80\}, k_{ij} = 2 \cdot 3^4 \\
&\alpha \quad -64x^4 - 64y - 146 \\
&\beta \quad -64x^7 + 256x^6 - 64x^5y - 128x^5 - 32x^4y^2 + 160x^4 + 46x^3 - 128x^2y^2 \\
&\quad + 128x^2y + 488x^2 + 46xy - 232x - 64y^3 - 41y^2 + 172y + 329 \\
&\gamma \quad 64x^7 + 32x^4y^2 + 128x^4y + 96x^4 - 46x^3 + 324x - 23y^2 + 200y + 483 \\
&\{i, j\} = \{42, 43\}, k_{ij} = 2^3 \cdot 3^2 \\
&\alpha \quad 2y + 9 \\
&\beta \quad -32x^7 + 16x^6 - 8x^5 + 8x^4y - 30x^3 + 4x^2y^2 - 12x^2 + 2xy^2 - 146x + 6y^3 \\
&\quad + 8y^2 - 35y - 63 \\
&\gamma \quad 32x^7 + 4x^4 + 30x^3 + 125x - 4y^3 - 5y^2 + 29y + 109 \\
&\{i, j\} = \{42, 47\}, k_{ij} = 2^5 \cdot 3^2 \cdot 7^2 \cdot 17 \\
&\alpha \quad -3300x^4 + 2200y + 33288 \\
&\beta \quad -19800x^7 + 39600x^6 + 9900x^5y - 9900x^5 + 3300x^4y^2 + 33000x^4y \\
&\quad + 17600x^4 + 28678x^3 + 13200x^2y^2 + 25850x^2y - 129681x^2 + 6600xy^2 - 1414xy \\
&\quad - 87924x + 11000y^3 + 32162y^2 - 56609y - 53273 \\
&\gamma \quad 39600x^7 - 6600x^4y^2 - 16500x^4y - 5500x^4 - 57356x^3 + 132201x - 8800y^3 \\
&\quad - 18674y^2 + 7973y + 153343 \\
&\{i, j\} = \{42, 54\}, k_{ij} = 2^4 \cdot 3 \cdot 7 \\
&\alpha \quad -304x^3 + 346 \\
&\beta \quad 912x^5 - 152x^3y^2 + 228x^3y - 38x^3 - 316x^2 + 247x + 173y^2 - 79y + 257 \\
&\gamma \quad 456x^4 + 152x^3y^2 + 228x^3y + 722x^3 + 89x - 173y^2 - 79y - 247 \\
&\{i, j\} = \{42, 55\}, k_{ij} = 2
\end{aligned}$$

α	$-112x^3 - 2$
β	$336x^5 - 56x^3y^2 + 140x^3y + 14x^3 + 104x^2 - 28x - y^2 + 27y + 23$
γ	$168x^4 + 56x^3y^2 + 28x^3y + 238x^3 + 24x + y^2 + 25y + 27$
$\{i, j\}$	$\{42, 60\}, k_{ij} = 7$
α	$-16y - 72$
β	$-128x^7 - 128x^6 - 32x^4y - 32x^4 - 32x^2y^2 - 32x^2y + 168x^2 - 16xy^2 - 16xy$ $-262x - 4y^2 - 12y - 361$
γ	$128x^7 - 32x^4y - 32x^4 + 346x - 16y^3 - 36y^2 + 72y + 141$
$\{i, j\}$	$\{42, 75\}, k_{ij} = 3 \cdot 7^3$
α	$88896x^4 - 26854y - 27883$
β	$-133344x^7 - 466704x^6 - 133344x^5y - 366696x^5 + 44448x^4y^2 - 33336x^4y$ $-125936x^4 - 77398x^3 + 161124x^2y^2 + 235204x^2y + 5518x^2 + 80562xy^2$ $+40204xy - 16364x + 26854y^3 + 118708y^2 + 57319y + 11970$
γ	$133344x^7 - 44448x^4y^2 - 266688x^4y - 290764x^4 + 77398x^3 - 58275x$ $+53708y^3 + 119737y^2 + 83144y + 40882$
$\{i, j\}$	$\{42, 95\}, k_{ij} = 2^3 \cdot 3$
α	$-54x^4 + 81y + 57$
β	$36x^7 + 216x^6 - 54x^5y + 54x^5 + 18x^4y^2 + 54x^4y + 54x^4 + 142x^3$ $+54x^2y^2 + 54x^2y + 258x^2 + 27xy^2 - 186xy - 194x + 9y^3 + 98y^2 + 49y - 115$
γ	$-72x^7 - 36x^4y^2 + 54x^4y + 162x^4 - 284x^3 + 220x + 36y^3 - 61y^2 + 28y + 74$
$\{i, j\}$	$\{42, 98\}, k_{ij} = 2^6 \cdot 3$
α	$24x^4 - 6y$
β	$16x^7 - 96x^6 - 24x^5y - 24x^5 - 8x^4y^2 - 96x^4y - 96x^4 - 26x^3$ $+12x^2y^2 - 54x^2y - 57x^2 + 6xy^2 + 12xy - 80x + 10y^3 - 18y^2 - 70y - 93$
γ	$-32x^7 + 16x^4y^2 + 72x^4y + 72x^4 + 52x^3 + 25x - 8y^3 + 41y + 99$
$\{i, j\}$	$\{43, 55\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 13$
α	$-704x^4 + 17094y + 17267$
β	$352x^7 + 2288x^6 + 352x^5y + 2464x^5 - 352x^4y^2 - 3960x^4y - 3168x^4$ $+105178x^3 - 4884x^2y^2 - 58058x^2y - 1676x^2 - 4884xy^2 + 47120xy + 195x$ $+7326y^3 - 13207y^2 + 21335y + 42431$
γ	$-352x^7 + 352x^4y^2 + 5456x^4y + 5632x^4 - 105178x^3 + 50718x - 9768y^3$ $-20706y^2 + 24947y - 16877$
$\{i, j\}$	$\{43, 77\}, k_{ij} = 2$
α	0
β	-1
γ	1
$\{i, j\}$	$\{43, 80\}, k_{ij} = 2 \cdot 17$
α	$-16x^3 - 20$
β	$48x^5 - 8x^3y^2 + 20x^3y + 20x^3 + 68x^2 - 10x - 10y^2 + 27y + 21$
γ	$48x^4 + 8x^3y^2 + 4x^3y + 28x^3 + 58x + 10y^2 + 7y + 27$
$\{i, j\}$	$\{44, 47\}, k_{ij} = 2^2 \cdot 3$
α	$-4y + 8$
β	$x^7 - 4x^6 + 8x^5 - 8x^4 + 3x^3 - 4x^2y^2 + 16x^2y - 48x^2 + 8xy^2 - 32xy + 59x$ $-8y^2 + 33y - 24$
γ	$-x^7 - 4x^4y - 4x^4 - 3x^3 + 37x - 4y^3 + 12y^2 - 33y + 28$
$\{i, j\}$	$\{44, 87\}, k_{ij} = 2^3$
α	27
β	$9x^3 - 189x^2 + 27xy + 174x + 9y^2 - 156y + 4$
γ	$-9x^3 + 186x - 9y^2 - 87y + 23$
$\{i, j\}$	$\{45, 47\}, k_{ij} = 2^3 \cdot 3$
α	3
β	$-x^5 + 3x^4 - 8x^3 - 8x^2 - 4x + 2y^2 - 7$
γ	$x^5 + 2x^3 + 3x^2y + 8x^2 + 10x + y^2 - 3y + 7$
$\{i, j\}$	$\{45, 66\}, k_{ij} = 2 \cdot 3^3$
α	$-3y - 12$
β	$x^7 - 3x^6 + 3x^5 - x^4y - 4x^4 + 18x^3 - 3x^2y^2 + 3x^2y + 3xy^2 - 3xy + 11x$ $-y^3 - 3y^2 + 5y - 18$
γ	$-x^7 - 2x^4y - 5x^4 - 18x^3 - 11x - 2y^3 - 3y^2 + 4y - 15$
$\{i, j\}$	$\{45, 82\}, k_{ij} = 2^2 \cdot 3^3$

α	$15x^3 - 12$
β	$-90x^5 - 30x^4 + 10x^3y^2 - 50x^3y - 40x^3 + 147x^2 + 94x - 8y^2 + 65y + 62$
γ	$60x^4 - 5x^3y^2 - 20x^3y - 115x^3 - 53x + 4y^2 + 41y + 122$
$\{i, j\}$	$\{45, 100\}, k_{ij} = 2^4 \cdot 3 \cdot 11$
α	$-9x^4 + 99y + 75$
β	$-3x^7 + 63x^6 - 9x^5y - 72x^5 - 3x^4y^2 + 18x^4y + 87x^4 - 287x^3$ $-99x^2y^2 - 738x^2y + 483x^2 + 99xy^2 - 123xy - 428x - 33y^3 - 665y^2 - 227y - 297$
γ	$3x^7 + 3x^4y^2 + 54x^4y + 111x^4 + 287x^3 - 916x - 66y^3 - 370y^2 - 643y - 141$
$\{i, j\}$	$\{46, 56\}, k_{ij} = 2 \cdot 3^2 \cdot 11$
α	$-6x^4 + 66y + 39$
β	$4x^7 + 42x^6 + 6x^5y + 48x^5 - 4x^4y^2 - 24x^4y - 24x^4 + 398x^3$ $-66x^2y^2 - 504x^2y + 399x^2 - 66xy^2 + 93xy - 234x + 22y^3 - 208y^2 + 314y - 89$
γ	$-2x^7 + 2x^4y^2 + 36x^4y + 78x^4 - 199x^3 + 615x - 44y^3 - 247y^2 - 415y + 19$
$\{i, j\}$	$\{46, 66\}, k_{ij} = 2 \cdot 3^2$
α	$-y - 4$
β	$x^7 - x^6 - x^5 - 3x^4y - 8x^4 + 18x^3 - x^2y^2 + x^2y - xy^2 + xy + 11x - 3y^3$ $-5y^2 + 9y - 18$
γ	$-x^7 + 2x^4y + 5x^4 - 18x^3 - 11x + 2y^3 + 3y^2 - 6y + 7$
$\{i, j\}$	$\{46, 74\}, k_{ij} = 3^4 \cdot 13$
α	$12x^4 - 156y - 429$
β	$-8x^7 - 84x^6 - 12x^5y - 108x^5 + 8x^4y^2 + 56x^4y + 60x^4 - 1074x^3$ $+156x^2y^2 + 1416x^2y + 1407x^2 + 156xy^2 - 195xy - 501x - 52y^3 + 550y^2 + 1400y$ $+496$
γ	$4x^7 - 4x^4y^2 - 76x^4y - 168x^4 + 537x^3 - 657x + 104y^3 + 667y^2 + 1322y + 1237$
$\{i, j\}$	$\{46, 80\}, k_{ij} = 2 \cdot 3^3$
α	$-21x^3 - 6$
β	$126x^5 - 42x^4 - 14x^3y^2 + 70x^3y + 56x^3 + 183x^2 - 124x - 4y^2 + 69y + 100$
γ	$84x^4 + 7x^3y^2 + 28x^3y + 161x^3 + 59x + 2y^2 + 57y + 130$
$\{i, j\}$	$\{46, 101\}, k_{ij} = 2 \cdot 3^5 \cdot 11$
α	$-93x^3 + 798$
β	$558x^5 + 186x^4 - 31x^3y^2 + 372x^3y + 713x^3 - 1905x^2 + 1659x + 266y^2 - 1270y$ $+20$
γ	$372x^4 + 31x^3y^2 + 186x^3y + 961x^3 - 123x - 266y^2 - 635y - 2294$
$\{i, j\}$	$\{47, 100\}, k_{ij} = 2^6 \cdot 3^2$
α	$-75x^3 + 147$
β	$450x^5 + 300x^4 - 25x^3y^2 + 300x^3y + 1025x^3 + 993x^2 - 188x + 49y^2 + 662y$ $+666$
γ	$600x^4 + 25x^3y^2 + 150x^3y + 325x^3 + 899x - 49y^2 + 331y - 237$
$\{i, j\}$	$\{48, 97\}, k_{ij} = 2^2 \cdot 5^3$
α	$38911x^4 - 1050430y - 1050930$
β	$-46927x^7 - 384601x^6 + 54943x^5y - 714259x^5 - 38911x^4y^2 + 557279x^4y$ $+2069297x^4 + 8426395x^3 - 1428518x^2y^2 + 35108577x^2y + 69245617x^2$ $-2857036xy^2 - 64097425xy - 102067907x + 1239474y^3 - 78046950y^2 - 241233318y$ -162296704
γ	$46927x^7 + 104208x^6 - 8016x^5y + 96192x^5 + 38911x^4y^2 - 631594x^4y$ $-3577808x^4 - 8426395x^3 + 761520x^2y^2 - 34637136x^2y - 59638872x^2$ $+761520xy^2 + 91251048xy + 172493874x - 1144952y^3 + 39023225y^2 + 120091694y$ $+80622637$
$\{i, j\}$	$\{48, 98\}, k_{ij} = 2^6 \cdot 5^3 \cdot 7$
α	$-121x^3 + 3379$
β	$726x^5 + 4356x^4 + 121x^3y^2 + 1694x^3y + 10043x^3 - 5633x^2 - 644x - 3379y^2$ $-18024y - 23816$
γ	$-2904x^4 - 121x^3y^2 - 968x^3y - 6413x^3 + 5955x + 3379y^2 + 12391y + 28805$
$\{i, j\}$	$\{50, 58\}, k_{ij} = 2^2 \cdot 3^5 \cdot 5$
α	$-29x^3 - 52$
β	$174x^5 + 522x^4 + 58x^3y^2 - 58x^3y + 464x^3 - 529x^2 - 688x + 104y^2 - 945y$ $+2456$
γ	$-348x^4 - 29x^3y^2 - 58x^3y - 319x^3 + 159x - 52y^2 + 737y - 514$
$\{i, j\}$	$\{50, 64\}, k_{ij} = 2^7 \cdot 3^5 \cdot 5$
α	$436x^4 - 52865y - 123010$

$$\begin{aligned}
&\beta \quad 109x^7 - 2507x^6 - 218x^5y + 2071x^5 - 218x^4y^2 + 7085x^4y + 10464x^4 \\
&\quad - 330666x^3 - 10573x^2y^2 + 275879x^2y + 370988x^2 + 10573xy^2 + 385453xy \\
&\quad + 1102543x + 31719y^3 - 124013y^2 - 644947y - 701760 \\
&\gamma \quad -109x^7 + 218x^4y^2 - 4360x^4y - 7521x^4 + 330666x^3 - 150867x - 21146y^3 \\
&\quad - 141293y^2 - 663252y - 841025 \\
&\{i, j\} = \{50, 70\}, k_{ij} = 2 \cdot 3^2 \\
&\alpha \quad y \\
&\beta \quad x^7 + x^6 - x^5 - 3x^4y - 4x^4 + 6x^3 + x^2y^2 - 3x^2y + 8x^2 - xy^2 + 3xy + 9x \\
&\quad - 3y^3 + 5y^2 - 11y + 2 \\
&\gamma \quad -x^7 + 2x^4y + 3x^4 - 6x^3 - 17x + 2y^3 - 3y^2 + 6y + 7 \\
&\{i, j\} = \{50, 76\}, k_{ij} = 2^2 \cdot 3^3 \\
&\alpha \quad 15x^3 - 12 \\
&\beta \quad -90x^5 - 30x^4 - 10x^3y^2 + 50x^3y - 20x^3 + 147x^2 + 94x + 8y^2 - 65y - 54 \\
&\gamma \quad 60x^4 + 5x^3y^2 + 20x^3y + 55x^3 - 53x - 4y^2 - 41y - 114 \\
&\{i, j\} = \{50, 106\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad 1 \\
&\beta \quad x^3 - 5x^2 + xy + 9x + y^2 + 9y + 36 \\
&\gamma \quad -x^3 - 5x - y^2 - 5y - 25 \\
&\{i, j\} = \{51, 78\}, k_{ij} = 2^7 \cdot 3 \\
&\alpha \quad -54x^4 + 54y - 30 \\
&\beta \quad 72x^7 + 108x^6 - 108x^5y - 216x^5 + 36x^4y^2 + 72x^4y + 126x^4 + 148x^3 \\
&\quad - 54x^2y^2 - 81x^2y + 330x^2 + 108xy^2 - 60xy + 28x - 18y^3 - 7y^2 - 259y - 348 \\
&\gamma \quad -36x^7 - 18x^4y^2 + 18x^4y + 207x^4 - 74x^3 - 344x + 36y^3 - 91y^2 - 16y - 33 \\
&\{i, j\} = \{51, 89\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \cdot 13^2 \\
&\alpha \quad 1770x^4 + 57230y - 52498 \\
&\beta \quad -9440x^7 - 15340x^6 - 15340x^5y + 46020x^5 + 1770x^4y^2 - 18880x^4y \\
&\quad + 16225x^4 + 12145106x^3 - 130390x^2y^2 + 2696005x^2y - 30466189x^2 \\
&\quad + 260780xy^2 + 23864256xy - 24076790x - 130390y^3 + 5309422y^2 - 66033314y + 60851916 \\
&\gamma \quad 9440x^7 + 23600x^6 + 5900x^5y - 5900x^5 - 1770x^4y^2 + 17700x^4y - 76405x^4 \\
&\quad - 12145106x^3 + 112100x^2y^2 - 2888050x^2y + 6352610x^2 - 17111160xy + 72864062x \\
&\quad + 36580y^3 - 2657077y^2 + 32985676y - 30459305 \\
&\{i, j\} = \{51, 96\}, k_{ij} = 2^4 \cdot 3^2 \\
&\alpha \quad 1131x^4 - 1170y + 1218 \\
&\beta \quad 169x^7 - 195x^6 + 39x^5y + 351x^5 - 377x^4y^2 - 8983x^4y + 6357x^4 - 4484x^3 \\
&\quad + 390x^2y^2 + 5343x^2y + 489x^2 - 780xy^2 + 4005xy - 5068x - 4966y^3 - 3370y^2 \\
&\quad + 12732y - 5534 \\
&\gamma \quad -169x^7 - 7098x^6 - 546x^5y - 6552x^5 + 377x^4y^2 - 5642x^4y + 390x^4 \\
&\quad + 4484x^3 - 3822x^2y^2 - 2730x^2y + 10101x^2 - 3822xy^2 - 3969xy + 739x - 3068y^3 \\
&\quad - 1661y^2 + 5805y - 3304 \\
&\{i, j\} = \{51, 104\}, k_{ij} = 2 \cdot 7 \cdot 13^2 \\
&\alpha \quad -13983x^4 - 713370y - 1413940 \\
&\beta \quad 21093x^7 + 141015x^6 + 28203x^5y - 197421x^5 - 13983x^4y^2 + 159975x^4y \\
&\quad - 932595x^4 - 37390808x^3 + 733278x^2y^2 - 22082949x^2y - 52552423x^2 \\
&\quad - 1466556xy^2 - 103490135xy - 93827346x + 9954y^3 - 47012740y^2 - 78530414y - 31158618 \\
&\gamma \quad -21093x^7 - 78210x^6 - 7110x^5y + 56880x^5 + 13983x^4y^2 - 193866x^4y \\
&\quad + 1567992x^4 + 37390808x^3 - 547470x^2y^2 + 24721470x^2y + 94372275x^2 \\
&\quad + 547470xy^2 + 85543755xy - 7612507x + 351708y^3 + 23866609y^2 - 7751087y - 14871156 \\
&\{i, j\} = \{52, 53\}, k_{ij} = 2^4 \cdot 3 \cdot 13 \\
&\alpha \quad -25086x^4 - 192326y + 2804726 \\
&\beta \quad 301032x^7 - 752580x^6 - 150516x^5y + 1204128x^5 + 50172x^4y^2 - 944906x^4 \\
&\quad - 5276084x^3 - 576978x^2y^2 + 2479333x^2y - 30112x^2 + 1153956xy^2 - 2320624xy \\
&\quad + 2743500x - 192326y^3 - 1014533y^2 + 221691y - 2804102 \\
&\gamma \quad -150516x^7 - 25086x^4y^2 + 526806x^4y - 355385x^4 + 2638042x^3 + 1296404x \\
&\quad + 384652y^3 - 1597867y^2 + 985168y - 29989 \\
&\{i, j\} = \{52, 58\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \cdot 7 \\
&\alpha \quad -22x^4 + 154y + 582 \\
&\beta \quad 88x^7 + 44x^6 - 44x^5y - 176x^5 + 44x^4y^2 + 176x^4y - 110x^4 + 4844x^3 \\
&\quad - 154x^2y^2 - 1177x^2y - 8204x^2 + 308xy^2 - 68xy - 14940x - 462y^3 - 955y^2 \\
&\quad - 7915y + 2778 \\
&\gamma \quad -44x^7 - 22x^4y^2 - 66x^4y + 209x^4 - 2422x^3 + 13252x + 308y^3 + 835y^2 + 8716y
\end{aligned}$$

$$\begin{aligned}
& +4401 \\
\{i, j\} &= \{52, 85\}, k_{ij} = 2^3 \cdot 3^2 \\
\alpha & -1562x^4 - 43594y - 22370 \\
\beta & 7668x^7 + 12212x^6 + 12212x^5y - 24424x^5 - 1562x^4y^2 + 8378x^4y + 12993x^4 \\
& -6077674x^3 + 79378x^2y^2 - 1444069x^2y + 12643806x^2 - 158756xy^2 \\
& -12335840xy - 6129380x + 79378y^3 - 2899006y^2 + 23839487y + 12654991 \\
\gamma & -7668x^7 - 18176x^6 - 4544x^5y + 1562x^4y^2 - 8662x^4y + 41819x^4 \\
& +6077674x^3 - 70432x^2y^2 + 1610848x^2y - 407488x^2 + 9146304xy - 19158232x \\
& -17892y^3 + 1429945y^2 - 11163416y - 12632549 \\
\{i, j\} &= \{52, 91\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \\
\alpha & 783x^4 + 16530y + 6492 \\
\beta & -1102x^7 - 7105x^6 - 1421x^5y + 14210x^5 + 783x^4y^2 - 8700x^4y + 9831x^4 \\
& +1723715x^3 - 42630x^2y^2 + 1315846x^2y - 3687936x^2 + 85260xy^2 + 4828754xy \\
& -3681504x - 13050y^3 + 2568062y^2 - 9575868y - 5430131 \\
\gamma & 1102x^7 + 3509x^6 + 319x^5y - 3509x^5 - 783x^4y^2 + 11310x^4y - 31146x^4 \\
& -1723715x^3 + 28710x^2y^2 - 1365958x^2y + 1945801x^2 - 28710xy^2 - 4370773xy \\
& +9111575x - 1740y^3 - 1279882y^2 + 4135579y + 5423699 \\
\{i, j\} &= \{53, 58\}, k_{ij} = 2^2 \cdot 11 \\
\alpha & -8y + 44 \\
\beta & -32x^7 - 64x^6 - 64x^5 - 32x^4 + 114x^3 - 16x^2y^2 + 56x^2y + 96x^2 - 16xy^2 \\
& +56xy + 30x - 8y^2 + 26y - 7 \\
\gamma & 32x^7 + 32x^4y - 32x^4 - 114x^3 + 66x + 8y^3 - 36y^2 - 18y - 29 \\
\{i, j\} &= \{53, 75\}, k_{ij} = 3^2 \cdot 7^2 \\
\alpha & 1024y - 3968 \\
\beta & 4096x^7 + 8192x^6 + 10240x^5 + 512x^4y + 5120x^4 - 5568x^3 + 2048x^2y^2 \\
& -5888x^2y - 1792x^2 + 2048xy^2 - 5888xy - 12x + 1024y^2 - 3384y + 143 \\
\gamma & -4096x^7 - 4608x^4y + 5120x^4 + 5568x^3 - 4564x - 1024y^3 + 3968y^2 - 1608y + 4409 \\
\{i, j\} &= \{54, 80\}, k_{ij} = 2^2 \cdot 3^4 \cdot 7 \\
\alpha & 2y + 27 \\
\beta & -32x^7 + 16x^6 - 40x^5 + 24x^4y - 64x^4 - 882x^3 + 4x^2y^2 + 44x^2y + 276x^2 \\
& +46xy^2 - 528xy - 726x - 6y^3 + 30y^2 - 105y - 1203 \\
\gamma & 32x^7 - 32x^4y + 52x^4 + 882x^3 - 44xy^2 + 550xy - 459x + 4y^3 - 71y^2 + 275y + 519 \\
\{i, j\} &= \{55, 81\}, k_{ij} = 2 \cdot 3^2 \\
\alpha & 6y + 3 \\
\beta & 32x^7 + 48x^6 - 24x^5 - 24x^4y - 32x^4 + 18x^3 + 12x^2y^2 + 12x^2y + 30x^2 \\
& +6xy^2 + 6xy - 4x - 2y^3 - 2y^2 - 12y - 1 \\
\gamma & -32x^7 - 4x^4 - 18x^3 - 8x - 4y^3 - 7y^2 - 6y - 8 \\
\{i, j\} &= \{55, 104\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \\
\alpha & 24x^4 - 18y - 102 \\
\beta & 16x^7 - 48x^6 + 24x^5y + 8x^4y^2 + 32x^4y + 24x^4 - 54x^3 - 12x^2y^2 \\
& +6x^2y + 201x^2 - 6xy^2 - 78xy + 67x + 2y^3 - 28y^2 - 168y - 88 \\
\gamma & -32x^7 - 16x^4y^2 + 8x^4y + 108x^3 - 95x + 8y^3 + 56y^2 + 51y + 137 \\
\{i, j\} &= \{56, 81\}, k_{ij} = 2 \cdot 7 \\
\alpha & 2y - 3 \\
\beta & 32x^7 + 16x^6 - 24x^5 - 40x^4y + 64x^4 + 18x^3 + 4x^2y^2 - 4x^2y + 6x^2 \\
& -2xy^2 + 2xy - 4x - 6y^3 + 14y^2 - 24y + 43 \\
\gamma & -32x^7 + 32x^4y - 44x^4 - 18x^3 - 8x + 4y^3 - 9y^2 + 18y - 30 \\
\{i, j\} &= \{56, 108\}, k_{ij} = 3^2 \\
\alpha & -6x^4 - 3y + 6 \\
\beta & -4x^7 + 12x^6 - 6x^5y - 2x^4y^2 - 2x^4y + 6x^4 + 2x^3 + 6x^2y^2 + 6x^2y \\
& -9x^2 - 3xy^2 + x - 5y^3 + 3y^2 + 10y - 7 \\
\gamma & 8x^7 + 4x^4y^2 - 2x^4y - 4x^3 + x + 4y^3 - 3y^2 - 5y + 2 \\
\{i, j\} &= \{58, 62\}, k_{ij} = 3 \cdot 5 \\
\alpha & -32x^4 - 12y + 32 \\
\beta & 64x^7 - 32x^6 - 64x^5y + 64x^5 + 16x^4y^2 + 24x^4y - 24x^4 - 56x^3 \\
& +24x^2y^2 + 16x^2y + 52x^2 - 24xy^2 + 40xy - 28x - 2y^2 - 36y - 7 \\
\gamma & -64x^7 - 16x^4y^2 + 56x^4y - 40x^4 + 56x^3 - 52x - 12y^3 + 18y^2 - 30y - 17 \\
\{i, j\} &= \{58, 63\}, k_{ij} = 2^2 \cdot 3 \\
\alpha & -48x^4 + 6y + 8 \\
\beta & 96x^7 - 48x^6 - 96x^5y + 24x^4y^2 + 60x^4y + 12x^4 - 28x^3 - 12x^2y^2
\end{aligned}$$

$$\begin{aligned}
& +24x^2y + 50x^2 + 12xy^2 + 4xy - 4x - 13y^2 - 23y - 24 \\
\gamma & -96x^7 - 24x^4y^2 + 60x^4y - 12x^4 + 28x^3 - 32x + 6y^3 - 11y^2 - 6y - 4 \\
\{i, j\} & = \{58, 73\}, k_{ij} = 3^6 \\
\alpha & -2y + 19 \\
\beta & 32x^7 - 16x^6 - 16x^5 + 8x^4y - 16x^4 - 66x^3 - 4x^2y^2 - 2x^2y + 8x^2 \\
& + 4xy^2 + 2xy - 200x - 6y^3 + y^2 - 50y + 426 \\
\gamma & -32x^7 + 32x^4 + 66x^3 + 258x + 8y^3 - 4y^2 + 44y - 355 \\
\{i, j\} & = \{58, 80\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
\alpha & 4y - 8 \\
\beta & -16x^7 + 32x^6 - 48x^5 + 4x^4y + 24x^4 - 12x^3 + 8x^2y^2 - 16x^2y + 24x^2 \\
& - 8xy^2 + 16xy - 42x + 4y^2 - 6y + 23 \\
\gamma & 16x^7 - 20x^4y + 24x^4 + 12x^3 + 6x - 4y^3 + 12y^2 - 22y + 25 \\
\{i, j\} & = \{58, 86\}, k_{ij} = 2^2 \cdot 3^2 \\
\alpha & 30x^3 + 36 \\
\beta & -90x^5 + 150x^4 + 10x^3y^2 + 60x^3y - 110x^3 - 33x^2 + 35x + 12y^2 + 22y - 12 \\
\gamma & -120x^4 - 20x^3y^2 - 30x^3y + 40x^3 - 34x - 24y^2 - 11y - 12 \\
\{i, j\} & = \{58, 100\}, k_{ij} = 2^4 \cdot 7 \\
\alpha & -1458x^4 - 5103y - 1674 \\
\beta & -5346x^7 + 5832x^6 - 11664x^5y - 13608x^5 - 486x^4y^2 - 28755x^4y + 31185x^4 \\
& + 48078x^3 - 3402x^2y^2 - 3321x^2y - 11286x^2 + 3402xy^2 + 11718xy - 154131x \\
& + 37827y^3 + 132876y^2 + 267249y + 105568 \\
\gamma & 10692x^7 + 29160x^6 + 7290x^5y + 63180x^5 + 972x^4y^2 + 31266x^4y + 62046x^4 \\
& - 96156x^3 - 27945x^2y^2 - 147825x^2y - 316440x^2 - 55890xy^2 - 168210xy \\
& - 205890x - 16362y^3 - 57420y^2 - 102363y + 1786 \\
\{i, j\} & = \{59, 63\}, k_{ij} = 2 \cdot 3^2 \\
\alpha & 8y + 8 \\
\beta & -32x^7 + 64x^6 - 32x^5 + 8x^4y + 8x^4 - 48x^3 + 16x^2y^2 - 16x^2y + 64x^2 \\
& - 16xy^2 + 16xy - 22x + 8y^2 + 2y + 3 \\
\gamma & 32x^7 + 24x^4y - 8x^4 + 48x^3 + 6x + 8y^3 - 8y^2 + 22y - 7 \\
\{i, j\} & = \{59, 71\}, k_{ij} = 2 \cdot 3^3 \cdot 5 \\
\alpha & -32x^4 + 20y + 108 \\
\beta & -64x^7 + 224x^6 - 64x^5y - 224x^5 - 16x^4y^2 + 48x^4y + 120x^4 + 148x^3 \\
& - 40x^2y^2 - 68x^2y - 660x^2 + 40xy^2 + 216xy + 414x + 17y^2 - 273y - 111 \\
\gamma & 64x^7 + 16x^4y^2 + 128x^4y - 8x^4 - 148x^3 + 246x - 20y^3 - 71y^2 - 239y + 273 \\
\{i, j\} & = \{59, 75\}, k_{ij} = 2 \cdot 5 \\
\alpha & -16x^3 - 4 \\
\beta & 48x^5 - 8x^3y^2 + 12x^3y + 16x^3 + 20x^2 - 2x - 2y^2 + 5y + 10 \\
\gamma & -48x^4 + 8x^3y^2 + 12x^3y + 8x^3 - 22x + 2y^2 + 5y + 4 \\
\{i, j\} & = \{59, 103\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \\
\alpha & 50x^3 + 80 \\
\beta & -150x^5 + 90x^4 + 10x^3y^2 - 50x^3y - 40x^3 - 115x^2 + 9x + 16y^2 - 30y + 46 \\
\gamma & 120x^4 - 20x^3y^2 - 50x^3y - 70x^3 + 62x - 32y^2 - 55y - 57 \\
\{i, j\} & = \{60, 70\}, k_{ij} = 2^3 \cdot 3^3 \cdot 7 \\
\alpha & 312x^4 + 234y + 300 \\
\beta & -208x^7 - 1248x^6 - 312x^5y + 104x^4y^2 - 416x^4y - 1560x^4 - 1786x^3 \\
& + 156x^2y^2 - 2262x^2y - 5919x^2 - 78xy^2 - 1548xy - 3644x + 130y^3 - 680y^2 \\
& - 2775y - 4594 \\
\gamma & 416x^7 - 208x^4y^2 - 104x^4y + 624x^4 + 3572x^3 + 2491x - 104y^3 - 512y^2 \\
& - 3345y - 3988 \\
\{i, j\} & = \{60, 78\}, k_{ij} = 3^3 \cdot 11 \\
\alpha & -6y - 45 \\
\beta & 32x^7 - 48x^6 + 24x^5 - 8x^4y - 24x^4 - 90x^3 - 12x^2y^2 - 36x^2y + 210x^2 \\
& + 6xy^2 + 18xy + 184x - 2y^3 - 12y^2 + 19y - 169 \\
\gamma & -32x^7 - 16x^4y - 36x^4 + 90x^3 - 289x - 4y^3 - 21y^2 + 41y - 2 \\
\{i, j\} & = \{61, 63\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \\
\alpha & 2y + 17 \\
\beta & -32x^7 + 16x^6 - 8x^5 + 8x^4y + 8x^4 + 90x^3 + 4x^2y^2 + 8x^2y - 86x^2 \\
& + 2xy^2 + 4xy - 388x + 6y^3 + 26y^2 - 58y - 183 \\
\gamma & 32x^7 + 4x^4 - 90x^3 + 390x - 4y^3 - 17y^2 + 25y + 293
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{61, 78\}, k_{ij} = 3^2 \cdot 11 \\
&\alpha \quad -2y - 15 \\
&\beta \quad 32x^7 - 16x^6 - 8x^5 - 24x^4y - 56x^4 - 90x^3 - 4x^2y^2 - 12x^2y + 70x^2 \\
&\quad - 2xy^2 - 6xy + 324x - 6y^3 - 32y^2 + 65y - 29 \\
&\gamma \quad -32x^7 + 16x^4y + 36x^4 + 90x^3 - 289x + 4y^3 + 21y^2 - 45y - 28 \\
&\{i, j\} = \{62, 71\}, k_{ij} = 2 \\
&\alpha \quad -16x^3 - 4 \\
&\beta \quad 48x^5 - 8x^3y^2 + 12x^3y + 16x^3 + 20x^2 - 2x - 2y^2 + 5y + 6 \\
&\gamma \quad 48x^4 + 8x^3y^2 + 12x^3y + 8x^3 + 18x + 2y^2 + 5y \\
&\{i, j\} = \{63, 71\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad 56x^3 - 48 \\
&\beta \quad -168x^5 + 28x^3y^2 - 70x^3y - 70x^3 - 52x^2 + 14x - 24y^2 + 11y - 73 \\
&\gamma \quad -168x^4 - 28x^3y^2 - 14x^3y - 182x^3 - 38x + 24y^2 - 37y + 23 \\
&\{i, j\} = \{64, 70\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \cdot 17 \\
&\alpha \quad -94x^3 - 680 \\
&\beta \quad 564x^5 - 47x^3y^2 + 376x^3y + 1457x^3 - 338x^2 - 423x - 340y^2 + 511y - 1868 \\
&\gamma \quad -1128x^4 + 47x^3y^2 + 188x^3y + 1927x^3 + 253x + 340y^2 - 849y + 1532 \\
&\{i, j\} = \{64, 93\}, k_{ij} = 2^2 \cdot 3^5 \cdot 5 \\
&\alpha \quad 17x^3 + 469 \\
&\beta \quad -102x^5 + 612x^4 - 17x^3y^2 - 221x^3y - 1564x^3 - 791x^2 + 360x - 469y^2 - 2051y \\
&\quad - 4031 \\
&\gamma \quad -408x^4 + 17x^3y^2 + 119x^3y + 952x^3 - 971x + 469y^2 + 1260y + 3671 \\
&\{i, j\} = \{64, 101\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \cdot 17 \\
&\alpha \quad -3x^4 + 225y + 594 \\
&\beta \quad -x^7 + 21x^6 - 3x^5y - 42x^5 - x^4y^2 - 7x^4y + 15x^4 - 1502x^3 - 75x^2y^2 \\
&\quad - 1500x^2y + 1197x^2 + 150xy^2 - 1506xy + 1261x + 25y^3 - 1052y^2 + 2181y - 3055 \\
&\gamma \quad x^7 + x^4y^2 + 34x^4y + 111x^4 + 1502x^3 - 3655x - 100y^3 - 898y^2 - 972y - 728 \\
&\{i, j\} = \{65, 72\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \cdot 11 \cdot 13 \\
&\alpha \quad 3x^4 - 153y + 1866 \\
&\beta \quad -2x^7 - 21x^6 - 3x^5y - 36x^5 + 2x^4y^2 + 29x^4y - 36x^4 - 2496x^3 \\
&\quad + 51x^2y^2 + 762x^2y - 14865x^2 + 102xy^2 - 2220xy + 1432x - 85y^3 + 1328y^2 \\
&\quad - 4965y + 12017 \\
&\gamma \quad x^7 - x^4y^2 - 28x^4y + 3x^4 + 1248x^3 - 11837x + 68y^3 - 232y^2 - 7932y + 4370 \\
&\{i, j\} = \{65, 81\}, k_{ij} = 2^4 \cdot 3 \cdot 11^2 \\
&\alpha \quad 150x^4 - 600y + 1728 \\
&\beta \quad -200x^7 - 1500x^6 - 300x^5y - 2400x^5 + 100x^4y^2 - 100x^4y - 2150x^4 \\
&\quad + 1796x^3 + 600x^2y^2 - 675x^2y - 480x^2 + 1200xy^2 + 1344xy - 1332x - 200y^3 \\
&\quad + 527y^2 + 1490y + 2464 \\
&\gamma \quad 100x^7 - 50x^4y^2 - 1000x^4y + 175x^4 - 898x^3 - 2508x + 400y^3 - y^2 + 1034y \\
&\quad + 424 \\
&\{i, j\} = \{66, 82\}, k_{ij} = 2^6 \cdot 3^5 \\
&\alpha \quad 150x^4 - 2400y - 6528 \\
&\beta \quad -200x^7 - 1500x^6 - 300x^5y - 3600x^5 + 100x^4y^2 + 700x^4y - 950x^4 \\
&\quad - 33596x^3 + 2400x^2y^2 + 22125x^2y - 54420x^2 + 4800xy^2 - 6144xy - 64692x \\
&\quad - 800y^3 + 11023y^2 - 6782y - 38456 \\
&\gamma \quad 100x^7 - 50x^4y^2 - 1400x^4y - 4625x^4 + 16798x^3 - 72468x + 1600y^3 \\
&\quad + 12751y^2 - 7838y - 27416 \\
&\{i, j\} = \{67, 69\}, k_{ij} = 2^4 \cdot 5 \cdot 11 \\
&\alpha \quad -4y + 88 \\
&\beta \quad -x^7 - 4x^6 - 8x^5 - 8x^4 + 267x^3 - 4x^2y^2 + 56x^2y + 492x^2 - 8xy^2 + 112xy \\
&\quad + 147x - 8y^2 + 111y - 668 \\
&\gamma \quad x^7 + 4x^4y - 16x^4 - 267x^3 + 837x + 4y^3 - 72y^2 - 267y + 272 \\
&\{i, j\} = \{67, 70\}, k_{ij} = 2^8 \\
&\alpha \quad -4y + 60 \\
&\beta \quad -x^7 - 4x^6 - 12x^5 - x^4y - 10x^4 + 123x^3 - 4x^2y^2 + 44x^2y + 156x^2 \\
&\quad - 8xy^2 + 88xy + 466x - 8y^2 + 210y + 235 \\
&\gamma \quad x^7 + 5x^4y - 22x^4 - 123x^3 + 338x + 4y^3 - 60y^2 - 102y + 333 \\
&\{i, j\} = \{67, 72\}, k_{ij} = 2^4 \cdot 3 \cdot 11 \\
&\alpha \quad -21x^4 - 35y + 118
\end{aligned}$$

β	$-126x^7 - 63x^6 + 63x^5y - 252x^5 + 42x^4y^2 - 315x^4y + 532x^4 + 1016x^3 + 105x^2y^2 + 154x^2y + 739x^2 + 210xy^2 - 200xy - 1928x + 175y^3 - 572y^2 + 709y - 243$
γ	$63x^7 - 21x^4y^2 + 252x^4y - 581x^4 - 508x^3 + 2211x - 140y^3 + 524y^2 - 770y + 38$
$\{i, j\}$	$\{68, 69\}, k_{ij} = 2^5 \cdot 3^2$
α	$-4y + 44$
β	$-x^7 - 4x^6 - 4x^5 + x^4y - 2x^4 + 87x^3 - 4x^2y^2 + 28x^2y + 92x^2 - 8xy^2 + 56xy - 32x - 8y^2 - 32y + 69$
γ	$x^7 + 3x^4y + 2x^4 - 87x^3 - 132x + 4y^3 - 28y^2 - 4y - 53$
$\{i, j\}$	$\{68, 70\}, k_{ij} = 2^5 \cdot 3^2$
α	$4y - 24$
β	$x^7 + 4x^6 + 8x^5 + 8x^4 - 27x^3 + 4x^2y^2 - 24x^2y + 20x^2 + 8xy^2 - 48xy + 69x + 8y^2 - 47y + 92$
γ	$-x^7 - 4x^4y + 27x^3 - 29x - 4y^3 + 24y^2 - 21y + 64$
$\{i, j\}$	$\{69, 77\}, k_{ij} = 2^4 \cdot 3 \cdot 11^2$
α	$-150x^4 - 600y + 1728$
β	$200x^7 + 300x^6 - 300x^5y + 100x^4y^2 - 100x^4y + 250x^4 - 3204x^3 + 600x^2y^2 + 3075x^2y - 456x^2 - 1200xy^2 - 1344xy - 3684x + 200y^3 - 177y^2 + 2794y - 880$
γ	$-100x^7 - 50x^4y^2 + 200x^4y + 175x^4 + 1602x^3 - 2508x - 400y^3 + 351y^2 + 3250y + 1160$
$\{i, j\}$	$\{70, 104\}, k_{ij} = 2^7 \cdot 5^2 \cdot 17$
α	$-81x^4 + 11745y + 25426$
β	$81x^7 + 405x^6 + 81x^5y - 810x^5 - 81x^4y^2 + 1215x^4y + 891x^4 - 375346x^3 + 2349x^2y^2 - 69984x^2y + 221251x^2 - 4698xy^2 - 235378xy - 14487x + 7047y^3 + 170092y^2 + 940055y + 845091$
γ	$-81x^7 + 81x^4y^2 - 1782x^4y - 81x^4 + 375346x^3 - 428015x - 9396y^3 - 95410y^2 - 550582y - 830604$
$\{i, j\}$	$\{71, 75\}, k_{ij} = 2^2 \cdot 5$
α	$8y - 12$
β	$32x^7 + 64x^6 + 64x^5 + 32x^4 + 6x^3 + 16x^2y^2 - 24x^2y + 32x^2 + 16xy^2 - 24xy + 40x + 8y^2 - 10y + 21$
γ	$-32x^7 - 32x^4y - 6x^3 - 8x - 8y^3 + 12y^2 - 18y + 11$
$\{i, j\}$	$\{72, 74\}, k_{ij} = 2 \cdot 3^2 \cdot 5^2 \cdot 17$
α	$4y + 14$
β	$32x^7 + 32x^6 + 80x^5 + 32x^4y - 44x^4 + 258x^3 + 8x^2y^2 + 32x^2y + 56x^2 + 4xy^2 + 16xy + 1230x + y^2 + 264y - 1402$
γ	$-32x^7 - 48x^4y + 116x^4 - 258x^3 - 686x - 4y^3 - 7y^2 - 252y + 2416$
$\{i, j\}$	$\{72, 75\}, k_{ij} = 2^2 \cdot 3 \cdot 13$
α	$-6y + 33$
β	$-32x^7 - 48x^6 - 72x^5 - 8x^4y - 8x^4 + 102x^3 - 12x^2y^2 + 36x^2y + 84x^2 - 6xy^2 + 18xy + 160x + 2y^3 - 12y^2 + 53y + 2$
γ	$32x^7 + 32x^4y - 76x^4 - 102x^3 + 35x + 4y^3 - 21y^2 - 50y + 127$
$\{i, j\}$	$\{73, 102\}, k_{ij} = 2 \cdot 3^5 \cdot 13$
α	$-42x^4 + 63y + 1158$
β	$28x^7 + 84x^6 + 42x^5y + 42x^5 - 14x^4y^2 - 28x^4y + 84x^4 - 968x^3 + 42x^2y^2 - 126x^2y - 1875x^2 - 21xy^2 - 1389xy - 1548x - 7y^3 + 526y^2 + 1131y - 1873$
γ	$-56x^7 + 28x^4y^2 - 70x^4y - 42x^4 + 1936x^3 - 837x - 28y^3 - 821y^2 + 750y + 707$
$\{i, j\}$	$\{74, 108\}, k_{ij} = 3^7$
α	$-120x^4 - 30y + 609$
β	$-80x^7 + 240x^6 - 120x^5y + 120x^5 - 40x^4y^2 - 120x^4y - 360x^4 + 476x^3 + 60x^2y^2 + 450x^2y - 573x^2 - 30xy^2 + 489xy - 1122x - 50y^3 - 217y^2 - 628y + 1045$
γ	$160x^7 + 80x^4y^2 + 120x^4y + 120x^4 - 952x^3 - 39x + 40y^3 - 106y^2 + 440y + 256$
$\{i, j\}$	$\{75, 78\}, k_{ij} = 5 \cdot 7 \cdot 13$
α	$-1056x^4 + 924y + 2348$
β	$2112x^7 - 1056x^6 - 2112x^5y - 2112x^5 + 528x^4y^2 + 2376x^4y + 2112x^4 + 4808x^3 - 1848x^2y^2 - 3960x^2y - 16660x^2 + 1848xy^2 - 848xy + 4332x + 278y^2 + 5756y + 5445$
γ	$-2112x^7 - 528x^4y^2 + 264x^4y + 528x^4 - 4808x^3 + 5116x + 924y^3 + 778y^2$

$$\begin{aligned}
& +5402y + 3669 \\
\{i, j\} &= \{75, 85\}, k_{ij} = 2^4 \cdot 5 \\
\alpha & 3844638x^4 + 8556717y + 4712099 \\
\beta & -2066166x^7 - 12565544x^6 - 3277968x^5y + 11472888x^5 + 1281546x^4y^2 \\
& -4395325x^4y - 7718336x^4 + 164735440x^3 - 24448178x^2y^2 + 240725775x^2y \\
& -133344217x^2 + 24448178xy^2 + 365091018xy + 176866565x - 12224089y^3 \\
& +522657668y^2 + 5765739y - 129634704 \\
\gamma & 4132332x^7 + 6195592x^6 + 357438x^5y + 119146x^5 - 2563092x^4y^2 \\
& +15520946x^4y - 1139152x^4 - 329470880x^3 + 23531335x^2y^2 - 537765471x^2y \\
& -86042190x^2 - 717427266xy + 77690744x + 1833686y^3 - 258489682y^2 - 139245221y \\
& +124922685 \\
\{i, j\} &= \{75, 104\}, k_{ij} = 2^3 \cdot 3 \\
\alpha & -30x^3 - 24 \\
\beta & 90x^5 - 150x^4 - 10x^3y^2 - 50x^3y + 40x^3 - 3x^2 - 15x - 8y^2 + 10y + 2 \\
\gamma & 120x^4 + 20x^3y^2 + 10x^3y + 10x^3 + 6x + 16y^2 - 17y + 23 \\
\{i, j\} &= \{76, 77\}, k_{ij} = 2^3 \cdot 3^4 \\
\alpha & 8y + 20 \\
\beta & -32x^7 + 64x^6 - 128x^5 - 16x^4y + 32x^4 - 126x^3 + 16x^2y^2 + 8x^2y + 64x^2 \\
& -16xy^2 - 8xy - 310x + 8y^2 - 61y + 27 \\
\gamma & 32x^7 + 48x^4y + 160x^4 + 126x^3 - 6x + 8y^3 + 28y^2 + 109y + 341 \\
\{i, j\} &= \{76, 94\}, k_{ij} = 2^2 \cdot 3^5 \cdot 5 \cdot 7 \\
\alpha & 174x^3 + 1716 \\
\beta & -522x^5 + 870x^4 - 58x^3y^2 - 464x^3y - 2030x^3 - 2625x^2 - 1077x - 572y^2 \\
& -2894y - 6158 \\
\gamma & -696x^4 + 116x^3y^2 + 406x^3y + 1972x^3 - 774x + 1144y^2 + 3163y + 9994 \\
\{i, j\} &= \{78, 81\}, k_{ij} = 2^8 \cdot 11 \\
\alpha & -288x^4 + 396y - 328 \\
\beta & 576x^7 + 2016x^6 + 576x^5y + 2016x^5 - 144x^4y^2 + 936x^4 + 1196x^3 \\
& -792x^2y^2 - 540x^2y + 7552x^2 - 792xy^2 + 656xy - 786x - 695y^2 + 3998y - 4644 \\
\gamma & -576x^7 + 144x^4y^2 + 1584x^4y + 3096x^4 - 1196x^3 + 8338x - 396y^3 - 1159y^2 \\
& -106y + 3072 \\
\{i, j\} &= \{79, 82\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \\
\alpha & 8y - 44 \\
\beta & 32x^7 + 64x^6 + 128x^5 + 16x^4y + 32x^4 - 18x^3 + 16x^2y^2 - 56x^2y - 32x^2 \\
& +16xy^2 - 56xy + 26x + 8y^2 - 35y - 27 \\
\gamma & -32x^7 - 48x^4y + 160x^4 + 18x^3 - 94x - 8y^3 + 52y^2 - 61y + 115 \\
\{i, j\} &= \{80, 88\}, k_{ij} = 3^2 \cdot 7 \\
\alpha & 15x - 12 \\
\beta & -45x^3 - 39x^2 + 5xy^2 + 30xy + 80x - 4y^2 - 24y + 34 \\
\gamma & 60x^2 - 10xy^2 - 15xy - 88x + 8y^2 + 12y + 7 \\
\{i, j\} &= \{83, 94\}, k_{ij} = 2 \\
\alpha & 0 \\
\beta & -1 \\
\gamma & 1 \\
\{i, j\} &= \{83, 101\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \\
\alpha & -16x^4 - 32y + 96 \\
\beta & 23x^7 - 1078x^6 + 44x^5y + 1056x^5 - 4x^4y^2 - 2204x^4y + 2264x^4 + 50226x^3 \\
& +4136x^2y^2 + 94424x^2y - 298452x^2 - 4136xy^2 + 6160xy + 147529x + 8256y^3 \\
& +180364y^2 - 684458y + 495694 \\
\gamma & -23x^7 + 1128x^6 + 48x^5y + 1104x^5 + 4x^4y^2 - 2204x^4y + 40x^4 - 50226x^3 \\
& -4128x^2y^2 - 94176x^2y + 197520x^2 - 4128xy^2 + 6144xy + 247991x + 8272y^3 \\
& +188588y^2 - 495854y - 192 \\
\{i, j\} &= \{84, 92\}, k_{ij} = 2^5 \cdot 3^2 \cdot 13 \cdot 17 \\
\alpha & -560x^4 + 56000y + 11168 \\
\beta & -35x^7 + 3290x^6 + 140x^5y - 70x^5 + 140x^4y^2 - 6720x^4y + 3045x^4 \\
& +91173x^3 + 14000x^2y^2 - 347900x^2y + 118698x^2 - 364692xy + 122876x - 28000y^3 \\
& +345108y^2 - 311544y - 26361 \\
\gamma & 35x^7 - 140x^4y^2 + 140x^4y + 245x^4 - 91173x^3 + 59470x + 364692y^2 - 273752y \\
& +26119 \\
\{i, j\} &= \{84, 98\}, k_{ij} = 2^7 \cdot 13
\end{aligned}$$

$$\begin{aligned}
&\alpha \quad -16x^4 + 1600y + 224 \\
&\beta \quad x^7 + 94x^6 - 4x^5y + 2x^5 + 4x^4y^2 - 192x^4y + 103x^4 - 2615x^3 \\
&\quad + 400x^2y^2 - 10004x^2y + 4014x^2 + 10460xy - 4644x - 800y^3 + 9948y^2 - 9544y + 701 \\
&\gamma \quad -x^7 - 4x^4y^2 + 4x^4y - 9x^4 + 2615x^3 - 586x + 10460y^2 - 8488y + 2141 \\
&\{i, j\} = \{84, 103\}, k_{ij} = 2^6 \cdot 3 \cdot 17 \\
&\alpha \quad -16y + 408 \\
&\beta \quad -x^7 - x^6 + x^5 + 4x^4y - 2x^4 + 2543x^3 - 4x^2y^2 + 98x^2y + 95x^2 - 329x \\
&\quad + 8y^3 - 200y^2 - 5194y + 7474 \\
&\gamma \quad x^7 - 2x^4y + x^4 - 2543x^3 - 2214x + 5102y - 5165 \\
&\{i, j\} = \{84, 106\}, k_{ij} = 2^5 \cdot 3^2 \cdot 17 \\
&\alpha \quad -32y + 816 \\
&\beta \quad -x^7 - 2x^6 - 2x^5 + 2x^4y - x^4 + 2543x^3 - 8x^2y^2 + 196x^2y + 190x^2 \\
&\quad + 7284x + 16y^3 - 400y^2 + 4886y + 253 \\
&\gamma \quad x^7 + 2x^4y - x^4 - 2543x^3 - 2198x - 5070y + 4333 \\
&\{i, j\} = \{85, 103\}, k_{ij} = 2^6 \cdot 3^4 \cdot 5^2 \\
&\alpha \quad -16y + 280 \\
&\beta \quad -x^7 - x^6 + 3x^5 + 8x^4y + 4x^4 + 1613x^3 - 4x^2y^2 + 74x^2y - 77x^2 + 9453x \\
&\quad + 8y^3 - 144y^2 - 9614y + 52686 \\
&\gamma \quad x^7 - 6x^4y - 3x^4 - 1613x^3 - 14292x + 9694y - 38317 \\
&\{i, j\} = \{85, 106\}, k_{ij} = 2 \cdot 3^3 \cdot 5 \\
&\alpha \quad -32y + 704 \\
&\beta \quad -x^7 - 2x^6 + 4x^4y + 2x^4 + 2018x^3 - 8x^2y^2 + 184x^2y - 190x^2 - 312x \\
&\quad + 16y^3 - 360y^2 + 180y - 82 \\
&\gamma \quad x^7 - 2018x^3 + 312x + 16y - 352 \\
&\{i, j\} = \{86, 104\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{89, 104\}, k_{ij} = 2^4 \cdot 3^3 \cdot 7 \cdot 13^2 \\
&\alpha \quad -16y + 320 \\
&\beta \quad -x^7 - x^6 - 4x^5 - 6x^4y + 6x^4 + 2036x^3 - 4x^2y^2 + 72x^2y + 144x^2 \\
&\quad + 40104x + 8y^3 - 152y^2 + 16128y - 111560 \\
&\gamma \quad x^7 + 8x^4y - 8x^4 - 2036x^3 - 31960x - 16272y + 143808 \\
&\{i, j\} = \{89, 108\}, k_{ij} = 2^5 \cdot 3^4 \\
&\alpha \quad -32y + 896 \\
&\beta \quad -x^7 - 2x^6 + 4x^4y - 4x^4 + 2900x^3 - 8x^2y^2 + 208x^2y + 416x^2 - 840x \\
&\quad + 16y^3 - 432y^2 - 432y - 400 \\
&\gamma \quad x^7 - 2900x^3 + 840x + 16y - 448 \\
&\{i, j\} = \{92, 94\}, k_{ij} = 2^4 \cdot 3^2 \\
&\alpha \quad -48y + 104 \\
&\beta \quad 3x^7 - 23x^6 + 69x^5 + 64x^4y + 126x^4 + 3051x^3 - 92x^2y^2 + 2254x^2y \\
&\quad + 2001x^2 + 184xy^2 - 4508xy + 5403x + 232y^3 - 4712y^2 + 6810y - 2306 \\
&\gamma \quad -3x^7 + 20x^6 - 40x^5 - 70x^4y - 241x^4 - 3051x^3 + 80x^2y^2 - 2120x^2y \\
&\quad - 5320x^2 - 80xy^2 + 2120xy - 1034x - 256y^3 + 4624y^2 - 4566y + 1173 \\
&\{i, j\} = \{92, 98\}, k_{ij} = 2^2 \cdot 13 \\
&\alpha \quad 32 \\
&\beta \quad -192x^2 - 4xy^2 - 98xy + 235x + 8y^2 + 196y - 94 \\
&\gamma \quad 4xy^2 + 98xy + 149x + 8y^2 - 188y + 114 \\
&\{i, j\} = \{93, 99\}, k_{ij} = 2^7 \cdot 5 \\
&\alpha \quad -72x^4 - 48y + 184 \\
&\beta \quad 87x^7 - 2028x^6 + 156x^5y + 4056x^5 - 18x^4y^2 - 4101x^4y + 2298x^4 \\
&\quad + 216435x^3 + 8268x^2y^2 + 197496x^2y - 905008x^2 - 16536xy^2 + 38428xy + 727190x \\
&\quad + 16512y^3 + 386302y^2 - 1140775y + 471558 \\
&\gamma \quad -87x^7 + 2112x^6 + 192x^5y + 4224x^5 + 18x^4y^2 - 4107x^4y - 1878x^4 \\
&\quad - 216435x^3 - 8256x^2y^2 - 196992x^2y + 38656x^2 - 16512xy^2 + 38336xy + 1160138x \\
&\quad + 16536y^3 + 402734y^2 - 746233y - 472014 \\
&\{i, j\} = \{94, 97\}, k_{ij} = 2^8 \cdot 3^2 \cdot 5 \cdot 7 \\
&\alpha \quad -32y + 832 \\
&\beta \quad x^7 - 2x^6 - 2x^5 - 4x^4y + 5x^4 - 2439x^3 - 8x^2y^2 + 212x^2y - 122x^2 \\
&\quad + 8xy^2 - 212xy - 5245x - 4y^2 + 9878y + 30258
\end{aligned}$$

γ	$-x^7 + x^4 + 2439x^3 + 15123x - 16y^3 + 436y^2 - 10334y - 9646$
$\{i, j\} = \{94, 101\}, k_{ij} = 2^4 \cdot 3$	
α	$16x^4 - 32y + 48$
β	$-41x^7 + 1960x^6 - 80x^5y - 1880x^5 + 4x^4y^2 + 3938x^4y - 3957x^4$ $-73019x^3 - 6880x^2y^2 - 149440x^2y + 312840x^2 + 6880xy^2 + 3440xy - 93911x$ $-13776y^3 - 285184y^2 + 778832y - 479862$
γ	$41x^7 - 1974x^6 - 84x^5y - 1890x^5 - 4x^4y^2 + 3926x^4y - 3921x^4 + 73019x^3$ $+6888x^2y^2 + 149520x^2y - 167034x^2 + 6888xy^2 + 3444xy - 93887x - 13760y^3$ $-285192y^2 + 778840y - 479862$
$\{i, j\} = \{97, 98\}, k_{ij} = 2^6 \cdot 5^3 \cdot 17$	
α	$-16y + 304$
β	$x^7 - x^6 + 2x^5 + 4x^4 - 1290x^3 - 4x^2y^2 + 98x^2y - 522x^2 - 4xy^2 + 98xy$ $-24705x - 24y^3 + 556y^2 - 10072y - 40273$
γ	$-x^7 - 2x^4y - 4x^4 + 1290x^3 + 20313x + 16y^3 - 372y^2 + 9322y + 55150$
$\{i, j\} = \{100, 102\}, k_{ij} = 2^5 \cdot 7^2$	
α	$16y - 360$
β	$x^7 + x^6 + 3x^5 + 8x^4y - 2x^4 - 2043x^3 + 4x^2y^2 - 94x^2y + 105x^2$ $+8xy^2 - 188xy + 929x + 24y^3 - 568y^2 - 3346y - 3946$
γ	$-x^7 - 10x^4y + 7x^4 + 2043x^3 - 2762x - 32y^3 + 768y^2 + 2854y + 2937$
$\{i, j\} = \{102, 105\}, k_{ij} = 2^4 \cdot 3^2$	
α	$-48y - 24$
β	$-x^7 - 18x^6 - 26x^4y + 109x^4 - 741x^3 - 72x^2y^2 + 1620x^2y - 900x^2$ $-72xy^2 + 1620xy + 1986x - 112y^3 + 3240y^2 - 414y - 515$
γ	$x^7 + 15x^6 - 45x^5 + 32x^4y + 32x^4 + 741x^3 + 60x^2y^2 - 1530x^2y$ $-1365x^2 - 120xy^2 + 3060xy + 2067x + 136y^3 - 3048y^2 - 4326y - 1198$
$\{i, j\} = \{103, 105\}, k_{ij} = 2^6$	
α	$-32y + 40$
β	$-2x^7 - 25x^6 - 38x^4y + 61x^4 - 3339x^3 - 100x^2y^2 + 2650x^2y + 9450x^2$ $+100xy^2 - 2650xy + 36923x - 136y^3 + 4644y^2 + 31022y - 16680$
γ	$2x^7 + 23x^6 - 23x^5 + 42x^4y - 42x^4 + 3339x^3 + 92x^2y^2 - 2530x^2y$ $-12949x^2 - 184xy^2 + 5060xy - 23814x + 152y^3 - 4724y^2 - 28724y + 33336$
$\{i, j\} = \{103, 108\}, k_{ij} = 2^6 \cdot 3 \cdot 7$	
α	$-16y + 464$
β	$-x^7 - x^6 + x^5 + 6x^4y - 8x^4 + 3057x^3 - 4x^2y^2 + 106x^2y + 267x^2$ $+4xy^2 - 106xy - 1508x + 24y^3 - 668y^2 - 770y + 118$
γ	$x^7 - 4x^4y + 5x^4 - 3057x^3 + 1241x - 16y^3 + 444y^2 + 554y - 558$
$\{i, j\} = \{104, 106\}, k_{ij} = 2^5 \cdot 3 \cdot 7$	
α	$16y - 304$
β	$x^7 + x^6 - 2x^5 - 8x^4y - 4x^4 - 1614x^3 + 4x^2y^2 - 86x^2y + 214x^2$ $-4xy^2 + 86xy + 2391x - 24y^3 + 500y^2 + 2304y + 1813$
γ	$-x^7 + 6x^4y + 4x^4 + 1614x^3 - 991x + 16y^3 - 332y^2 - 2646y - 2650$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 10$ and for $x \leq -11$.

i	k_i	$\varphi_i(x)$	$x \geq 10$	$x \leq -11$
0	-1	$-6x^4 - x - 1$	+	+
1	-1	$-6x^4 - x + 1$	+	+
2	-1	$-6x^4 - 8x^3 - 3x - 1$	+	+
3	-1	$-6x^4 + x^3 + 3x^2 + 3x + 1$	+	+
4	-1	$-6x^4 - 8x^3 + 12x^2 - 9x + 1$	+	+
5	-1	$-6x^4 - x^3 + 6x^2 - 14x + 9$	+	+
6	-1	$-6x^4 + x^3 - 6x^2 + 12x - 11$	+	+
7	-1	$-6x^4 + x^3 + 6x^2 + 12x + 9$	+	+
8	-1	$-6x^4 + 8x^3 - 12x^2 + 7x - 3$	+	+
9	-1	$-6x^4 + 8x^3 + 12x^2 + 7x + 1$	+	+
10	-1	$-6x^4 - 8x^3 + 24x^2 - 27x + 9$	+	+
11	-1	$-6x^4 + 8x^3 + 24x^2 + 25x + 9$	+	+
12	-2	$-48x^4 - 8x - 3$	+	+
13	-2	$-48x^4 - x^3 - 3x^2 - 15x - 13$	+	+

i	k_i	$\varphi_i(x)$	$x \geq 10$	$x \leq -11$
14	-2	$-48x^4 - x^3 + 3x^2 - 15x - 3$	+	+
15	-2	$-48x^4 + x^3 + 3x^2 - x - 3$	+	+
16	-2	$-48x^4 - 8x^3 + 12x^2 - 22x - 3$	+	+
17	-2	$-48x^4 - x^3 + 6x^2 - 24x + 8$	+	+
18	-2	$-48x^4 + 8x^3 + 12x^2 + 6x - 3$	+	+
19	-1	$-8x^6 - 6x^4 - 2x^2 - x - 1$	+	+
20	-1	$-x^6 - 9x^4 - 4x^2 - x - 3$	+	+
21	-1	$-x^6 - 3x^4 - 4x^2 - x + 1$	+	+
22	-1	$-x^6 + 3x^5 - 9x^4 + x^3 - x^2 - 1$	+	+
23	1	$-x^6 + 3x^5 + 3x^4 + x^3 - x^2 + 2x + 1$	+	+
24	1	$-x^6 + 9x^4 - 4x^2 + x + 3$	+	+
25	1	$-x^6 - 3x^5 + 3x^4 - x^3 - x^2 + 1$	+	+
26	1	$-8x^6 + 6x^4 - 2x^2 + x + 1$	+	+
27	-1	$-8x^6 - 18x^4 - 8x^2 - x - 3$	+	+
28	-1	$-8x^6 + 6x^4 - 8x^2 - x + 1$	+	+
29	-1	$-8x^6 + 12x^5 - 12x^4 + x^3 - 2x^2 - 1$	+	+
30	-1	$-x^6 + 3x^5 - 12x^4 + 7x^3 - 7x^2 + 3x - 3$	+	+
31	-1	$-x^6 + 3x^5 - 6x^4 - 5x^3 - x^2 + 3x + 1$	+	+
32	-1	$-x^6 + 6x^5 - 18x^4 + 8x^3 - x^2 + x - 1$	+	+
33	1	$-x^6 + 6x^5 - 6x^4 + 8x^3 - x^2 + 3x + 1$	+	+
34	1	$-x^6 + 3x^5 + 6x^4 - 5x^3 - x^2 + 5x + 3$	+	+
35	1	$-x^6 + 3x^5 + 7x^3 - 7x^2 + 5x - 1$	+	+
36	1	$-8x^6 + 12x^5 + x^3 - 2x^2 + 2x + 1$	+	+
37	1	$-8x^6 + 18x^4 - 8x^2 + x + 3$	+	+
38	1	$-8x^6 - 12x^5 - x^3 - 2x^2 + 1$	+	+
39	-1	$-8x^6 - 12x^5 + 11x^3 - 5x^2 - 5x + 1$	+	+
40	-1	$-8x^6 - 30x^4 - 26x^2 - x - 11$	+	+
41	-1	$-8x^6 + 18x^4 - 26x^2 - x + 9$	+	+
42	-1	$-8x^6 + 12x^5 - 11x^3 - 5x^2 + 3x + 1$	+	+
43	-1	$-8x^6 + 24x^5 - 30x^4 + 8x^3 - 2x^2 + x - 1$	+	+
44	-1	$-x^6 - 6x^5 - 21x^4 - 20x^3 - 16x^2 - 9x - 3$	+	+
45	-1	$-x^6 - 3x^5 - 3x^4 + 11x^3 - 7x^2 - 14x + 9$	+	+
46	-1	$-x^6 + 3x^5 - 3x^4 - 11x^3 - 7x^2 + 12x + 9$	+	+
47	-1	$-x^6 + 6x^5 - 21x^4 + 20x^3 - 16x^2 + 7x - 3$	+	+
48	-1	$-x^6 + 6x^5 - 15x^4 - 4x^3 + 8x^2 + 7x + 1$	+	+
49	1	$-x^6 + 6x^5 - 3x^4 - 4x^3 + 8x^2 + 9x + 3$	+	+
50	1	$-x^6 - 3x^5 - 3x^4 - 13x^3 - 19x^2 - 12x - 9$	+	+
51	1	$-x^6 - 6x^5 - 3x^4 + 4x^3 + 8x^2 - 7x + 3$	+	+
52	1	$-x^6 - 6x^5 - 9x^4 - 20x^3 - 16x^2 - 7x - 1$	+	+
53	1	$-8x^6 + 24x^5 - 18x^4 + 8x^3 - 2x^2 + 3x + 1$	+	+
54	1	$-8x^6 + 12x^5 + 12x^4 - 11x^3 - 5x^2 + 5x + 3$	+	+
55	1	$-8x^6 + 12x^5 - 12x^4 + 13x^3 - 11x^2 + 5x - 1$	+	+
56	1	$-8x^6 - 12x^5 + 12x^4 + 11x^3 - 5x^2 - 3x + 3$	+	+
57	1	$-8x^6 - 12x^5 - 12x^4 - 13x^3 - 11x^2 - 3x - 1$	+	+
58	1	$-8x^6 - 24x^5 - 18x^4 - 8x^3 - 2x^2 - x + 1$	+	+
59	-1	$-8x^6 - 24x^5 - 42x^4 - 32x^3 - 20x^2 - 9x - 3$	+	+
60	-1	$-8x^6 - 12x^5 + 12x^4 + 23x^3 - 20x^2 - 14x + 9$	+	+
61	-1	$-8x^6 + 12x^5 + 12x^4 - 23x^3 - 20x^2 + 12x + 9$	+	+
62	-1	$-8x^6 + 24x^5 - 42x^4 + 32x^3 - 20x^2 + 7x - 3$	+	+
63	-1	$-8x^6 + 24x^5 - 18x^4 - 16x^3 + 4x^2 + 7x + 1$	+	+
64	-1	$-x^6 - 6x^5 - 12x^4 + 16x^3 + 11x^2 - 27x + 9$	+	+
65	-1	$-x^6 + 6x^5 - 24x^4 + 32x^3 - 37x^2 + 25x - 11$	+	+
66	-1	$-x^6 + 6x^5 - 12x^4 - 16x^3 + 11x^2 + 25x + 9$	+	+
67	1	$-x^6 + 6x^5 - 16x^3 + 11x^2 + 27x + 11$	+	+
68	1	$-x^6 + 6x^5 - 12x^4 + 32x^3 - 37x^2 + 27x - 9$	+	+
69	1	$-x^6 - 6x^5 + 16x^3 + 11x^2 - 25x + 11$	+	+
70	1	$-x^6 - 6x^5 - 12x^4 - 32x^3 - 37x^2 - 25x - 9$	+	+
71	1	$-8x^6 + 24x^5 - 30x^4 + 32x^3 - 20x^2 + 9x - 1$	+	+
72	1	$-8x^6 + 12x^5 + 24x^4 - 23x^3 - 20x^2 + 14x + 11$	+	+
73	1	$-8x^6 - 12x^5 + 24x^4 + 23x^3 - 20x^2 - 12x + 11$	+	+
74	1	$-8x^6 - 12x^5 - 24x^4 - 25x^3 - 32x^2 - 12x - 9$	+	+
75	1	$-8x^6 - 24x^5 - 30x^4 - 32x^3 - 20x^2 - 7x - 1$	+	+
76	-1	$-8x^6 - 24x^5 - 6x^4 + 40x^3 - 2x^2 - 27x + 9$	+	+

i	k_i	$\varphi_i(x)$	$x \geq 10$	$x \leq -11$
77	-1	$-8x^6 + 24x^5 - 54x^4 + 56x^3 - 50x^2 + 25x - 11$	+	+
78	-1	$-8x^6 + 24x^5 - 6x^4 - 40x^3 - 2x^2 + 25x + 9$	+	+
79	1	$-8x^6 + 24x^5 + 6x^4 - 40x^3 - 2x^2 + 27x + 11$	+	+
80	1	$-8x^6 + 24x^5 - 42x^4 + 56x^3 - 50x^2 + 27x - 9$	+	+
81	1	$-8x^6 - 24x^5 + 6x^4 + 40x^3 - 2x^2 - 25x + 11$	+	+
82	1	$-8x^6 - 24x^5 - 42x^4 - 56x^3 - 50x^2 - 25x - 9$	+	+
83	-2	$-x^6 - 3x^5 - 51x^4 - x^3 - 4x^2 - 12x - 8$	+	+
84	2	$-x^6 + 51x^4 - 7x^2 + 8x + 13$	+	+
85	2	$-x^6 + 45x^4 - 7x^2 + 8x + 3$	+	+
86	2	$-x^6 - 3x^5 + 45x^4 - x^3 - 4x^2 + 4x + 8$	+	+
87	2	$-x^6 + 3x^5 + 48x^4 - 5x^3 - 4x^2 + 15x + 13$	+	+
88	2	$-x^6 + 3x^5 + 42x^4 + 7x^3 - 10x^2 + 15x + 3$	+	+
89	2	$-x^6 + 54x^4 - 16x^2 + 8x + 24$	+	+
90	2	$-x^6 + 42x^4 - 16x^2 + 8x - 8$	+	+
91	2	$-x^6 - 3x^5 + 42x^4 - 7x^3 - 10x^2 + x + 3$	+	+
92	-2	$-x^6 - 6x^5 - 63x^4 - 20x^3 - 19x^2 - 22x - 13$	+	+
93	-2	$-x^6 - 6x^5 - 57x^4 + 4x^3 + 5x^2 - 22x - 3$	+	+
94	-2	$-x^6 - 3x^5 - 57x^4 - 13x^3 - 22x^2 - 24x - 24$	+	+
95	-2	$-x^6 - 3x^5 - 45x^4 + 11x^3 - 10x^2 - 24x + 8$	+	+
96	-2	$-x^6 + 3x^5 - 57x^4 + 13x^3 - 22x^2 + 8x - 24$	+	+
97	-2	$-x^6 + 3x^5 - 45x^4 - 11x^3 - 10x^2 + 8x + 8$	+	+
98	-2	$-x^6 + 6x^5 - 63x^4 + 20x^3 - 19x^2 + 6x - 13$	+	+
99	2	$-x^6 + 6x^5 + 39x^4 - 4x^3 + 5x^2 + 22x + 13$	+	+
100	2	$-x^6 + 6x^5 + 33x^4 + 20x^3 - 19x^2 + 22x + 3$	+	+
101	2	$-x^6 + 3x^5 + 51x^4 - 11x^3 - 10x^2 + 24x + 24$	+	+
102	2	$-x^6 + 3x^5 + 39x^4 + 13x^3 - 22x^2 + 24x - 8$	+	+
103	2	$-x^6 - 3x^5 + 51x^4 + 11x^3 - 10x^2 - 8x + 24$	+	+
104	2	$-x^6 - 3x^5 + 39x^4 - 13x^3 - 22x^2 - 8x - 8$	+	+
105	2	$-x^6 - 6x^5 + 39x^4 + 4x^3 + 5x^2 - 6x + 13$	+	+
106	2	$-x^6 - 6x^5 + 33x^4 - 20x^3 - 19x^2 - 6x + 3$	+	+
107	-2	$-x^6 + 6x^5 - 54x^4 - 16x^3 + 8x^2 + 24x + 8$	+	+
108	2	$-x^6 - 6x^5 + 42x^4 + 16x^3 + 8x^2 - 24x + 24$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p .

p	k	B_p	classes (a, b)
2	6	-1	(13, 12), (27, 11), (27, 27), (27, 43), (27, 59), (29, 60), (45, 44), (49, 8), (55, 5), (55, 37), (61, 28)
2	7	-1	(1, 120), (3, 30), (7, 74), (17, 104), (19, 78), (23, 13), (23, 25), (23, 45), (23, 57), (23, 77), (23, 89), (23, 109), (23, 121), (23, 122), (27, 15), (27, 47), (27, 79), (27, 102), (27, 111), (31, 18), (33, 88), (35, 126), (39, 42), (51, 46), (55, 90), (55, 97), (59, 23), (59, 35), (59, 55), (59, 70), (59, 87), (59, 99), (59, 119), (63, 114), (65, 56), (67, 94), (71, 10), (81, 40), (83, 14), (87, 9), (87, 29), (87, 41), (87, 58), (87, 61), (87, 73), (87, 93), (87, 105), (87, 125), (91, 31), (91, 38), (91, 63), (91, 95), (91, 127), (95, 82), (97, 24), (99, 62), (103, 106), (115, 110), (119, 26), (119, 81), (123, 6), (123, 7), (123, 19), (123, 39), (123, 51), (123, 71), (123, 83), (123, 103), (123, 115), (127, 50)
2	8	-1	(5, 20), (9, 48), (11, 182), (21, 132), (41, 16), (53, 100), (55, 1), (55, 65), (55, 129), (55, 193), (73, 240), (75, 118), (85, 68), (105, 208), (117, 36), (119, 49), (119, 113), (119, 177), (119, 241), (137, 176), (139, 54), (149, 4), (169, 144), (175, 194), (181, 228), (183, 33), (183, 161), (201, 112), (203, 246), (213, 196), (217, 224), (233, 80), (245, 164), (247, 17), (247, 21), (247, 85), (247, 145), (247, 149), (247, 213), (249, 192)

p	k	B_p	classes (a, b)
2	9	-1	(25, 160), (37, 244), (43, 22), (47, 322), (57, 384), (69, 468), (89, 96), (107, 214), (111, 2), (119, 53), (119, 181), (119, 309), (119, 437), (121, 320), (133, 404), (153, 32), (165, 116), (171, 406), (185, 256), (197, 340), (235, 86), (239, 386), (281, 416), (293, 500), (299, 278), (303, 66), (313, 128), (315, 3), (315, 131), (315, 259), (315, 387), (325, 212), (345, 352), (363, 470), (367, 258), (375, 117), (375, 245), (375, 373), (375, 501), (377, 64), (389, 148), (409, 288), (421, 372), (427, 150), (441, 0), (453, 84), (491, 342), (495, 130)
2	10	-1	(15, 738), (59, 67), (59, 323), (59, 579), (59, 835), (79, 930), (101, 692), (207, 290), (229, 564), (335, 674), (357, 436), (463, 34), (485, 308), (571, 195), (571, 451), (571, 707), (571, 963), (591, 418), (613, 180), (719, 802), (741, 52), (847, 162), (869, 948), (911, 354), (975, 546), (997, 820)
2	11	-1	(143, 1122), (271, 1506), (399, 1890), (527, 226), (655, 610), (783, 994), (1167, 98), (1295, 482), (1423, 866), (1551, 1250), (1679, 1634), (1807, 2018)
3	3	-1	(0, 7), (1, 11), (7, 14), (8, 12), (9, 25), (13, 17), (26, 21)
3	4	-1	(2, 63), (4, 80), (5, 78), (6, 19), (10, 2), (14, 42), (15, 37), (16, 59), (19, 47), (20, 72), (21, 22), (22, 8), (23, 6), (24, 55), (29, 36), (31, 53), (32, 51), (38, 0), (41, 15), (42, 10), (43, 32), (44, 30), (46, 20), (48, 76), (49, 62), (51, 28), (56, 9), (57, 13), (58, 26), (64, 29), (65, 54), (68, 69), (69, 64), (70, 5), (71, 3), (72, 43), (73, 74), (75, 49), (79, 50)
3	5	-1	(3, 67), (11, 108), (12, 166), (17, 138), (18, 16), (30, 121), (33, 235), (37, 137), (39, 220), (45, 70), (47, 207), (50, 222), (52, 77), (59, 186), (60, 46), (66, 31), (74, 99), (76, 197), (77, 114), (78, 163), (84, 229), (92, 27), (93, 85), (98, 57), (99, 178), (106, 104), (111, 40), (114, 154), (118, 56), (120, 139), (126, 232), (131, 141), (133, 239), (140, 105), (141, 208), (147, 193), (155, 18), (157, 116), (158, 33), (159, 82), (165, 148), (173, 189), (174, 4), (179, 219), (180, 97), (187, 23), (192, 202), (195, 73), (199, 218), (201, 58), (209, 45), (212, 60), (214, 158), (222, 127), (228, 112), (236, 180), (238, 35), (239, 195)
3	6	-1	(25, 428), (128, 369), (207, 637), (240, 1), (268, 185), (371, 126), (450, 394), (464, 267), (483, 487), (511, 671), (614, 612), (693, 151), (707, 24), (726, 244)
3	7	-1	(221, 1239), (1679, 1968)
3	8	-1	(950, 510), (3137, 4884), (5324, 2697)
5	2	-1	(8, 23), (11, 11), (13, 8)
5	3	-1	(1, 111), (3, 113), (6, 11), (8, 40), (8, 112), (13, 47), (13, 120), (16, 86), (18, 0), (18, 57), (18, 68), (21, 11), (23, 17), (23, 53), (23, 55), (26, 111), (28, 35), (28, 38), (28, 52), (31, 11), (33, 65), (38, 20), (38, 97), (41, 86), (43, 25), (43, 107), (43, 118), (46, 11), (48, 67), (48, 80), (48, 103), (51, 111), (53, 88), (53, 102), (56, 11), (58, 90), (63, 22), (63, 45), (66, 86), (68, 32), (68, 43), (68, 50), (73, 28), (73, 105), (73, 117), (76, 111), (78, 13), (78, 27), (78, 85), (81, 11), (83, 115), (88, 70), (88, 72), (91, 86), (93, 75), (93, 82), (93, 93), (96, 11), (98, 5), (98, 42), (98, 78), (101, 111), (103, 63), (103, 77), (106, 11), (108, 15), (113, 95), (113, 122), (116, 86), (118, 7), (118, 18), (118, 100), (121, 11), (123, 30), (123, 92)
5	4	-1	(3, 252), (3, 385), (33, 37), (53, 435), (58, 462), (71, 261), (83, 262), (103, 485), (108, 62), (123, 3), (128, 502), (128, 510), (158, 287), (178, 560), (183, 87), (196, 261), (208, 512), (228, 610), (233, 312), (248, 253), (253, 10), (253, 127), (283, 537), (303, 60), (308, 337), (321, 261), (333, 137), (353, 110), (358, 562), (373, 503), (378, 135), (378, 377), (408, 162), (428, 185), (433, 587), (446, 261), (458, 387), (478, 235), (483, 187), (498, 128), (503, 2), (503, 260), (533, 412), (553, 310), (558, 212), (583, 12), (603, 360), (608, 437), (623, 378)
5	5	-1	(1196, 1511), (1821, 1511), (2446, 1511), (3071, 1511)

p	k	B_p	classes (a, b)
5	6	-1	(571, 14011), (3696, 14011), (9946, 14011), (13071, 14011)
5	7	-1	(6821, 76511), (22446, 76511), (38071, 76511), (53696, 76511), (69321, 76511)
7	2	1	(3, 0), (10, 14), (17, 28), (24, 42), (31, 7), (38, 21), (45, 35)
7	3	1	(5, 191), (12, 226), (19, 261), (33, 331), (40, 23), (47, 58), (54, 93), (61, 128), (68, 163), (82, 233), (89, 268), (96, 303), (103, 338), (110, 30), (117, 65), (131, 135), (138, 170), (145, 205), (152, 240), (159, 275), (166, 310), (180, 37), (187, 72), (194, 107), (201, 142), (208, 177), (215, 212), (229, 282), (236, 317), (243, 9), (250, 44), (257, 79), (264, 114), (278, 184), (285, 219), (292, 254), (299, 289), (306, 324), (313, 16), (327, 86), (334, 121), (341, 156)
7	4	1	(26, 1668), (75, 1913), (124, 2158), (173, 2), (222, 247), (271, 492), (320, 737), (369, 982), (418, 1227), (467, 1472), (516, 1717), (565, 1962), (614, 2207), (663, 51), (712, 296), (761, 541), (810, 786), (859, 1031), (908, 1276), (957, 1521), (1006, 1766), (1055, 2011), (1104, 2256), (1153, 100), (1202, 345), (1251, 590), (1300, 835), (1349, 1080), (1398, 1325), (1447, 1570), (1496, 1815), (1545, 2060), (1594, 2305), (1643, 149), (1692, 394), (1741, 639), (1790, 884), (1839, 1129), (1888, 1374), (1937, 1619), (1986, 1864), (2035, 2109), (2084, 2354), (2133, 198), (2182, 443), (2231, 688), (2280, 933), (2329, 1178), (2378, 1423)
11	1	1	(1, 8), (8, 0)
11	2	1	(0, 57), (2, 22), (11, 112), (13, 77), (22, 46), (24, 11), (33, 101), (35, 66), (46, 0), (55, 90), (57, 55), (59, 35), (66, 24), (68, 110), (77, 79), (79, 44), (81, 68), (88, 13), (90, 99), (92, 24), (99, 68), (101, 33), (110, 2), (112, 88), (114, 57)
11	3	1	(4, 134), (15, 90), (26, 1256), (37, 970), (44, 519), (48, 563), (70, 717), (103, 706), (125, 981), (136, 937), (147, 772), (158, 486), (165, 1124), (169, 79), (191, 233), (224, 222), (246, 497), (257, 453), (268, 288), (279, 2), (286, 398), (290, 926), (312, 1080), (345, 1069), (367, 13), (378, 1300), (389, 1135), (400, 849), (407, 1003), (411, 442), (433, 596), (466, 585), (488, 860), (499, 816), (510, 651), (521, 365), (528, 277), (532, 1289), (554, 112), (587, 101), (609, 376), (620, 332), (631, 167), (642, 1212), (649, 882), (653, 805), (675, 959), (708, 948), (730, 1223), (741, 1179), (752, 1014), (763, 728), (770, 156), (774, 321), (796, 475), (829, 464), (851, 739), (862, 695), (873, 530), (884, 244), (891, 761), (895, 1168), (917, 1322), (950, 1311), (972, 255), (983, 211), (994, 46), (1005, 1091), (1012, 35), (1016, 684), (1038, 838), (1071, 827), (1093, 1102), (1104, 1058), (1115, 893), (1126, 607), (1133, 640), (1137, 200), (1159, 354), (1192, 343), (1214, 618), (1225, 574), (1236, 409), (1247, 123), (1254, 1245), (1258, 1047), (1280, 1201), (1313, 1190)
13	1	1	(3, 3), (6, 3), (7, 11), (8, 12), (10, 2)
13	2	1	(0, 72), (5, 123), (7, 43), (9, 97), (10, 35), (11, 23), (13, 137), (18, 162), (20, 4), (22, 32), (23, 48), (24, 166), (26, 33), (31, 32), (33, 134), (35, 136), (36, 61), (37, 140), (39, 98), (44, 71), (46, 95), (48, 71), (49, 74), (50, 114), (52, 163), (57, 110), (59, 56), (61, 6), (63, 88), (65, 59), (70, 149), (72, 17), (74, 110), (75, 100), (76, 62), (78, 124), (83, 19), (85, 147), (87, 45), (88, 113), (89, 36), (91, 20), (96, 58), (98, 108), (100, 149), (101, 126), (102, 10), (104, 85), (109, 97), (111, 69), (113, 84), (114, 139), (115, 153), (117, 150), (122, 136), (124, 30), (126, 19), (127, 152), (128, 127), (130, 46), (135, 6), (137, 160), (139, 123), (140, 165), (141, 101), (143, 111), (148, 45), (150, 121), (152, 58), (153, 9), (154, 75), (156, 7), (161, 84), (163, 82), (166, 22), (167, 49)

p	k	B_p	classes (a, b)
13	3	1	(62, 763), (165, 669), (231, 932), (334, 2021), (400, 1101), (503, 1176), (569, 1270), (672, 331), (738, 1439), (841, 1683), (907, 1608), (1010, 838), (1076, 1777), (1179, 2190), (1245, 1946), (1348, 1345), (1414, 2115), (1517, 500), (1583, 87), (1686, 1852), (1752, 256), (1855, 1007), (1921, 425), (2024, 162), (2090, 594), (2193, 1514)
17	1	1	(0, 11), (4, 12), (5, 6), (9, 3), (10, 11), (16, 5)
17	2	1	(7, 154), (11, 288), (12, 5), (15, 26), (15, 44), (15, 219), (24, 205), (28, 135), (32, 146), (32, 202), (32, 230), (41, 256), (45, 271), (46, 107), (49, 145), (49, 185), (49, 248), (58, 18), (62, 118), (63, 158), (66, 60), (66, 61), (66, 168), (75, 69), (79, 254), (80, 209), (83, 151), (83, 163), (83, 264), (92, 120), (96, 101), (97, 260), (100, 134), (100, 179), (100, 265), (109, 171), (113, 237), (114, 22), (117, 78), (117, 94), (117, 117), (126, 222), (130, 84), (131, 73), (134, 9), (134, 100), (134, 180), (143, 273), (147, 220), (148, 124), (151, 83), (151, 213), (151, 282), (160, 35), (164, 67), (165, 175), (168, 66), (168, 95), (168, 128), (177, 86), (181, 203), (182, 226), (185, 43), (185, 49), (185, 197), (194, 137), (198, 50), (199, 277), (202, 10), (202, 32), (202, 247), (211, 188), (215, 186), (216, 39), (219, 15), (219, 112), (219, 162), (228, 239), (232, 33), (233, 90), (236, 77), (236, 214), (236, 287), (245, 1), (249, 169), (250, 141), (253, 27), (253, 270), (253, 281), (262, 52), (266, 16), (267, 192), (270, 129), (270, 196), (270, 253), (279, 103), (283, 152), (284, 243), (287, 111), (287, 231), (287, 236)
17	3	1	(29, 345), (318, 1212), (607, 2079), (896, 2946), (1185, 3813), (1474, 4680), (1763, 634), (2052, 1501), (2341, 2368), (2630, 3235), (2919, 4102), (3208, 56), (3497, 923), (3786, 1790), (4075, 2657), (4364, 3524), (4653, 4391)